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QUANTUM COMPUTING

From Linear Algebra to Physical Realizations

Mikio Nakahara and Tetsuo Ohmi

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Dedication

To our families

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Preface

One of the authors (MN) had an opportunity to give a series of lectures on quantum computing at Materials Physics Laboratory, Helsinki University of Technology, Finland during the 2001-2002 Winter term. The audience included advanced undergraduate students, postgraduate students and researchers in physics, mathematics, information science, computer science and electrical engineering among others. The host scientist, Professor Martti M. Salomaa, suggested that the lectures, mostly devoted to theoretical aspects of quantum computing, be published with additional chapters on physical realization. In fact Martti himself was willing to contribute to the physical realization part, but his unexpected early death made it impossible. After Martti passed away, MN asked his longstanding collaborator, TO, to coauthor the book. This is how this book was created. Part I, the theory part, was written by MN, while Part II, the physical realization part, was written jointly by MN and TO. Both authors have reviewed the final manuscript carefully and are equally responsible for the whole content.

Quantum computing and quantum information processing are emerging disciplines in which the principles of quantum physics are employed to store and process information. We use classical digital technology at almost every moment in our lives: computers, mobile phones, mp3 players, just to name a few. Even though quantum mechanics is used in the design of devices such as LSI, the logic is purely classical. This means that an AND circuit, for example, produces *definitely* 1 when the inputs are 1 and 1. One of the most remarkable aspects of the principles of quantum physics is the *superposition principle* by which a quantum system can take several different states *simultaneously*. The input for a quantum computing device may be a superposition of many possible inputs, and accordingly the output is also a superposition of the corresponding input states. Another aspect of quantum physics, which is far beyond the classical description, is *entanglement*. Given several objects in a classical world, they can be described by specifying each object separately. Given a group of five people, for example, this group can be described by specifying the height, color of eyes, personality and so on of each constituent person. In a quantum world, however, only a very small subset of all possible states can be described by such individual specifications. In other words, most quantum states cannot be described by such individual specifications, thereby being called "entangled." Why and how these two features give rise to the enormous computational power in quantum computing will be explained in this book.

Part I is devoted to theoretical aspects of quantum computing, starting with Chapter 1 in which a brief summary of linear algebra is given. Some subjects in this chapter, such as spectral decomposition, singular value decomposition and tensor product, may not be taught in a standard physics curriculum. The principles of quantum mechanics are outlined in Chapter 2. Some examples introduced in this chapter are important for understanding some parts in Part II. Qubit, the quantum counterpart of bit in classical information processing, is introduced in Chapter 3. Here we illustrate the first application of quantum information processing, namely quantum key distribution. By making use of the theory of measurement, a cryptosystem that is 100% secure can be realized. Quantum gates, the important parameters for quantum computing and quantum information processing, are introduced in Chapter 4, where the universality theorem is proved. Quantum gates are quantum counterparts of the elementary logic gates such as AND, NOT, OR, NAND, XOR and NOR in a classical logic circuit. In fact, it will be shown that all these classical gates can be reproduced with the quantum gates as special cases. A few simple but elucidating examples of quantum algorithms are introduced in Chapter 5. They employ the principle of quantum physics to achieve outstanding efficiency compared to their classical counterparts. Chapter 6 is devoted to the explanation of quantum circuits that implement integral transforms, which play central roles in several practical quantum algorithms, such as Grover's database search algorithm (Chapter 7) and Shor's factorization algorithm (Chapter 8). Chapter 9 describes a disturbing issue of decoherence, which is one of the obstacles against the physical realization of a working quantum computer. A quantum system gradually loses its coherence through interactions with its environment, a phenomenon known as decoherence. Quantum error correcting codes (QECC) introduced in Chapter 10 are designed to overcome certain kinds of decoherence. We will illustrate QECC with several important examples.

Part II starts with Chapter 11, where the DiVincenzo criteria, the criteria that any physical system has to satisfy to be a candidate for a working quantum computer, are introduced. The subsequent chapters introduce physical systems wherein the DiVincenzo criteria are evaluated for respective realizations. Liquid state NMR, the subject of Chapter 12, is introduced first since it is one of the well-understood systems. The subject of liquid state NMR has a long history, and numerous theoretical techniques have been developed for understanding the system. The liquid state NMR system has, however, several drawbacks and cannot be the ultimate candidate for a scalable quantum computer — at least not in its current form. The molecular Hamiltonian for the liquid state NMR system is determined very precisely, and the agreement between the theory and experiments is remarkable. Chapters 13 and 14 are devoted to ionic and atomic qubits, respectively. The ion trap quantum computer is one of the most promising systems: the largest quantum register with 8 qubits has been reported. Atomic qubits trapped in an optical lattice are expected to have very small decoherence due to their charge neutrality.

Chapters 15 and 16 describe solid state realization of a quantum computer. Chapter 15 introduces several types of Josephson junction qubits. The interaction among them is analyzed in detail. Chapter 16 describes quantum dots realization of qubits. There are two types: charge qubits and spin qubits, and they are treated separately.

Suggestion to readers and instructors: The whole book may be used for a one-year course on quantum computing. Using Part I for a semester and Part II for the subsequent semester is ideal. Alternatively, Part I may be used for a single semester course for physics, mathematics or information science graduate students. It may not be a good idea to use only Part II for lectures. Instead, Chapters 1 through 4 followed by Part II may be reasonable course materials for physics graduate students. An instructor may choose chapters in Part II depending on his/her preference. Chapters in Part II are only loosely related with each other.

MN used various parts of the book for lectures at several universities including Kinki University, Helsinki University of Technology, Shizuoka University, Kyoto University, Osaka City University, Ehime University, Kobe University and Kumamoto University. He would like to thank Yoshimasa Nakano, Martti M Salomaa, Mikko Paalanen, Jukka Pekola, Akihiko Matsuyama, Takao Mizusaki, Tohru Hata, Katsuhiro Nakamura, Ayumu Sugita, Taro Kashiwa, Yukio Fukuda, Toshiro Kohmoto and Masaharu Mitsunaga for giving him opportunities to improve the manuscript and also for the warm hospitality extended to him.

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Part I

From Linear Algebra to Quantum Computing

1

Basics of Vectors and Matrices

The set of natural numbers $\{1, 2, 3, \ldots\}$ is denoted by $\mathbb{N}$. The set of integers $\{\ldots, -2, -1, 0, 1, 2, \ldots\}$ is denoted by $\mathbb{Z}$. $\mathbb{Q}$ denotes the set of rational numbers. Finally $\mathbb{R}$ and $\mathbb{C}$ denote the sets of real numbers and complex numbers, respectively. Observe that

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

The vector spaces encountered in physics are mostly real vector spaces and complex vector spaces. Classical mechanics and electrodynamics are formulated mainly in real vector spaces while quantum mechanics (and hence this book) is founded on complex vector spaces. In the rest of this chapter, we briefly summarize vector spaces and matrices (linear maps), taking applications to quantum mechanics into account.

The **Pauli matrices**, also known as the spin matrices, are defined by

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They are also referred to as σ_1, σ_2 and σ_3, respectively.

The symbol I_n denotes the **unit matrix** of order n with ones on the diagonal and zeros off the diagonal. The subscript n will be dropped when the dimension is clear from the context. The arrow $\rightarrow$ often indicates logical implication. We use e^x and $\exp(x)$ interchangeably to denote the exponential function.

For any two matrices A and B of the same dimension, their commutator, or commutation relation, is a matrix defined as

$$[A, B] \equiv AB - BA,$$

while the anticommutator, or anticommutation relation, is

$$\{A, B\} \equiv AB + BA.$$

The symbol $\blacksquare$ denotes the end of a proof.

1.1 Vector Spaces

Let K be a field, which is a set where ordinary addition, substraction, multiplication and division are well-defined. The sets $\mathbb{R}$ and $\mathbb{C}$ are the only fields which we will be concerned with in this book. A **vector space** is a set where the addition of two vectors and a multiplication by an element of K, so-called a **scalar**, are defined.

DEFINITION 1.1 A vector space V is a set with the following properties;

(0-1) For any $u, v \in V$, their sum $u + v \in V$.

(0-2) For any $u \in V$ and $c \in K$, their scalar multiple $cu \in V$.

(1-1) $(u + v) + w = u + (v + w)$ for any $u, v, w \in V$.

(1-2) $u + v = v + u$ for any $u, v \in V$.

(1-3) There exists an element $0 \in V$ such that $u + 0 = u$ for any $u \in V$. This element 0 is called the **zero-vector**.

(1-4) For any element $u \in V$, there exists an element $v \in V$ such that $u+v = 0$. The vector v is called the **inverse** of u and denoted by $-u$.

(2-1) $c(x + y) = cx + cy$ for any $c \in K, u, v \in V$.

(2-2) $(c + d)u = cu + du$ for any $c, d \in K, u \in V$.

(2-3) $(cd)u = c(du)$ for any $c, d \in K, u \in V$.

(2-4) Let 1 be the unit element of K. Then $1u = u$ for any $u \in V$.

It is assumed that the reader is familiar with the above properties. We will be concerned mostly with the **complex vector space** $\mathbb{C}^n$ in the following. There are, however, occasional instances where the **real vector space** $\mathbb{R}^n$ is considered.

An element of $V = \mathbb{C}^n$ will be denoted by $|x\rangle$, instead of u, and expressed as a column of n complex numbers x_i $(1 \leq i \leq n)$ as

$$|x\rangle = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad x_i \in \mathbb{C} \tag{1.1}$$

It is often written as a **transpose** of a row vector, as $|x\rangle = (x_1, x_2, \ldots, x_n)^t$, to save space. The integer $n \in \mathbb{N}$ is called the **dimension** of the vector space. In some literature, $\mathbb{C}^n$ is denoted by $V(n, \mathbb{C})$. Similary we define the real vector space $\mathbb{R}^n = V(n, \mathbb{R})$ as the set of column vectors with real entries. An element $|x\rangle$ is also called a **ket vector** or simply a **ket**. We will later introduce another kind of vector called a bra vector, which, combined with a ket vector, yields the bracket (see Eq. (1.6)). For $|x\rangle, |y\rangle \in \mathbb{C}^n$ and $a \in \mathbb{C}$, vector addition and scalar multiplication are defined as

$$|x\rangle = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \ |y\rangle = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \Rightarrow |x\rangle + |y\rangle = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix}, \ a|x\rangle = \begin{pmatrix} ax_1 \\ ax_2 \\ \vdots \\ ax_n \end{pmatrix}, \tag{1.2}$$

respectively. All the components of the **zero-vector** $|0\rangle$ are zero. The zero-vector is also written as 0 in a less strict manner. The reader should verify that these definitions satisfy all the axioms in the definition of a vector space. Note, in particular, that any **linear combination** $c_1|x\rangle + c_2|y\rangle$ of vectors $|x\rangle, |y\rangle \in \mathbb{C}^n$ with $c_1, c_2 \in \mathbb{C}$ is also an element of $\mathbb{C}^n$.

1.2 Linear Dependence and Independence of Vectors

Let us consider a set of k vectors $\{|x_1\rangle, \ldots, |x_k\rangle\}$ in $V = \mathbb{C}^n$. This set is said to be **linearly dependent** if the equation

$$\sum_{i=1}^{k} c_i|x_i\rangle = |0\rangle \tag{1.3}$$

has a solution $c_1, \ldots, c_k$, at least one of which is non-vanishing. In other words, vectors $\{|x_i\rangle\}$ are linearly dependent if one of the vectors is expressed as a linear combination of the other vectors. This definition implies that any set containing the zero-vector $|0\rangle$ is linearly dependent.

If, in contrast, the trivial solution $c_i = 0$ $(1 \le i \le k)$ is the only solution of Eq. (1.3), the set is said to be **linearly independent**.

EXERCISE 1.1 Find the condition under which two vectors

$$|v_1\rangle = \begin{pmatrix} x \\ y \\ 3 \end{pmatrix}, \ |v_2\rangle = \begin{pmatrix} 2 \\ x - y \\ 1 \end{pmatrix} \in \mathbb{R}^3$$

are linearly independent.

THEOREM 1.1 If a set of k vectors in $\mathbb{C}^n$ is linearly independent, then the number k satisifies $k \le n$. The set is always linearly dependent if $k > n$.

The proof is left as an exercise for the readers. Suppose there are n linearly independent vectors $\{|v_i\rangle\}$ in $\mathbb{C}^n$. Then any $|x\rangle \in \mathbb{C}^n$ can be expressed uniquely as a linear combination of these n vectors;

$$|x\rangle = \sum_{i=1}^{n} c_i |v_i\rangle, \quad c_i \in \mathbb{C}.$$

The set of n linearly independent vectors is called a **basis** of $\mathbb{C}^n$ and the vectors are called **basis vectors**. The vector space spanned by a basis $\{|v_i\rangle\}$ is often denoted as $\mathrm{Span}(\{|v_i\rangle\})$.

EXERCISE 1.2 Show that a set of vectors

$$|v_1\rangle = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad |v_2\rangle = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad |v_3\rangle = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$$

is a basis of $\mathbb{C}^3$.

1.3 Dual Vector Spaces

A function $f : \mathbb{C}^n \to \mathbb{C}$ ($f : |x\rangle \mapsto f(|x\rangle) \in \mathbb{C}$) satisfing the linearity condition

$$\begin{gathered} f(c_1|x\rangle + c_2|y\rangle) = c_1 f(|x\rangle) + c_2 f(|y\rangle), \\ \forall |x\rangle, |y\rangle \in \mathbb{C}^n, \forall c_1, c_2 \in \mathbb{C} \end{gathered} \tag{1.4}$$

is called a **linear function**. To express f in a component form, let us introduce a row vector $\langle\alpha|$,

$$\langle\alpha| = (\alpha_1, \ldots, \alpha_n), \quad \alpha_i \in \mathbb{C} \tag{1.5}$$

A row vector is called a **bra vector** or simply a **bra** in the following. Let us define the **inner product** of a bra vector $\langle\alpha|$ and a ket vector $|x\rangle$ by

$$\langle\alpha|x\rangle = \sum_{i=1}^{n} \alpha_i x_i \tag{1.6}$$

Note that this product is nothing but an ordinary matrix multiplication of a $1 \times n$ matrix and an $n \times 1$ matrix.

A bra vector with the above inner product induces a linear function $\langle\alpha|(|x\rangle) = \langle\alpha|x\rangle$. In fact,

$$\begin{aligned} \langle\alpha|(c_1|x\rangle + c_2|y\rangle) &= \sum_i \alpha_i(c_1 x_i + c_2 y_i) = c_1 \sum_i \alpha_i x_i + c_2 \sum_i \alpha_i y_i \\ &= c_1\langle\alpha|x\rangle + c_2\langle\alpha|y\rangle. \end{aligned}$$

Conversely, any linear function can be expressed as a linear function induced by a bra vector. The bra vector is explicitly constructed once a dual basis is introduced as we will see below.

The vector space of linear functions on a vector space V ($\mathbb{C}^n$ in the present case) is called the **dual vector space**, or simply the **dual space**, of V and denoted by V^*. The symbol $*$ here denotes the dual and should not be confused with complex conjugation. As mentioned above, we may identify the set of all bra vectors with

$$\mathbb{C}^{n*} = \{\langle\alpha| = (\alpha_1, \dots, \alpha_n)|\alpha_i \in \mathbb{C}\}\,. \tag{1.7}$$

The reader is encouranged to verify directly that $\mathbb{C}^{n*}$ indeed satisfies the axioms of a vector space.

An important linear function is a bra vector obtained from a ket vector. Given a vector $|x\rangle = (x_1, \dots, x_n)^t \in \mathbb{C}^n$, define a bra vector $\langle x|$ associated to $|x\rangle$ by

$$|x\rangle \mapsto \langle x| = (x_1^*, \dots, x_n^*) \in \mathbb{C}^{n*}, \tag{1.8}$$

Note that each component is complex-conjugated under this correspondence. When a norm of a vector $|x\rangle$ is defined by

$$\||x\rangle\| = \sqrt{\langle x|x\rangle}, \tag{1.9}$$

it takes a non-negative real value due to this convention. In fact, observe that

$$\sqrt{\langle x|x\rangle} = \left[\sum_{i=1}^{n} x_i^* x_i\right]^{1/2} = \left[\sum_{i=1}^{n} |x_i|^2\right]^{1/2} \geq 0.$$

Given vectors $|x\rangle, |y\rangle \in \mathbb{C}^n$, their inner product is given by

$$\langle x|y\rangle = \sum_{i=1}^{n} x_i^* y_i. \tag{1.10}$$

In the mathematical literature, complex conjugation is taken rather with respect to the y_i. In the present book, however, we stick to physicists' convention (1.10), which should not be confused with Eq. (1.6).

Note the following sesquilinearity:*

$$\langle x|c_1y_1 + c_2y_2\rangle = c_1\langle x|y_1\rangle + c_2\langle x|y_2\rangle \tag{1.11}$$
$$\langle c_1x_1 + c_2x_2|y\rangle = c_1^*\langle x_1|y\rangle + c_2^*\langle x_2|y\rangle, \tag{1.12}$$

where $|c_1y_1 + c_2y_2\rangle \equiv c_1|y_1\rangle + c_2|y_2\rangle$.

*sesqui = 1.5.

EXERCISE 1.3 Let

$$|x\rangle = \begin{pmatrix} 1 \\ i \\ 2+i \end{pmatrix}, \quad |y\rangle = \begin{pmatrix} 2-i \\ 1 \\ 2+i \end{pmatrix}.$$

Find $\| |x\rangle \|$, $\langle x|y\rangle$ and $\langle y|x\rangle$.

EXERCISE 1.4 Prove that

$$\langle x|y\rangle = \langle y|x\rangle^*. \tag{1.13}$$

1.4 Basis, Projection Operator and Completeness Relation

1.4.1 Orthonormal Basis and Completeness Relation

Any set of n linearly independent vectors $\{|v_1\rangle, \ldots, |v_n\rangle\}$ in $\mathbb{C}^n$ is called the **basis**, and an arbitrary vector $|x\rangle \in \mathbb{C}^n$ is expressed uniquely as a linear combination of these basis vectors as $|x\rangle = \sum_{i=1}^n c_i|v_i\rangle$. The n complex numbers c_i are called the **components** of $|x\rangle$ with respect to the basis $\{|v_i\rangle\}$.

A basis $\{|e_i\rangle\}$ that satisfies

$$\langle e_i|e_j\rangle = \delta_{ij} \tag{1.14}$$

is called an **orthonormal basis**. Clearly the choice of $\{|e_i\rangle\}$ which satisfies the above condition is far from unique. It turns out that orthonormal bases are convenient for many purposes.

Let $|x\rangle = \sum_{i=1}^n c_i|e_i\rangle$. The inner product of $|x\rangle$ and $\langle e_j|$ yields

$$\langle e_j|x\rangle = \sum_{i=1}^n c_i\langle e_j|e_i\rangle = \sum_{i=1}^n c_i\delta_{ji} = c_j \to c_j = \langle e_j|x\rangle.$$

Substituting this result into the expansion of $|x\rangle$, we obtain

$$|x\rangle = \sum_{i=1}^n \langle e_i|x\rangle|e_i\rangle = \sum_{i=1}^n |e_i\rangle\langle e_i|x\rangle = \left(\sum_{i=1}^n |e_i\rangle\langle e_i|\right)|x\rangle.$$

Since $|x\rangle$ is arbitrary, we finally obtain the **completeness relation**

$$\sum_{i=1}^n |e_i\rangle\langle e_i| = I. \tag{1.15}$$

The completeness relation is quite useful and will be frequently made use of in the following.

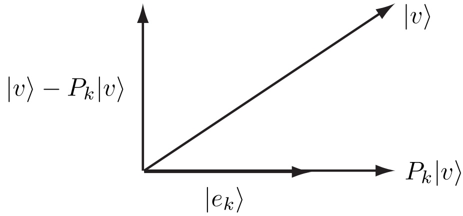

FIGURE 1.1
A vector $|v\rangle$ is projected to the direction defined by a unit vector $|e_k\rangle$ by the action of $P_k = |e_k\rangle\langle e_k|$. The difference $|v\rangle - P_k|v\rangle$ is orthogonal to $|e_k\rangle$.

1.4.2 Projection Operators

The *matrix*

$$P_k \equiv |e_k\rangle\langle e_k| \tag{1.16}$$

introduced above is called a **projection operator** in the direction defined by $|e_k\rangle$. This projects a vector $|v\rangle$ to a vector parallel to $|e_k\rangle$ in such a way that $|v\rangle - P_k|v\rangle$ is orthogonal to $|e_k\rangle$ (see Fig. 1.1).

The set $\{P_k = |e_k\rangle\langle e_k|\}$ satisfies the conditions

$$\text{(i)} \qquad P_k^2 = P_k, \tag{1.17}$$

$$\text{(ii)} \qquad P_k P_j = 0 \quad (k \neq j), \tag{1.18}$$

$$\text{(iii)} \qquad \sum_k P_k = I \quad \text{(completeness relation)}. \tag{1.19}$$

The conditions (i) and (ii) are obvious from the orthonormality $\langle e_j|e_k\rangle = \delta_{jk}$.

EXAMPLE 1.1 Let

$$|e_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, |e_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

They define an orthonormal basis as is easily verified. Projection operators are

$$P_1 = |e_1\rangle\langle e_1| = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \ P_2 = |e_2\rangle\langle e_2| = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

They satisfy the completeness relation

$$\sum_k P_k = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

and the orthogonality condition

$$P_1 P_2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

The reader should verify that $P_k^2 = P_k$.

EXERCISE 1.5 Let $\{|e_k\rangle\}$ be as in Example 1.1 and let

$$|v\rangle = \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \sum c_k |e_k\rangle.$$

Find the coefficients c_1 and c_2.

1.4.3 Gram-Schmidt Orthonormalization

Let us construct a set of k orthonormal vectors, given a linearly independent set of k vectors $\{|v_i\rangle\}$ in $\mathbb{C}^n$ $(k \le n)$. The first step is to define a vector

$$|e_1\rangle = \frac{|v_1\rangle}{\||v_1\rangle\|},$$

which is clearly normalized; $\||e_1\rangle\| = 1$. Before we proceed further, let us recall that the component of a vector $|u\rangle$ along $|e_k\rangle$ is given by $\langle e_k|u\rangle$. Then we define, in the next step, a vector

$$|f_2\rangle = |v_2\rangle - |e_1\rangle\langle e_1|v_2\rangle,$$

which is clearly orthogonal to $|e_1\rangle$; $\langle e_1|f_2\rangle = \langle e_1|v_2\rangle - \langle e_1|e_1\rangle\langle e_1|v_2\rangle = 0$. This vector must be normalized as

$$|e_2\rangle = \frac{|f_2\rangle}{\||f_2\rangle\|}.$$

Similarly we find, in the jth step, the vector

$$|e_j\rangle = \frac{|v_j\rangle - \sum_{i=1}^{j-1}\langle e_i|v_j\rangle|e_i\rangle}{\||v_j\rangle - \sum_{i=1}^{j-1}\langle e_i|v_j\rangle|e_i\rangle\|} \quad (1 \le j \le k).$$

By construction, $\{|e_1\rangle, |e_2\rangle, \ldots, |e_k\rangle\}$ is an orthonormal set, which spans a k-dimensional subspace in $\mathbb{C}^n$. This is called the **Gram-Schmidt orthonormalization**. When $k = n$, it spans the whole vector space $\mathbb{C}^n$.

EXAMPLE 1.2 Let

$$|v_1\rangle = \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad |v_2\rangle = \begin{pmatrix} 2i \\ 4 \end{pmatrix}.$$

Then we obtain

$$|e_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}.$$

Moreover

$$|f_2\rangle = \begin{pmatrix} 2i \\ 4 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 1 \\ i \end{pmatrix}(1, -i)\begin{pmatrix} 2i \\ 4 \end{pmatrix} = \begin{pmatrix} 3i \\ 3 \end{pmatrix},$$

from which we find

$$|e_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} i \\ 1 \end{pmatrix}.$$

EXERCISE 1.6 (1) Use the Gram-Schmidt orthonormalization to find an orthonormal basis $\{|e_k\rangle\}$ from a linearly independent set of vectors

$$|v_1\rangle = (-1, 2, 2)^t, \quad |v_2\rangle = (2, -1, 2)^t, \quad |v_3\rangle = (3, 0, -3)^t.$$

(2) Let

$$|u\rangle = (1, -2, 7)^t = \sum_k c_k |e_k\rangle.$$

Find the coefficients c_k.

EXERCISE 1.7 Let

$$|v_1\rangle = (1, i, 1)^t, \quad |v_2\rangle = (3, 1, i)^t.$$

Find an orthonormal basis for a two-dimensional subspace spanned by $\{|v_1\rangle, |v_2\rangle\}$.

1.5 Linear Operators and Matrices

A map $A : \mathbb{C}^n \to \mathbb{C}^n$ is a **linear operator** if

$$A(c_1|x\rangle + c_2|y\rangle) = c_1 A|x\rangle + c_2 A|y\rangle \tag{1.20}$$

is satified for arbitrary $|x\rangle, |y\rangle \in \mathbb{C}^n$ and $c_k \in \mathbb{C}$. Let us choose an arbitrary orthonormal basis $\{|e_k\rangle\}$. It is shown below that A is expressed as an $n \times n$ matrix.

Let $|v\rangle = \sum_{k=1}^n v_k|e_k\rangle$ be an arbitrary vector in $\mathbb{C}^n$. Linearity implies that $A|v\rangle = \sum_k v_k A|e_k\rangle$. Therefore, the action of A on an arbitrary vector is fixed provided that its action on the basis vectors is given. Since $A|e_k\rangle \in \mathbb{C}^n$, it can be expanded as

$$A|e_k\rangle = \sum_{i=1}^n |e_i\rangle A_{ik}.$$

By taking the inner product between $\langle e_j|$ and the above equation, we obtain

$$A_{jk} = \langle e_j|A|e_k\rangle. \tag{1.21}$$

This is the matrix element of A given an orthonormal basis $\{|e_k\rangle\}$.

It is easy then to show that

$$A = \sum_{j,k} A_{jk}|e_j\rangle\langle e_k| \tag{1.22}$$

since by multiplying the completeness relation $I = \sum_{i=1}^n |e_i\rangle\langle e_i|$ from the left and the right on A simultaneously, we obtain

$$A = IAI = \sum_{j,k} |e_j\rangle\langle e_j|A|e_k\rangle\langle e_k| = \sum_{j,k} A_{jk}|e_j\rangle\langle e_k|.$$

1.5.1 Hermitian Conjugate, Hermitian and Unitary Matrices

Hermitian matrices play important role in many areas in mathematics and physics. To define a Hermitian matrix, we need to introduce the **Hermitian conjugate** operation, denoted $\dagger$.[†]

DEFINITION 1.2 (**Hermitian conjugate**) Given a linear operator $A : \mathbb{C}^n \to \mathbb{C}^n$, its Hermitian conjugate $A^\dagger$ is defined by

$$\langle u|A|v\rangle \equiv \langle A^\dagger u|v\rangle = \langle v|A^\dagger|u\rangle^*, \tag{1.23}$$

where $|u\rangle, |v\rangle$ are arbitrary vectors in $\mathbb{C}^n$.

The above definition shows that $\langle e_j|A|e_k\rangle = \langle e_k|A^\dagger|e_j\rangle^*$. Therefore, we find the relation $A_{jk} = (A^\dagger)^*_{kj}$, namely

$$(A^\dagger)_{jk} = A^*_{kj}. \tag{1.24}$$

In other words, the matrix elements of $A^\dagger$ are obtained by the transpose and the complex conjugation of A.

This definition also applies to a ket vector $|x\rangle$. We have

$$|x\rangle^\dagger = (x_1^*, \ldots, x_n^*) = \langle x|.$$

Namely, the procedure to produce a bra vector from a ket vector is regarded as a Hermitian conjugation of the ket vector.

EXERCISE 1.8 Let A and B be $n \times n$ matrices and $c \in \mathbb{C}$. Show that

$$(cA)^\dagger = c^* A^\dagger, \quad (A+B)^\dagger = A^\dagger + B^\dagger, \quad (AB)^\dagger = B^\dagger A^\dagger. \tag{1.25}$$

DEFINITION 1.3 (**Hermitian matrix**) A matrix $A : \mathbb{C}^n \to \mathbb{C}^n$ is said to be a **Hermitian matrix** if it satisifies $A^\dagger = A$.

Let $\{|e_1\rangle, \ldots, |e_n\rangle\}$ be an orthonormal basis in $\mathbb{C}^n$. Suppose a matrix $U : \mathbb{C}^n \to \mathbb{C}^n$ satisifes $U^\dagger U = I$. By operating U on $\{|e_k\rangle\}$, we obtain a vector $|f_k\rangle = U|e_k\rangle$. These vectors are again orthonormal since

$$\langle f_j|f_k\rangle = \langle e_j|U^\dagger U|e_k\rangle = \langle e_j|e_k\rangle = \delta_{jk}. \tag{1.26}$$

Note that $|\det U| = 1$ since $\det U^\dagger U = \det U^\dagger \det U = |\det U|^2 = 1$.

[†]Mathematicians tend to use $*$ to denote Hermitian conjugate. We will follow the physicists' convention here.

DEFINITION 1.4 (**Unitary matrix**) Let $U : \mathbb{C}^n \to \mathbb{C}^n$ be a matrix which satisfies $U^\dagger = U^{-1}$. Then U is called a **unitary matrix**. Moreover, if U is unimodular, namely $\det U = 1$, U is said to be a **special unitary matrix**.

The set of unitary matrices is a group called the **unitary group**, while that of the special unitary matrices is a group called the **special unitary group**. They are denoted by $\mathrm{U}(n)$ and $\mathrm{SU}(n)$, respectively.

Remarks: If a real matrix $A : \mathbb{R}^n \to \mathbb{R}^n$ satisfies $A^t = A^{-1}$, A is called an orthogonal matrix. From $\det(AA^t) = \det A \det A^t = (\det A)^2 = \det I = 1$, we find that $\det A = \pm 1$. If A is unimodular, $\det A = 1$, it is called a special orthogonal matrix. The set of orthogonal (special orthogonal) matrices is a group called the **orthogonal group** (**special orthogonal group**) and denoted by $\mathrm{O}(n)$ ($\mathrm{SO}(n)$).

1.6 Eigenvalue Problems

Suppose we operate a matrix A on a vector $|v\rangle \in \mathbb{C}^n$, where $|v\rangle \neq |0\rangle$. The result $A|v\rangle$ is not proportional to $|v\rangle$ in general. If, however, $|v\rangle$ is properly chosen, we may end up with $A|v\rangle$, which is a scalar multiple of $|v\rangle$;

$$A|v\rangle = \lambda|v\rangle, \quad \lambda \in \mathbb{C}. \tag{1.27}$$

Then λ is called an **eigenvalue** of A, while $|v\rangle$ is called the corresponding **eigenvector**. The above equation being a linear equation, the norm of the eigenvector cannot be fixed. Of course, it is always possible to normalize $|v\rangle$ such that $\||v\rangle\| = 1$. We often use the symbol $|\lambda\rangle$ for an eigenvector corresponding to an eigenvalue λ to save symbols.

Let $\{|e_k\rangle\}$ be an orthonormal basis in $\mathbb{C}^n$ and let $\langle e_i|A|e_j\rangle = A_{ij}$ and $v_i = \langle e_i|v\rangle$ be the components of A and $|v\rangle$ with respect to the basis. Then the component expression for the above equation is obtained from

$$A|v\rangle = \sum_{i,j} |e_i\rangle\langle e_i|A|e_j\rangle\langle e_j|v\rangle = \sum_{i,j} A_{ij} v_j |e_i\rangle$$

as

$$\sum_j A_{ij} v_j = \lambda v_i. \tag{1.28}$$

Let us find the eigenvalue λ next. Note first that the eigenvalue equation is rewritten as

$$\sum_j (A - \lambda I)_{ij} v_j = 0.$$

This equation in v_j has nontrivial solutions if and only if the matrix $A - \lambda I$ has no inverse, namely

$$D(\lambda) \equiv \det(A - \lambda I) = 0. \tag{1.29}$$

If it had the inverse, then $|v\rangle = (A - \lambda I)^{-1}|0\rangle = 0$ would be the unique solution. This equation (1.29) is called the **characteristic equation** or the **eigen equation** of A.

Let A be an $n \times n$ matrix. Then the characteristic equation has n solutions, including the multiplicity, which we write as $\{\lambda_1, \lambda_2, \ldots, \lambda_n\}$. The function $D(\lambda)$ is also written as

$$\begin{aligned} D(\lambda) &= \prod_{i=1}^{n} (\lambda_i - \lambda) \\ &= (-\lambda)^n + \sum_i \lambda_i (-\lambda)^{(n-1)} + \ldots + \prod_{i=1}^{n} \lambda_i \\ &= (-\lambda)^n + \operatorname{tr} A (-\lambda)^{(n-1)} + \ldots + \det A, \end{aligned} \tag{1.30}$$

where use has been made of the facts $\operatorname{tr} A = \sum_i \lambda_i$ and $\det A = \prod_i \lambda_i$.

1.6.1 Eigenvalue Problems of Hermitian and Normal Matrices

The eigenvalue problems of Hermitian matrices and unitary matrices are particularly important in practical applications.

THEOREM 1.2 All the eigenvalues of a Hermitian matrix are real numbers. Moreover, two eigenvectors corresponding to different eigenvalues are orthogonal.

Proof. Let A be a Hermitian matrix and let $A|\lambda\rangle = \lambda|\lambda\rangle$. The Hermitian conjugate of this equation is $\langle\lambda|A = \lambda^*\langle\lambda|$. From these equations we obtain $\langle\lambda|A|\lambda\rangle = \lambda\langle\lambda|\lambda\rangle = \lambda^*\langle\lambda|\lambda\rangle$, which proves $\lambda = \lambda^*$ since $\langle\lambda|\lambda\rangle \neq 0$.

Let $A|\mu\rangle = \mu|\mu\rangle$ $(\mu \neq \lambda)$, next. Then $\langle\mu|A = \mu\langle\mu|$ since $\mu \in \mathbb{R}$. From $\langle\mu|A|\lambda\rangle = \lambda\langle\mu|\lambda\rangle$ and $\langle\mu|A|\lambda\rangle = \mu\langle\mu|\lambda\rangle$, we obtain $0 = (\lambda - \mu)\langle\mu|\lambda\rangle$. Since $\mu \neq \lambda$, we must have $\langle\mu|\lambda\rangle = 0$. ∎

Suppose λ is k-fold degenerate. Then there are k independent eigenvectors corresponding to λ. We may invoke to the Gram-Schmidt orthonormalization, for example, to obtain an orthonormal basis in this k-dimensional space. Accordingly, the set of eigenvectors of a Hermitian matrix is always chosen to be orthonormal. Therefore, the set of eigenvectors $\{|\lambda_k\rangle\}$ of a Hermitian matrix A may be made into a complete set

$$\sum_{k=1}^{n} |\lambda_k\rangle\langle\lambda_k| = I$$

EXAMPLE 1.3 The Pauli matrix

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

is Hermitian. Let us find its eigenvalues and corresponding eigenvectors. From

$$\det(\sigma_y - \lambda I) = \lambda^2 - 1 = 0,$$

we find the eigenvalues $\lambda_1 = 1$ and $\lambda_2 = -1$. Let $|\lambda_1\rangle = (x, y)^t$ be an eigenvector corresponding to λ_1;

$$\sigma_y|\lambda_1\rangle = |\lambda_1\rangle \;\to\; \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \;\to\; \begin{cases} -iy = x \\ ix = y \end{cases}.$$

These relations are satified with $x = 1, y = i$. Thus the normalized eigenvector is

$$|\lambda_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}.$$

Similarly we obtain

$$|\lambda_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} i \\ 1 \end{pmatrix}.$$

It is easy to verify that they are orthogonal

$$\langle\lambda_1|\lambda_2\rangle = \frac{1}{2}(1, -i)\begin{pmatrix} i \\ 1 \end{pmatrix} = 0$$

and satisfy the completeness relation

$$\sum_k |\lambda_k\rangle\langle\lambda_k| = \frac{1}{2}\begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} = I.$$

Finally let us find a unitary matrix U which diagonalizes σ_y as

$$U^\dagger \sigma_y U = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}.$$

Let us consider a matrix

$$U = (|\lambda_1\rangle, |\lambda_2\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix} i & 1 \\ 1 & i \end{pmatrix}.$$

Then

$$\sigma_y U = (\sigma_y|\lambda_1\rangle, \sigma_y|\lambda_2\rangle) = (\lambda_1|\lambda_1\rangle, \lambda_2|\lambda_2\rangle)$$

from which we find

$$U^\dagger \sigma_y U = \begin{pmatrix} \langle\lambda_1| \\ \langle\lambda_2| \end{pmatrix}(\lambda_1|\lambda_1\rangle, \lambda_2|\lambda_2\rangle) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

as promised. Note that the unitarity of U is attributed to the orthonormality of $\{|\lambda_k\rangle\}$.

EXAMPLE 1.4 (1) The eigenvalues and the corresponding eigenvectors of σ_x are found in a similar way as the above example as $\lambda_1 = 1, \lambda_2 = -1$ and

$$|\lambda_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |\lambda_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

(2) Let us consider the eigenvalue problem of a matrix

$$A = \begin{pmatrix} 1\,0\,0\,0 \\ 0\,1\,0\,0 \\ 0\,0\,0\,1 \\ 0\,0\,1\,0 \end{pmatrix}.$$

Note that this matrix is block diagonal with diagonal blocks I and σ_x. It is found from this observation that the eigenvalues are $1, 1, 1$ and -1. The corresponding eigenvectors are obtained by making use of the result of (1) as

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix}.$$

(3) Let us consider the eigenvalue problem of a matrix

$$B = \begin{pmatrix} 0\,0\,0\,1 \\ 0\,1\,0\,0 \\ 0\,0\,1\,0 \\ 1\,0\,0\,0 \end{pmatrix}.$$

Although this matrix is not block diagonal, change of the order of basis vectors from $|e_1\rangle, |e_2\rangle, |e_3\rangle, |e_4\rangle$ to $|e_3\rangle, |e_2\rangle, |e_1\rangle, |e_4\rangle$ maps the matrix B to A in (2). Therefore the eivenvalues of B are the same as those of A. (Note that the characteristic equation is left unchanged under a permutation of basis vectors.) By putting back the order of the basis vectors, the eigenvectors of A are mapped to those of B as

$$\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}.$$

EXERCISE 1.9 Let

$$A = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1+i \\ 1-i & 0 \end{pmatrix}.$$

Find the eigenvalues and the corresponding normalized eigenvectors. Show that the eigenvectors are mutually orthogonal and that they satisfy the completeness relation. Find a unitary matrix which diagonalizes A.

It has been shown above that eigenvalues of a Hermitian matrix are real. Note that converse is not true. For example,

$$A = \begin{pmatrix} a & b \\ -b & -a \end{pmatrix} \qquad a, b \in \mathbb{R}$$

has real eigenvalues $\pm\sqrt{a^2 - b^2}$ for $|a| \geq |b|$. How about the orthonormality of the eigenvectors?

A matrix A is **normal** if it satisfies

$$AA^\dagger = A^\dagger A. \tag{1.31}$$

THEOREM 1.3 Let A be a normal matrix. Then its eigenvectors corresponding to different eigenvalues are orthogonal.

Proof. Let us write the eigenvalue equation as $(A - \lambda_j)|\lambda_j\rangle = 0$. Then we find, from the assumed condition $[A, A^\dagger] = 0$, that

$$\langle\lambda_j|(A^\dagger - \lambda_j^*)(A - \lambda_j)|\lambda_j\rangle = \langle\lambda_j|(A - \lambda_j)(A^\dagger - \lambda_j^*)|\lambda_j\rangle = 0,$$

which implies $\langle\lambda_j|A = \lambda_j\langle\lambda_j|$. Then it follows that

$$\langle\lambda_k|A|\lambda_j\rangle = \lambda_k\langle\lambda_k|\lambda_j\rangle = \lambda_j\langle\lambda_k|\lambda_j\rangle,$$

which proves that $\langle\lambda_k|\lambda_j\rangle = 0$ for $\lambda_j \neq \lambda_k$. ∎

If some of the eigenvalues are degenerate, we may use the Gram-Schmidt procedure to make the corresponding eigenvectors orthonormal. Therefore it is always possible to assume the set of eigenvectors of a normal matrix satisfies the completeness relation.

Important examples of normal matrices are Hermitian matrices, unitary matrices and skew-Hermitian matrices; see the next exercise.

EXERCISE 1.10 (1) Suppose A is skew-Hermitian, namely $A^\dagger = -A$. Show that all the eigenvalues are pure imaginary.

(2) Let U be a unitary matrix. Show that all the eigenvalues are unimodular, namely $|\lambda_j| = 1$.

(3) Let A be a normal matrix. Show that A is Hermitian if and only if all the eigenvalues of A are real.

EXERCISE 1.11 Let

$$U = \begin{pmatrix} 0 & 0 & i \\ 0 & i & 0 \\ i & 0 & 0 \end{pmatrix}.$$

Find the eigenvalues (without calculation if possible) and the corresponding eigenvectors.

EXERCISE 1.12 Let H be a Hermitian matrix. Show that

$$U = (I + iH)(I - iH)^{-1}$$

is unitary. This transformation is called the **Cayley transformation**.

1.7 Pauli Matrices

Let us consider spin 1/2 particles, such as an electron or a proton. These particles have an internal degree of freedom: the spin-up and spin-down states. (To be more precise, these are expressions that are relevant when the z-component of an angular momentum S_z is diagonalized. If S_x is diagonalized, for example, these two quantum states can be either "spin-right" or "spin-left.") Since the spin-up and spin-down states are orthogonal, we can take their components to be

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{1.32}$$

Verify that they are eigenvectors of σ_z satisfying $\sigma_z|\uparrow\rangle = |\uparrow\rangle$ and $\sigma_z|\downarrow\rangle = -|\downarrow\rangle$. In quantum information, we often use the notations $|0\rangle = |\uparrow\rangle$ and $|1\rangle = |\downarrow\rangle$. Moreover, the states $|0\rangle$ and $|1\rangle$ are not necessarily associated with spins. They may represent any two mutually orthogonal states, such as horizontally and vertically polarized photons. Thus we are free from any physical system, even though the terminology of spin algebra may be employed.

For electrons and protons, the spin angular momentum operator is conveniently expressed in terms of the Pauli matrices σ_k as $S_k = (\hbar/2)\sigma_k$. We often employ natural units in which $\hbar = 1$. Note the tracelessness property $\operatorname{tr}\sigma_k = 0$ and the Hermiticity $\sigma_k^\dagger = \sigma_k$.[‡] In addition to the Pauli matrices, we introduce the unit matrix I in the algebra, which amounts to expanding the Lie algebra $\mathfrak{su}(2)$ to $\mathfrak{u}(2)$. The Pauli matrices satisfy the anticommutation relations

$$\{\sigma_i, \sigma_j\} = \sigma_i\sigma_j + \sigma_j\sigma_i = 2\delta_{ij} I. \tag{1.33}$$

Therefore, the eigenvalues of σ_k are found to be ± 1.

The commutation relations between the Pauli matrices are

$$[\sigma_i, \sigma_j] = \sigma_i\sigma_j - \sigma_j\sigma_i = 2i \sum_k \varepsilon_{ijk}\sigma_k, \tag{1.34}$$

[‡]Mathematically speaking, these two properties imply that $i\sigma_k$ are generators of the $\mathfrak{su}(2)$ Lie algebra associated with the Lie group SU(2).

where ε_{ijk} is the totally antisymmetric tensor of rank 3, also known as the **Levi-Civita symbol**,

$$\varepsilon_{ijk} = \begin{cases} 1, & (i,j,k) = (1,2,3),(2,3,1),(3,1,2) \\ -1 & (i,j,k) = (2,1,3),(1,3,2),(3,2,1) \\ 0 & \text{otherwise.} \end{cases}$$

The commutation relations, together with the anticommutation relations, yield

$$\sigma_i \sigma_j = i \sum_{k=1}^{3} \varepsilon_{ijk} \sigma_k + \delta_{ij} I \, . \tag{1.35}$$

The spin-flip ("ladder") operators are defined by

$$\sigma_+ = \frac{1}{2}(\sigma_x + i\sigma_y) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \sigma_- = \frac{1}{2}(\sigma_x - i\sigma_y) = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \tag{1.36}$$

Verify that $\sigma_+ | \uparrow\rangle = \sigma_- | \downarrow\rangle = 0$, $\sigma_+ | \downarrow\rangle = | \uparrow\rangle$, $\sigma_- | \uparrow\rangle = | \downarrow\rangle$. The projection operators to the eigenspaces of σ_z with the eigenvalues ± 1 are

$$\begin{aligned} P_+ &= | \uparrow\rangle\langle\uparrow | = \tfrac{1}{2}(I + \sigma_z) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \\ P_- &= | \downarrow\rangle\langle\downarrow | = \tfrac{1}{2}(I - \sigma_z) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned} \tag{1.37}$$

In fact, it is straightforward to show

$$P_+ | \uparrow\rangle = | \uparrow\rangle, \;\; P_+ | \downarrow\rangle = 0, \;\; P_- | \uparrow\rangle = 0, \;\; P_- | \downarrow\rangle = | \downarrow\rangle \;\; .$$

Finally, we note the following identities:

$$\sigma_\pm^2 = 0, \quad P_\pm^2 = P_\pm, \quad P_+ P_- = 0. \tag{1.38}$$

1.8 Spectral Decomposition

Spectral decomposition of a normal matrix is quite a powerful technique in several applications.

THEOREM 1.4 Let A be a normal matrix with eigenvalues $\{\lambda_i\}$ and eigenvectors $\{|\lambda_i\rangle\}$, which are assumed to be orthonormal. Then A is decomposed as

$$A = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i|,$$

which is called the **spectral decomposition** of A.

Proof. This is a straightforward consequence of the completeness relation

$$I = \sum_{i=1}^{n} |\lambda_i\rangle\langle\lambda_i|.$$

If we operate A on the above equation from the left, we obtain

$$A = AI = \sum_{i=1}^{n} A|\lambda_i\rangle\langle\lambda_i| = \sum_{i=1}^{n} \lambda_i|\lambda_i\rangle\langle\lambda_i|,$$

which proves the theorem. ∎

Let us recall that $P_i = |\lambda_i\rangle\langle\lambda_i|$ is a projection operator onto the direction of $|\lambda_i\rangle$. Then the spectral decomposition claims that the operation of A in the one-dimensional subspace spanned by $|\lambda_i\rangle$ is equivalent with a multiplication by a scalar λ_i. This observation reveals a neat way to obtain the spectral decomposition of a normal matrix. Let A be a normal matrix and let $\{\lambda_\alpha\}$ and $\{|\lambda_{\alpha,p}\rangle\ (1 \le p \le g_\alpha)\}$ be the sets of eigenvalues and eigenvectors, respectively. Here we use subscripts $\alpha, \beta, \ldots$ to denote distinct eigenvalues, while g_α denotes the degeneracy of the eigenvalue λ_α, namely λ_α has g_α linearly independent eigenvectors, which are indexed by p. Therefore we have

$$\sum_\alpha 1 \le n, \quad \sum_\alpha g_\alpha = \sum_i 1 = n.$$

Now consider the following expression:

$$P_\alpha = \frac{\prod_{\beta\neq\alpha}(A - \lambda_\beta I)}{\prod_{\gamma\neq\alpha}(\lambda_\alpha - \lambda_\gamma)}. \tag{1.39}$$

This is a projection operator onto the g_α-dimensional space corresponding to the eigenvalue λ_α. In fact, it is straightforward to verify that

$$P_\alpha|\lambda_{\alpha,p}\rangle = \frac{\prod_{\beta\neq\alpha}(\lambda_\alpha - \lambda_\beta)}{\prod_{\gamma\neq\alpha}(\lambda_\alpha - \lambda_\gamma)}|\lambda_{\alpha,p}\rangle = |\lambda_{\alpha,p}\rangle \quad (1 \le p \le g_\alpha)$$

and

$$P_\alpha|\lambda_{\delta,q}\rangle = \frac{\prod_{\beta\neq\alpha}(\lambda_\delta - \lambda_\beta)}{\prod_{\gamma\neq\alpha}(\lambda_\alpha - \lambda_\gamma)}|\lambda_{\delta,q}\rangle = 0 \quad (\delta \neq \alpha,\ 1 \le q \le g_\delta)$$

since one of $\beta(\neq \alpha)$ is equal to $\delta(\neq \alpha)$ in the numerator. Therefore, we conclude that P_α is a projection operator

$$P_\alpha = \sum_{p=1}^{g_\alpha} |\lambda_{\alpha,p}\rangle\langle\lambda_{\alpha,p}| \tag{1.40}$$

onto the g_α-dimensional subspace correcponding to the eigenvalue λ_α. It follows from Eq. (1.40) that rank $P_\alpha = g_\alpha$. Note also that

$$AP_\alpha = \lambda_\alpha P_\alpha. \tag{1.41}$$

The above method is particularly suitable when the eigenvalues are degenerate. It is also useful when eigenvectors are difficult to obtain or unnecessary.

EXAMPLE 1.5 Let us take σ_y as an example. We found in Example 1.3 that the eigenvalues are $\lambda_1 = +1$ and $\lambda_2 = -1$, from which we obtain the projection operators directly by using Eq. (1.39) as

$$P_1 = \frac{(\sigma_y - (-I))}{(1-(-1))} = \frac{1}{2}\begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}, \quad P_2 = \frac{(\sigma_y - I)}{(-1-1)} = \frac{1}{2}\begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}.$$

We find the spectral decomposition of σ_y as

$$\sigma_y = \sum_i \lambda_i P_i = \frac{1}{2}\begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix} + (-1)\frac{1}{2}\begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}.$$

One of the advantages of the spectral decomposition is that a function of a matrix is evaluated quite easily. Let us prove the following formula.

PROPOSITION 1.1 Let A be a normal matrix in the above theorem. Then for an arbitrary $n \in \mathbb{N}$, we obtain

$$A^n = \sum_\alpha \lambda_\alpha^n P_\alpha. \tag{1.42}$$

If, furthermore, A^{-1} exists, the above formula may be extended to $n \in \mathbb{Z}$ by noting that λ_α^{-1} is an eigenvalue of A^{-1}.

Proof. Let $n \in \mathbb{N}$. Then

$$A^n P_\alpha = \lambda_\alpha A^{n-1} P_\alpha = \ldots = \lambda_\alpha^{n-1} A P_\alpha = \lambda_\alpha^n P_\alpha,$$

from which we obtain

$$A^n = A^n \sum_\alpha P_\alpha = \sum_\alpha A^n P_\alpha = \sum_\alpha \lambda_\alpha^n P_\alpha.$$

To prove the second half of the proposition, we only need to show that A^{-1} has an eigenvalue λ_α^{-1}, provided that A^{-1} exists (and hence $\lambda_\alpha \neq 0$), and the corresponding projection operator is P_α. We find

$$|\lambda_{\alpha,p}\rangle = A^{-1}A|\lambda_{\alpha,p}\rangle = \lambda_\alpha A^{-1}|\lambda_{\alpha,p}\rangle \to A^{-1}|\lambda_{\alpha,p}\rangle = \lambda_\alpha^{-1}|\lambda_{\alpha,p}\rangle.$$

Therefore the projection operator corresponding to the eivengalue λ_α^{-1} is P_α. The case $n = 0$, $I = \sum_\alpha P_\alpha$, is nothing but the completeness relation. Now we have proved that Eq. (1.42) applies to an arbitrary $n \in \mathbb{Z}$. ∎

From the above proposition, we obtain for a normal matrix A and an arbitrary analytic function $f(x)$,

$$f(A) = \sum_\alpha f(\lambda_\alpha) P_\alpha. \tag{1.43}$$

Even when $f(x)$ does not admit a series expansion, we may still formally define $f(A)$ by Eq. (1.43). Let $f(x) = \sqrt{x}$ and $A = \sigma_y$, for example. Then we obtain from Example 1.5 that

$$\sqrt{\sigma_y} = (\pm 1)P_1 + (\pm i)P_2.$$

It is easy to show that the RHS squares to σ_y. However, there are four possible $\sqrt{\sigma}$ depending on the choice of $\pm$ for each eigenvalue. Therefore the spectral decomposition is not unique in this case. Of course this ambiguity originates in the choice of the branch in the definition of $\sqrt{x}$.

EXAMPLE 1.6 Let us consider σ_y again. It follows directly from Example 1.5 that

$$\exp(i\alpha\sigma_y) \equiv \sum_{k=0}^{\infty} \frac{(i\alpha\sigma_y)^k}{k!} = e^{i\alpha}P_1 + e^{-i\alpha}P_2 = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix}.$$

EXERCISE 1.13 Suppose a 2×2 matrix A has eigenvalues $-1, 3$ and the corresponding eigenvectors

$$|e_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ i \end{pmatrix}, \quad |e_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix},$$

respectively. Find A.

EXERCISE 1.14 Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

(1) Find the eigenvalues and the corresponding normalized eigenvectors of A.
(2) Write down the spectral decomposition of A.
(3) Find $\exp(i\alpha A)$.

EXERCISE 1.15 Let

$$A = \begin{pmatrix} 5 & -2 & -4 \\ -2 & 2 & 2 \\ -4 & 2 & 5 \end{pmatrix}.$$

(1) Find the eigenvalues and the corresponding eigenvectors of A.
(2) Find the spectral decomposition of A.
(3) Find the inverse of A by making use of the spectral decomposition.

Now we prove a formula which will turn out to be very useful in the following. This is a generalization of Example 1.6

PROPOSITION 1.2 Let $\hat{\boldsymbol{n}} \in \mathbb{R}^3$ be a unit vector and $\alpha \in \mathbb{R}$. Then

$$\exp(i\alpha\hat{\boldsymbol{n}} \cdot \boldsymbol{\sigma}) = \cos\alpha I + i(\hat{\boldsymbol{n}} \cdot \boldsymbol{\sigma}) \sin\alpha, \tag{1.44}$$

where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$.

Proof. Let

$$A = \boldsymbol{n} \cdot \boldsymbol{\sigma} = \begin{pmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{pmatrix}.$$

The eigenvalues of A are $\lambda_1 = +1$ and $\lambda_2 = -1$. It then follows that

$$P_1 = \frac{(A+I)}{2} = \frac{1}{2}\begin{pmatrix} 1+n_z & n_x - in_y \\ n_x + in_y & 1-n_z \end{pmatrix},$$
$$P_2 = \frac{(A-I)}{-2} = \frac{1}{2}\begin{pmatrix} 1-n_z & -n_x + in_y \\ -n_x - in_y & 1+n_z \end{pmatrix},$$

from which we readily find

$$\begin{aligned} e^{i\alpha A} &= \frac{e^{i\alpha}}{2}\begin{pmatrix} 1+n_z & n_x - in_y \\ n_x + in_y & 1-n_z \end{pmatrix} + \frac{e^{-i\alpha}}{2}\begin{pmatrix} 1-n_z & -n_x + in_y \\ -n_x - in_y & 1+n_z \end{pmatrix} \\ &= \cos\alpha I + i(\boldsymbol{n} \cdot \boldsymbol{\sigma}) \sin\alpha. \end{aligned}$$

∎

EXERCISE 1.16 Let $f : \mathbb{C} \to \mathbb{C}$ be an analytic function. Let $\hat{\boldsymbol{n}}$ be a real three-dimensional unit vector and α be a real number. Show that

$$f(\alpha\hat{\boldsymbol{n}} \cdot \boldsymbol{\sigma}) = \frac{f(\alpha) + f(-\alpha)}{2} I + \frac{f(\alpha) - f(-\alpha)}{2} \hat{\boldsymbol{n}} \cdot \boldsymbol{\sigma}. \tag{1.45}$$

(c.f., Proposition 1.2.)

1.9 Singular Value Decomposition (SVD)

A subject somewhat related to the eigenvalue problem is the singular value decomposition. In a sense, it is a generalization of the eigenvalue problem to arbitrary matrices.

THEOREM 1.5 Let A be an $m \times n$ matrix with complex entries. Then it is possible to decompose A as

$$A = U\Sigma V^\dagger, \tag{1.46}$$

where $U \in \mathrm{U}(m), V \in \mathrm{U}(n)$ and Σ is an $m \times n$ matrix whose diagonals are nonnegative real numbers, called the **singular values**, while all the off diagonal components are zero. The matrix Σ is called the **singular value matrix**.

The decompostion (1.46) is called the **singular value decomosition** and is often abbreviated as SVD.

We now sketch the proof of the decomposition. Let us assume $m > n$ for definiteness. Consider the eigenvalue problem of an $n \times n$ Hermitian matrix $A^\dagger A$;

$$A^\dagger A|\lambda_i\rangle = \lambda_i|\lambda_i\rangle \quad (1 \le i \le n),$$

where λ_i is a nonnegative real number, where nonnegativity follows from the observation $\lambda_i = \langle\lambda_i|\lambda_i|\lambda_i\rangle = \langle\lambda_i|A^\dagger A|\lambda_i\rangle = \|A|\lambda_i\rangle\|^2 \ge 0$. Note that the set $\{|\lambda_i\rangle\}$ satisfies the completeness relation

$$\sum_{i=1}^{n} |\lambda_i\rangle\langle\lambda_i| = I_n$$

if they are made orthonormal by the Gram-Schmidt orthonormalization. We assume r of the eigenvalues are strictly positive and $n - r$ are zero. The set $\{\lambda_i\}$ is arranged in nonincreasing order $\lambda_1 \ge \lambda_2 \ge \ldots \ge \lambda_r > 0$ while $\lambda_{r+1} = \ldots = \lambda_n = 0$. Now define

$$V = (|\lambda_1\rangle, |\lambda_2\rangle, \ldots, |\lambda_r\rangle, |\lambda_{r+1}\rangle, \ldots, |\lambda_n\rangle)\,,$$

$$\Sigma = \begin{pmatrix} \sqrt{\lambda_1} & & & & & & \\ & \sqrt{\lambda_2} & & & & & \\ & & \ddots & & & & \\ & & & \sqrt{\lambda_r} & & & \\ & & & & 0 & & \\ & & & & & 0 & \\ & & & & & & \ddots \end{pmatrix},$$

and

$$U = (|\mu_1\rangle, |\mu_2\rangle, \ldots, |\mu_r\rangle, |\mu_{r+1}\rangle, \ldots, |\mu_m\rangle)\,,$$

where

$$|\mu_k\rangle = \frac{1}{\sqrt{\lambda_k}} A|\lambda_k\rangle \quad (1 \le k \le r),$$

while other orthonormal vectors $|\mu_{r+1}\rangle, \ldots, |\mu_m\rangle$ are taken to be orthogonal to $|\mu_k\rangle$ $(1 \le k \le r)$. Note that $V \in \mathrm{U}(n)$ and $U \in \mathrm{U}(m)$ by construction.

Then we find

$U\Sigma V^\dagger$

$$= \left(\frac{1}{\sqrt{\lambda_1}} A|\lambda_1\rangle, \frac{1}{\sqrt{\lambda_2}} A|\lambda_2\rangle, \ldots \frac{1}{\sqrt{\lambda_r}} A|\lambda_r\rangle, |\mu_{r+1}\rangle, \ldots, |\mu_m\rangle \right) \begin{pmatrix} \sqrt{\lambda_1}\langle\lambda_1| \\ \sqrt{\lambda_2}\langle\lambda_2| \\ \vdots \\ \sqrt{\lambda_r}\langle\lambda_r| \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$= A \sum_{i=1}^{r} |\lambda_i\rangle\langle\lambda_i| = A \sum_{i=1}^{n} |\lambda_i\rangle\langle\lambda_i| = A,$$

where we noted that $A|\lambda_i\rangle = 0$ for $r+1 \leq i \leq n$ and use has been made of the completeness relation of $\{|\lambda_i\rangle\}$. The reader should examine the case in which $m \leq n$. ∎

EXAMPLE 1.7 Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ i & i \end{pmatrix}$$

for which

$$A^\dagger A = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}.$$

The eigenvalues of $A^\dagger A$ are $\lambda_1 = 4$ and $\lambda_2 = 0$ with the corresponding eigenvectors

$$|\lambda_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |\lambda_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Unitary matrix V and the singular value matrix Σ are found from these data as

$$V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \text{ and } \Sigma = \begin{pmatrix} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

To construct U, we need

$$|\mu_1\rangle = \frac{1}{2} A|\lambda_1\rangle = \frac{1}{\sqrt{2}} (1, 0, i)^t$$

and two other vectors orthogonal to $|\mu_1\rangle$. By inspection, we find

$$|\mu_2\rangle = (0, 1, 0)^t \text{ and } |\mu_3\rangle = \frac{1}{\sqrt{2}} (1, 0, -i)^t,$$

for example. From these vectors we construct U as

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ i & 0 & -i \end{pmatrix}.$$

The reader should verify that $U\Sigma V^\dagger$ really reproduces A.

EXERCISE 1.17 Find the SVD of

$$A = \begin{pmatrix} 1 & 0 & i \\ i & 0 & 1 \end{pmatrix}.$$

1.10 Tensor Product (Kronecker Product)

DEFINITION 1.5 Let A be an $m \times n$ matrix and let B be a $p \times q$ matrix. Then

$$A \otimes B = \begin{pmatrix} a_{11}B, a_{12}B, \ldots, a_{1n}B \\ a_{21}B, a_{22}B, \ldots, a_{2n}B \\ \cdots \\ a_{m1}B, a_{m2}B, \ldots, a_{mn}B \end{pmatrix} \tag{1.47}$$

is an $(mp) \times (nq)$ matrix called the **tensor product** (**Kronecker product**) of A and B.

It should be noted that not all $(mp) \times (nq)$ matrices are tensor products of an $m \times n$ matrix and a $p \times q$ matrix. In fact, an $(mp) \times (np)$ matrix has $mnpq$ degrees of freedom, while $m \times n$ and $p \times q$ matrices have $mn + pq$ in total. Observe that $mnpq \gg mn + pq$ for large enough m, n, p and q. This fact is ultimately related to the power of quantum computing compared to its classical counterpart.

EXAMPLE 1.8

$$\sigma_x \otimes \sigma_z = \begin{pmatrix} 0 & \sigma_z \\ \sigma_z & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}.$$

EXAMPLE 1.9 We can also apply the tensor product to vectors as a special case. Let

$$|u\rangle = \begin{pmatrix} a \\ b \end{pmatrix}, \quad |v\rangle = \begin{pmatrix} c \\ d \end{pmatrix}.$$

Then we obtain

$$|u\rangle \otimes |v\rangle = \begin{pmatrix} a|v\rangle \\ b|v\rangle \end{pmatrix} = \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix}.$$

The tensor product $|u\rangle \otimes |v\rangle$ is often abbreviated as $|u\rangle|v\rangle$ or $|uv\rangle$ when it does not cause confusion.

EXERCISE 1.18 Let A and B be as above and let C be an $n \times r$ matrix and D be a $q \times s$ matrix. Show that

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD). \tag{1.48}$$

It similarly holds that

$$(A_1 \otimes B_1)(A_2 \otimes B_2)(A_3 \otimes B_3) = (A_1 A_2 A_3) \otimes (B_1 B_2 B_3),$$

and its generalizations whenever the dimensions of the matrices match so that the products make sense.

EXERCISE 1.19 Show that

$$A \otimes (B + C) = A \otimes B + A \otimes C \tag{1.49}$$
$$(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger \tag{1.50}$$
$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1} \tag{1.51}$$

whenever the matrix operations are well-defined.

Show, from the above observations, that the tensor product of two unitary matrices is also unitary and that the tensor product of two Hermitian matrices is also Hermitian.

EXERCISE 1.20 Let A and B be an $m \times m$ matrix and a $p \times p$ matrix, respectively. Show that

$$\begin{aligned} \mathrm{tr}(A \otimes B) &= (\mathrm{tr} A)(\mathrm{tr} B), \\ \det(A \otimes B) &= (\det A)^p (\det B)^m. \end{aligned}$$

EXERCISE 1.21 Let $|a\rangle, |b\rangle, |c\rangle, |d\rangle \in \mathbb{C}^n$. Show that

$$(|a\rangle\langle b|) \otimes (|c\rangle\langle d|) = (|a\rangle \otimes |c\rangle)(\langle b| \otimes \langle d|) = |ac\rangle\langle bd|.$$

THEOREM 1.6 Let A be an $m \times m$ matrix and B be a $p \times p$ matrix. Let A have the eigenvalues $\lambda_1, \ldots, \lambda_m$ with the corresponding eigenvectors $|u_1\rangle, \ldots, |u_m\rangle$ and let B have the eigenvalues $\mu_1, \ldots, \mu_p$ with the corresponding eigenvectors $|v_1\rangle, \ldots, |v_p\rangle$. Then $A \otimes B$ has mp eigenvalues $\{\lambda_j \mu_k\}$ with the corresponding eigenvectors $\{|u_j v_k\rangle\}$.

Proof. We show that $|u_j v_k\rangle$ is an eigenvector. In fact,

$$\begin{aligned} (A \otimes B)(|u_j v_k\rangle) &= (A|u_j\rangle) \otimes (B|v_k\rangle) = (\lambda_j |u_j\rangle) \otimes (\mu_k |v_k\rangle) \\ &= \lambda_j \mu_k (|u_j v_k\rangle) \ . \end{aligned}$$

Therefore, the eigenvalue is $\lambda_j \mu_k$ with the corresponding eigenvector $|u_j v_k\rangle$. Since there are mp eigenvectors, the vectors $|u_j v_k\rangle$ exhaust all of them. ∎

EXERCISE 1.22 Let A and B be as above. Show that $A \otimes I_p + I_m \otimes B$ has the eigenvalues $\{\lambda_j + \mu_k\}$ with the corresponding eigenvectors $\{|u_j v_k\rangle\}$, where I_p is the $p \times p$ unit matrix.

2

Framework of Quantum Mechanics

Quantum mechanics is founded on several postulates, which cannot be proven theoretically. They are justified only through an empirical fact that they are consistent with all the known experimental results. The choice of the postulates depends heavily on authors' taste. Here we give one that turns out to be the most convenient in the study of quantum information and computation. For a general introduction to quantum mechanics, we recommend [1, 2, 3, 4], for example. [5] and [6] contain more advanced subjects than those treated in this chapter.

2.1 Fundamental Postulates

Quantum mechanics was discovered roughly a century ago. In spite of its long history, the interpretation of the wave function remains an open question. Here we adopt the most popular one, called the **Copenhagen interpretation**.

A 1 A pure state in quantum mechanics is represented in terms of a normalized vector $|\psi\rangle$ in a Hilbert space $\mathcal{H}$ (a complex vector space with an inner product): $\langle\psi|\psi\rangle = 1$. Suppose two states $|\psi_1\rangle$ and $|\psi_2\rangle$ are physical states of the system. Then their linear superposition $c_1|\psi_1\rangle + c_2|\psi_2\rangle$ ($c_k \in \mathbb{C}$) is also a possible state of the same system. This is called the **superposition principle**.

A 2 For any physical quantity (i.e., **observable**) a, there exists a corresponding Hermitian operator A acting on the Hilbert space $\mathcal{H}$. When we make a measurement of a, we obtain one of the eigenvalues λ_j of the operator A. Let λ_1 and λ_2 be two eigenvalues of A: $A|\lambda_i\rangle = \lambda_i|\lambda_i\rangle$. Suppose the system is in a superposition state $c_1|\lambda_1\rangle + c_2|\lambda_2\rangle$. If we measure a in this state, then the state undergoes an abrupt change to one of the eigenstates corresponding to the observed eigenvalue: If the observed eigenvalue is λ_1 (λ_2), the system undergoes a **wave function collapse** as follows: $c_1|\lambda_1\rangle + c_2|\lambda_2\rangle \to |\lambda_1\rangle$ ($|\lambda_2\rangle$), and the state immediately after the measurement is $|\lambda_1\rangle$ ($|\lambda_2\rangle$). Suppose we prepare many copies of the state $c_1|\lambda_1\rangle + c_2|\lambda_2\rangle$. The probability of collapsing to the state $|\lambda_k\rangle$ is given by $|c_k|^2$ $(k = 1, 2)$. In this sense, the complex coefficient c_i is called the **probability amplitude**. It should be noted that a measurement produces one outcome λ_i and the probability of obtaining it is experimentally evaluated only after repeating measurements with many copies of the same state. These statements are easily generalized to states in a superposition of more than two states.

A 3 The time dependence of a state is governed by the **Schrödinger equation**

$$i\hbar\frac{\partial|\psi\rangle}{\partial t} = H|\psi\rangle, \tag{2.1}$$

where $\hbar$ is a physical constant known as the **Planck constant** and H is a Hermitian operator (matrix) corresponding to the energy of the system and is called the **Hamiltonian**.

Several comments are in order.

- In Axiom A 1, the phase of the vector may be chosen arbitrarily; $|\psi\rangle$ in fact represents the "ray" $\{e^{i\alpha}|\psi\rangle \; |\alpha \in \mathbb{R}\}$. This is called the **ray representation**. In other words, we can totally igonore the phase of a vector since it has no observable consequence. Note, however, that the *relative* phase of two different states is meaningful. Although $|\langle\phi|e^{i\alpha}\psi\rangle|^2$ is independent of α, $|\langle\phi|\psi_1 + e^{i\alpha}\psi_2\rangle|^2$ does depend on α.

- Axiom A 2 may be formulated in a different but equivalent way as follows. Suppose we would like to measure an observable a. Let $A = \sum_i \lambda_i|\lambda_i\rangle\langle\lambda_i|$ be the corresponding operator, where $A|\lambda_i\rangle = \lambda_i|\lambda_i\rangle$. Then the expectation value $\langle A\rangle$ of a after measurements with respect to many copies of a state $|\psi\rangle$ is

$$\langle A\rangle = \langle\psi|A|\psi\rangle. \tag{2.2}$$

Let us expand $|\psi\rangle$ in terms of $|\lambda_i\rangle$ as $|\psi\rangle = \sum_i c_i|\lambda_i\rangle$ to show the equivalence between two formalisms. According to A 2, the probability of observing λ_i upon measurement of a is $|c_i|^2$, and therefore the expectation value after many measurements is $\sum_i \lambda_i|c_i|^2$. If, conversely, Eq. (2.2) is employed, we will obtain the same result since

$$\langle\psi|A|\psi\rangle = \sum_{i,j} c_j^* c_i\langle\lambda_j|A|\lambda_i\rangle = \sum_{i,j} c_j^* c_i\lambda_i\delta_{ij} = \sum_i \lambda_i|c_i|^2.$$

This measurement is called the **projective measurement**. Any particular outcome λ_i will be found with the probability

$$|c_i|^2 = \langle\psi|P_i|\psi\rangle, \tag{2.3}$$

where $P_i = |\lambda_i\rangle\langle\lambda_i|$ is the projection operator, and the state immediately after the measurement is $|\lambda_i\rangle$ or equivalently

$$\frac{P_i|\psi\rangle}{\sqrt{\langle\psi|P_i|\psi\rangle}}, \tag{2.4}$$

where the overall phase has been ignored.

- The Schrödinger equation (2.1) in Axiom A 3 is formally solved to yield

$$|\psi(t)\rangle = e^{-iHt/\hbar}|\psi(0)\rangle, \tag{2.5}$$

if the Hamiltonian H is time-independent, while

$$|\psi(t)\rangle = \mathcal{T}\exp\left[-\frac{i}{\hbar}\int_0^t H(t)dt\right]|\psi(0)\rangle \tag{2.6}$$

if H depends on t, where $\mathcal{T}$ is the time-ordering operator defined by

$$\mathcal{T}[A(t_1)B(t_2)] = \begin{cases} A(t_1)B(t_2), & t_1 > t_2 \\ B(t_2)A(t_1), & t_2 \geq t_1 \end{cases},$$

for a product of two operators. Generalization to products of more than two operators should be obvious. We write Eqs. (2.5) and (2.6) as $|\psi(t)\rangle = U(t)|\psi(0)\rangle$. The operator $U(t) : |\psi(0)\rangle \mapsto |\psi(t)\rangle$, which we call the **time-evolution operator**, is unitary. Unitarity of $U(t)$ guarantees that the norm of $|\psi(t)\rangle$ is conserved:

$$\langle\psi(0)|U^\dagger(t)U(t)|\psi(0)\rangle = \langle\psi(0)|\psi(0)\rangle = 1.$$

EXERCISE 2.1 (Uncertainty Principle)
(1) Let A and B be Hermitian operators and $|\psi\rangle$ be some quantum state on which A and B operate. Show that

$$|\langle\psi|[A,B]|\psi\rangle|^2 + |\langle\psi|\{A,B\}|\psi\rangle|^2 = 4|\langle\psi|AB|\psi\rangle|^2.$$

(2) Prove the Cauchy-Schwarz inequality

$$|\langle\psi|AB|\psi\rangle|^2 \leq \langle\psi|A^2|\psi\rangle\langle\psi|B^2|\psi\rangle.$$

(3) Show that

$$|\langle\psi|[A,B]|\psi\rangle|^2 \leq 4\langle\psi|A^2|\psi\rangle\langle\psi|B^2|\psi\rangle.$$

(4) Show that

$$\Delta(A)\Delta(B) \geq \frac{1}{2}|\langle\psi|[A,B]|\psi\rangle|, \tag{2.7}$$

where $\Delta(A) \equiv \sqrt{\langle\psi|A^2|\psi\rangle - \langle\psi|A|\psi\rangle^2}$.
(5) Suppose $A = Q$ and $B = P \equiv \dfrac{\hbar}{i}\dfrac{d}{dQ}$. Deduce from the above arguments that

$$\Delta(Q)\Delta(P) \geq \frac{\hbar}{2}.$$

The uncertaintly principle in terms of standard deviation has been formulated first in [7] and [8].

2.2 Some Examples

We now give some examples to clarify the axioms introduced in the previous section. They turn out to have relevance to certain physical realizations of a quantum computer.

EXAMPLE 2.1 Let us consider a time-independent Hamiltonian

$$H = -\frac{\hbar}{2}\omega\sigma_x. \tag{2.8}$$

Suppose the system is in the eigenstate of σ_z with the eigenvalue $+1$ at time $t = 0$;

$$|\psi(0)\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The wave function $|\psi(t)\rangle$ $(t > 0)$ is then found from Eq. (2.5) to be

$$|\psi(t)\rangle = \exp\left(i\frac{\omega}{2}\sigma_x t\right)|\psi(0)\rangle. \tag{2.9}$$

The matrix exponential function in this equation is evaluated with the help of Eq. (1.44) and we find

$$|\psi(t)\rangle = \begin{pmatrix} \cos\omega t/2 & i\sin\omega t/2 \\ i\sin\omega t/2 & \cos\omega t/2 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cos\omega t/2 \\ i\sin\omega t/2 \end{pmatrix}. \tag{2.10}$$

Suppose we measure the observable σ_z. Note that $|\psi(t)\rangle$ is expanded in terms of the eigenvectors of σ_z as

$$|\psi(t)\rangle = \cos\frac{\omega}{2}t|\sigma_z = +1\rangle + i\sin\frac{\omega}{2}t|\sigma_z = -1\rangle.$$

Therefore we find the spin is in the spin-up state with the probability $P_\uparrow(t) = \cos^2(\omega t/2)$ and in the spin-down state with the probability $P_\downarrow(t) = \sin^2(\omega t/2)$ as depicted in Fig. 2.1. Of course, the total probability is independent of time since $\cos^2(\omega t/2) + \sin^2(\omega t/2) = 1$. This result is consistent with classical spin dynamics. The Hamiltonian (2.8) depicts a spin under a magnetic field along the x-axis. Our initial condition signifies that the spin points the z-direction at $t = 0$. Then the spin starts precession around the x-axis, and the z-component of the spin oscillates sinusoidally as is shown above.

Next let us take the initial state

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

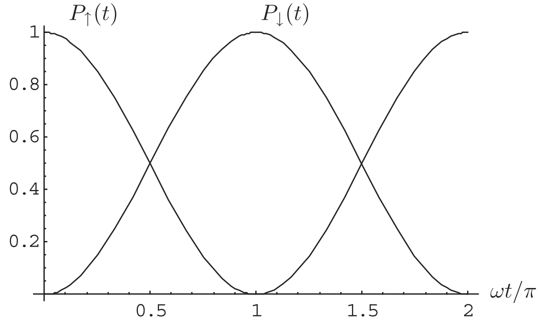

FIGURE 2.1
Probability $P_\uparrow(t)$ with which a spin is observed in the $\uparrow$-state and $P_\downarrow(t)$ observed in the $\downarrow$-state.

which is an eigenvector of σ_x (and hence the Hamiltonian) with the eigenvalue $+1$. We find $|\psi(t)\rangle$ in this case as

$$|\psi(t)\rangle = \begin{pmatrix} \cos\omega t/2 & i\sin\omega t/2 \\ i\sin\omega t/2 & \cos\omega t/2 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{e^{i\omega t/2}}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{2.11}$$

Therefore the state remains in its initial state at an arbitrary $t > 0$. This is an expected result since the system at $t = 0$ is an eigenstate of the Hamiltonian.

EXERCISE 2.2 Let us consider a Hamiltonian

$$H = -\frac{\hbar}{2}\omega\sigma_y. \tag{2.12}$$

Suppose the initial state of the system is

$$|\psi(0)\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{2.13}$$

(1) Find the wave function $|\psi(t)\rangle$ at later time $t > 0$.
(2) Find the probability for the system to have the outcome $+1$ upon measurement of σ_z at $t > 0$.
(3) Find the probability for the system to have the outcome $+1$ upon measurement of σ_x at $t > 0$.

Now let us formulate Example 2.1 and Exercise 2.2 in the most general form. Consider a Hamiltonian

$$H = -\frac{\hbar}{2}\omega\hat{\boldsymbol{n}} \cdot \boldsymbol{\sigma}, \tag{2.14}$$

where $\hat{\boldsymbol{n}}$ is a unit vector in $\mathbb{R}^3$. The time-evolution operator is readily obtained, by making use of the result of Proposition 1.2, as

$$U(t) = \exp(-iHt/\hbar) = \cos\frac{\omega}{2}t\, I + i(\hat{\boldsymbol{n}} \cdot \boldsymbol{\sigma})\sin\frac{\omega}{2}t. \tag{2.15}$$

Suppose the initial state is

$$|\psi(0)\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

for example. Then we find

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle = \begin{pmatrix} \cos(\omega t/2) + in_z \sin(\omega t/2) \\ i(n_x + in_y)\sin(\omega t/2) \end{pmatrix}. \tag{2.16}$$

The reader should verify that $|\psi(t)\rangle$ is normalized at any instant of time $t > 0$.

EXAMPLE 2.2 (Rabi oscillation) This example is often employed for a quantum gate implementation as will be shown later. We will take the natural unit $\hbar = 1$ to simplify our notation throughout this example. Let us consider a spin-1/2 particle in a magnetic field along the z-axis, whose Hamiltonian is given by

$$H_0 = -\frac{\omega_0}{2}\sigma_z. \tag{2.17}$$

Suppose the particle is irradiated by an oscillating magnetic field of angular frequency ω, which introduces transitions between two energy eigenstates of H_0. Then the perturbed Hamiltonian is modelled as

$$H = -\frac{\omega_0}{2}\sigma_z + \frac{\omega_1}{2}\begin{pmatrix} 0 & e^{i\omega t} \\ e^{-i\omega t} & 0 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} -\omega_0 & \omega_1 e^{i\omega t} \\ \omega_1 e^{-i\omega t} & \omega_0 \end{pmatrix}, \tag{2.18}$$

where $\omega_1 > 0$ is a parameter proportional to the amplitude of the oscillating field. Let us evaluate the wave function $|\psi(t)\rangle$ at time $t > 0$ assuming that the system is in the ground state of the unperturbed Hamiltonian

$$|\psi(0)\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{2.19}$$

at $t = 0$. Note that we cannot simply exponentiate the Hamiltonian since it is time-dependent. Surprisingly, however, the following trick makes it time-independent. Let us consider the following "gauge transformation":

$$|\phi(t)\rangle = e^{-i\omega\sigma_z t/2}|\psi(t)\rangle. \tag{2.20}$$

A straightforward calculation shows that $|\phi(t)\rangle$ satisfies

$$i\frac{d}{dt}|\phi(t)\rangle = \tilde{H}|\phi(t)\rangle, \tag{2.21}$$

where

$$\begin{aligned} \tilde{H} &= e^{-i\omega\sigma_z t/2} H e^{i\omega\sigma_z t/2} - ie^{-i\omega\sigma_z t/2}\frac{d}{dt}e^{i\omega\sigma_z t/2} = \frac{1}{2}\begin{pmatrix} -\omega_0 + \omega & \omega_1 \\ \omega_1 & \omega_0 - \omega \end{pmatrix} \\ &= -\frac{\delta}{2}\sigma_z + \frac{\omega_1}{2}\sigma_x \end{aligned} \tag{2.22}$$

is, in fact, time-independent. Here $\delta = \omega_0 - \omega$ stands for the "detuning" between ω and ω_0. Note that the Hamiltonian $\tilde{H}$ can be put into the form (2.14) as

$$\tilde{H} = \frac{\Delta}{2}\left(\frac{\omega_1}{\Delta}\sigma_x - \frac{\delta}{\Delta}\sigma_z\right), \quad \Delta \equiv \sqrt{\delta^2 + \omega_1^2}. \tag{2.23}$$

Now it is easy to solve Eq. (2.21). The time evolution operator is obtained using Eq. (2.15) as

$$\begin{aligned}\tilde{U}(t) &= \cos\frac{\Delta t}{2} I - i\left(\frac{\omega_1}{\Delta}\sigma_x - \frac{\delta}{\Delta}\sigma_z\right)\sin\frac{\Delta t}{2} \\ &= \begin{pmatrix} \cos\frac{\Delta t}{2} + i\frac{\delta}{\Delta}\sin\frac{\Delta t}{2} & -i\frac{\omega_1}{\Delta}\sin\frac{\Delta t}{2} \\ -i\frac{\omega_1}{\Delta}\sin\frac{\Delta t}{2} & \cos\frac{\Delta t}{2} - i\frac{\delta}{\Delta}\sin\frac{\Delta t}{2} \end{pmatrix}.\end{aligned} \tag{2.24}$$

The wave function $|\phi(t)\rangle$ with the initial condtion $|\phi(0)\rangle = (1,0)^t$ is

$$|\phi(t)\rangle = \tilde{U}(t)|\phi(0)\rangle = \begin{pmatrix} \cos\frac{\Delta t}{2} + i\frac{\delta}{\Delta}\sin\frac{\Delta t}{2} \\ -i\frac{\omega_1}{\Delta}\sin\frac{\Delta t}{2} \end{pmatrix}. \tag{2.25}$$

We find $|\psi(t)\rangle$ from Eq. (2.20) as

$$|\psi(t)\rangle = e^{i\omega\sigma_z t/2}|\phi(t)\rangle = \begin{pmatrix} e^{i\omega t/2}\left(\cos\frac{\Delta t}{2} + i\frac{\delta}{\Delta}\sin\frac{\Delta t}{2}\right) \\ -ie^{-i\omega t/2}\frac{\omega_1}{\Delta}\sin\frac{\Delta t}{2} \end{pmatrix}. \tag{2.26}$$

Suppose the applied field is in resonance with the energy difference of two levels, namely $\omega = \omega_0$. We obtain $\delta = 0$ and $\Delta = \omega_1$ in this case. The wave function $|\psi(t)\rangle$ at later time $t > 0$ is

$$|\psi(t)\rangle = e^{i\omega\sigma_z t/2}|\phi(t)\rangle = \begin{pmatrix} e^{i\omega_0 t/2}\cos\frac{\omega_1 t}{2} \\ -ie^{-i\omega_0 t/2}\sin\frac{\omega_1 t}{2} \end{pmatrix}. \tag{2.27}$$

The probability with which the system is found in the ground (excited) state of H_0 is given by

$$P_0 = \cos^2\omega_1 t/2 \quad (P_1 = \sin^2\omega_1 t/2). \tag{2.28}$$

This oscillatory behavior is called the **Rabi oscillation**. The frequency ω_1 is called the **Rabi frequency**, while Δ in Eq. (2.23) is called the **generalized Rabi frequency**

2.3 Multipartite System, Tensor Product and Entangled State

So far, we have assumed implictly that the system is made of a single component. Suppose a system is made of two components; one lives in a Hilbert space $\mathcal{H}_1$ and the other in another Hilbert space $\mathcal{H}_2$. A system composed of two separate components is called **bipartite**. Then the system as a whole lives in a Hilbert space $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$, whose general vector is written as

$$|\psi\rangle = \sum_{i,j} c_{ij} |e_{1,i}\rangle \otimes |e_{2,j}\rangle, \tag{2.29}$$

where $\{|e_{a,i}\rangle\}$ $(a = 1, 2)$ is an orthonormal basis in $\mathcal{H}_a$ and $\sum_{i,j} |c_{ij}|^2 = 1$.

A state $|\psi\rangle \in \mathcal{H}$ written as a tensor product of two vectors as $|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$, $(|\psi_a\rangle \in \mathcal{H}_a)$ is called a **separable state** or a **tensor product state**. A separable state admits a classical interpretation such as "The first system is in the state $|\psi_1\rangle$, while the second system is in $|\psi_2\rangle$." It is clear that the set of separable states has dimension $\dim\mathcal{H}_1 + \dim\mathcal{H}_2$. Note however that the total space $\mathcal{H}$ has different dimensions since we find, by counting the number of coefficients in (2.29), that $\dim\mathcal{H} = \dim\mathcal{H}_1 \dim\mathcal{H}_2$. This number is considerably larger than the dimension of the sparable states when $\dim\mathcal{H}_a$ $(a = 1, 2)$ are large. What are the missing states then? Let us consider a spin state

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle \otimes |\uparrow\rangle + |\downarrow\rangle \otimes |\downarrow\rangle) \tag{2.30}$$

of two separated electrons. Suppose $|\psi\rangle$ may be decomposed as

$$\begin{aligned} |\psi\rangle &= (c_1|\uparrow\rangle + c_2|\downarrow\rangle) \otimes (d_1|\uparrow\rangle + d_2|\downarrow\rangle) \\ &= c_1 d_1 |\uparrow\rangle \otimes |\uparrow\rangle + c_1 d_2 |\uparrow\rangle \otimes |\downarrow\rangle + c_2 d_1 |\downarrow\rangle \otimes |\uparrow\rangle + c_2 d_2 |\downarrow\rangle \otimes |\downarrow\rangle. \end{aligned}$$

However this decomposition is not possible since we must have

$$c_1 d_2 = c_2 d_1 = 0, \; c_1 d_1 = c_2 d_2 = \frac{1}{\sqrt{2}}$$

simultaneously, and it is clear that the above equations have no common solution. Therefore the state $|\psi\rangle$ is not separable.

Such non-separable states are called **entangled** in quantum theory [9]. The fact

$$\dim\mathcal{H}_1 \dim\mathcal{H}_2 \gg \dim\mathcal{H}_1 + \dim\mathcal{H}_2$$

tells us that most states in a Hilbert space of a bipartite system are entangled when the constituent Hilbert spaces are higher dimensional. These entangled states refuse classical descriptions. Entanglement will be used extensively as a powerful computational resource in quantum information processing and quantum computation.

Suppose a bipartite state (2.29) is given. We are interested in when the state is separable and when entangled. The criterion is given by the Schmidt decomposition of $|\psi\rangle$.

PROPOSITION 2.1 Let $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$ be the Hilbert space of a bipartite system. Then a vector $|\psi\rangle \in \mathcal{H}$ admits the **Schmidt decomposition**

$$|\psi\rangle = \sum_{i=1}^{r} \sqrt{s_i}|f_{1,i}\rangle \otimes |f_{2,i}\rangle \text{ with } \sum_i s_i = 1, \tag{2.31}$$

where $s_i > 0$ are called the **Schmidt coefficients** and $\{|f_{a,i}\rangle\}$ is an orthonormal set of $\mathcal{H}_a$. The number $r \in \mathbb{N}$ is called the **Schmidt number** of $|\psi\rangle$.

Proof. This is a direct consequence of SVD introduced in §1.9. Let $|\psi\rangle$ be expanded as in Eq. (2.29). Note that the coefficients c_{ij} form a $\dim\mathcal{H}_1 \times \dim\mathcal{H}_2$ matrix C. We apply the SVD to obtain $C = U\Sigma V^\dagger$, where U and V are unitary matrices and Σ is a matrix whose diagonal elements are nonnegative real numbers while all the off-diagonal elements vanish. Now $|\psi\rangle$ of Eq. (2.29) is put in the form

$$|\psi\rangle = \sum_{i,j,k} U_{ik}\Sigma_{kl}V^*_{jl}|e_{1,i}\rangle \otimes |e_{2,j}\rangle.$$

Now define $|f_{1,k}\rangle = \sum_i U_{ik}|e_{1,i}\rangle$ and $|f_{2,k}\rangle = \sum_j V^*_{jk}|e_{2,j}\rangle$. Unitarity of U and V guarantees that they are orthonormal bases of $\mathcal{H}_1$ and $\mathcal{H}_2$, respectively. By noting that $\Sigma_{kl} = d_k\delta_{kl}$, we obtain

$$|\psi\rangle = \sum_{i=1}^{r} d_i|f_{1,i}\rangle \otimes |f_{2,i}\rangle,$$

where r is the number of nonvanishing diagonal elements in Σ. The wave function (2.31) is obtained by replacing the positive number d_i by $d_i = \sqrt{s_i}$. Moreover, the normalization condition implies $\langle\psi|\psi\rangle = \sum_i s_i = 1$. ∎

It follows from the above proposition that a bipartite state $|\psi\rangle$ is separable if and only if its Schmidt number r is 1.

EXAMPLE 2.3 Consider a bipartite state

$$|\psi\rangle = \frac{1}{2}(|e_{1,1}\rangle|e_{2,1}\rangle + |e_{1,1}\rangle|e_{2,2}\rangle + i|e_{1,3}\rangle|e_{2,1}\rangle + i|e_{1,3}\rangle|e_{2,2}\rangle),$$

whose coefficients form a matrix

$$C = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 0 & 0 \\ i & i \end{pmatrix}.$$

Note that this is essentially the same matrix whose SVD was analyzed in Example 1.7. By making use of the result obtained there, we find

$$|\psi\rangle = |f_{1,1}\rangle|f_{2,1}\rangle,$$

where

$$|f_{1,1}\rangle = \sum_{i=1}^{3} U_{i1}|e_{1,i}\rangle = \frac{1}{\sqrt{2}}(|e_{1,1}\rangle + i|e_{1,3}\rangle)$$

and

$$|f_{2,1}\rangle = \sum_{j=1}^{2} V_{j1}^{*}|e_{2,j}\rangle = \frac{1}{\sqrt{2}}(|e_{2,1}\rangle + |e_{2,2}\rangle).$$

Therefore the Schmidt number is 1 and the state is separable.

Generalization to a system with more components, i.e., a **multipartite system**, should be obvious. A system composed of N components has a Hilbert space

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \ldots \otimes \mathcal{H}_N, \tag{2.32}$$

where $\mathcal{H}_a$ is the Hilbert space to which the ath component belongs. Classification of entanglement in a multipartite system is far from obvious, and an analogue of the Schmidt decompostion is not known to date for $N \geq 3$.*

2.4 Mixed States and Density Matrices

It might happen in some cases that a quantum system under considertation is in the state $|\psi_i\rangle$ with a probability p_i. In other words, we cannot say definitely which state the system is in. Therefore some random nature comes into the description of the system. This random nature should not be confused with a probabilistic behavior of a quantum system. Such a system is said to be in a **mixed state**, while a system whose vector is uniquely specified is in a **pure state**. A pure state is a special case of a mixed state in which $p_i = 1$ for some i and $p_j = 0$ $(j \neq i)$.

Mixed states may happen in the following cases, for example.

- Suppose we observe a beam of totally unpolarized light and measure whether photons are polarized vertically or horizontally. The measurement outcome of a particular photon is *either* horizontal *or* vertical. Therefore when the beam passes through a linear polarizer, the intensity is halved. The beam is a mixture of horizontally polarized photons and vertically polarized photons.

*See, however, [10, 11].

- A particle source emits a particle in a state $|\psi_i\rangle$ with a probability p_i $(1 \le i \le N)$.
- Let us consider a canonical ensemble. If we pick up one of the members in the ensemble, it is in a state $|\psi_i\rangle$ with energy E_i with a probability $p_i = e^{-E_i/k_BT}/Z(T)$, where $Z(T) = \mathrm{Tr} e^{-H/k_BT}$ is the partition function.

In each of these examples, a particular state $|\psi_i\rangle \in \mathcal{H}$ appears with probability p_i, in which case the expectation value of the observable a is $\langle\psi_i|A|\psi_i\rangle$, where we assume $|\psi_i\rangle$ is normalized; $\langle\psi_i|\psi_i\rangle = 1$. The mean value of a is then given by

$$\langle A\rangle = \sum_{i=1}^{N} p_i \langle\psi_i|A|\psi_i\rangle, \tag{2.33}$$

where N is the number of available states. Let us introduce the **density matrix** by

$$\rho = \sum_{i=1}^{N} p_i |\psi_i\rangle\langle\psi_i|. \tag{2.34}$$

Then Eq. (2.33) is rewritten in a compact form as

$$\langle A\rangle = \mathrm{Tr}(\rho A). \tag{2.35}$$

EXERCISE 2.3 Let A be a Hermitian matrix. A is called positive-semidefinite if $\langle\psi|A|\psi\rangle \ge 0$ for any $|\psi\rangle$ in the relevant Hilbert space $\mathcal{H}$. Show that all the eigenvalues of a positive-semidefinite Hermitian matrix are non-negative.

Conversely, show that a Hermitian matrix A, whose eigenvalues are all non-negative, satisfies $\langle\psi|A|\psi\rangle \ge 0$ for any $|\psi\rangle \in \mathcal{H}$.

Properties which a density matrix ρ satisfies are very much like axioms for pure states.

A 1′ A physical state of a system, whose Hilbert space is $\mathcal{H}$, is completely specified by its associated density matrix $\rho : \mathcal{H} \to \mathcal{H}$. A density matrix is a positive semi-definite Hermitian operator with $\mathrm{tr}\,\rho = 1$ (see remarks below).

A 2′ The mean value of an observable a is given by

$$\langle A\rangle = \mathrm{tr}\,(\rho A). \tag{2.36}$$

A 3′ The temporal evolution of the density matrix is given by the **Liouville-von Neumann equation**,

$$i\hbar \frac{d}{dt}\rho = [H, \rho], \tag{2.37}$$

where H is the system Hamiltonian (see remarks below).

Several remarks are in order.

- We assume $\{|\psi_i\rangle\}$ is not necessarily an orthogonal basis of $\mathcal{H}$, although it is assumed $\langle\psi_i|\psi_i\rangle = 1$. The density matrix (2.34) is Hermitian since $p_i \in \mathbb{R}$. It is positive semi-definite since

$$\langle\phi|\rho|\phi\rangle = \sum_i p_i\langle\phi|\psi_i\rangle\langle\psi_i|\phi\rangle = \sum_i p_i|\langle\psi_i|\phi\rangle|^2 \geq 0,$$

where $|\phi\rangle$ is an arbitrary vector. We also find

$$\begin{aligned}\operatorname{tr}\rho &= \sum_k \langle e_k|\rho|e_k\rangle = \sum_{i,k}\langle e_k|p_i|\psi_i\rangle\langle\psi_i|e_k\rangle \\ &= \sum_i p_i\langle\psi_i|\left(\sum_k |e_k\rangle\langle e_k|\right)|\psi_i\rangle = \sum_i p_i\langle\psi_i|\psi_i\rangle = 1,\end{aligned}$$

where $\{|e_k\rangle\}$ is an orthonormal basis of $\mathcal{H}$.

- Each $|\psi_i\rangle$ follows the Schrödinger equation

$$i\hbar\frac{d}{dt}|\psi_i\rangle = H|\psi_i\rangle$$

in a closed quantum system. Its Hermitian conjugate is

$$-i\hbar\frac{d}{dt}\langle\psi_i| = \langle\psi_i|H.$$

We prove the Liouville-von Neumann equation from these equations as

$$\begin{aligned}i\hbar\frac{d}{dt}\rho = i\hbar\frac{d}{dt}\sum_i p_i|\psi_i\rangle\langle\psi_i| &= \sum_i p_i H|\psi_i\rangle\langle\psi_i| - \sum_i p_i|\psi_i\rangle\langle\psi_i|H \\ &= [H,\rho].\end{aligned}$$

We denote the set of all possible density matrices as $\mathcal{S}(\mathcal{H})$, where $\mathcal{H}$ is the Hilbert space associated with a system under consideration. It is easy to verify that $t\rho_1 + (1-t)\rho_2$ for $\rho_{1,2} \in \mathcal{S}(\mathcal{H})$ is also a density matrix, which shows that $\mathcal{S}(\mathcal{H})$ is a convex set.

EXAMPLE 2.4 A pure state $|\psi\rangle$ is a special case in which the corresponding density matrix is

$$\rho = |\psi\rangle\langle\psi|. \tag{2.38}$$

Therefore ρ in this case is nothing but the projection operator onto the state $|\psi\rangle$. Observe that

$$\langle A\rangle = \operatorname{tr}\rho A = \sum_i \langle\psi_i|\psi\rangle\langle\psi|A|\psi_i\rangle = \sum_i \langle\psi|A|\psi_i\rangle\langle\psi_i|\psi\rangle = \langle\psi|A|\psi\rangle.$$

A general density matrix is a convex combination of pure states.

Let us consider a beam of photons. We take a horizontally polarized state $|e_1\rangle = |\leftrightarrow\rangle$ and a vertically polarized state $|e_2\rangle = |\updownarrow\rangle$ as orthonormal basis vectors. If the photons are a totally uniform mixture of two polarized states, the density matrix is given by

$$\rho = \frac{1}{2}|e_1\rangle\langle e_1| + \frac{1}{2}|e_2\rangle\langle e_2| = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{2}I.$$

This state is a uniform mixture of $|\updownarrow\rangle$ and $|\leftrightarrow\rangle$ and called a **maximally mixed state**.

If photons are in a pure state $|\psi\rangle = (|e_1\rangle + |e_2\rangle)/\sqrt{2}$, the density matrix, with $\{|e_i\rangle\}$ as basis, is

$$\rho = |\psi\rangle\langle\psi| = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

If $|\psi\rangle$ itself is used as a basis vector, the other vector being $|\phi\rangle = (|e_1\rangle - |e_2\rangle)/\sqrt{2}$, the density matrix with respect to the basis $\{|\psi\rangle, |\phi\rangle\}$ has a component expression

$$\rho = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Verify that they all satisfy Hermiticity, positive semi-definitness and $\mathrm{tr}\,\rho = 1$.

Let $A = \sum_a \lambda_a |\lambda_a\rangle\langle\lambda_a|$ be the spectral decomposition of an observable A and let $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ be an arbitrary state. Then the measurement outcome of A is λ_a with the probability

$$p(a) = \sum_i p_i |\langle\lambda_a|\psi_i\rangle|^2 = \langle\lambda_a|\rho|\lambda_a\rangle = \mathrm{tr}\,(P_a\rho), \tag{2.39}$$

where $P_a = |\lambda_a\rangle\langle\lambda_a|$ is the projection operator. The state changes to a pure state $|\lambda_a\rangle\langle\lambda_a|$ immediately after the measurement with the outcome λ_a. This change is written as $\rho \mapsto P_a\rho P_a/p(a)$.

Now, we are interested in when ρ represents a pure state or a mixed state.

THEOREM 2.1 A state ρ is pure if and only if $\rho^2 = \rho$.

Proof. Since ρ is Hermitian, all its eigenvalues λ_i $(1 \le i \le \dim\mathcal{H})$ are real and the corresponding eigenvectors $\{|\lambda_i\rangle\}$ are made orthonormal. Let $\rho = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i|$ be the spectral decomposition of ρ. Suppose $\rho^2 = \sum_i \lambda_i^2 |\lambda_i\rangle\langle\lambda_i| = \rho$. Then the eigenvalue λ_i satisfies $\lambda_i^2 = \lambda_i$ for any i. Therefore λ_i is either 0 or 1. It follows from $\mathrm{tr}\,\rho = \sum_i \lambda_i = 1$ that $\lambda_p = 1$ for some p and $\lambda_i = 0$ for $i \neq p$, namely, ρ is a pure state $|\lambda_p\rangle\langle\lambda_p|$.

The converse is trivial. ∎

EXERCISE 2.4 Show that a state ρ is pure if and only if $\mathrm{tr}\,\rho^2 = 1$.

We classify mixed states into three classes, similar to the classification of pure states into separable states and entangled states. We use a bipartite system in the definition, but generalization to multipartite systems should be obvious.

DEFINITION 2.1 A state ρ is called **uncorrelated** if it is written as

$$\rho = \rho_1 \otimes \rho_2. \tag{2.40}$$

It is called **separable** if it is written in the form

$$\rho = \sum_i p_i \rho_{1,i} \otimes \rho_{2,i}, \tag{2.41}$$

where $0 \le p_i \le 1$ and $\sum_i p_i = 1$. It is called **inseparable** if ρ does not admit the decompostion (2.41),

It is important to realize that only inseparable states have quantum correlations analogous to that of an entangled pure state. However, it does not necessarily imply separable states have no non-classical correlation. It was pointed out that useful non-classical correlation exists in the subset of separable states [12].

In the next subsection, we discuss how to find whether a given bipartite density matrix is separable or inseparable.

2.4.1 Negativity

Let ρ be a bipartite state and define the **partial transpose** ρ^{pt} of ρ with respect to the second Hilbert space as

$$\rho_{ij,kl} \to \rho_{il,kj}, \tag{2.42}$$

where

$$\rho_{ij,kl} = (\langle e_{1,i}| \otimes \langle e_{2,j}|)\ \rho\ (|e_{1,k}\rangle \otimes |e_{2,l}\rangle).$$

Here $\{|e_{1,k}\rangle\}$ is the basis of the first system, while $\{|e_{2,k}\rangle\}$ is the basis of the second system. Suppose ρ takes a separable form (2.41). Then the partial transpose yields

$$\rho^{\mathrm{pt}} = \sum_i p_i \rho_{1,i} \otimes \rho_{2,i}^t. \tag{2.43}$$

Note here that ρ^t for any density matrix ρ is again a density matrix since it is still a positive semi-definite Hermitian with unit trace. Therefore the partial transposed density matrix (2.43) is another density matrix. It was conjectured by Peres [6] and subsequently proved by the Hordecki family [14] that positivity of the partially transposed density matrix is a necessary and sufficient condition for ρ to be separable in the cases of $\mathbb{C}^2 \otimes \mathbb{C}^2$ systems and $\mathbb{C}^2 \otimes \mathbb{C}^3$ systems. Conversely, if the partial transpose of ρ of these systems is not a density matrix, then ρ is inseparable. Instead of giving the proof, we look at the following example.

EXAMPLE 2.5 Let us consider the Werner state

$$\rho = \begin{pmatrix} \frac{1-p}{4} & 0 & 0 & 0 \\ 0 & \frac{1+p}{4} & -\frac{p}{2} & 0 \\ 0 & -\frac{p}{2} & \frac{1+p}{4} & 0 \\ 0 & 0 & 0 & \frac{1-p}{4} \end{pmatrix}, \tag{2.44}$$

where $0 \le p \le 1$. Here the basis vectors are arranged in the order

$$|e_{1,1}\rangle|e_{2,1}\rangle, |e_{1,1}\rangle|e_{2,2}\rangle, |e_{1,2}\rangle|e_{2,1}\rangle, |e_{1,2}\rangle|e_{2,2}\rangle.$$

Partial transpose of ρ yields

$$\rho^{\text{pt}} = \begin{pmatrix} \frac{1-p}{4} & 0 & 0 & -\frac{p}{2} \\ 0 & \frac{1+p}{4} & 0 & 0 \\ 0 & 0 & \frac{1+p}{4} & 0 \\ -\frac{p}{2} & 0 & 0 & \frac{1-p}{4} \end{pmatrix}.$$

Note that we need to consider off-diagonal matrix elements only when we partically transpose the elements. We have, for example,

$$\begin{aligned} \rho_{12,21} &= (\langle e_{1,1}| \otimes \langle e_{2,2}|)\ \rho\ (|e_{1,2}\rangle \otimes |e_{2,1}\rangle) \\ &\to (\langle e_{1,1}| \otimes \langle e_{2,1}|)\ \rho\ (|e_{1,2}\rangle \otimes |e_{2,2}\rangle) = \rho^{\text{pt}}_{11,22}. \end{aligned}$$

For ρ^{pt} to be a physically acceptable state, it must have non-negative eigenvalues. The characteristic equation of ρ^{pt} is

$$D(\lambda) = \det(\rho^{\text{pt}} - \lambda I) = \left(\lambda - \frac{p+1}{4}\right)^3 \left(\lambda - \frac{1-3p}{4}\right) = 0.$$

There are threefold degenerate eigenvalues $\lambda = (1+p)/4$ and a nondegenerate eigenvalue $\lambda = (1-3p)/4$. This shows that ρ^{pt} is an unphysical state for $1/3 < p \le 1$. If this is the case, ρ is inseparable.

From the above observation, inseparable states are characterized by nonvanishing **negativity** defined as

$$N(\rho) \equiv \frac{\sum_i |\lambda_i| - 1}{2}. \tag{2.45}$$

Note that negativity vanishes if and only if all the eigenvalues of ρ^{pt} are nonnegative.

Negativity is one of the so-called entanglement monotones [15], which also include concurrence, entanglement of formation and entropy of entanglement.

EXAMPLE 2.6 It was mentioned above that vanishing negativity is equivalent with separability only for $\mathbb{C}^2 \otimes \mathbb{C}^2$ systems and $\mathbb{C}^2 \otimes \mathbb{C}^3$ systems. A counter example in a $\mathbb{C}^2 \otimes \mathbb{C}^4$ system has been given in Horodecki [16]. Let us consider

$$\rho = \frac{1}{7b+1}\begin{pmatrix} b\,0\,0\,0 & 0 & b\,0 & 0 \\ 0\,b\,0\,0 & 0 & 0\,b & 0 \\ 0\,0\,b\,0 & 0 & 0\,0 & b \\ 0\,0\,0\,b & 0 & 0\,0 & 0 \\ 0\,0\,0\,0 & \frac{1+b}{2} & 0\,0 & \frac{\sqrt{1-b^2}}{2} \\ b\,0\,0\,0 & 0 & b\,0 & 0 \\ 0\,b\,0\,0 & 0 & 0\,b & 0 \\ 0\,0\,b\,0 & \frac{\sqrt{1-b^2}}{2} & 0\,0 & \frac{1+b}{2} \end{pmatrix} \quad (0 \le b \le 1), \tag{2.46}$$

which is known to be inseparable. The partial transposed matrix with respect to the second system is

$$\rho^{\mathrm{pt}} = \frac{1}{7b+1}\begin{pmatrix} b\,0\,0\,0 & 0 & 0\,0 & 0 \\ 0\,b\,0\,0 & b & 0\,0 & 0 \\ 0\,0\,b\,0 & 0 & b\,0 & 0 \\ 0\,0\,0\,b & 0 & 0\,b & 0 \\ 0\,b\,0\,0 & \frac{1+b}{2} & 0\,0 & \frac{\sqrt{1-b^2}}{2} \\ 0\,0\,b\,0 & 0 & b\,0 & 0 \\ 0\,0\,0\,b & 0 & 0\,b & 0 \\ 0\,0\,0\,0 & \frac{\sqrt{1-b^2}}{2} & 0\,0 & \frac{1+b}{2} \end{pmatrix}. \tag{2.47}$$

The eigenvalues of ρ^{pt} are

$$0, 0, 0, \frac{b}{7b+1}, \frac{2b}{7b+1}, \frac{2b}{7b+1}$$

$$\frac{1 + 14b^2 + 9b - \sqrt{98b^4 - 70b^3 + 23b^2 + 12b + 1}}{2\,(49b^2 + 14b + 1)},$$

$$\frac{1 + 14b^2 + 9b + \sqrt{98b^4 - 70b^3 + 23b^2 + 12b + 1}}{2\,(49b^2 + 14b + 1)}.$$

It can be shown that the seventh eigenvalue takes the maximum value $(25 - 2\sqrt{10})/130$ at $b = (47 - 10\sqrt{10})/31$ and the minimum value 0 at $b = 0$, and hence all the eigenvalues are non-negative for $0 \le b \le 1$ in spite of inseparability of ρ.

EXERCISE 2.5 Verify that

$$\rho_1 = \begin{pmatrix} \frac{1+p}{4} & 0 & 0 & \frac{p}{2} \\ 0 & \frac{1-p}{4} & 0 & 0 \\ 0 & 0 & \frac{1-p}{4} & 0 \\ \frac{p}{2} & 0 & 0 & \frac{1+p}{4} \end{pmatrix} \qquad (0 \le p \le 1) \tag{2.48}$$

is a density matrix. Show that the negativity does not vanish for $p > 1/3$.

EXERCISE 2.6 Verify that

$$\rho_2 = \begin{pmatrix} \frac{p}{2} & 0 & 0 & \frac{p}{2} \\ 0 & \frac{1-p}{2} & \frac{1-p}{2} & 0 \\ 0 & \frac{1-p}{2} & \frac{1-p}{2} & 0 \\ \frac{p}{2} & 0 & 0 & \frac{p}{2} \end{pmatrix} \qquad (0 \le p \le 1) \tag{2.49}$$

is a density matrix. Show that the negativity vanishes only for $p = 1/2$.

2.4.2 Partial Trace and Purification

Let $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$ be a Hilbert space of a bipartite system made of components 1 and 2 and let A be an arbitrary operator acting on $\mathcal{H}$. The **partial trace** of A over $\mathcal{H}_2$ is an operator acting on $\mathcal{H}_1$ defined as

$$A_1 = \mathrm{tr}\,_2 A \equiv \sum_k (I \otimes \langle k|) A (I \otimes |k\rangle). \tag{2.50}$$

Let $\rho = |\psi\rangle\langle\psi| \in \mathcal{S}(\mathcal{H})$ be a density matrix of a pure state $|\psi\rangle$. Suppose we are interested only in the first system and have no access to the second system. Then the partial trace allows us to "forget" about the second system. In other words, the partial trace quantifies our ignorance of the second sytem.

To be more concrete, let us consider a pure state of a two-dimensional system

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|e_1\rangle|e_1\rangle + |e_2\rangle|e_2\rangle),$$

where $\{|e_i\rangle\}$ is an orthonormal basis. The corresponding density matrix is

$$\rho = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix},$$

where the basis vectors are ordered as $\{|e_1\rangle|e_1\rangle, |e_1\rangle|e_2\rangle, |e_2\rangle|e_1\rangle, |e_2\rangle|e_2\rangle\}$. The partial trace of ρ is

$$\rho_1 = \mathrm{tr}\,_2 \rho = \sum_{i=1,2} (I \otimes \langle e_i|) \rho (I \otimes |e_i\rangle) = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{2.51}$$

Note that a pure state $|\psi\rangle$ is mapped to a maximally mixed state ρ_1.

Observe that

$$\mathrm{tr}\,(\rho_1 A) = \mathrm{tr}\,(\rho(A \otimes I)) \tag{2.52}$$

for an observable A acting on the first Hilbert space. The expectation value of A under that state ρ is equally obtained by using ρ_1.

EXERCISE 2.7 Let

$$|\psi'\rangle = \frac{1}{\sqrt{2}}(|e_1\rangle|e_2\rangle - |e_2\rangle|e_1\rangle).$$

Find the corresponding density matrix. Then partial-trace it over the first Hilbert space to find a density matrix of the second system.

We have seen above that the partial trace of a pure state density matrix of a bipartite system over one of the constituent Hilbert spaces yields a mixed state. How about the converse? Given a mixed state density matrix, is it always possible to find a pure state density matrix whose partial trace over the extra Hilbert space yields the given density matrix? The answer is yes and the process to find the pure state is called the **purification**. Let $\rho_1 = \sum_k p_k |\psi_k\rangle\langle\psi_k|$ be a general density matrix of a system 1 with the Hilbert space $\mathcal{H}_1$. Now let us introduce the second Hilbert space $\mathcal{H}_2$ whose dimension is the same as that of $\mathcal{H}_1$. Then formally introduce a normalized vector

$$|\Psi\rangle = \sum_k \sqrt{p_k}|\psi_k\rangle \otimes |\phi_k\rangle, \tag{2.53}$$

where $\{|\phi_k\rangle\}$ is an orthonormal basis of $\mathcal{H}_2$. We find

$$\begin{aligned}\mathrm{tr}\,_2|\Psi\rangle\langle\Psi| &= \sum_{i,j,k}(I \otimes \langle\phi_i|)\left[\sqrt{p_j p_k}|\psi_j\rangle|\phi_j\rangle\langle\psi_k|\langle\phi_k|\right](I \otimes |\phi_i\rangle) \\ &= \sum_k p_k|\psi_k\rangle\langle\psi_k| = \rho_1. \end{aligned}\tag{2.54}$$

Thus it is always possible to purify a mixed state by tensoring an extra Hilbert space of the same dimension as that of the original Hilbert space. It is easy to see, by construction, that purification is far from unique. In fact, there are an infinite number of purifications of a given mixed state density matrix; see Exercise 2.9.

EXERCISE 2.8 Let

$$\rho_1 = \frac{1}{4}\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$$

be a density matrix with a basis $\{|\psi_i\rangle\}$. Find a purification of ρ_1.

EXERCISE 2.9 Let

$$|\Psi\rangle = \sum_k \sqrt{p_k}|\psi_k\rangle \otimes |\phi_k\rangle$$

be a purification of $\rho_1 = \sum_k p_k|\psi_k\rangle\langle\psi_k| \in \mathcal{S}(\mathcal{H})$. Show that

$$|\Psi'\rangle = \sum_k \sqrt{p_k}|\psi_k\rangle \otimes U|\phi_k\rangle$$

is another purification of ρ_1, where U is an arbitrary unitary matrix in $\mathrm{U}(\dim\mathcal{H})$.

2.4.3 Fidelity

It often happens that one has to compare two density matrices and tell how much they are close to each other. An experimentalist, for example, conducts an experiment and then wants to compare the resulting quantum state with the theoretical prediction. A good measure for this purpose is **fidelity**, which we now define [17].

DEFINITION 2.2 Let ρ_1 and ρ_2 be two density matrices belonging to the same state space $\mathcal{S}(\mathcal{H})$. Then the fidelity is defined by

$$F(\rho_1, \rho_2) = \mathrm{tr}\left(\sqrt{\sqrt{\rho_1}\rho_2\sqrt{\rho_1}}\right), \tag{2.55}$$

where $\sqrt{\rho_1}$ is chosen such that all the squre-roots of the eivenvalues are positive-semidefinite.

A few comments are in order.

- Let $\rho_1 = \sum_i p_i |p_i\rangle\langle p_i|$ be the spectral decomposition of ρ_1. Then the requirement in the definition claims that $\sqrt{\rho_1} = \sum_i \sqrt{p_i}|p_i\rangle\langle p_i|$.
- $F(\rho, \rho) = 1$ since

$$F(\rho, \rho) = \mathrm{tr}\left(\sqrt{\sqrt{\rho}\rho\sqrt{\rho}}\right) = \mathrm{tr}\,\rho = 1.$$

- F is non-negative by definition and $F(\rho_1, \rho_2) < 1$ for $\rho_1 \neq \rho_2$. See [17] for the proof.

EXAMPLE 2.7 Let $\rho_1 = |\psi_1\rangle\langle\psi_1|$ and $\rho_2 = |\psi_2\rangle\langle\psi_2|$. We note $\sqrt{\rho_i} = |\psi_i\rangle\langle\psi_i| = \rho_i$ by definition. Then the fidelity for them is

$$\begin{aligned} F(\rho_1, \rho_2) &= \mathrm{tr}\sqrt{|\psi_1\rangle\langle\psi_1||\psi_2\rangle\langle\psi_2||\psi_1\rangle\langle\psi_1|} \\ &= |\langle\psi_1|\psi_2\rangle|\mathrm{tr}\,|\psi_1\rangle\langle\psi_1| = |\langle\psi_1|\psi_2\rangle|. \end{aligned}$$

Let $[\rho_1, \rho_2] = 0$. Then they are simultaneously diagonalizable. Let $\rho_i = \sum_i p_i|i\rangle\langle i|$ and $\rho_2 = \sum_i q_i|i\rangle\langle i|$ be their spectral decompositions, where $\{|i\rangle\}$ is the set of simultaneous eigenvectors, which is assumed to be an orthnormal set. Then the fidelity is

$$F(\rho_1, \rho_2) = \mathrm{tr}\sqrt{\sum_{ijk}\sqrt{p_i p_k}q_j|i\rangle\langle i|j\rangle\langle j|k\rangle\langle k|} = \sum_j \sqrt{p_j q_j}.$$

EXERCISE 2.10 Let U be a unitary operator acting on ρ_1 and ρ_2. Show that

$$F(U\rho_1 U^\dagger, U\rho_2 U^\dagger) = F(\rho_1, \rho_2). \tag{2.56}$$

EXERCISE 2.11 Let

$$\rho_1 = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \rho_2 = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

Find the fidelity $F(\rho_1, \rho_2)$.

Now we are ready to proceed to the world of quantum information and quantum computation. Variations on the themes introduced here and in the previous chapter will appear repeatedly in the following chapters.

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3

Qubits and Quantum Key Distribution

3.1 Qubits

A (Boolean) **bit** assumes two distinct values, 0 and 1. Bits constitute the building blocks of the classical information theory founded by C. Shannon. Quantum information theory, on the other hand, is based on **qubits**.

General references for this chapter are [1] and [2].

3.1.1 One Qubit

A qubit is a (unit) vector in the vector space $\mathbb{C}^2$, whose basis vectors are denoted as

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.1}$$

What these vectors physically mean depends on the physical realization employed for quantum-information processing.

- In some cases, $|0\rangle$ stands for a vertically polarized photon $|\updownarrow\rangle$, while $|1\rangle$ represents a horizontally polarized photon $|\leftrightarrow\rangle$. Alternatively they might correspond to photons polarized in different directions. For example, $|0\rangle$ may represent a polarization state

  $$|\nearrow\!\!\!\!\swarrow\rangle = \frac{1}{\sqrt{2}}(|\updownarrow\rangle + |\leftrightarrow\rangle),$$

  while $|1\rangle$ represents a state

  $$|\nwarrow\!\!\!\!\searrow\rangle = \frac{1}{\sqrt{2}}(|\updownarrow\rangle - |\leftrightarrow\rangle).$$

  Note that if $|\updownarrow\rangle$ ($|\leftrightarrow\rangle$) corresponds to an eigenstate of σ_z with the eigenvalue $+1$ (-1), respectively, then $|\nearrow\!\!\!\!\swarrow\rangle$ ($|\nwarrow\!\!\!\!\searrow\rangle$) corresponds to an eigenstate of σ_x with the eigenvalue $+1$ (-1), respectively.

  Similarly, the states

  $$|\sigma^+\rangle = \frac{1}{\sqrt{2}}(|\updownarrow\rangle + i|\leftrightarrow\rangle), \quad |\sigma^-\rangle = \frac{1}{\sqrt{2}}(|\updownarrow\rangle - i|\leftrightarrow\rangle)$$

correspond to the eigenstates of σ_y with the eigenvalues ± 1 and represent circularly polarized photons.

- They may represent spin states of an electron, $|0\rangle = |\uparrow\rangle$ and $|1\rangle = |\downarrow\rangle$. Electrons are replaced by nuclei with spin 1/2 in NMR quantum computing.

- Truncated two states from many levels may also be employed as a qubit. Take the ground state and the first excited state of ionic energy levels or atomic energy levels, for example. We may assign $|0\rangle$ to the ground state and $|1\rangle$ to the first excited state.

In any case, we have to fix a set of basis vectors when we carry out quantum information processing. All the physics should be described with respect to this basis. In the following, the basis is written in an abstract form as $\{|0\rangle, |1\rangle\}$, unless otherwise stated.

A remark is in order. The third example of a qubit above suggests that a quantum system with more than two states may be employed for information storage and information processing. If a quantum system admits three different states, it is called a **qutrit**, while if it takes d different states, it is called a **qudit**. A spin S particle, for example, takes $d = 2S + 1$ spin states and works as a qudit. The significance of qutrits and qudits in information processing is still to be explored.

It is convenient to assume the vector $|0\rangle$ corresponds to the classical value 0, while $|1\rangle$ to 1 in quantum computation. Moreover it is possible for a qubit to be in a superposition state:

$$|\psi\rangle = a|0\rangle + b|1\rangle \text{ with } \; a, b \in \mathbb{C}, \; |a|^2 + |b|^2 = 1. \tag{3.2}$$

The fundamental requirement of quantum mechanics is that if we make measurement on $|\psi\rangle$ to see whether it is in $|0\rangle$ or $|1\rangle$, the outcome will be 0 (1) with the probability $|a|^2$ ($|b|^2$), and the state immediately after the measurement is $|0\rangle$ ($|1\rangle$).

Although a qubit may take infinitely many different states, it should be kept in mind that we can extract from it as the same amount of information as that of a classical bit. Information can be extracted only through measurements. When we make measurement on a qubit, the state vector "collapses" to the eigenvector that corresponds to the eigenvalue observed. Suppose that a spin is in the state $a|0\rangle + b|1\rangle$. If we observe that the z-component of the spin is $+1/2$, the system immediately after the measurement is *definitely* in the state $|0\rangle$. This happens with probability $\langle\psi|0\rangle\langle 0|\psi\rangle = |a|^2$. The outcome of a measurement on a qubit is always one of the eigenvalues, which we call abstractly 0 and 1, just like for a classical bit. We are tempted to think that by making measurements of a large number of copies of this system, we may be able to determine the coefficients a and b (or, at least, $|a|$ and $|b|$) of the wavefunction. But this is not the case due to the "no-cloning theorem" proved later. It is impossible to duplicate an unknown quantum system with a unitary transformation.

3.1.2 Bloch Sphere

It is useful, for many purposes, to express a state of a single qubit graphically. Let us parameterize a one-qubit state $|\psi\rangle$ with θ and ϕ as

$$|\psi(\theta,\phi)\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle. \tag{3.3}$$

We are not interested in the overall phase, and the phase of $|\psi\rangle$ is fixed in such a way that the coefficient of $|0\rangle$ is real. Now we show that $|\psi(\theta,\phi)\rangle$ is an eigenstate of $\hat{\boldsymbol{n}}(\theta,\phi)\cdot\boldsymbol{\sigma}$ with the eigenvalue $+1$. Here $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ and $\hat{\boldsymbol{n}}(\theta,\phi)$ is a real unit vector called the **Bloch vector** with components

$$\hat{\boldsymbol{n}}(\theta,\phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)^t.$$

In fact, a straightforward calculation shows that

$$\begin{aligned}
\hat{\boldsymbol{n}}(\theta,\phi)\cdot\boldsymbol{\sigma}|\psi(\theta,\phi)\rangle &= \begin{pmatrix} \cos\theta & \sin\theta e^{-i\phi} \\ \sin\theta e^{i\phi} & -\cos\theta \end{pmatrix}\begin{pmatrix} \cos\frac{\theta}{2} \\ e^{i\phi}\sin\frac{\theta}{2} \end{pmatrix} \\
&= \begin{pmatrix} \cos\frac{\theta}{2}\cos\theta + \sin\frac{\theta}{2}\sin\theta \\ e^{i\phi}\left(\cos\frac{\theta}{2}\sin\theta - \cos\theta\sin\frac{\theta}{2}\right) \end{pmatrix} \\
&= \begin{pmatrix} \cos\frac{\theta}{2} \\ e^{i\phi}\sin\frac{\theta}{2} \end{pmatrix} = |\psi(\theta,\phi)\rangle.
\end{aligned}$$

It is therefore natural to assign a unit vector $\hat{\boldsymbol{n}}(\theta,\phi)$ to a state vector $|\psi(\theta,\phi)\rangle$. Namely, a state $|\psi(\theta,\phi)\rangle$ is expressed as a unit vector $\hat{\boldsymbol{n}}(\theta,\phi)$ on the surface of the unit sphere, called the **Bloch sphere**. This correspondence is one-to-one if the ranges of θ and ϕ are restricted to $0 \le \theta \le \pi$ and $0 \le \phi < 2\pi$.

EXERCISE 3.1 Let $|\psi(\theta,\phi)\rangle$ be the state given by Eq. (3.3). Show that

$$\langle\psi(\theta,\phi)|\boldsymbol{\sigma}|\psi(\theta,\phi)\rangle = \hat{\boldsymbol{n}}(\theta,\phi), \tag{3.4}$$

where $\hat{\boldsymbol{n}}$ is the unit vector defined above.

It is possible to express a density matrix ρ of a qubit using a unit ball this time. Since ρ is a positive semi-definite Hermitian matrix with unit trace, its most general form is

$$\rho = \frac{1}{2}\left(I + \sum_{i=x,y,z} u_i\sigma_i\right), \tag{3.5}$$

where u_i are components of a real vector $\boldsymbol{u}$ with $|\boldsymbol{u}| \le 1$. The reality follows from the Hermiticity requirement, and $\operatorname{Tr}\rho = 1$ is easy to check. The eigenvalues of ρ are

$$\lambda_+ = \frac{1}{2}\left(1 + \sqrt{|\boldsymbol{u}|}\right),\ \lambda_- = \frac{1}{2}\left(1 - \sqrt{|\boldsymbol{u}|}\right) \tag{3.6}$$

and therefore non-negative. In case $|\boldsymbol{u}| = 1$, the eigenvalue λ_- vanishes and rank $\rho = 1$. Therefore the surface of the unit ball corresponds to pure states. The converse is also shown easily. In contrast, all the points $\boldsymbol{u}$ inside a unit ball correspond to mixed states. The ball is called the **Bloch ball**, also called the Bloch sphere in a mathematically less strict sense, and the vector $\boldsymbol{u}$ is also called the Bloch vector. The normalized vector $\boldsymbol{n}$ of the Bloch sphere is a special case of $\boldsymbol{u}$ restricted in pure states.

EXERCISE 3.2 Find the density matrix of a pure state (3.3) and write it in the form of Eq. (3.5).

EXERCISE 3.3 Let ρ be given by Eq. (3.5). Show that

$$\langle \boldsymbol{\sigma} \rangle = \mathrm{tr}\,(\rho \boldsymbol{\sigma}) = \boldsymbol{u}. \tag{3.7}$$

3.1.3 Multi-Qubit Systems and Entangled States

Let us consider a group of many (n) qubits next. Such a system behaves quite differently from a classical one, and this difference gives a distinguishing aspect to quantum information theory. An n-qubit system is often called a (quantum) **register** in the context of quantum computing.

Consider a classical system made of several components. The state of this system is completely determined by specifying the state of each component. This is *not* the case for a quantum system. A quantum system made of many components is not necessarily described by specifying the state of each component as we have learned in §2.3.

As an example, let us consider an n-qubit register. Suppose we specify the state of each qubit separately in analogy with a classical case. Each of the qubits is then described by a two-dimensional complex vector of the form $a_i|0\rangle + b_i|1\rangle$, and we need $2n$ complex numbers $\{a_i, b_i\}_{1\le i\le n}$ to specify the state. This corresponds the the tensor product state

$$(a_1|0\rangle + b_1|1\rangle) \otimes (a_2|0\rangle + b_2|1\rangle) \otimes \ldots \otimes (a_n|0\rangle + b_n|1\rangle)$$

introduced in §2.3. If the system is treated in a fully quantum-mechanical way, however, we have to include superposition of such tensor product states, which is not necessarily decomposable into a tensor product form. Such a state is **entangled** (see §2.3). A general state vector of the register is represented as

$$|\psi\rangle = \sum_{i_k=0,1} a_{i_1 i_2 \ldots i_n} |i_1\rangle \otimes |i_2\rangle \otimes \ldots \otimes |i_n\rangle$$

and lives in a 2^n-dimensional complex vector space. Note that $2^n \gg 2n$ for a large number n. The ratio $2^n/2n$ is $\sim 6.3 \times 10^{27}$ for $n = 100$ and $\sim 5.4 \times 10^{297}$ for $n = 1000$. These astronomical numbers tell us that most quantum states in a Hilbert space with large n are entangled, i.e., they do not have classical analogy which tensor product states have. Entangled states that have no classical counterparts are extremely powerful resources for quantum computation and quantum communication as we will show later.

Let us consider a system of two qubits for definiteness. The combined system has a basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. More generally, a basis for a system of n qubits may be taken to be $\{|b_{n-1}b_{n-2}\ldots b_0\rangle\}$, where $b_{n-1}, b_{n-2}, \ldots, b_0 \in \{0, 1\}$. It is also possible to express the basis in terms of the decimal system. We write $|x\rangle$, instead of $|b_{n-1}b_{n-2}\ldots b_0\rangle$, where $x = b_{n-1}2^{n-1} + b_{n-2}2^{n-2} + \ldots + b_0$ is the decimal expression of the binary number $b_{n-1}b_{n-2}\ldots b_0$. Thus the basis for a two-qubit system may be written also as $\{|0\rangle, |1\rangle, |2\rangle, |3\rangle\}$ with this decimal notation. Whether the binary system or the decimal system is employed should be clear from the context. An n-qubit system has $2^n = \exp(n \ln 2)$ basis vectors.

The set

$$\begin{aligned}\{|\Phi^+\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad |\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle), \\ |\Psi^+\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \quad |\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)\}\end{aligned} \tag{3.8}$$

is an orthonormal basis of a two-qubit system and is called the **Bell basis**. Each vector is called the **Bell state** or the **Bell vector**. Note that all the Bell states are entangled.

EXERCISE 3.4 The Bell basis is obtained from the binary basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ by a unitary transformation. Write down the unitary transformation explicitly.

Among three-qubit entangled states, the following two states are important for various reasons and hence deserve special names. The state

$$|\mathrm{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle) \tag{3.9}$$

is called the **Greenberger-Horne-Zeilinger state** and is often abbreviated as the **GHZ state**[3]. Another important three-qubit state is the **W state** [4],

$$|\mathrm{W}\rangle = \frac{1}{\sqrt{3}}(|100\rangle + |010\rangle + |001\rangle). \tag{3.10}$$

EXERCISE 3.5 Find the expectation value of $\sigma_x \otimes \sigma_z$ measured in each of the Bell states.

3.1.4 Measurements

Classical information theory is formulated independently of measurements of the system under consideration. This is because the readout of the result is always the same for anyone and at any time, provided that the system processes the same information. This is completely different in quantum information processing. Measurement is an essential part of the theory as we see below.

By making a measurement on a system, we *project* the state vector to one of the basis vectors that the measurement equipment defines.* Suppse we have a state vector $|\psi\rangle = a|0\rangle + b|1\rangle$ and measure it to see if it is in the state $|0\rangle$ or $|1\rangle$. Depending on the system, this means if a spin points up or down or a photon is polarized horizontally or vertically, for example. The result is either 0 or 1. In the first case, the state "collapses" to $|0\rangle$ while in the second case, to $|1\rangle$. We find, after many measurements, the probability of obtaining outcome 0 (1) is $|a|^2$ ($|b|^2$).

To be more formal, we construct a **measurement operator** M_m such that the probability of obtaining the outcome m in the state $|\psi\rangle$ is

$$p(m) = \langle\psi|M_m^\dagger M_m|\psi\rangle, \tag{3.11}$$

and the state immediately after the measurement is

$$|m\rangle = \frac{M_m|\psi\rangle}{\sqrt{p(m)}}. \tag{3.12}$$

In the above example, the measurement operators are nothing but projection operators; $M_0 = |0\rangle\langle 0|$ and $M_1 = |1\rangle\langle 1|$. In fact, we have

$$p(0) = \langle\psi|M_0^\dagger M_0|\psi\rangle = \langle\psi|0\rangle\langle 0|\psi\rangle = |a|^2,$$

and

$$\frac{M_0|\psi\rangle}{\sqrt{p(0)}} = \frac{a}{|a|}|0\rangle \simeq |0\rangle,$$

and similarly for the other case M_1. It should be noted that a quantum state is defined up to a phase and hence $a/|a|$ does not play any role. [*Remark*: See [2] for the difference between a general measurement operator and a projective measurement operator.]

Suppose we are given many copies of a particular state $|\psi\rangle$. If we measure an observable M in each of the copies, the expectation value of M is given, in terms of the projection operators, by

$$\begin{aligned} E(M) &= \sum_m m\, p(m) = \sum_m m\langle\psi|P_m|\psi\rangle \\ &= \langle\psi|\sum_m mP_m|\psi\rangle = \langle\psi|M|\psi\rangle, \end{aligned} \tag{3.13}$$

*This is called a projective measurement as was noted in Chapter 2. We will be concerned only with projective measurements, unless stated otherwise.

where use has been made of the spectral decomposition $M = \sum mP_m$. The standard deviation is given by

$$\Delta(M) = \sqrt{\langle (M - \langle M \rangle)^2 \rangle} = \sqrt{\langle M^2 \rangle - \langle M \rangle^2}. \tag{3.14}$$

Let us analyze measurements in a two-qubit system in some detail. An arbitrary state is written as

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle, \quad |a|^2 + |b|^2 + |c|^2 + |d|^2 = 1,$$

where $a, b, c, d \in \mathbb{C}$. We make a measurement of the first qubit with respect to the basis $\{|0\rangle, |1\rangle\}$. To this end, we rewrite the state as

$$\begin{aligned}
&a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle \\
&= |0\rangle \otimes (a|0\rangle + b|1\rangle) + |1\rangle \otimes (c|0\rangle + d|1\rangle) \\
&= u|0\rangle \otimes \left(\frac{a}{u}|0\rangle + \frac{b}{u}|1\rangle\right) + v|1\rangle \otimes \left(\frac{c}{v}|0\rangle + \frac{d}{v}|1\rangle\right),
\end{aligned}$$

where $u = \sqrt{|a|^2 + |b|^2}$ and $v = \sqrt{|c|^2 + |d|^2}$. The measurement operators acting on the first qubit are

$$M_0 = |0\rangle\langle 0| \otimes I, \quad M_1 = |1\rangle\langle 1| \otimes I. \tag{3.15}$$

Note that we need to specify $\otimes I$ explicitly since we are working in a two-qubit Hilbert space $\mathbb{C}^4$. Upon a measurement of the first qubit, we obtain 0 with the probability

$$\langle \psi | M_0 | \psi \rangle = u^2 = |a|^2 + |b|^2,$$

projecting the state to

$$\frac{M_0|\psi\rangle}{\sqrt{p(0)}} = |0\rangle \otimes \left(\frac{a}{u}|0\rangle + \frac{b}{u}|1\rangle\right),$$

while we obtain $|1\rangle$ with the probability $v^2 = |c|^2 + |d|^2$, projecting the state to $|1\rangle \otimes \left(\frac{c}{v}|0\rangle + \frac{d}{v}|1\rangle\right)$. Note that the state after the measurement has unit norm in both cases. The measurement of the second qubit can be carried out similarly. Measurements on an n-qubit system can be carried out by repeating one-qubit measurement n times.

In the two-qubit example above, the Hilbert space for the system is separated into a direct sum of $\mathcal{H}_0$, where the first qubit is in the state $|0\rangle$, and $\mathcal{H}_1$, where it is in $|1\rangle$: $\mathcal{H} = \mathcal{H}_0 \oplus \mathcal{H}_1$. An arbitrary two-qubit state $|\psi\rangle$ is uniquely decomposed into two vectors, each of which belongs to $\mathcal{H}_0$ or $\mathcal{H}_1$ as

$$(|0\rangle\langle 0| \otimes I)|\psi\rangle \in \mathcal{H}_0, \quad (|1\rangle\langle 1| \otimes I)|\psi\rangle \in \mathcal{H}_1,$$

where normalization has been ignored. More generally, an observation of k qubits in an n-qubit system yields 2^k possible outcomes m_i $(1 \le i \le 2^k)$.

Accordingly, the 2^n-dimensional Hilbert space of the system is separated into the direct sum of mutually orthogonal subspaces $\mathcal{H}_{m_1}, \mathcal{H}_{m_2}, \ldots, \mathcal{H}_{m_{2^k}}$ as $\mathcal{H} = \mathcal{H}_{m_1} \oplus \mathcal{H}_{m_2} \oplus \ldots \oplus \mathcal{H}_{m_{2^k}}$. When the result of the observation of the k qubits is m_i, the state after the observation is projected to the subspace $\mathcal{H}_{m_i}$. It should be clear from the construction that each subspace $\mathcal{H}_{m_i}$ has dimension $2^n/2^k = 2^{n-k}$. The measurement device projects the state before the observation

$$|\psi\rangle = c_{m_1}|\psi_{m_1}\rangle + c_{m_2}|\psi_{m_2}\rangle + \ldots + c_{m_{2^k}}|\psi_{m_{2^k}}\rangle, \quad (|\psi_{m_i}\rangle \in \mathcal{H}_{m_i})$$

into one of the subspaces $\mathcal{H}_{m_i}$ randomly with the probability $|c_{m_i}|^2$.

Measurement gives an alternative viewpoint to entangled states. A state is not entangled if a measurement of a qubit does not affect the state of the other qubits. Suppose the first qubit of the state

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

was measured to be 0 (1). Then the outcome of the measurement of the second qubit is *definitely* 0 (1). Therefore the measurement of the first qubit affects the outcome of the measurement on the second qubit, which shows that the initial state is an entangled state. In other words, there exists a strong correlation between the two qubits. This correlation may be used for information processing as will be shown later. In contrast with this, the state $\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle)$ is not entangled since it can be written as

$$\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle) = |0\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle).$$

Irrespectively of the measurement of the second qubit, the measurement of the first qubit definitely yields 0. Moreover, the second qubit is measured to be 0 (1) with the probability 1/2, independently of whether the first qubit is measured or not.

EXERCISE 3.6 In many quantum algorithms, the result of an action of a function f on x is encoded into the form

$$U_f : \mathcal{N}\sum_x |x\rangle|0\rangle \to \mathcal{N}\sum_x |x\rangle|f(x)\rangle,$$

where $|x\rangle$ stands for the tensor product state $|b_{n-1}b_{n-2}\ldots b_0\rangle$ with $x = b_{n-1}2^{n-1} + b_{n-2}2^{n-2} + \ldots b_0$ and $\mathcal{N}$ is the normalization constant. The first register is for the input x, while the second one is for the corresponding output $f(x)$. Note that U_f acts on *all* possible states simultaneously.

Let $f(x) = a^x \bmod N$, where a and N are coprime, and consider the state

$$U_f\left[\frac{1}{\sqrt{512}}\sum_{x=0}^{511}|x\rangle|0\rangle\right] = \frac{1}{\sqrt{512}}\sum_{x=0}^{511}|x\rangle|a^x \bmod N\rangle$$

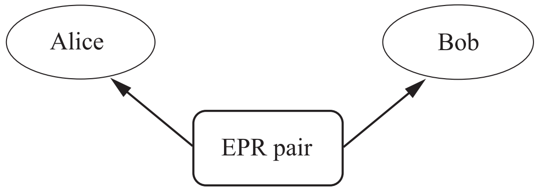

FIGURE 3.1
EPR pair produced by a source in the middle. One particle is sent to Alice and the other to Bob.

with $a = 6$ and $N = 91$. Suppose the measurement of the first register results in (1) $x = 11$, (2) $x = 23$ and (3) $x = 35$. What is the state immediately after each measurement?

3.1.5 Einstein-Podolsky-Rosen (EPR) Paradox

Einstein, Podolsky and Rosen (EPR) proposed a Gedanken experiment which, at first glance, shows that an entangled state violates an axiom of the special theory of relativity [5]. Suppose a particle source produces the so-called EPR pair in the state

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

and it sends one particle to Alice and the other to Bob, who may be separated far away (see Fig. 3.1).[†] Alice measures her particle and obtains her reading $|0\rangle$ or $|1\rangle$. Depending on her reading, the EPR state is projected to $|01\rangle$ ($|10\rangle$), and Bob will *definitely* observe $|1\rangle$ ($|0\rangle$) in his measurement. The change of the state

$$\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \rightarrow |01\rangle \quad \text{or} \quad |10\rangle \tag{3.16}$$

takes place instantaneously even when they are separated by a large distance. It seems that Alice's measurement propageted to Bob's qubit instantaneously, and it violates the special theory of relativity. This is the very point EPR proposed to defeat quantum mechanics.

Note, however, that nothing has propagated from Alice to Bob and *vice versa*, upon Alice's measurement. Clearly no energy has propagated. What about information? It is impossible for Alice to control her and hence Bob's readings. Therefore it is impossible to use EPR pairs to send a sensible message from Alice to Bob. If they could, the message would be sent instantaneously, which certainly violates the special theory of relativity. If a large number of EPR pairs are sent to Alice and Bob and they independently

[†]Alice and Bob are names frequently used in information theory.

measure their qubits, they will observe random sequences of 0 and 1. They notice that their readings are strongly correlated only after they exchange their sequences by means of classical communication, which can be done at most with the speed of light.

3.2 Quantum Key Distribution (BB84 Protocol)

A large number of qubits and gate operations on them are involved in most practical applications of quantum computing. We will postpone these applications in the subsequent chapters. Is there any practical use of single qubits then? There is a suprisingly secure way of distributing a cryptographic key using a sequence of individual qubits [6] called BB84 protocol, which is already available commercially [7]. **Quantum key distribution** (QKD) is a secure way of distributing an encryption and decryption key by making use of qubits. The sender and the receiver can detect a possible third party eavesdropping their communication by comparing the sequence sent with that of the received one.

One-time pad is an absolutely secure cryptosystem if and only if the key for encoding and decoding is shared only between the sender and the receiver and used once for all. Suppose we want to send a message 1100101001 in a binary form using a key 1001010011, for example. The message is encrypted by adding the message to the key bitwise modulo 2, which we denote by $i \oplus j$. We have explicitly $0 \oplus 0 = 0$, $0 \oplus 1 = 1$, $1 \oplus 0 = 1$, $1 \oplus 1 = 0$. For the above case, we have the encrypted message 0101111010. For decryption, the receiver is required only to add the same key bitwise again since $(i \oplus j) \oplus j = i$. Decryption of an encrypted message is impossible without the key since there are 2^n possible keys for an n-bit string and many of them yield sensible messages. This cryptosystem is not secure any more if the same key is used many times.

A key must be sent from the sender to the receiver, or in the opposite direction, each time this cryptosystem is used. If the key is sent through a classical communication channel, there always exists a possibility of eavesdropping. However, this problem is completely solved if a quantum channel is employed as we show now. Suppose Alice wants to send Bob a one-time pad key to encode and decode her secret message. They can communicate with each other using a bidirectional classical channel. There also exists a quantum channel that is unidirectional from Alice to Bob. See Fig. 3.2. There is a possibility that their communication is being eavesdropped by a third party, which we call Eve. Alice sends Bob many qubits, one by one, and Bob measures the states of each of the qubits he receives. To make our discussion concrete, we assume qubits are made of polarized photons.

Alice employs two coding systems when she sends photons to Bob. The

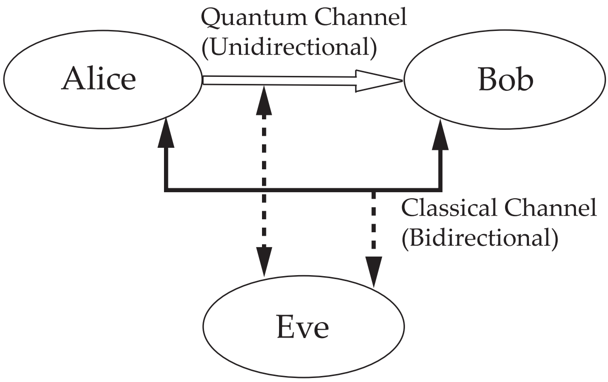

FIGURE 3.2
Quantum key distribution protocol BB84.

coding systems are

$$\begin{array}{lll} \text{coding system (1)} & 0 \mapsto |\updownarrow\rangle, & 1 \mapsto |\leftrightarrow\rangle, \\ \text{coding system (2)} & 0 \mapsto |\searrow\rangle, & 1 \mapsto |\nearrow\rangle. \end{array}$$

They are chosen *at random* for each photon. Bob also chooses coding systems (1) and (2) randomly, independently of Alice, to measure the polarization of each photon Alice sends. Suppose $4N$ photons are sent from Alice to Bob. After all the photons have been sent, Alice and Bob exchange the sequence of the coding systems they employed using the classical communication channel (so this is not 100 % secure), without disclosing the bits (0 or 1) Alice sent and Bob received. They will know, as a result, for which photons they employed the same coding systems. They discard all the cases for which they employed different coding systems since the sent bits and the received bits agree only with probability 1/2 in these cases. Now $\sim 2N$ photons, on average, should be correctly transmitted and they share $\sim 2N$ bits of binary numbers in their hands. To make sure that no one eavesdrops their quantum channel, they choose N cases randomly out of $2N$ cases with the same coding systems employed and exchange N bits (0 or 1) associated with these N cases over the classical channel. If there are no eavesdroppers operating, they should have the same bits for all the N cases. After verifying that they are free from eavesdroppers, they discard these N cases (since the classical channel may be eavesdropped) and use the remaining N bits to generate a one-time pad key.

Suppose Eve is in action. After eavesdropping the photons, she immediately sends Bob her results in order to hide her presence. Note that Bob will immediately notice the presence of an eavesdropper from missing photons unless Eve sends some photons to Bob. Eve's coding system is different from Alice's also with probability 1/2, and she sends Bob the results of her measurement with the same coding system as she used to eavesdrop. Then there exist cases in which the photon Alice sends disagrees with the one Bob receives, even when Alice and Bob employ the same coding system. This happens with probability 1/4 as is shown now. Suppose both Alice and Bob happen to employ the coding system (1) and Alice sends Bob 0. Eve will use the coding system (1) with probability 1/2, in which case Eve measures 0 and sends Bob $|\updownarrow\rangle$. Bob, also employing the coding system (1), will obtain 0 with probability 1. If, in contrast, Eve employs coding system (2), which happens with probability 1/2, then Eve measures 0 or 1 with probability 1/2 for each photon and sends Bob her result with coding system (2). Then Bob, with coding system (1), will obtain 0 or 1 both with probability 1/2. In the end of the day, Bob obtains 0 with probability 3/4 and 1 with probability 1/4, even though Alice and Bob employ the same coding system. Suppose $4N$ photons are sent from Alice to Bob, as before. They find their codings agree in $2N$ cases and discard the remaining cases. Comparing N cases to check if Eve is in action, they find approximately $N/4$ bits disagree, from which they detect there is an eavesdropper in the quantum channel. They may try different quantum channels until they find one whose security is verified.

EXAMPLE 3.1 Suppose the sent and received sequences are

$$\begin{array}{cccccccccccccc}
\text{Alice sends} & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & \ldots \\
\text{Alice's code} & (1) & (2) & (1) & (2) & (2) & (1) & (2) & (1) & (2) & (2) & (1) & (1) & \ldots \\
\text{Bob's code} & (1) & (2) & (2) & (1) & (2) & (2) & (1) & (2) & (1) & (2) & (2) & (1) & \ldots \\
\text{Bob reads} & 0 & 1 & ? & ? & 1 & ? & ? & ? & ? & 0 & ? & 0 & \ldots
\end{array} \tag{3.17}$$

where ? stands for 0 or 1. Alice and Bob keep the sequence $0, 1, 1, 0, 0, \ldots$ to check the security of the channel and to generate a key.

Suppose Eve eavesdrops their communication. Then their readings may, for example, be

$$\begin{array}{cccccccccccccc}
\text{Alice sends} & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & \ldots \\
\text{Alice's code} & (1) & (2) & (1) & (2) & (2) & (1) & (2) & (1) & (2) & (2) & (1) & (1) & \ldots \\
\text{Eve's code} & (1) & (2) & (1) & (2) & (1) & (2) & (1) & (2) & (1) & (2) & (1) & (2) & \ldots \\
\text{Eve reads} & 0 & 1 & 0 & 0 & ? & ? & ? & ? & ? & 0 & 1 & ? & \ldots \\
\text{Bob's code} & (1) & (2) & (2) & (1) & (2) & (2) & (1) & (2) & (1) & (2) & (2) & (1) & \ldots \\
\text{Bob reads} & 0 & 1 & ? & ? & \underline{?} & ? & ? & ? & ? & 0 & ? & \underline{?} & \ldots
\end{array} \tag{3.18}$$

The 5th and 12th bits Bob obtains may not be the correct ones, even though Alice and Bob employ the same coding system.

Other QKD protols are E91 protocol [8] and B92 protocol [9].

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4

Quantum Gates, Quantum Circuit and Quantum Computation

4.1 Introduction

Now that we have introduced qubits to store information, it is time to consider operations acting on them. If they are simple, these operations are called gates, or more precisely **quantum gates**, in analogy with those in classical logic circuits. More complicated **quantum circuits** are composed of these simple gates. A collection of quantum circuits for executing a complicated algorithm, a **quantum algorithm**, is a part of a quantum computation.

DEFINITION 4.1 (**Quantum Computation**) A quantum computation is a collection of the following three elements:

(1) A register or a set of registers,

(2) A unitary matrix U, which is taylored to execute a given quantum algorithm, and

(3) Measurements to extract information we need.

More formally, we say a quantum computation is the set $\{\mathcal{H}, U, \{M_m\}\}$, where $\mathcal{H} = \mathbb{C}^{2^n}$ is the Hilbert space of an n-qubit register, $U \in \mathrm{U}(2^n)$ represents the quantum algorithm and $\{M_m\}$ is the set of measurement operators.
The hardware (1) along with equipment to control the qubits is called a quantum computer.

Suppose the register is set to a fiducial initial state, $|\psi_{\text{in}}\rangle = |00\ldots 0\rangle$, for example. A unitary matrix U_{alg} is designed to represent an algorithm which we want to execute. Operation of U_{alg} on $|\psi_{\text{in}}\rangle$ yields the output state $|\psi_{\text{out}}\rangle = U_{\text{alg}}|\psi_{\text{in}}\rangle$. Information is extracted from $|\psi_{\text{out}}\rangle$ by appropriate measurements.
Actual quantum computation processes are very different from those of a classical counterpart. In a classical computer, we input the data from a keyboard or other input devices and the signal is sent to the I/O port of the computer, which is then stored in the memory, then fed into the microprocessor, and the result is stored in the memory before it is printed or it is

displayed on the screen. Thus information travels around the circuit. In contrast, information in quantum computation is stored in a register, first of all, and then external fields, such as oscillating magnetic fields, electric fields or laser beams are applied to produce gate operations on the register. These external fields are designed so that they produce desired gate operation, i.e., unitary matrix acting on a particular set of qubits. Therefore the information sits in the register and they are updated each time the gate operation acts on the register.

One of the other distinctions between classical computation and quantum computation is that the former is based upon digital processing and the latter upon hybrid (digital + analogue) processing. A qubit may take an arbitrary superposition of $|0\rangle$ and $|1\rangle$, and hence their coefficients are continuous complex numbers. A gate is also an element of a relevant unitary group, which contains continuous parameters. An operation such as "rotate a specified spin around the x-axis by an angle π" is implemented by applying a particular pulse of specified amplitude, angle and duration. These parameters are continuous numbers and always contain errors. These aspects might cause challenging difficulties in a physical realization of a quantum computer.

Parts of this chapter depend on [1, 2] and [3].

4.2 Quantum Gates

We have so far studied the change of a state upon measurements. When measurements are not made, the time evolution of a state is described by the Schrödinger equation. The system preserves the norm of the state vector during time evolution. Thus the time development is unitary. Let U be such a time-evolution operator; $UU^\dagger = U^\dagger U = I$. We will be free from the Schrödinger equation in the following and assume there exist unitary matrices which we need. Physical implementation of these unitary matrices is another important area of quantum information processing and is a subject of the second part of this book.

One of the important conclusions derived from the unitarity of gates is that the computational process is reversible.

4.2.1 Simple Quantum Gates

Examples of quantum gates which transform a one-qubit state are given below. We call them one-qubit gates in the following. Linearity guarantees that the action of a gate is completely specified as soon as its action on the basis $\{|0\rangle,\ |1\rangle\}$ is given. Let us consider the gate I whose action on the basis vectors are defined by $I:\ |0\rangle \to |0\rangle,\ |1\rangle \to |1\rangle$. The matrix expression of this gate is easily found as

$$I = |0\rangle\langle 0| + |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{4.1}$$

Similarly we introduce $X: \ |0\rangle \to |1\rangle, \ |1\rangle \to |0\rangle$, $Y: \ |0\rangle \to -|1\rangle, \ |1\rangle \to |0\rangle$, and $Z: \ |0\rangle \to |0\rangle, \ |1\rangle \to -|1\rangle$, whose matrix representations are

$$X = |1\rangle\langle 0| + |0\rangle\langle 1| = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \sigma_x, \tag{4.2}$$

$$Y = |0\rangle\langle 1| - |1\rangle\langle 0| = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -i\sigma_y, \tag{4.3}$$

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma_z. \tag{4.4}$$

The transformation I is the trivial (identity) transformation, while X is the negation (NOT), Z the phase shift and $Y = XZ$ the combination of them. It is easily verified that these gates are unitary.

The **CNOT** (**controlled-NOT**) gate is a two-qubit gate, which plays quite an important role in quantum computation. The gate flips the second qubit (the **target qubit**) when the first qubit (the **control qubit**) is $|1\rangle$, while leaving the second bit unchanged when the first qubit state is $|0\rangle$. Let $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ be a basis for the two-qubit system. In the following, we use the standard basis vectors with components

$$|00\rangle = (1,0,0,0)^t, \ |01\rangle = (0,1,0,0)^t, \ |10\rangle = (0,0,1,0)^t, \ |11\rangle = (0,0,0,1)^t.$$

The action of the CNOT gate, whose matrix expression will be written as U_{CNOT}, is

$$U_{\text{CNOT}} : |00\rangle \mapsto |00\rangle, \ |01\rangle \mapsto |01\rangle, \ |10\rangle \mapsto |11\rangle, \ |11\rangle \mapsto |10\rangle.$$

It has two equivalent expressions

$$\begin{aligned} U_{\text{CNOT}} &= |00\rangle\langle 00| + |01\rangle\langle 01| + |11\rangle\langle 10| + |10\rangle\langle 11| \\ &= |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X, \end{aligned} \tag{4.5}$$

having a matrix form

$$U_{\text{CNOT}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{4.6}$$

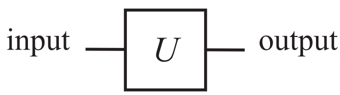
The second expression of the RHS in Eq. (4.5) shows that the action of U_{CNOT} on the target qubit is I when the control qubit is in the state $|0\rangle$, while it is σ_x when the control qubit is in $|1\rangle$. Verify that U_{CNOT} is unitary and, moreover, idempotent, i.e., $U_{\text{CNOT}}^2 = I$.

Let $\{|i\rangle\}$ be the basis vectors, where $i \in \{0, 1\}$. The action of CNOT on the input state $|i\rangle|j\rangle$ is written as $|i\rangle|i \oplus j\rangle$, where $i \oplus j$ is an addition mod 2, that is, $0 \oplus 0 = 0, 0 \oplus 1 = 1, 1 \oplus 0 = 1$ and $1 \oplus 1 = 0$.

EXERCISE 4.1 Show that the U_{CNOT} cannot be written as a tensor product of two one-qubit gates.

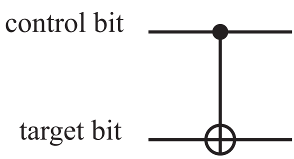
EXERCISE 4.2 Let $(a|0\rangle + b|1\rangle) \otimes |0\rangle$ be an input state to a CNOT gate. What is the output state?

It is convenient to introduce graphical representations of quantum gates. A one-qubit gate whose unitary matrix representation is U is depicted as



The left horizontal line is the input qubit state, while the right horizontal line is the output qubit state. Therefore the time flows from the left to the right.

A CNOT gate is expressed as

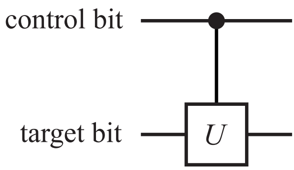


where $\bullet$ denotes the control bit, while $\bigoplus$ denotes the conditional negation. There may be many control bits (see CCNOT gate below).

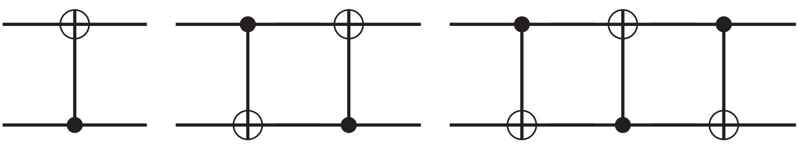
More generally, we consider a controlled-U gate,

$$V = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U, \tag{4.7}$$

in which the target bit is acted on by a unitary transformation U only when the control bit is $|1\rangle$. This gate is denoted graphically as



EXERCISE 4.3 (1) Find the matrix representation of the "upside down" CNOT gate (a) in the basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$.



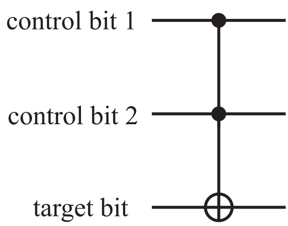
(a) (b) (c)

(2) Find the matrix representation of the circuit (b).
(3) Find the matrix representation of the circuit (c). Find the action of the circuit on a tensor product state $|\psi_1\rangle \otimes |\psi_2\rangle$.

The **CCNOT** (**Controlled-Controlled-NOT**) gate has three inputs, and the third qubit flips when and only when the first two qubits are both in the state $|1\rangle$. The explicit form of the CCNOT gate is

$$U_{\mathrm{CCNOT}} = (|00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10|) \otimes I + |11\rangle\langle 11| \otimes X. \tag{4.8}$$

This gate is graphically expressed as



The CCNOT gate is also known as the **Toffoli gate**.

4.2.2 Walsh-Hadamard Transformation

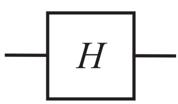
The **Hadamard gate** or the **Hadamard transformation** H is an important unitary transformation defined by

$$\begin{aligned} U_{\mathrm{H}} : |0\rangle &\to \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \\ |1\rangle &\to \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle). \end{aligned} \tag{4.9}$$

It is used to generate a superposition state from $|0\rangle$ or $|1\rangle$. The matrix representation of H is

$$U_{\mathrm{H}} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\langle 0| + \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\langle 1| = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{4.10}$$

A Hadamard gate is depicted as



There are numerous important applications of the Hadamard transformation. All possible 2^n states are generated, when U_{H} is applied on each qubit of the state $|00\dots0\rangle$:

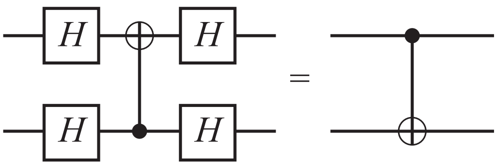
$$
\begin{aligned}
&(H \otimes H \otimes \dots \otimes H)|00\dots0\rangle \\
&= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \dots \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \\
&= \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle. \qquad (4.11)
\end{aligned}
$$

Therefore, we produce a superposition of all the states $|x\rangle$ with $0 \leq x \leq 2^n - 1$ simultaneously. This action of H on an n-qubit system is called the **Walsh transformation**, or **Walsh-Hadamard transformation**, and denoted as W_n. Note that

$$W_1 = U_{\mathrm{H}}, \quad W_{n+1} = U_{\mathrm{H}} \otimes W_n. \qquad (4.12)$$

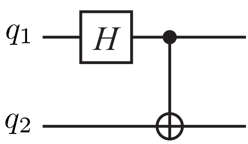
EXERCISE 4.4 Show that W_n is unitary.

EXERCISE 4.5 Show that the two circuits below are equivalent:



This exercise shows that the control bit and the target bit in a CNOT gate are interchangeable by introducing four Hadamard gates.

EXERCISE 4.6 Let us consider the following quantum circuit



(4.13)

where q_1 denotes the first qubit, while q_2 denotes the second. What are the outputs for the inputs $|00\rangle, |01\rangle, |10\rangle$ and $|11\rangle$?

4.2.3 SWAP Gate and Fredkin Gate

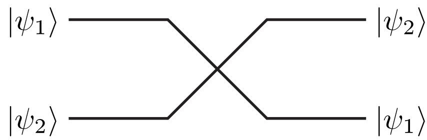
The SWAP gate acts on a tensor product state as

$$U_{\mathrm{SWAP}}|\psi_1, \psi_2\rangle = |\psi_2, \psi_1\rangle. \qquad (4.14)$$

The explict form of U_{SWAP} is given by

$$U_{\mathrm{SWAP}} = |00\rangle\langle 00| + |01\rangle\langle 10| + |10\rangle\langle 01| + |11\rangle\langle 11| = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{4.15}$$

Needless to say, it works as a linear operator on a superposition of states. The SWAP gate is expressed as



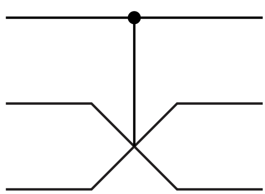
Note that the SWAP gate is a special gate which maps an arbitrary tensor product state to a tensor product state. In contrast, most two-qubit gates map a tensor product state to an entangled state.

EXERCISE 4.7 Show that the above U_{SWAP} is written as

$$U_{\mathrm{SWAP}} = (|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X)(I \otimes |0\rangle\langle 0| + X \otimes |1\rangle\langle 1|) (|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X). \tag{4.16}$$

This shows that the SWAP gate is implemented with three CNOT gates as given in Exercise 4.3 (3).

The controlled-SWAP gate



is also called the **Fredkin gate**. It flips the second (middle) and the third (bottom) qubits when and only when the first (top) qubit is in the state $|1\rangle$. Its explicit form is

$$U_{\mathrm{Fredkin}} = |0\rangle\langle 0| \otimes I_4 + |1\rangle\langle 1| \otimes U_{\mathrm{SWAP}}. \tag{4.17}$$

4.3 Correspondence with Classical Logic Gates

Before we proceed further, it is instructive to show that all the elementary logic gates, NOT, AND, XOR, OR and NAND, in classical logic circuits can be implemented with quantum gates. In this sense, quantum information processing contains the classical one.

4.3.1 NOT Gate

Let us consider the **NOT gate** first. It is defined by the following logic function,

$$\mathrm{NOT}(x) = \neg x = \begin{cases} 0 & x = 1 \\ 1 & x = 0 \end{cases} \tag{4.18}$$

where $\neg x$ stands for the **negation** of x. Under the correspondence $0 \leftrightarrow |0\rangle$, $1 \leftrightarrow |1\rangle$, we have already seen in Eq. (4.2) that the gate X negates the basis vectors as

$$X|x\rangle = |\neg x\rangle = |\mathrm{NOT}(x)\rangle, \ (x = 0, 1). \tag{4.19}$$

Now let us measure the output state. We employ the following measurement operator:

$$M_1 = |1\rangle\langle 1|. \tag{4.20}$$

M_1 has eigenvalues 0 and 1 with the eigenvectors $|0\rangle$ and $|1\rangle$, respectively. When the input is $|0\rangle$, the output is $|1\rangle$ and the measurement gives the value 1 with the probability 1. If, on the other hand, the input is $|1\rangle$, the output is $|0\rangle$ and the measurement yields 1 with probability 0, or in other words, it yields 0 with probability 1. It should be kept in mind that the operator X acts on an arbitrary linear combination $|\psi\rangle = a|0\rangle + b|1\rangle$, which is classically impossible. The output state is then $X|\psi\rangle = a|1\rangle + b|0\rangle$.

We show in the following that the CCNOT gate implements all classical logic gates. The first and the second input qubits are set to $|1\rangle$ to obtain the NOT gate as

$$U_{\mathrm{CCNOT}}|1, 1, x\rangle = |1, 1, \neg x\rangle. \tag{4.21}$$

4.3.2 XOR Gate

Since a quantum gate has to be reversible, we cannot construct a unitary gate corresponding to the classcial **XOR gate** whose function is $x, y \mapsto x \oplus y$ ($x, y \in \{0, 1\}$), where $x \oplus y$ is an addition mod 2; $0 \oplus 0 = 0$, $0 \oplus 1 = 1 \oplus 0 = 1$, $1 \oplus 1 = 0$. Clearly this operation has no inverse. This operation may be made reversible if we keep the first bit x during the gate operation, namely, if we define

$$f(x, y) = (x, x \oplus y), \quad x, y \in \{0, 1\}. \tag{4.22}$$

We call this function f, also the XOR gate. The quantum gate that does this operation is nothing but the CNOT gate defined by Eq. (4.5),

$$U_{\mathrm{XOR}} = U_{\mathrm{CNOT}} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X. \tag{4.23}$$

Note that the XOR gate may be also obtained from the CCNOT gate. Suppose the first qubit of the CCNOT gate is fixed to $|1\rangle$. Then it is easy to verify that

$$U_{\text{CCNOT}}|1, x, y\rangle = |1, x, x \oplus y\rangle. \tag{4.24}$$

Thus the CCNOT gate can be used to construct the XOR gate.

4.3.3 AND Gate

The logical **AND gate** is defined by

$$\text{AND}(x, y) \equiv x \wedge y \equiv \begin{cases} 1 & x = y = 1 \\ 0 & \text{otherwise} \end{cases} \quad x, y \in \{0, 1\}. \tag{4.25}$$

Clearly this operation is not reversible and we have to introduce the same sort of prescription which we employed in the XOR gate.

Let us define the logic function

$$f(x, y, 0) \equiv (x, y, x \wedge y), \tag{4.26}$$

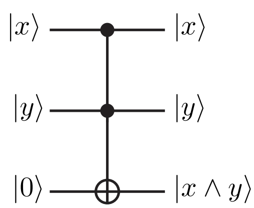
which we also call AND. Note that we have to keep both x and y for f to be reversible since $x = x \wedge y = 0$ implies *both* $x = y = 0$ and $x = 0, y = 1$. The unitary matrix that computes f is

$$\begin{aligned} U_{\text{AND}} &= (|00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10|) \otimes I \\ &\quad + |11\rangle\langle 11| \otimes X. \end{aligned} \tag{4.27}$$

It is readily verified that

$$U_{\text{AND}}|x, y, 0\rangle = |x, y, x \wedge y\rangle, \quad x, y \in \{0, 1\}. \tag{4.28}$$

Observe that the third qubit in the RHS is 1 if and only if $x = y = 1$ and 0 otherwise. Thus the CCNOT gate may be employed to implement the AND gate. It follows from Eq. (4.28) that the AND gate is denoted graphically as



4.3.4 OR Gate

The **OR gate** represents the logical function

$$\text{OR}(x, y) = x \vee y = \begin{cases} 0 & x = y = 0 \\ 1 & \text{otherwise} \end{cases} \quad x, y \in \{0, 1\}. \tag{4.29}$$

This function OR is not reversible either and special care must be taken.

Let us define

$$f(x, y, 0) \equiv (\neg x, \neg y, x \vee y), \quad x, y \in \{0, 1\}, \tag{4.30}$$

which we also call OR. Although the first and the second bits are negated, it is not essential in the construction of the OR gate. These negations appear due to our construction of the OR gate based on the de Morgan theorem

$$x \vee y = \neg(\neg x \wedge \neg y). \tag{4.31}$$

They may be removed by adding extra NOT gates if necessary.

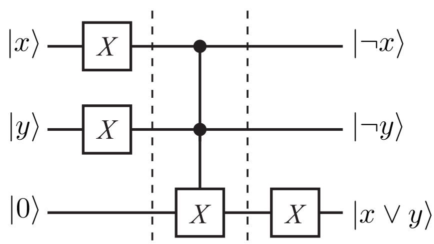
Let $|x, y, 0\rangle$ be the input state. The unitary matrix that represents f is

$$U_{\mathrm{OR}} = |00\rangle\langle 11| \otimes X + |01\rangle\langle 10| \otimes X + |10\rangle\langle 01| \otimes X + |11\rangle\langle 00| \otimes I. \tag{4.32}$$

EXERCISE 4.8 Verify that the above matrix U_{OR} indeed satisfies

$$U_{\mathrm{OR}}|x, y, 0\rangle = |\neg x, \neg y, x \vee y\rangle, \quad x, y \in \{0, 1\}. \tag{4.33}$$

Now it is obvious why negations in the first and the second qubits appear in the OR gate. Since we have already constructed the NOT gate and AND gate, we take advantage of this in the construction of the OR gate. The equality (4.31) leads us to the following diagram:



Accordingly, the first and the second qubits are negated. The unitary matrix obtained from this diagram is

$$\begin{aligned} U_{\mathrm{OR}} = {} & (I \otimes I \otimes X) \\ & \cdot(|00\rangle\langle 00| \otimes I + |01\rangle\langle 01| \otimes I + |10\rangle\langle 10| \otimes I + |11\rangle\langle 11| \otimes X) \\ & \cdot(X \otimes X \otimes I). \end{aligned} \tag{4.34}$$

The matrix products are readily evaluated to yield

$$\begin{aligned} U_{\mathrm{OR}} &= (|00\rangle\langle 00| \otimes X + |01\rangle\langle 01| \otimes X + |10\rangle\langle 10| \otimes X + |11\rangle\langle 11| \otimes I) \\ &\quad \cdot(X \otimes X \otimes I) \\ &= |00\rangle\langle 11| \otimes X + |01\rangle\langle 10| \otimes X + |10\rangle\langle 01| \otimes X + |11\rangle\langle 00| \otimes I, \end{aligned}$$

which verifies Eq. (4.32).

Observe that the OR gate is implemented with the X and the CCNOT gates and, moreover, the X gate is obtained from the CCNOT gate by putting the first and the second bits to $|1\rangle$.

If we want to have a gate $V_{\text{OR}}|x, y, 0\rangle = |x, y, x \vee y\rangle$, we may multiply $X \otimes X \otimes I$ to U_{OR} from the left so that $V_{\text{OR}} = (X \otimes X \otimes I)U_{\text{OR}}$.

EXERCISE 4.9 Show that the NAND gate can be obtained from the CCNOT gate. Here NAND is defined by the function

$$\text{NAND}(x, y) = \neg(x \wedge y) = \begin{cases} 0 & x = y = 1 \\ 1 & \text{otherwise} \end{cases} \quad x, y \in \{0, 1\}. \tag{4.35}$$

In summary, we have shown that all the classical logic gates, NOT, AND, OR, XOR and NAND gates, may be obtained from the CCNOT gate. Thus all the classical computation may be carried out with a quantum computor. Note, however, that these gates belong to a tiny subset of the set of unitary matrices.

In the next section, we show that copying unknown information is impossible in quantum computing. However, it is also shown that this does not restrict the superiority of quantum computing over the classical counterpart.

4.4 No-Cloning Theorem

We copy classical data almost every day. In fact, this is amongst the most common functions with digital media. (Of course we should not copy media that are copyright protected.) This cannot be done in quantum information theory! We cannot clone an unknown quantum state with unitary operations.

THEOREM 4.1 (Wootters and Zurek [4], Dieks [5]) An unknown quantum system cannot be cloned by unitary transformations.

Proof. Suppose there would exist a unitary transformation U that makes a clone of a quantum system. Namely, suppose U acts, for any state $|\varphi\rangle$, as

$$U : |\varphi 0\rangle \to |\varphi\varphi\rangle. \tag{4.36}$$

Let $|\varphi\rangle$ and $|\phi\rangle$ be two states that are linearly independent. Then we should have $U|\varphi 0\rangle = |\varphi\varphi\rangle$ and $U|\phi 0\rangle = |\phi\phi\rangle$ by definition. Then the action of U on $|\psi\rangle = \dfrac{1}{\sqrt{2}}(|\varphi\rangle + |\phi\rangle)$ yields

$$U|\psi 0\rangle = \frac{1}{\sqrt{2}}(U|\varphi 0\rangle + U|\phi 0\rangle) = \frac{1}{\sqrt{2}}(|\varphi\varphi\rangle + |\phi\phi\rangle).$$

If U were a cloning transformation, we must also have

$$U|\psi 0\rangle = |\psi\psi\rangle = \frac{1}{2}(|\varphi\varphi\rangle + |\varphi\phi\rangle + |\phi\varphi\rangle + |\phi\phi\rangle),$$

which contradicts the previous result. Therefore, there does not exist a unitary cloning transformation. ∎

Clearly, there is no way to clone a state by measurements. A measurement is probabilistic and non-unitary, and it gets rid of the component of the state which is in the orthogonal complement of the observed subspace.

EXERCISE 4.10 Suppose U is a cloning unitary transformation, such that

$$\begin{aligned} |\Psi\rangle &\equiv U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle \\ |\Phi\rangle &\equiv U|\phi\rangle|0\rangle = |\phi\rangle|\phi\rangle \end{aligned}$$

for arbitrary $|\psi\rangle$ and $|\phi\rangle$.
(1) Write down $\langle\Psi|\Phi\rangle$ in all possible ways.
(2) Show, by inspecting the result of (1), that such U does not exist.

It was mentioned in the end of the previous section that a quantum computer can simulate arbitrary classical logic circuits. Then how about copying data? It should be kept in mind that the no-cloning theorem states that we cannot copy an *arbitrary* state $|\psi\rangle = a|0\rangle + b|1\rangle$. The loophole is that the theorem does not apply if the states to be cloned are limited to $|0\rangle$ and $|1\rangle$. For these cases, the copying operator U should work as

$$U : |00\rangle \mapsto |00\rangle, \quad : |10\rangle \mapsto |11\rangle.$$

We can assign arbitrary action of U on a state whose second input is $|1\rangle$ since this case does not happen. What we have to keep in our mind is only that U be unitary. An example of such U is

$$U = (|00\rangle\langle 00| + |11\rangle\langle 10|) + (|01\rangle\langle 01| + |10\rangle\langle 11|), \tag{4.37}$$

where the first set of operators renders U the cloning operator and the second set is added just to make U unitary. We immediately notice that U is nothing but the CNOT gate introduced in §4.2.

Therefore, if the data under consideration are limited within $|0\rangle$ and $|1\rangle$, we can copy the qubit states even in a quantum computer. This fact is used to construct quantum error correcting codes.

4.5 Dense Coding and Quantum Teleportation

Now we are ready to introduce two simple applications of qubits and quantum gates: **dense coding** and **quantum teleportation**. The Bell state has been delivered beforehand, and one of the qubits carries two classical bits of information in the dense coding system. In the quantum teleportation, on the other hand, two classical bits are used to transmit a single qubit. At first glance, the quantum teleportation may seem to be in contradiction with the no-cloning theorem. However, this is not the case since the original state is destroyed.

Entanglement is the keyword in both applications. The setting is common for both cases. Suppose Alice wants to send Bob information. Each of them has been sent each of the qubits of the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \tag{4.38}$$

in advance. Suppose Alice has the first qubit and Bob has the second.

4.5.1 Dense Coding

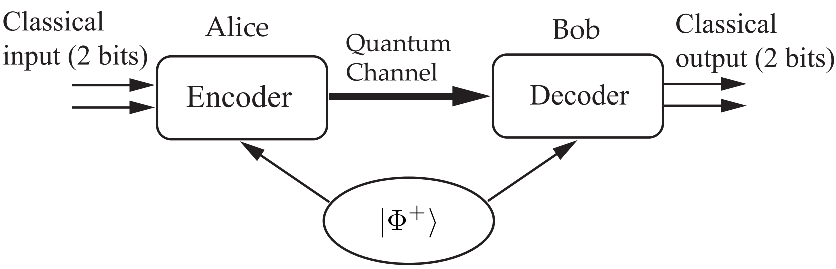


FIGURE 4.1
Communication from Alice to Bob using dense coding. Each qubit of the Bell state $|\Phi^+\rangle$ has been distributed to each of them beforehand. Then two bits of classical information can be transmitted by sending a single qubit through the quantum channel.

Alice: Alice wants to send Bob a binary number $x \in \{00, 01, 10, 11\}$. She picks up one of $\{I, X, Y, Z\}$ according to x she has chosen and applies the transformation on her qubit (the first qubit of the Bell state). Applying the transformation to only her qubit means she applies an identity transformation to the second qubit which Bob keeps with him. This results in

$$\begin{array}{ccl}
x & \text{transformation } U & \text{state after transformation} \\
0=00 & I\otimes I & |\psi_0\rangle = \dfrac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \\
1=01 & X\otimes I & |\psi_1\rangle = \dfrac{1}{\sqrt{2}}(|10\rangle + |01\rangle) \\
2=10 & Y\otimes I & |\psi_2\rangle = \dfrac{1}{\sqrt{2}}(|10\rangle - |01\rangle) \\
3=11 & Z\otimes I & |\psi_3\rangle = \dfrac{1}{\sqrt{2}}(|00\rangle - |11\rangle).
\end{array} \tag{4.39}$$

Alice sends Bob her qubit after the transformation given above is applied. Note that the set of four states in the rightmost column is nothing but the four Bell basis vectors.

Bob: Bob applies CNOT to the entangled pair in which the first qubit, the received qubit, is the control bit, while the second one, which Bob keeps, is the target bit. This results in a tensor-product state:

$$\begin{array}{cccc}
\text{Received state} & \text{Output of CNOT} & \text{1st qubit} & \text{2nd qubit} \\
|\psi_0\rangle & \dfrac{1}{\sqrt{2}}(|00\rangle + |10\rangle) & \dfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle) & |0\rangle \\
|\psi_1\rangle & \dfrac{1}{\sqrt{2}}(|11\rangle + |01\rangle) & \dfrac{1}{\sqrt{2}}(|1\rangle + |0\rangle) & |1\rangle \\
|\psi_2\rangle & \dfrac{1}{\sqrt{2}}(|11\rangle - |01\rangle) & \dfrac{1}{\sqrt{2}}(|1\rangle - |0\rangle) & |1\rangle \\
|\psi_3\rangle & \dfrac{1}{\sqrt{2}}(|00\rangle - |10\rangle) & \dfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle) & |0\rangle
\end{array} \tag{4.40}$$

Note that Bob can measure the first and second qubits independently since the output is a tensor-product state. The number x is either 00 or 11 if the measurement outcome of the second qubit is $|0\rangle$, while it is either 01 or 10 if the meansurement outcome is $|1\rangle$.

Finally, a Hadamard transformation H is applied on the first qubit. Bob obtains

$$\begin{array}{ccc}
\text{Received state} & \text{1st qubit} & U_{\rm H}|\text{1st qubit}\rangle \\
|\psi_0\rangle & \dfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle) & |0\rangle \\
|\psi_1\rangle & \dfrac{1}{\sqrt{2}}(|1\rangle + |0\rangle) & |0\rangle \\
|\psi_2\rangle & \dfrac{1}{\sqrt{2}}(|1\rangle - |0\rangle) & -|1\rangle \\
|\psi_3\rangle & \dfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle) & |1\rangle
\end{array} \tag{4.41}$$

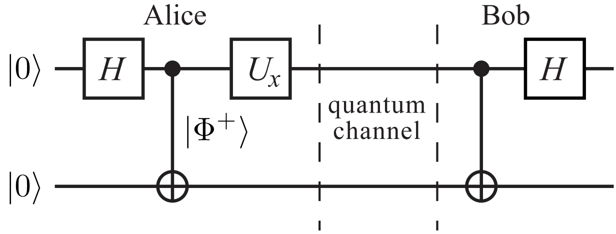


FIGURE 4.2
Quantum circuit implementation of the dense coding system. The leftmost Hadamard gate and the next CNOT gate generate the Bell state. Then a unitary gate U, depending on the bits Alice wants to send, is applied to the first qubit. Bob applies the rightmost CNOT gate and the Hadamard gate to decode Alice's message.

The number x is either 00 or 01 if the measurement of the first qubit results in $|0\rangle$, while it is either 10 or 11 if it is $|1\rangle$. Therefore, Bob can tell what x is in every case.

Quantum circuit implementation for the dense coding is given in Fig. 4.2

4.5.2 Quantum Teleportation

The purpose of **quantum teleportation** is to transmit an unknown quantum *state* of a qubit using two classical bits such that the recipient reproduces exactly the same state as the original qubit state. Note that the qubit itself is not transported but the information required to reproduce the quantum state is transmitted. The original state is destroyed such that quantum teleportation should not be in contradiction with the no-cloning theorem. Quantum teleportation has already been realized under laboratory conditions using photons [6, 7, 8, 9], coherent light field [10], NMR [11], and trapped ions [12, 13]. The teleportation scheme introduced in this section is due to [11]. Figure 4.3 shows the schematic diagram of quantum teleportation, which will be described in detail below.

Alice: Alice has a qubit, whose state she does not know. She wishes to send Bob the quantum state of this qubit through a classical communication channel. Let

$$|\phi\rangle = a|0\rangle + b|1\rangle \tag{4.42}$$

be the state of the qubit. Both of them have been given one of the qubits of the entangled pair

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

as in the case of the dense coding.

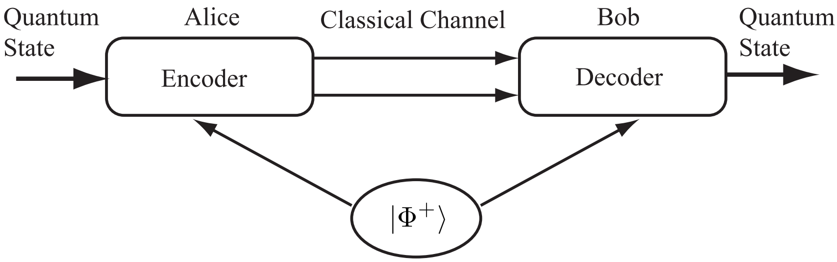


FIGURE 4.3
In quantum teleportation, Alice sends Bob two classical bits so that Bob reproduces a qubit state Alice used to have.

Alice applies the decoding step in the dense coding to the qubit $|\phi\rangle = a|0\rangle + b|1\rangle$ to be sent and her qubit of the entangled pair. They start with the state

$$\begin{aligned} |\phi\rangle \otimes |\Phi^+\rangle &= \frac{1}{\sqrt{2}}\big[a|0\rangle \otimes (|00\rangle + |11\rangle) + b|1\rangle \otimes (|00\rangle + |11\rangle)\big] \\ &= \frac{1}{\sqrt{2}}\left(a|000\rangle + a|011\rangle + b|100\rangle + b|111\rangle\right), \end{aligned} \tag{4.43}$$

where Alice has the first two qubits while Bob has the third. Alice applies $U_{\text{CNOT}} \otimes I$ followed by $U_{\text{H}} \otimes I \otimes I$ to this state, which results in

$$\begin{aligned} &(U_{\text{H}} \otimes I \otimes I)(U_{\text{CNOT}} \otimes I)(|\phi\rangle \otimes |\Phi^+\rangle) \\ &= (U_{\text{H}} \otimes I \otimes I)(U_{\text{CNOT}} \otimes I)\frac{1}{\sqrt{2}}\left(a|000\rangle + a|011\rangle + b|100\rangle + b|111\rangle\right) \\ &= \frac{1}{2}\left[a(|000\rangle + |011\rangle + |100\rangle + |111\rangle) + b(|010\rangle + |001\rangle - |110\rangle - |101\rangle)\right] \\ &= \frac{1}{2}[|00\rangle(a|0\rangle + b|1\rangle) + |01\rangle(a|1\rangle + b|0\rangle) \\ &\qquad + |10\rangle(a|0\rangle - b|1\rangle) + |11\rangle(a|1\rangle - b|0\rangle)]. \end{aligned} \tag{4.44}$$

If Alice measures the two qubits in her hand, she will obtain one of the states $|00\rangle, |01\rangle, |10\rangle$ or $|11\rangle$ with equal probability $1/4$. Bob's qubit (a qubit from the Bell state initially) collapses to $a|0\rangle + b|1\rangle, a|1\rangle + b|0\rangle, a|0\rangle - b|1\rangle$ or $a|1\rangle - b|0\rangle$, respectively, depending on the result of Alice's measurement. Alice then sends Bob her result of the measurement using two classical bits.

Notice that Alice has totally destroyed the initial qubit $|\phi\rangle$ upon her measurement. This makes quantum teleportation consistent with the no-cloning theorem.

Bob: After receiving two classical bits, Bob knows the state of the qubit in his hand;

$$
\begin{array}{ccc}
\text{Received bits} & \text{Bob's state} & \text{Decoding} \\
00 & a|0\rangle + b|1\rangle & I \\
01 & a|1\rangle + b|0\rangle & X \\
10 & a|0\rangle - b|1\rangle & Z \\
11 & a|1\rangle - b|0\rangle & Y
\end{array}
\tag{4.45}
$$

Bob reconstructs the intial state $|\phi\rangle$ by applying the decoding process shown above. Suppose Alice sends Bob the classical bits 10, for example. Then Bob applies Z to his state to reconstruct $|\phi\rangle$ as follows:

$$Z: \ (a|0\rangle - b|1\rangle) \mapsto (a|0\rangle + b|1\rangle) = |\phi\rangle.$$

Figure 4.4 shows the actual quantum circuit for quantum teleportation.

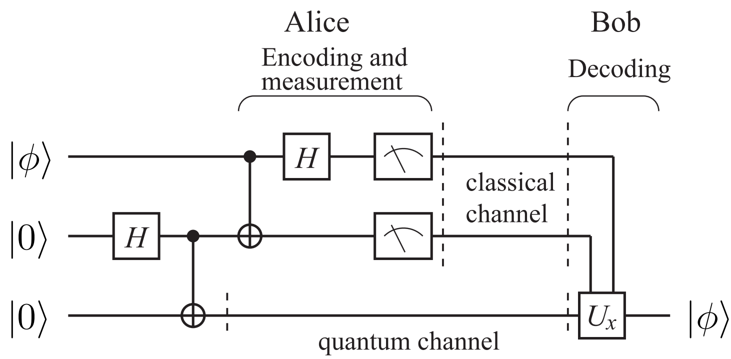


FIGURE 4.4
Quantum circuit implementation of quantum teleportation. Alice operates gates in the left side. The first Hadamard gate and the next CNOT gates generate the Bell state $|\Phi^+\rangle$ from $|00\rangle$. The bottom qubit is sent to Bob through a quantum channel while the first and the second qubits are measured after applying the second set of the CNOT gate and the Hadamard gate on them. The measurement outcome x is sent to Bob through a classical channel. Bob operates a unitary operation U_x, which depends on the received message x, on his qubit.

EXERCISE 4.11 Let $|\psi\rangle = a|00\rangle + b|11\rangle$ be a two-qubit state. Apply a Hadamard gate to the first qubit and then measure the first qubit. Find the second qubit state after the measurement corresponding to the outcome of the first qubit measurement.

4.6 Universal Quantum Gates

It can be shown that any classical logic gate can be constructed by using a small set of gates, AND, NOT and XOR, for example. Such a set of gates is called the *universal* set of classical gates. Since the CCNOT gate can simulate these classical gates, quantum circuits simulate any classical circuits. It should be noted that the set of quantum gates is much larger than those classical gates which can be simulated by quantum gates. Thus we want to find a universal set of *quantum* gates from which any quantum circuits, i.e., any unitary matrix, can be constructed.

In the following, it will be shown that
(1) the set of single qubit gates and
(2) CNOT gate
form a universal set of quantum circuits (**universality theorem**).

We will prove the following Lemma before stating the main theorem. Let us start with a definition. A **two-level unitary matrix** is a unitary matrix which acts non-trivially only on two vector components. Suppose V is a two-level unitary matrix. Then V has the same matrix elements as those of the unit matrix except for certain four elements V_{aa}, V_{ab}, V_{ba} and V_{bb}. An example of a two-level unitary matrix is

$$V = \begin{pmatrix} \alpha^* & 0 & 0 & \beta^* \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\beta & 0 & 0 & \alpha \end{pmatrix}, \quad (|\alpha|^2 + |\beta|^2 = 1),$$

where $a = 1$ and $b = 4$.

LEMMA 4.1 Let U be a unitary matrix acting on $\mathbb{C}^d$. Then there are $N \le d(d-1)/2$ two-level unitary matrices $U_1, U_2, \ldots, U_N$ such that

$$U = U_1 U_2 \ldots U_N. \tag{4.46}$$

Proof. The proof requires several steps. It is instructive to start with the case $d = 3$. Let

$$U = \begin{pmatrix} a & d & g \\ b & e & h \\ c & f & j \end{pmatrix}$$

be a unitary matrix. We want to find two-level unitary matrices U_1, U_2, U_3 such that

$$U_3 U_2 U_1 U = I.$$

Then it follows that

$$U = U_1^\dagger U_2^\dagger U_3^\dagger.$$

(Never mind the daggers! If U_k is two-level unitary, $U_k^\dagger$ is also two-level unitary.) We prove the above decomposition by constructing U_k explicitly.

(i) Let

$$U_1 = \begin{pmatrix} \frac{a^*}{u} & \frac{b^*}{u} & 0 \\ -\frac{b}{u} & \frac{a}{u} & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

where $u = \sqrt{|a|^2 + |b|^2}$. Verify that U_1 is unitary. Then we obtain

$$U_1 U = \begin{pmatrix} a' & d' & g' \\ 0 & e' & h' \\ c' & f' & j' \end{pmatrix},$$

where $a', \ldots, j'$ are some complex numbers, whose details are not necessary. Observe that, with this choice of U_1, the first component of the second row vanishes.

(ii) Let

$$U_2 = \begin{pmatrix} \frac{a'^*}{u'} & 0 & \frac{c'^*}{u'} \\ 0 & 1 & 0 \\ -\frac{c'}{u'} & 0 & \frac{a'}{u'} \end{pmatrix} = \begin{pmatrix} a'^* & 0 & c'^* \\ 0 & 1 & 0 \\ -c' & 0 & a' \end{pmatrix},$$

where $u' = \sqrt{|a'|^2 + |c'|^2} = 1$. Then

$$U_2 U_1 U = \begin{pmatrix} 1 & d'' & g'' \\ 0 & e'' & h'' \\ 0 & f'' & j'' \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & e'' & h'' \\ 0 & f'' & j'' \end{pmatrix},$$

where the equality $d'' = g'' = 0$ follows from the fact that $U_2 U_1 U$ is unitary, and hence the first row must be normalized.

(iii) Finally let

$$U_3 = (U_2 U_1 U)^\dagger = \begin{pmatrix} 1 & 0 & 0 \\ 0 & e''^* & f''^* \\ 0 & h''^* & j''^* \end{pmatrix}.$$

Then, by definition, $U_3 U_2 U_1 U = I$ is obvious. This completes the proof for $d = 3$.

Suppose U is a unitary matrix acting on $\mathbb{C}^d$ with a general dimension d. Then by repeating the above arguments, we find two-level unitary matrices $U_1, U_2, \ldots, U_{d-1}$ such that

$$U_{d-1} \ldots U_2 U_1 U = \begin{pmatrix} 1 & 0 & 0 & \ldots & 0 \\ 0 & * & * & \ldots & * \\ 0 & * & * & \ldots & * \\ & & \ldots & \ldots & \\ 0 & * & * & \ldots & * \end{pmatrix},$$

namely the $(1, 1)$ component is unity and other components of the first row and the first column vanish. The number of matrices $\{U_k\}$ to achieve this form is the same as the number of zeros in the first column, hence $(d-1)$.

We then repeat the same procedure to the $(d-1) \times (d-1)$ block unitary matrix using $(d-2)$ two-level unitary matrices. After repeating this, we finally decompose U into a product of two-level unitary matrices

$$U = V_1 V_2 \dots V_N,$$

where $N \le (d-1) + (d-2) + \dots + 1 = d(d-1)/2$. ∎

EXERCISE 4.12 Let U be a general 4×4 unitary matrix. Find two-level unitary matrices U_1, U_2 and U_3 such that

$$U_3 U_2 U_1 U = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{pmatrix}.$$

EXERCISE 4.13 Let

$$U = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}. \tag{4.47}$$

Decompose U into a product of two-level unitary matrices.

Let us consider a unitary matrix acting on an n-qubit system. Then this unitary matrix is decomposed into a product of at most $2^n(2^n-1)/2 = 2^{n-1}(2^n-1)$ two-level unitary matrices. Now we are in a position to state the main theorem.

THEOREM 4.2 (Barenco *et al.*)[14] The set of single qubit gates and CNOT gate are universal. Namely, any unitary gate acting on an n-qubit register can be implemented with single qubit gates and CNOT gates.

Proof. We closely follow [1] for the proof here. Thanks to the previous Lemma, it suffices to prove the theorem for a two-level unitary matrix. Let U be a two-level unitary matrix acting nontrivially only on $|s\rangle$ and $|t\rangle$ basis vectors of an n-qubit system, where $s = s_{n-1}2^{n-1} + \dots + s_1 2 + s_0$ and $t = t_{n-1}2^{n-1} + \dots + t_1 2 + t_0$ are binary expressions for decimal numbers s and t. This means that matrix elements U_{ss}, U_{st}, U_{ts} and U_{tt} are different from those of the unit matrix, while all the others are the same, where $|s\rangle$ stands for $|s_{n-1}\rangle|s_{n-2}\rangle \dots |s_0\rangle$, for example. We can construct $\tilde{U}$, the non-trivial 2×2 unitary submatrix of U. $\tilde{U}$ may be thought of as a unitary matrix acting on a single qubit, whose basis is $\{|s\rangle, |t\rangle\}$.

STEP 1: U is reduced to $\tilde{U} \in \mathrm{U}(2)$.

The basis vectors $|s\rangle$ and $|t\rangle$ may be put together to form a basis for a single qubit using the following trick. This is done by introducing **Gray codes**. For two binary numbers $s = s_{n-1} \dots s_1 s_0$ and $t = t_{n-1} \dots t_1 t_0$, a Gray code connecting s and t is a sequence of binary numbers $\{g_1, \dots, g_m\}$ where the adjacent numbers, g_k and g_{k+1}, differ in exactly one bit. Moreover, the sequence satisfies the boundary conditions $g_1 = s$ and $g_m = t$.

Suppose $s = 100101$ and $t = 110110$, for example. An example of a Gray code connecting s and t is

$$\begin{aligned} s = g_1 &= 100101 \\ g_2 &= 1\hat{1}0101 \\ g_3 &= 1101\hat{1}1 \\ g_4 &= 11011\hat{0} = t, \end{aligned}$$

where the digit with $\hat{}$ has been renewed. It is clear from this construction that if s and t differ in p bits, the shortest Gray code is made of $p+1$ elements. It should be also clear that if s and t are of n digits, then $m \leq (n+1)$ since s and t differ at most in n bits.

With these preparations, we consider the implementation of U. The strategy is to find gates providing the sequence of state changes

$$|s\rangle = |g_1\rangle \to |g_2\rangle \to \dots \to |g_{m-1}\rangle. \tag{4.48}$$

Then g_{m-1} and g_m differ only in one bit, which is identified with the single qubit on which $\tilde{U}$ acts. In the example above, we have $|g_3\rangle = |11011\rangle \otimes |1\rangle$ and $|t\rangle = |g_4\rangle = |11011\rangle \otimes |0\rangle$. Now the operator $\tilde{U}$ may be introduced so that it acts on a two-dimensional subspace of the total Hilbert space, in which the first five qubits are in the state $|11011\rangle$. Then we undo the sequence (4.48) so that $|g_{m-1}\rangle \to |g_{m-2}\rangle \to \dots \to |g_1\rangle = |s\rangle$. Each of these steps can be easily implemented using simple gates that have been introduced previously (see below).

Let us consider the following example of a three-qubit system, whose basis is $\{|000\rangle, |001\rangle, \dots, |111\rangle\}$. Let

$$U = \begin{pmatrix} a & 0 & 0 & 0 & 0 & 0 & 0 & c \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ b & 0 & 0 & 0 & 0 & 0 & 0 & d \end{pmatrix}, \quad (a, b, c, d \in \mathbb{C}) \tag{4.49}$$

be a two-level unitary matrix which we wish to implement. Note that U acts non-trivially only in the subspace spanned by $|000\rangle$ and $|111\rangle$. The unitarity

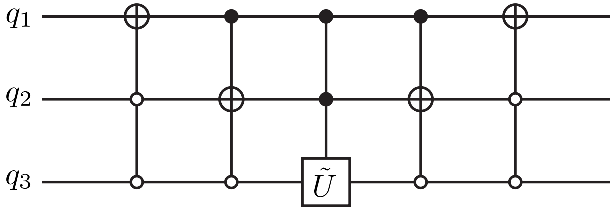


FIGURE 4.5
Example of circuit implementing the gate U.

of U ensures that the matrix

$$\tilde{U} = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \tag{4.50}$$

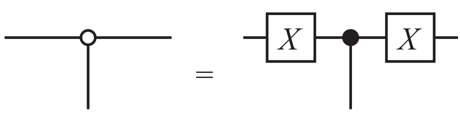
is also unitary. An example of a Gray code connecting 000 and 111 is

$$\begin{array}{rcccc} & & q_1 & q_2 & q_3 \\ g_1 & = & 0 & 0 & 0 \\ g_2 & = & 1 & 0 & 0 \\ g_3 & = & 1 & 1 & 0 \\ g_4 & = & 1 & 1 & 1 \end{array} \tag{4.51}$$

Since g_3 and g_4 differ only in the third qubit, which we call q_3, we have to bring g_1 to g_3 and then operate $\tilde{U}$ on the qubit q_3 provided that the first and the second qubits are in the state $|11\rangle$. (Namely we have a controlled-$\tilde{U}$ gate with the target bit q_3 and the control bits q_1 and q_2.) After this controlled operation is done, we have to put $|g_3\rangle = |110\rangle$ back to the state $|000\rangle$ as

$$|110\rangle \to |100\rangle \to |000\rangle.$$

This operation is graphically shown in Fig. 4.5. Here $\circ$ denotes the negated control node. This means that the unitary gate acts on the target bit only when the control bit is in the state $|0\rangle$. This is easily implemented by adding two X gates as



It is easy to see that this gate indeed implements U. Suppose the input is $|101\rangle$, for example. Figure 4.6 shows that the gate has no effect on this basis vector since U should act as a unit matrix on this vector. The operation of U

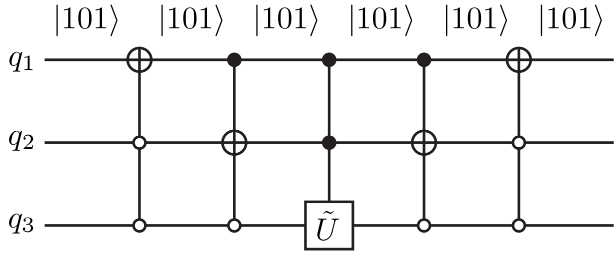


FIGURE 4.6
U-gate has no effect on the vector $|101\rangle$.

on the input $\alpha|000\rangle + \beta|111\rangle$ is

$$U(\alpha|000\rangle + \beta|111\rangle) = \begin{pmatrix} a & 0 & 0 & 0 & 0 & 0 & 0 & c \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ b & 0 & 0 & 0 & 0 & 0 & 0 & d \end{pmatrix} \begin{pmatrix} \alpha \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha a + \beta c \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \alpha b + \beta d \end{pmatrix}. \tag{4.52}$$

If we use the circuit shown in Fig. 4.5, we produce the same result as shown in Fig. 4.7

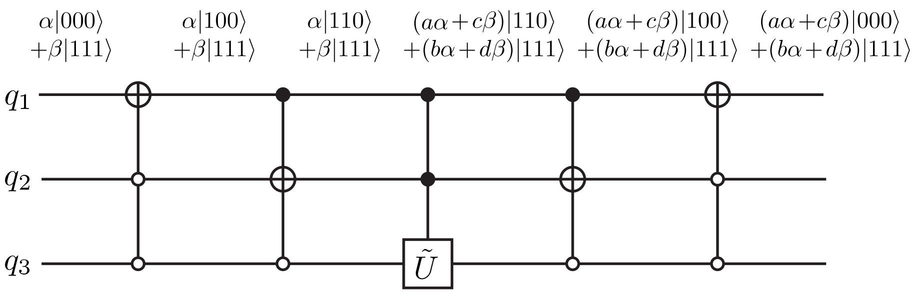


FIGURE 4.7
U-gate acting on $\alpha|000\rangle + \beta|111\rangle$ yields the desired output $(a\alpha + c\beta)|000\rangle + (b\alpha + d\beta)|111\rangle$.

This construction is easily generalized to any two-level unitary matrix $U \in \mathrm{U}(2^n)$. It will be shown below that all the gates in the above circuit can be implemented with single-qubit gates and CNOT gates, which proves the universality of these gates.

EXERCISE 4.14 (1) Find the shortest Gray code which connects 000 with 110.
(2) Use this result to find a quantum circuit, such as Fig. 4.5, implementing a two-level unitary gate

$$U = \begin{pmatrix} a & 0 & 0 & 0 & 0 & 0 & c & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ b & 0 & 0 & 0 & 0 & 0 & d & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad \tilde{U} \equiv \begin{pmatrix} a & c \\ b & d \end{pmatrix} \in U(2).$$

You may use various controlled-NOT gates and controlled-$\tilde{U}$ gates.

STEP 2: Two-level unitary gates are decomposed into single-qubit gates and CNOT gates.

A controlled-U gate can be constructed from at most four single-qubit gates and two CNOT gates for any single-qubit unitary $U \in \mathrm{U}(2)$. Let us prove several Lemmas before we prove this statement.

LEMMA 4.2 Let $U \in \mathrm{SU}(2)$. Then there exist $\alpha, \beta, \gamma \in \mathbb{R}$ such that $U = R_z(\alpha)R_y(\beta)R_z(\gamma)$, where

$$R_z(\alpha) = \exp(i\alpha\sigma_z/2) = \begin{pmatrix} e^{i\alpha/2} & 0 \\ 0 & e^{-i\alpha/2} \end{pmatrix},$$

$$R_y(\beta) = \exp(i\beta\sigma_y/2) = \begin{pmatrix} \cos(\beta/2) & \sin(\beta/2) \\ -\sin(\beta/2) & \cos(\beta/2) \end{pmatrix}.$$

Proof. After some calculation, we obtain

$$R_z(\alpha)R_y(\beta)R_z(\gamma) = \begin{pmatrix} e^{i(\alpha+\gamma)/2}\cos(\beta/2) & e^{i(\alpha-\gamma)/2}\sin(\beta/2) \\ -e^{i(-\alpha+\gamma)/2}\sin(\beta/2) & e^{-i(\alpha+\gamma)/2}\cos(\beta/2) \end{pmatrix}. \tag{4.53}$$

Any $U \in \mathrm{SU}(2)$ may be written in the form

$$U = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix} = \begin{pmatrix} \cos\theta e^{i\lambda} & \sin\theta e^{i\mu} \\ -\sin\theta e^{-i\mu} & \cos\theta e^{-i\lambda} \end{pmatrix}, \tag{4.54}$$

where we used the fact that $\det U = |a|^2 + |b|^2 = 1$. Now we obtain $U = R_z(\alpha)R_y(\beta)R_z(\gamma)$ by making identifications

$$\theta = \frac{\beta}{2}, \lambda = \frac{\alpha+\gamma}{2}, \mu = \frac{\alpha-\gamma}{2}. \tag{4.55}$$

∎

LEMMA 4.3 Let $U \in \mathrm{SU}(2)$. Then there exist $A, B, C \in \mathrm{SU}(2)$ such that $U = AXBXC$ and $ABC = I$, where $X = \sigma_x$.

Proof. Lemma 4.2 states that $U = R_z(\alpha)R_y(\beta)R_z(\gamma)$ for some $\alpha, \beta, \gamma \in \mathbb{R}$. Let

$$A = R_z(\alpha)R_y\left(\frac{\beta}{2}\right), B = R_y\left(-\frac{\beta}{2}\right)R_z\left(-\frac{\alpha+\gamma}{2}\right), C = R_z\left(-\frac{\alpha-\gamma}{2}\right).$$

Then

$$\begin{aligned} AXBXC &= R_z(\alpha)R_y\left(\frac{\beta}{2}\right)XR_y\left(-\frac{\beta}{2}\right)R_z\left(-\frac{\alpha+\gamma}{2}\right)XR_z\left(-\frac{\alpha-\gamma}{2}\right) \\ &= R_z(\alpha)R_y\left(\frac{\beta}{2}\right)\left[XR_y\left(-\frac{\beta}{2}\right)X\right]\left[XR_z\left(-\frac{\alpha+\gamma}{2}\right)X\right]R_z\left(-\frac{\alpha-\gamma}{2}\right) \\ &= R_z(\alpha)R_y\left(\frac{\beta}{2}\right)R_y\left(\frac{\beta}{2}\right)R_z\left(\frac{\alpha+\gamma}{2}\right)R_z\left(-\frac{\alpha-\gamma}{2}\right) \\ &= R_z(\alpha)R_y(\beta)R_z(\gamma) = U, \end{aligned}$$

where use has been made of the identities $X^2 = I$ and $X\sigma_{y,z}X = -\sigma_{y,z}$.

It is also verified that

$$\begin{aligned} ABC &= R_z(\alpha)R_y\left(\frac{\beta}{2}\right)R_y\left(-\frac{\beta}{2}\right)R_z\left(-\frac{\alpha+\gamma}{2}\right)R_z\left(-\frac{\alpha-\gamma}{2}\right) \\ &= R_z(\alpha)R_y(0)R_z(-\alpha) = I. \end{aligned}$$

This proves the Lemma. ∎

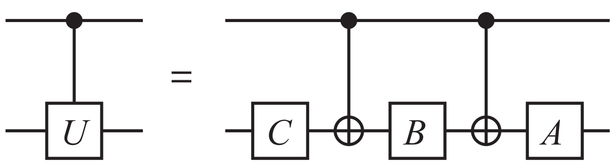


FIGURE 4.8
Controlled-U gate is made of at most three single-qubit gates and two CNOT gates for any $U \in \mathrm{SU}(2)$.

LEMMA 4.4 Let $U \in \mathrm{SU}(2)$ be factorized as $U = AXBXC$ as in the previous Lemma. Then the controlled-U gate can be implemented with at most three single-qubit gates and two CNOT gates (see Fig. 4.8).

Proof. The proof is almost obvious. When the control bit is 0, the target bit $|\psi\rangle$ is operated by C, B and A in this order so that

$$|\psi\rangle \mapsto ABC|\psi\rangle = |\psi\rangle,$$

while when the control bit is 1, we have

$$|\psi\rangle \mapsto AXBXC|\psi\rangle = U|\psi\rangle.$$

∎

So far, we have worked with $U \in \mathrm{SU}(2)$. To implement a general U-gate with $U \in \mathrm{U}(2)$, we have to deal with the phase. Let us first recall that any $U \in \mathrm{U}(2)$ is decomposed as $U = e^{i\alpha}V,\ V \in \mathrm{SU}(2), \alpha \in \mathbb{R}$.

LEMMA 4.5 Let

$$\Phi(\phi) = e^{i\phi}I = \begin{pmatrix} e^{i\phi} & 0 \\ 0 & e^{i\phi} \end{pmatrix}$$

and

$$D = R_z(-\phi)\Phi\left(\frac{\phi}{2}\right) = \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix}\begin{pmatrix} e^{i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}.$$

Then the controlled-$\Phi(\phi)$ gate is expressed as a tensor product of single qubit gates as

$$U_{\mathrm{C}\Phi(\phi)} = D \otimes I. \tag{4.56}$$

.

Proof. The LHS is

$$\begin{aligned} U_{\mathrm{C}\Phi(\phi)} &= |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes \Phi(\phi) = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes e^{i\phi}I \\ &= |0\rangle\langle 0| \otimes I + e^{i\phi}|1\rangle\langle 1| \otimes I, \end{aligned}$$

while the RHS is

$$\begin{aligned} D \otimes I &= \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \otimes I \\ &= \left[|0\rangle\langle 0| + e^{i\phi}|1\rangle\langle 1|\right] \otimes I = U_{\mathrm{C}\Phi(\phi)}, \end{aligned}$$

which proves the lemma. ∎

Figure 4.9 shows the statement of the above lemma.

EXERCISE 4.15 Let us consider the controlled-V_1 gate $U_{\mathrm{C}V_1}$ and the controlled-V_2 gate $U_{\mathrm{C}V_2}$. Show that the controlled-V_1 gate followed by the controlled-V_2 gate is the controlled-V_2V_1 gate $U_{\mathrm{C}(V_2V_1)}$ as shown in Fig. 4.10.

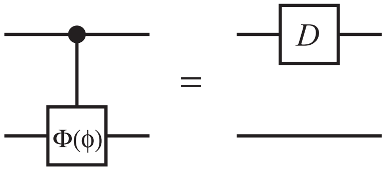


FIGURE 4.9
Equality $U_{\mathrm{C}\Phi(\phi)} = D \otimes I$.

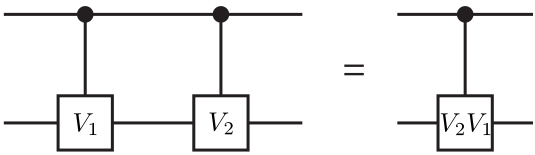


FIGURE 4.10
Equality $U_{\mathrm{C}V_2} U_{\mathrm{C}V_1} = U_{\mathrm{C}(V_2 V_1)}$.

Now we are ready to prove the main proposition.

PROPOSITION 4.1 Let $U \in \mathrm{U}(2)$. Then the controlled-U gate $U_{\mathrm{C}U}$ can be constructed by at most four single-qubit gates and two CNOT gates.

Proof. Let $U = \Phi(\phi)AXBXC$. According to the exercise above, the controlled-U gate is written as a product of the controlled-$\Phi(\phi)$ gate and the controlled-$AXBXC$ gate. Moreover, Lemma 4.5 states that the controlled-$\Phi(\phi)$ gate may be replaced by a single-qubit phase gate acting on the first qubit. The rest of the gate, the controlled-$AXBXC$ gate is implemented with three SU(2) gates and two CNOT gates as proved in Lemma 4.3. Therefore we have the following decomposition:

$$U_{\mathrm{C}U} = (D \otimes A)U_{\mathrm{CNOT}}(I \otimes B)U_{\mathrm{CNOT}}(I \otimes C), \tag{4.57}$$

where

$$D = R_z(-\phi)\Phi(\phi/2)$$

and use has been made of the identity $(D \otimes I)(I \otimes A) = D \otimes A$. ∎

Figure 4.11 shows the statement of the proposition.

STEP 3: CCNOT gate and its variants are implemented with CNOT gates and their variants.

Now our final task is to prove that controlled-U gates with $n - 1$ control bits are also constructed using single-qubit gates and CNOT gates. Let us start with the simplest case, in which $n = 3$.

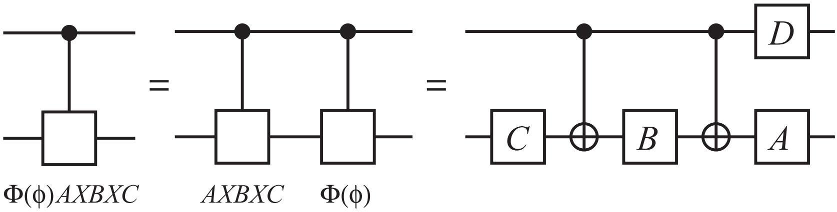


FIGURE 4.11
Controlled-U gate is implemented with at most four single-qubit gates and two CNOT gates.

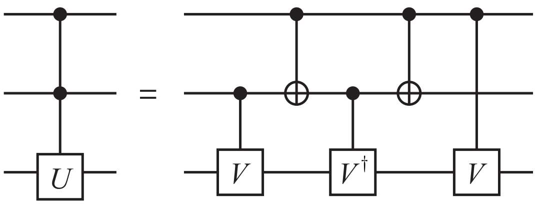


FIGURE 4.12
Controlled-controlled-U gate is equivalent to the gate made of controlled-V gates with $U = V^2$ and CNOT gates.

LEMMA 4.6 The two quantum circuits in Fig. 4.12 are equivalent, where $U = V^2$.

Proof. If both the first and the second qubits are 0 in the RHS, all the gates are ineffective and the third qubit is unchanged; the gate in this subspace acts as $|00\rangle\langle 00| \otimes I$. In case the first qubit is 0 and the second is 1, the third qubit is mapped as $|\psi\rangle \mapsto V^\dagger V|\psi\rangle = |\psi\rangle$; the gate is then $|01\rangle\langle 01| \otimes I$. When the first qubit is 1 and the second is 0, the third qubit is mapped as $|\psi\rangle \mapsto VV^\dagger|\psi\rangle = |\psi\rangle$; hence the gate in this subspace is $|10\rangle\langle 10| \otimes I$. Finally let both the first and the second qubits be 1. Then the action of the gate on the third qubit is $|\psi\rangle \mapsto VV|\psi\rangle = U|\psi\rangle$; namely the gate in this subspace is $|11\rangle\langle 11| \otimes U$. Thus it has been proved that the RHS of Fig. 4.12 is

$$(|00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10|) \otimes I + |11\rangle\langle 11| \otimes U, \tag{4.58}$$

namely the controlled-controlled-U gate. ∎

This decomposition is explained intuitively as follows. The first V operates on the third qubit $|\psi\rangle$ if and only if the second qubit is 1. $V^\dagger$ is in action if and only if $x_1 \oplus x_2 = 1$, where x_k is the input bit of the kth qubit. The second V operation is applied if and only if the first qubit is 1. Thus the action of this gate on the third qubit is $V^2 = U$ only when $x_1 \wedge x_2 = 1$ and

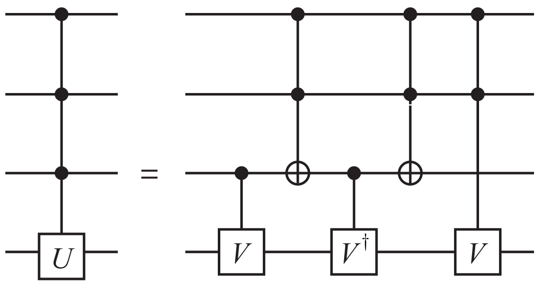


FIGURE 4.13
Decomposition of the C^3U gate.

I otherwise. This intuitive picture is of help when we implement the U gate with more control qubits.

EXERCISE 4.16 Prove Lemma 4.6 by writing down the action of each gate in the RHS of Fig. 4.12 explicitly using bras, kets and $I, U, V, V^\dagger$. (For example, $U_{\text{CNOT}} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X$ for a two-qubit system.)

A simple generalization of the above construction is applied to a controlled-U gate with three control bits as the following exercise shows.

EXERCISE 4.17 Show that the circuit in Fig. 4.13 is a controlled-U gate with three control bits, where $U = V^2$.

Now it should be clear how these examples are generalized to gates with more control bits.

PROPOSITION 4.2 The quantum circuit in Fig. 4.14 with $U = V^2$ is a decomposition of the controlled-U gate with $n-1$ control bits.

The proof of the above proposition is very similar to that of Lemma 4.6 and Exercise 4.17 and is left as an exercise to the readers.
Theorem 4.2 has been now proved. ∎

Other types of gates are also implemented with single-qubit gates and the CNOT gates. See Barenco *et al.* [14] for further details. A few remarks are in order. The above controlled-U gate with $(n-1)$ control bits requires $\Theta(n^2)$ elementary gates.*† Let us write the number of the elementary gates required

*We call single-qubit gates and the CNOT gates elementary gates from now on.

†We will be less strict in the definition of "the order of." In the theory of computational complexity, people use three types of "order of magnitude." One writes "$f(n)$ is $O(g(n))$" if there exist $n_0 \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $f(n) \leq cg(n)$ for $n \geq n_0$. In other words, O sets the asymptotic upper bound of $f(n)$. A function $f(n)$ is said to be $\Omega(g(n))$ if there exist

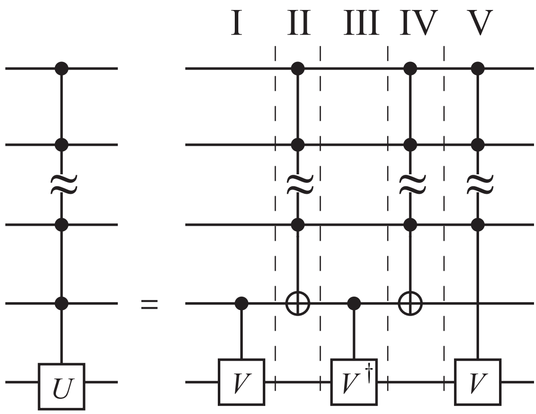


FIGURE 4.14
Decomposition of the $C^{(n-1)}U$ gate. The number on the top denotes the layer refered to in the text.

to construct the gate in Fig. 4.14 by $C(n)$. Construction of layers I and III requires elementary gates whose number is independent of n. It can be shown that the number of the elementary gates required to construct the controlled NOT gate with $(n-2)$ control bits is $\Theta(n)$ [14]. Therefore layers II and IV require $\Theta(n)$ elementary gates. Finally the layer V, a controlled-V gate with $(n-2)$ control bits, requires $C(n-1)$ basic gates by definition. Thus we obtain a recursion relation

$$C(n) - C(n-1) = \Theta(n). \tag{4.59}$$

The solution to this recursion relation is

$$C(n) = \Theta(n^2). \tag{4.60}$$

Therefore, implementation of a controlled-U gate with $U \in \mathrm{U}(2)$ and $(n-1)$ control bits requires $\Theta(n^2)$ elementary gates.

$n_0 \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $f(n) \geq cg(n)$ for $n \geq n_0$. In other words, Ω sets the asymptotic lower bound of $f(n)$. Finally $f(n)$ is said to be $\Theta(f(n))$ if $f(n)$ behaves asymptotically as $g(n)$, namely if $f(n)$ is both $O(g(n))$ and $\Omega(g(n))$.

4.7 Quantum Parallelism and Entanglement

Given an input x, a typical quantum computer computes $f(x)$ in such a way as

$$U_f : |x\rangle|0\rangle \mapsto |x\rangle|f(x)\rangle, \tag{4.61}$$

where U_f is a unitary matrix that implements the function f.

Suppose U_f acts on the input which is a superposition of many states. Since U_f is a linear operator, it acts simultaneously on all the vectors that constitute the superposition. Thus the output is also a superposition of all the results;

$$U_f : \sum_x |x\rangle|0\rangle \mapsto \sum_x |x\rangle|f(x)\rangle. \tag{4.62}$$

Namely, when the input is a superposition of n states, U_f computes n values $f(x_k)$ $(1 \le k \le n)$ simultaneously. This feature, called the *quantum parallelism*, gives a quantum computer an enormous power. A quantum computer is advantageous compared to a classical counterpart in that it makes use of this quantum parallelism and also entanglement.

A unitary transformation acts on a superposition of all possible states in most quantum algorithms. This superposition is prepared by the action of the Walsh-Hadamard transformation on an n-qubit register in the state $|00\ldots 0\rangle = |0\rangle \otimes |0\rangle \otimes \ldots \otimes |0\rangle$ resulting in

$$\frac{1}{\sqrt{2^n}}(|00\ldots 0\rangle + |00\ldots 1\rangle + \ldots |11\ldots 1\rangle) = \frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}|x\rangle. \tag{4.63}$$

This state is a superposition of vectors encoding all the integers between 0 and $2^n - 1$. Then the linearity of U_f leads to

$$U_f\left(\frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}|x\rangle|0\rangle\right) = \frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}U_f|x\rangle|0\rangle = \frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}|x\rangle|f(x)\rangle. \tag{4.64}$$

Note that the superposition is made of $2^n = e^{n\ln 2}$ states, which makes quantum computation exponentially faster than the classical counterpart in a certain kind of computation.

What about the limitation of a quantum computer? Let us consider the CCNOT gate, for example. This gate flips the third qubit if and only if the first and the second qubits are both in the state $|1\rangle$, while it leaves the third qubit unchanged otherwise. Let us fix the third input qubit to $|0\rangle$. It was shown in §4.3.3 that the third output is $|x \wedge y\rangle$, where $|x\rangle$ and $|y\rangle$ are the first and the second input qubit states, respectively. Suppose the input state is a superposition of all possible states while the third qubit is fixed to $|0\rangle$. This can be achieved by the Walsh-Hadamard transformation as

$$\begin{aligned} U_{\rm H}|0\rangle \otimes U_{\rm H}|0\rangle \otimes |0\rangle &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle \\ &= \frac{1}{2}(|000\rangle + |010\rangle + |100\rangle + |110\rangle). \end{aligned} \tag{4.65}$$

By operating CCNOT on this state, we obtain

$$U_{\rm CCNOT}(U_{\rm H}|0\rangle \otimes U_{\rm H}|0\rangle \otimes |0\rangle) = \frac{1}{2}(|000\rangle + |010\rangle + |100\rangle + |111\rangle). \tag{4.66}$$

This output may be thought of as the truth table of AND: $|x, y, x \wedge y\rangle$. It is extremely important to note that the output is an entangled state and the measurement projects the state to *one line* of the truth table, i.e., a single term in the RHS of Eq. (4.66). The order of the measurements of the three qubits does not matter at all. The measurement of the third qubit projects the state to the superposition of the states with the given value of the third qubit. Repeating the measurements on the rest of the qubits leads to the collapse of the output state to one of $|x, y, x \wedge y\rangle$.

There is no advantage of quantum computation over classical at this stage. This is because only *one* result may be obtained by a single set of measurements. What is worse, we cannot choose a specific vector $|x, y, x \wedge y\rangle$ at our will! Thus any quantum algorithm should be programmed so that the particular vector we want to observe should have larger probability to be measured compared to other vectors. This step has no classical analogy and is very special in quantum computation. The programming strategies to deal with this feature are [2]

1. to amplify the amplitude, and hence the probability, of the vector that we want to observe. This strategy is employed in the Grover's database search algorithm.

2. to find a common property of all the $f(x)$. This idea was employed in the quantum Fourier transform to find the order[‡] of f in the Shor's factoring algorithm.

Now we consider the power of entanglement. Suppose we have an n-qubit register, whose Hilbert space is 2^n-dimensional. Since each qubit has two basis vectors $|0\rangle$ and $|1\rangle$, there are $2n$ basis vectors (n $|0\rangle$'s and n $|1\rangle$'s) involved to span this 2^n-dimensional Hilbert space. Imagine that we have a single quantum system, instead, which has the same Hilbert space. One might think that the system may do the same quantum computation as the n-qubit register does. One possible problem is that one cannot measure the "kth digit"

[‡]Let $m, N \in \mathbb{N}$ $(m < N)$ be numbers coprime to each other. Then there exists $P \in \mathbb{N}$ such that $m^P \equiv 1 \pmod N$. The smallest such number P is called the **period** or the **order**. It is easily seen that $m^{x+P} \equiv m^x \pmod N$, $\forall x \in \mathbb{N}$.

leaving other digits unaffected. Even worse, consider how many different basis vectors are required for this system. This single system must have an enormous number, 2^n, of basis vectors! Let us consider 20 spin-1/2 particles in a magnetic field. We can employ the spin-up and spin-down energy eigenstates of each particle as the qubit basis vectors. Then there are merely 40 energy eigenvectors involved. Suppose we use energy eigenstates of a certain molecule to replace this register. Then we have to use $2^{20} \sim 10^6$ eigenstates. Separation and control of so many eigenstates are certainly beyond current technology. These simple consideration shows that multipartite implementation of a quantum algorithm requires an exponentially smaller number of basis vectors than monopartite implementation since the former makes use of entanglement as a computational resource.

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5

Simple Quantum Algorithms

Before we start presenting "useful" but rather complicated quantum algorithms, we introduce a few simple quantum algorithms which will be of help for readers to understand how quantum algorithms are different from and superior to classical algorithms. We follow closely Meglicki [1].

5.1 Deutsch Algorithm

The **Deutsch algorithm** is one of the first quantum algorithms which showed quantum algorithms may be more efficient than their classical counterparts. In spite of its simplicty, full use of the superposition principle has been made here.

Let $f : \{0, 1\} \to \{0, 1\}$ be a binary function. Note that there are only four possible f, namely

$$\begin{aligned} &f_1 : 0 \mapsto 0,\ 1 \mapsto 0, \quad &&f_2 : 0 \mapsto 1,\ 1 \mapsto 1, \\ &f_3 : 0 \mapsto 0,\ 1 \mapsto 1, \quad &&f_4 : 0 \mapsto 1,\ 1 \mapsto 0. \end{aligned}$$

The first two cases, f_1 and f_2, are called *constant*, while the rest, f_3 and f_4, are *balanced*. If we only have classical resources, we need to evaluate f twice to tell if f is constant or balanced. There is a quantum algorithm, however, with which it is possible to tell if f is constant or balanced with a single evaluation of f, as was shown by Deutsch [2].

Let $|0\rangle$ and $|1\rangle$ correspond to classical bits 0 and 1, respectively, and consider the state $|\psi_0\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$. We apply f on this state in terms of the unitary operator $U_f : |x, y\rangle \mapsto |x, y \oplus f(x)\rangle$, where $\oplus$ is an addition mod 2. To be explicit, we obtain

$$\begin{aligned} |\psi_1\rangle &= U_f|\psi_0\rangle \\ &= \frac{1}{2}(|0, f(0)\rangle - |0, 1 \oplus f(0)\rangle + |1, f(1)\rangle - |1, 1 \oplus f(1)\rangle) \\ &= \frac{1}{2}(|0, f(0)\rangle - |0, \neg f(0)\rangle + |1, f(1)\rangle - |1, \neg f(1)\rangle), \end{aligned}$$

where $\neg$ stands for negation. Therefore this operation is nothing but the CNOT gate with the control bit $f(x)$; the target bit y is flipped if and only if

$f(x) = 1$ and left unchanged otherwise. Subsequently we apply a Hadamard gate on the first qubit to obtain

$$\begin{aligned}|\psi_2\rangle &= (U_{\mathrm{H}} \otimes I)|\psi_1\rangle \\ &= \frac{1}{2\sqrt{2}}\left[(|0\rangle + |1\rangle)(|f(0)\rangle - |\neg f(0)\rangle) + (|0\rangle - |1\rangle)(|f(1)\rangle - |\neg f(1)\rangle)\right].\end{aligned}$$

The wave function reduces to

$$|\psi_2\rangle = \frac{1}{\sqrt{2}}|0\rangle(|f(0)\rangle - |\neg f(0)\rangle) \tag{5.1}$$

in case f is constant, for which $|f(0)\rangle = |f(1)\rangle$, and

$$|\psi_2\rangle = \frac{1}{\sqrt{2}}|1\rangle(|f(0)\rangle - |\neg f(0)\rangle) \tag{5.2}$$

if f is balanced, for which $|\neg f(0)\rangle = |f(1)\rangle$. Therefore the measurement of the first qubit tells us whether f is constant or balanced.

Let us consider a quantum circuit which implements the Deutsch algorithm. We first apply Walsh-Hadamard transformation $W_2 = U_{\mathrm{H}} \otimes U_{\mathrm{H}}$ on $|01\rangle$ to obtain $|\psi_0\rangle$. We need to introduce a conditional gate U_f, i.e., the controlled-NOT gate with the control bit $f(x)$, whose action is $U_f : |x, y\rangle \to |x, y \oplus f(x)\rangle$. Then a Hadamard gate is applied on the first qubit before it is measured. Figure 5.1 depicts this implementation.

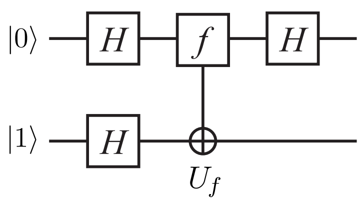

FIGURE 5.1
Implementation of the Deutsch algorithm.

In the quauntum circuit, we assume the gate U_f is a black box for which we do not ask the explicit implementation. We might think it is a kind of subroutine. Such a black box is often called an **oracle**. The gate U_f is called the Deutsch oracle. Its implementation is given only after f is specified.

Then what is the merit of the Deutsch algorithm? Suppose your friend gives you a unitary matrix U_f and asks you to tell if f is constant or balanced. Instead of applying $|0\rangle$ and $|1\rangle$ separately, you may construct the circuit in Fig. 5.1 with the given matrix U_f and apply the circuit on the input state $|01\rangle$. Then you can tell your friend whether f is constant or balanced with a single use of U_f.

5.2 Deutsch-Jozsa Algorithm and Bernstein-Vazirani Algorithm

The Deutsch algorithm introduced in the previous section may be generalized to the **Deutsch-Jozsa algorithm** [3].

Let us first define the **Deutsch-Jozsa problem**. Suppose there is a binary function

$$f : S_n \equiv \{0, 1, \ldots, 2^n - 1\} \to \{0, 1\}. \tag{5.3}$$

We require that f be either *constant* or *balanced* as before. When f is constant, it takes a constant value 0 or 1 irrespetive of the input value x. When it is balanaced the value $f(x)$ for the half of $x \in S_n$ is 0, while it is 1 for the rest of x. In other words, $|f^{-1}(0)| = |f^{-1}(1)| = 2^{n-1}$, where $|A|$ denotes the number of elements in a set A, known as the cardinality of A. Although there are functions which are neither constant nor balanced, we will not consider such cases here. Our task is to find an algorithm which tells if f is constant or balanced with the least possible number of evaluations of f.

It is clear that we need at least $2^{n-1} + 1$ steps, in the worst case with classical manipulations, to make sure if $f(x)$ is constant or balanced with 100% confidence. It will be shown below that the number of steps reduces to a single step if we are allowed to use a quantum algorithm.

The algorithm is divided into the following steps:

1. Prepare an $(n + 1)$-qubit register in the state $|\psi_0\rangle = |0\rangle^{\otimes n} \otimes |1\rangle$. First n qubits work as input qubits, while the $(n + 1)$st qubit serves as a "scratch pad." Such qubits, which are neither input qubits nor output qubits, but work as a scratch pad to store temporary information are called **ancillas** or **ancillary qubits**.

2. Apply the Walsh-Hadamard transforamtion to the register. Then we have the state

$$\begin{aligned} |\psi_1\rangle = U_{\mathrm{H}}^{\otimes n+1}|\psi_0\rangle &= \frac{1}{\sqrt{2^n}}(|0\rangle + |1\rangle)^{\otimes n} \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\ &= \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle). \end{aligned} \tag{5.4}$$

3. Apply the $f(x)$-controlled-NOT gate on the register, which flips the $(n + 1)$st qubit if and only if $f(x) = 1$ for the input x. Therefore we need a U_f gate which evaluates $f(x)$ and acts on the register as $U_f|x\rangle|c\rangle = |x\rangle|c \oplus f(x)\rangle$, where $|c\rangle$ is the one-qubit state of the $(n+1)$st qubit. Observe that $|c\rangle$ is flipped if and only if $f(x) = 1$ and left

unchanged otherwise. We then obtain a state

$$\begin{aligned}|\psi_2\rangle &= U_f|\psi_1\rangle \\ &= \frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}|x\rangle\frac{1}{\sqrt{2}}(|f(x)\rangle - |\neg f(x)\rangle) \\ &= \frac{1}{\sqrt{2^n}}\sum_x(-1)^{f(x)}|x\rangle\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle). \end{aligned} \tag{5.5}$$

Although the gate U_f is applied once for all, it is applied to *all* the n-qubit states $|x\rangle$ simultaneously.

4. The Walsh-Hadamard transformation (4.11) is applied on the first n qubits next. We obtain

$$|\psi_3\rangle = (W_n \otimes I)|\psi_2\rangle = \frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}(-1)^{f(x)}U_{\mathrm{H}}^{\otimes n}|x\rangle\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle). \tag{5.6}$$

It is instructive to write the action of the one-qubit Hadamard gate in the following form,

$$U_{\mathrm{H}}|x\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^x|1\rangle) = \frac{1}{\sqrt{2}}\sum_{y\in\{0,1\}}(-1)^{xy}|y\rangle,$$

where $x \in \{0,1\}$, to find the resulting state. The action of the Walsh-Hadamard transformation on $|x\rangle = |x_{n-1}\dots x_1x_0\rangle$ yields

$$\begin{aligned}W_n|x\rangle &= (U_{\mathrm{H}}|x_{n-1}\rangle)(U_{\mathrm{H}}|x_{n-2}\rangle)\dots(U_{\mathrm{H}}|x_0\rangle) \\ &= \frac{1}{\sqrt{2^n}}\sum_{y_{n-1},y_{n-2},\dots,y_0\in\{0,1\}}(-1)^{x_{n-1}y_{n-1}+x_{n-2}y_{n-2}+\dots+x_0y_0} \\ &\quad\times|y_{n-1}y_{n-2}\dots y_0\rangle \\ &= \frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}(-1)^{x\cdot y}|y\rangle, \end{aligned} \tag{5.7}$$

where $x\cdot y = x_{n-1}y_{n-1}\oplus x_{n-2}y_{n-2}\oplus\dots\oplus x_0y_0$. Substituting this result into Eq. (5.6), we obtain

$$|\psi_3\rangle = \frac{1}{2^n}\left(\sum_{x,y=0}^{2^n-1}(-1)^{f(x)}(-1)^{x\cdot y}|y\rangle\right)\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle). \tag{5.8}$$

5. The first n qubits are measured. Suppose $f(x)$ is constant. Then $|\psi_3\rangle$ is put in the form

$$|\psi_3\rangle = \frac{1}{2^n}\sum_{x,y}(-1)^{x\cdot y}|y\rangle\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

up to an overall phase. Now let us consider the summation

$$\frac{1}{2^n}\sum_{x=0}^{2^n-1}(-1)^{x\cdot y}$$

with a fixed $y \in S_n$. Clearly it vanishes since $x \cdot y$ is 0 for half of x and 1 for the other half of x unless $y = 0$. Therefore the summation yields δ_{y0}. Now the state reduces to

$$|\psi_3\rangle = |0\rangle^{\otimes n}\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle),$$

and the measurement outcome of the first n qubits is always $00\ldots0$. Suppose $f(x)$ is balanced next. The probability amplitude of $|y=0\rangle$ in $|\psi_3\rangle$ is proportional to

$$\sum_{x=0}^{2^n-1}(-1)^{f(x)}(-1)^{x\cdot 0} = \sum_{x=0}^{2^n-1}(-1)^{f(x)} = 0.$$

Therefore the probability of obtaining measurement outcome $00\ldots0$ for the first n qubits vanishes. In conclusion, the function f is constant if we obtain $00\ldots0$ upon the meaurement of the first n qubits in the state $|\psi_3\rangle$, and it is balanced otherwise.

EXERCISE 5.1 Let us take $n = 2$ for definiteness. Consider the following cases and find the final wave function $|\psi_3\rangle$ and evaluate the measurement outcomes and their probabilities for each case.
(1) $f(x) = 1\ \forall x \in S_2$.
(2) $f(00) = f(01) = 1, f(10) = f(11) = 0$.
(3) $f(00) = 0, f(01) = f(10) = f(11) = 1$. (This function is neither constant nor balanced.)

The above exercise shows that the measurement gives $|00\rangle$ with probability 1 if f is constant and with probability 0 if balanced. If f is neither constant nor balanced $|\psi_3\rangle$ is a superposition of several states including $|00\rangle$, which is attributed to "incomplete" interference.

A quantum circuit which implements the Deutsch-Jozsa algorithm is given in Fig. 5.2. The gate U_f is called the **Deutsch-Jozsa oracle**.

The **Bernstein-Vazirani algorithm** is a special case of the Deutsch-Jozsa algorithm, in which $f(x)$ is given by $f(x) = c \cdot x$, where $c = c_{n-1}c_{n-2}\ldots c_0$ is an n-bit binary number [4]. Our aim is to find c with the smallest number of evaluations of f. If we apply the Deutsch-Jozsa algorithm with this f, we obtain

$$|\psi_3\rangle = \frac{1}{2^n}\left[\sum_{x,y=0}^{2^n-1}(-1)^{c\cdot x}(-1)^{x\cdot y}|y\rangle\right]\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

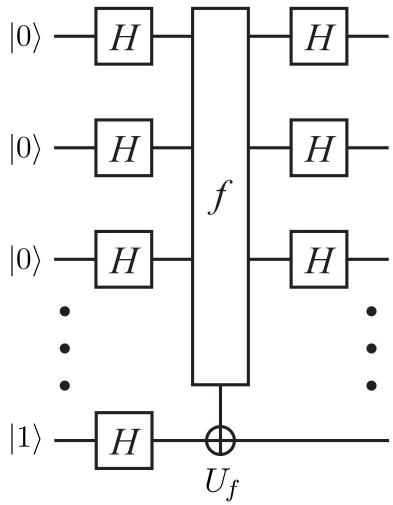

FIGURE 5.2
Quantum circuit implementing the Deutsch-Jozsa algorithm. The gate U_f is the Deutsch-Jozsa oracle.

Let us fix y first. If we take $y = c$, we obtain

$$\sum_x (-1)^{c \cdot x}(-1)^{x \cdot c} = \sum_x (-1)^{2c \cdot x} = 2^n.$$

If $y \neq c$, on the other hand, there will be the same number of x such that $c \cdot x = 0$ and x such that $c \cdot x = 1$ in the summation over x and, as a result, the probability amplitude of $|y \neq c\rangle$ vanishes. By using these results, we end up with

$$|\psi_3\rangle = |c\rangle \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle). \tag{5.9}$$

We are able to tell what c is by measuring the first n qubits. Note that this is done by a single application of the circuit in Fig. 5.2.

EXERCISE 5.2 Consider the Bernstein-Vazirani algorithm with $n = 3$ and $c = 101$. Work out the quantum circuit depicted in Fig. 5.2 to show that the measurement outcome of the first three qubits is $c = 101$.

5.3 Simon's Algorithm

The final example of simple quantum algorithms is **Simon's algorithm**. Let us consider a function (oracle) $f : \{0,1\}^n \to \{0,1\}^n$ such that

1. f is 2 to 1; namely, for any x_1, there is one and only one $x_2 \neq x_1$ such that $f(x_1) = f(x_2)$.
2. f is periodic; namely, there exists $p \in \{0,1\}^n$ such that $f(x \oplus p) = f(x)$, $\forall x \in \{0,1\}^n$, where $\oplus$ is a bitwise addition mod 2.

The function f is made of n component functions $f_k : \{0,1\}^n \to \{0,1\}$ as $f = (f_1, f_2, \ldots, f_n)$.

Suppose we want to find the period p, given an unknown oracle f. Since p can be any number between $00\ldots0$ and $11\ldots1$, we have to try $\sim 2^n$ possibilities classically before we hit the right number. It is shown below that the number of trials required to find p is reduced to $O(n)$ if Simon's algorithm is employed.

The algorithm is decomposed into the following steps:

1. Prepare two sets of n-qubit regiters in the state $|\psi_0\rangle = |0\rangle|0\rangle$. Then the Walsh-Hadamard transformation W_n is applied on the first register to yield
$$|\psi_1\rangle = (W_n \otimes I)|\psi_0\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle|0\rangle.$$
2. Introduce n controlled-NOT gates with control qubits $f_k(x)$ $(1 \leq k \leq n)$ and the target bit is the kth qubit of the second register. We write
$$U_f : |x\rangle|0\rangle \mapsto |x\rangle|f(x)\rangle,$$
where $|0\rangle$ is an n-qubit register state and $|f(x)\rangle = |f_1(x)\rangle|f_2(x)\rangle \ldots |f_n(x)\rangle$. Linearity implies the state $|\psi_2\rangle$ after the U_f gate operation on $|\psi_1\rangle$ is
$$|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle|f(x)\rangle. \tag{5.10}$$
3. Now we measure the second register. In fact, we do not need to know the measurement outcome. What we have to do is to project the second register to a certain state $|f(x_0)\rangle$, for example. After one of these operations, the state is now projected to
$$|\psi_3\rangle = \frac{1}{\sqrt{2}}(|x_0\rangle + |x_0 \oplus p\rangle)|f(x_0)\rangle, \tag{5.11}$$
where we noted that there are exactly two states $|x_0\rangle$ and $|x_0 \oplus p\rangle$ that give the second register state $|f(x_0)\rangle$ in step 2.

4. Now we apply W_n again on the first register to obtain

$$
\begin{aligned}
|\psi_4\rangle &= \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}\left[(-1)^{x_0\cdot y}+(-1)^{(x_0\oplus p)\cdot y}\right]|y\rangle\otimes|f(x_0)\rangle \\
&= \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}(-1)^{x_0\cdot y}\left[1+(-1)^{p\cdot y}\right]|y\rangle\otimes|f(x_0)\rangle. \qquad (5.12)
\end{aligned}
$$

The inner product $p\cdot y$ takes two values 0 and 1. We immediately notice that such y which satisfies $1+(-1)^{p\cdot y}=0$, namely, $p\cdot y=1$ does not contribute to the summation in Eq. (5.12). Now we are left with

$$
|\psi_4\rangle = \frac{2}{\sqrt{2}}\frac{1}{\sqrt{2^n}}\left[\sum_{p\cdot y=0}(-1)^{x_0\cdot y}|y\rangle\right]\otimes|f(x_0)\rangle. \qquad (5.13)
$$

5. Finally we measure the first register. Upon this measurement, we obtain $|y\rangle$ such that $p\cdot y=0$. Of course, this equation is not enough to identify the period p. Now we repeat the algorithm many times to obtain

$$
p\cdot y_1 = p\cdot y_2 = \ldots = p\cdot y_m = 0. \qquad (5.14)
$$

It should be clear that we need at least n trials since not all equations are linearly independent. For a sufficiently large number of trials m, we are able to solve Eq. (5.14) for p classically. The number of trials necessary for this is $O(n)$ with a good probability.

Figure 5.3 shows the quantum circuit to implement Simon's algorithm for the case $n=3$.

Simon's algorithm has been improved so that it may be executed in deterministic polynomial time [6].

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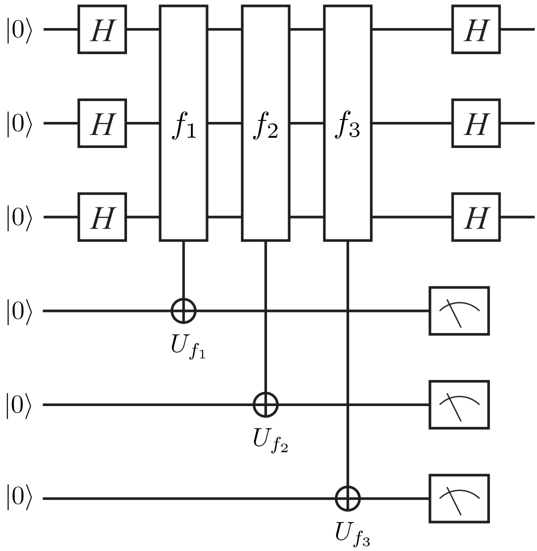

FIGURE 5.3
Quantum circuit implementing Simon's algorithm. The second register is measured only for projection purposes, and reading the outcome is not necessary.

6

Quantum Integral Transforms

We demonstrated in the previous chapter that there are some quantum algorithms superior to their classical counterparts. It is, however, rather difficult to find any practical use of these algorithms. There are two quantum algorithms, known to date, which are potentially useful: Grover's search algorithm and Shor's prime number factorization algorithm. Both of them depend on quantum integral transforms, which will be introduced in the present chapter. We mainly follow [1] in our presentation.

6.1 Quantum Integral Transforms

DEFINITION 6.1 (**Discrete Integral Transform**) Let $n \in \mathbb{N}$ and $S_n = \{0, 1, \ldots, 2^n - 1\}$ be a set of integers. Consider a map

$$K : S_n \times S_n \to \mathbb{C}. \tag{6.1}$$

For any function $f : S_n \to \mathbb{C}$, its **discrete integral transform** (DIT) $\tilde{f} : S_n \to \mathbb{C}$ with the **kernel** K is defined as:

$$\tilde{f}(y) = \sum_{x=0}^{2^n-1} K(y, x) f(x). \tag{6.2}$$

The transformation $f \to \tilde{f}$ is also called the discrete integral transform.

We define $N \equiv 2^n$ to simplify our notations. The kernel K is expressed as a matrix,

$$K = \begin{pmatrix} K(0,0) & \ldots & K(0, N-1) \\ K(1,0) & \ldots & K(1, N-1) \\ \ldots & \ldots & \ldots \\ K(N-1,0) & \ldots & K(N-1, N-1), \end{pmatrix} \tag{6.3}$$

and the function f as a vector,

$$f = (f(0), f(1), \ldots f(N-1))^t \,.$$

The definition of DIT then reduces to the ordinary multiplication of a matrix on a vector as

$$\tilde{f} = Kf.$$

PROPOSITION 6.1 Suppose the kernel K is unitary: $K^\dagger = K^{-1}$. Then the inverse transform $\tilde{f} \to f$ of a DIT exists and is given by

$$f(x) = \sum_{y=0}^{N-1} K^\dagger(x,y)\tilde{f}(y). \tag{6.4}$$

Proof. By substituting Eq. (6.2) into Eq. (6.4), we prove

$$\begin{aligned}\sum_{y=0}^{N-1} K^\dagger(x,y)\tilde{f}(y) &= \sum_{y=0}^{N-1} K^\dagger(x,y)\left[\sum_{z=0}^{N-1} K(y,z)f(z)\right] \\ &= \sum_{z=0}^{N-1}\left[\sum_{y=0}^{N-1} K^\dagger(x,y)K(y,z)\right] f(z) \\ &= \sum_{z=0}^{N-1} \delta_{xz} f(z) = f(x).\end{aligned}$$

∎

Let U be an $N \times N$ unitary matrix which acts on the n-qubit space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$. Let $\{|x\rangle = |x_{n-1}, x_{n-2} \ldots, x_0\rangle\}$ ($x_k \in \{0,1\}$) be the standard binary basis of $\mathcal{H}$, where $x = x_{n-1}2^{n-1} + x_{n-2}2^{n-2} + \ldots + x_0 2^0$. Then

$$U|x\rangle = \sum_{y=0}^{N-1} |y\rangle\langle y|U|x\rangle = \sum_{y=0}^{N-1} U(y,x)|y\rangle. \tag{6.5}$$

The complex number $U(x,y) = \langle x|U|y\rangle$ is the (x,y)-component of U in this basis.

PROPOSITION 6.2 Let U be a unitary transformation, acting on $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$. Suppose U acts on a basis vector $|x\rangle$ as

$$U|x\rangle = \sum_{y=0}^{N-1} K(y,x)|y\rangle. \tag{6.6}$$

Then U *computes** the DIT $\tilde{f}(y) = \sum_{x=0}^{N-1} K(y,x)f(x)$ for any $y \in S_n$, in the sense that

$$U\left[\sum_{x=0}^{N-1} f(x)|x\rangle\right] = \sum_{y=0}^{N-1} \tilde{f}(y)|y\rangle. \tag{6.7}$$

*The proposition claims that U maps a state with the probability amplitude $f(x)$ to another state with the probability amplitude $\tilde{f}(y)$ that is related with $f(x)$ through the kernel K.

Here $|x\rangle$ and $|y\rangle$ are basis vectors of $\mathcal{H}$.

Proof. In fact,

$$\begin{aligned}
U\left[\sum_{x=0}^{N-1} f(x)|x\rangle\right] &= \sum_{x=0}^{N-1} f(x)U|x\rangle \\
&= \sum_{x=0}^{N-1} f(x)\left[\sum_{y=0}^{N-1} K(y,x)|y\rangle\right] = \sum_{y=0}^{N-1}\left[\sum_{x=0}^{N-1} K(y,x)f(x)\right]|y\rangle \\
&= \sum_{y=0}^{N-1} \tilde{f}(y)|y\rangle. \qquad (6.8)
\end{aligned}$$

∎

Note that the unitary matrix U computes the discrete integral transform $\tilde{f}(y)$ for *all* variables y by a single operation if it acts on the superposition state $\sum_x f(x)|x\rangle$. There are exponentially large numbers 2^n of y for an n-qubit register, and this fact provides a quantum computer with exponentially fast computing power for a certain kind of computations compared to classical alternatives.

The unitary matrix U implementing a discrete integral transform as in Eq. (6.7) is called the **quantum integral transform** (**QIT**).

EXERCISE 6.1 Let $f \to \tilde{f}$ be a DFT with a unitary kernel K. Prove **Parseval's theorem**

$$\sum_{x=0}^{N-1} |f(x)|^2 = \sum_{y=0}^{N-1} |\tilde{f}(y)|^2. \qquad (6.9)$$

6.2 Quantum Fourier Transform (QFT)

One of the most important quantum integral transforms is the quantum Fourier transform. Let ω_n be the Nth primitive root of 1;

$$\omega_n = e^{2\pi i/N}, \qquad (6.10)$$

where $N = 2^n$ as before. The complex number ω_n defines a kernel K by

$$K(x,y) = \frac{1}{\sqrt{N}}\omega_n^{-xy}. \qquad (6.11)$$

The discrete integral transform with the kernel K,

$$\tilde{f}(y) = \frac{1}{\sqrt{N}}\sum_{x=0}^{N-1} \omega_n^{-xy} f(x), \qquad (6.12)$$

is called the **discrete Fourier transform** (**DFT**).

The kernel K is unitary since

$$(KK^\dagger)(x,y) = \langle x|K \sum_z |z\rangle\langle z|K^\dagger|y\rangle = \sum_z K(x,z)K^\dagger(z,y)$$
$$= \frac{1}{N}\sum_z \omega_n^{-xz}\omega_n^{yz} = \frac{1}{N}\sum_z \omega^{-(x-y)z} = \delta_{xy}.$$

The quantum integral transform defined with this kernel is called the **quantum Fourier transform** (**QFT**).

The kernel for $n = 1$ is

$$K_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & e^{2\pi i/2} \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \tag{6.13}$$

which is nothing but our familiar Hadamard gate. For $n = 2$, we have $\omega_2 = e^{2\pi i/4} = i$ and

$$K_2 = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & \omega_2^{-1} & \omega_2^{-2} & \omega_2^{-3} \\ 1 & \omega_2^{-2} & \omega_2^{-4} & \omega_2^{-6} \\ 1 & \omega_2^{-3} & \omega_2^{-6} & \omega_2^{-9} \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{pmatrix}. \tag{6.14}$$

The inverse DFT is given by

$$f(x) = \frac{1}{\sqrt{N}}\sum_{y=0}^{N-1} \omega_n^{xy}\tilde{f}(y). \tag{6.15}$$

It is important to note that

$$U_{\mathrm{QFT}n}|0\rangle = \frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1} |y\rangle, \tag{6.16}$$

where $U_{\mathrm{QFT}n}$ is the n-qubit QFT gate. This equality shows that the QFT of $f(x) = \delta_{x0}$ is $\tilde{f}(y) = 1/\sqrt{2^n}$, which is similar to the FT of the Dirac delta function $\delta(x)$. Observe that a single application of $U_{\mathrm{QFT}n}$ on the state $|0\rangle$ has produced the superposition of all the basis vectors of $\mathcal{H}$.

EXERCISE 6.2 Let

$$|\psi\rangle = \mathcal{N}\sum_{x=0}^{N-1} \cos\left(\frac{2\pi x}{N}\right)|x\rangle$$

be an n-qubit state.
(1) Normalize $|\psi\rangle$.
(2) Find $U_{\mathrm{QFT}n}|\psi\rangle$.

TABLE 6.1
Coefficient of a vector $|x\rangle|f(y)\rangle$. Only the diagonal combination $|x\rangle|f(x)\rangle$ has nonvanishing amplitude in the initial state $|\Psi\rangle$. Moreover all the non-vanishing coefficients have vanishing phase.

$\|f(7)\rangle$	0	0	0	0	0	0	0	$\frac{1}{\sqrt{8}}$
$\|f(6)\rangle$	0	0	0	0	0	0	$\frac{1}{\sqrt{8}}$	0
$\|f(5)\rangle$	0	0	0	0	0	$\frac{1}{\sqrt{8}}$	0	0
$\|f(4)\rangle$	0	0	0	0	$\frac{1}{\sqrt{8}}$	0	0	0
$\|f(3)\rangle$	0	0	0	$\frac{1}{\sqrt{8}}$	0	0	0	0
$\|f(2)\rangle$	0	0	$\frac{1}{\sqrt{8}}$	0	0	0	0	0
$\|f(1)\rangle$	0	$\frac{1}{\sqrt{8}}$	0	0	0	0	0	0
$\|f(0)\rangle$	$\frac{1}{\sqrt{8}}$	0	0	0	0	0	0	0
	$\|0\rangle$	$\|1\rangle$	$\|2\rangle$	$\|3\rangle$	$\|4\rangle$	$\|5\rangle$	$\|6\rangle$	$\|7\rangle$

6.3 Application of QFT: Period-Finding

There is a cool application of QFT, which is essential in Shor's factorization algorithm. The following example is taken from [2]. Let $|\text{REG}_1\rangle \in \mathcal{H}_1$ be the input register and $|\text{REG}_2\rangle \in \mathcal{H}_2$ be the output register. Each register is a 3-qubit system, to make our argument concrete, and the total system Hilbert space is $\mathcal{H}_1 \otimes \mathcal{H}_2$. Let the initial state of $|\text{REG}_1\rangle$ be

$$\frac{1}{\sqrt{2^3}}(|000\rangle + |001\rangle + \ldots + |111\rangle) = \frac{1}{\sqrt{8}}(|0\rangle + |1\rangle + \ldots + |7\rangle). \tag{6.17}$$

Let $S_3 = \{0, 1, \ldots, 2^3 - 1 = 7\}$ and let $f : S_3 \to S_3$ be a function. Apply f on the initial state to obtain

$$|\Psi\rangle = U_f \frac{1}{\sqrt{8}} \sum_x |x\rangle|0\rangle = \frac{1}{\sqrt{8}} \sum_x |x, f(x)\rangle = \frac{1}{\sqrt{8}}(|0, f(0)\rangle + \ldots + |7, f(7)\rangle). \tag{6.18}$$

It is interesting to visualize the coefficient of each vector as in Table. 6.1.

TABLE 6.2
Coefficient of a vector $|y\rangle|f(x)\rangle$ in $|\psi'\rangle$. The amplitude is 1/8 for all the coefficients. The arrow denotes the phase associated with each coefficient. $\nearrow$ denotes $e^{i\pi/4}$, for example.

$\lvert f(7)\rangle$	$\rightarrow$	$\searrow$	$\downarrow$	$\swarrow$	$\leftarrow$	$\nwarrow$	$\uparrow$	$\nearrow$
$\lvert f(6)\rangle$	$\rightarrow$	$\downarrow$	$\leftarrow$	$\uparrow$	$\rightarrow$	$\downarrow$	$\leftarrow$	$\uparrow$
$\lvert f(5)\rangle$	$\rightarrow$	$\swarrow$	$\uparrow$	$\searrow$	$\leftarrow$	$\nearrow$	$\downarrow$	$\nwarrow$
$\lvert f(4)\rangle$	$\rightarrow$	$\leftarrow$	$\rightarrow$	$\leftarrow$	$\rightarrow$	$\leftarrow$	$\rightarrow$	$\leftarrow$
$\lvert f(3)\rangle$	$\rightarrow$	$\nwarrow$	$\downarrow$	$\nearrow$	$\leftarrow$	$\searrow$	$\uparrow$	$\swarrow$
$\lvert f(2)\rangle$	$\rightarrow$	$\uparrow$	$\leftarrow$	$\downarrow$	$\rightarrow$	$\uparrow$	$\leftarrow$	$\downarrow$
$\lvert f(1)\rangle$	$\rightarrow$	$\nearrow$	$\uparrow$	$\nwarrow$	$\leftarrow$	$\swarrow$	$\downarrow$	$\searrow$
$\lvert f(0)\rangle$	$\rightarrow$	$\rightarrow$	$\rightarrow$	$\rightarrow$	$\rightarrow$	$\rightarrow$	$\rightarrow$	$\rightarrow$
	$\lvert 0\rangle$	$\lvert 1\rangle$	$\lvert 2\rangle$	$\lvert 3\rangle$	$\lvert 4\rangle$	$\lvert 5\rangle$	$\lvert 6\rangle$	$\lvert 7\rangle$

Let us apply the following QFT,

$$|x\rangle \rightarrow \frac{1}{\sqrt{8}}\sum_{y=0}^{7} e^{-2\pi ixy/8}|y\rangle, \tag{6.19}$$

on the first register. Then we obtain

$$\begin{aligned}
|\Psi'\rangle &= \frac{1}{8}\sum_{x,y} e^{-2\pi ixy/8}|y, f(x)\rangle \\
&= \frac{1}{8}|0\rangle \otimes [|f(0)\rangle + |f(1)\rangle + \ldots + |f(7)\rangle] \quad (y=0) \\
&\quad +\frac{1}{8}|1\rangle \otimes \left[|f(0)\rangle + e^{-2\pi i/8}|f(1)\rangle + \ldots + e^{-2\pi i 7/8}|f(7)\rangle\right] \quad (y=1) \\
&\quad \ldots \\
&\quad +\frac{1}{8}|7\rangle \otimes \left[|f(0)\rangle + e^{-14\pi i/8}|f(1)\rangle + \ldots + e^{-14\pi i 7/8}|f(7)\rangle\right]. \quad (y=7)
\end{aligned} \tag{6.20}$$

The coefficient of each component $|y\rangle|f(x)\rangle$ is shown in Table 6.2.

Suppose $f(x)$ is a periodic function satisfying $f(x+P) = f(x)$, $P \in \mathbb{N}$. This period is found from the measurement outcome of $|\mathrm{REG}_1\rangle$. Let $P = 2$, for example. Then it follows that

$$f(0) = f(2) = f(4) = f(6) = a, \quad f(1) = f(3) = f(5) = f(7) = b,$$

where $a, b \in S_3$ and $a \neq b$. The state $|\Psi'\rangle$ now reduces to

$$|\Psi'\rangle = \frac{1}{8}\sum_{x,y\in S_3} e^{-2\pi ixy/8}|y, f(x)\rangle$$

TABLE 6.3
Coefficient of a vector $|y\rangle|f(x)\rangle$ in the state $|\Psi'\rangle$ in which $f(0) = f(2) = f(4) = f(6) = a$ and $f(1) = f(3) = f(5) = f(7) = b$. The amplitude of all the non-vanishing coefficients is 1/2.

$\vert b\rangle$	$\rightarrow$	0	0	0	$\leftarrow$	0	0	0
$\vert a\rangle$	$\rightarrow$	0	0	0	$\rightarrow$	0	0	0
	$\vert 0\rangle$	$\vert 1\rangle$	$\vert 2\rangle$	$\vert 3\rangle$	$\vert 4\rangle$	$\vert 5\rangle$	$\vert 6\rangle$	$\vert 7\rangle$

$$\begin{aligned}
&= \frac{1}{2}|0\rangle \otimes [|a\rangle + |b\rangle] \\
&+\frac{1}{8}|1\rangle \otimes \Big[|a\rangle \left(1 + e^{-1\cdot 2\cdot 2\pi i/8} + e^{-1\cdot 4\cdot 2\pi i/8} + e^{-1\cdot 6\cdot 2\pi i/8}\right) \\
&+|b\rangle \left(e^{-1\cdot 1\cdot 2\pi i/8} + e^{-1\cdot 3\cdot 2\pi i/8} + e^{-1\cdot 5\cdot 2\pi i/8} + e^{-1\cdot 7\cdot 2\pi i/8}\right)\Big] \\
&\ldots
\end{aligned} \tag{6.21}$$

As a result, all the vectors but

$$|0, a\rangle, |0, b\rangle, |4, a\rangle, |4, b\rangle$$

cancel out to vanish, and we are left with

$$|\Psi'\rangle = \frac{1}{2}\left(|0, a\rangle + |0, b\rangle + |4, a\rangle + e^{-i\pi}|4, b\rangle\right) \tag{6.22}$$

(see Table 6.3).

If we measure the first register $|\mathrm{REG}_1\rangle$, the result is either 0 or 4, which is the direct consequence of the periodicity $P = 2$ of $f(x)$.

EXERCISE 6.3 Suppose each register above is an n-qubit system. Let $f(x)$ be a periodic function with the period P. Show that the observed value of the first register is one of

$$0, \frac{1\cdot 2^n}{P}, \frac{2\cdot 2^n}{P}, \frac{3\cdot 2^n}{P}, \ldots \frac{(P-1)2^n}{P}, \tag{6.23}$$

where it is assumed that $2^n/P \in \mathbb{N}$.

The cancellation observed above is extensively made use of in Shor's factorization algorithm.

6.4 Implementation of QFT

We now consider a quantum circuit $U_{\text{QFT}n}$ which implements the n-qubit QFT. The circuit $U_{\text{QFT}n}$ maps a state $\sum_x f(x)|x\rangle$ to a state $\sum_y \tilde{f}(y)|y\rangle$, where

$$\tilde{f}(y) = \frac{1}{\sqrt{N}} \sum_x \omega_n^{-xy} f(x), \quad \omega_n = e^{2\pi i/N}.$$

Thus for $f(x') = \delta_{x'x}$, we obtain $\tilde{f}(y) = \omega_n^{-xy}/\sqrt{N}$, namely

$$U_{\text{QFT}n}|x\rangle = \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} e^{-2\pi i xy/N}|y\rangle.$$

Let us start our implmentation of QFT with $n = 1, 2$ and 3 to familiarize ourselves with the problem.

$\underline{n = 1}$

Eq. (6.13) shows that the kernel for $n = 1$ QFT is the Hadamard gate H, whose action on $|x\rangle$, $x \in \{0, 1\}$, is concisely written as

$$U_{\text{H}}|x\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^x|1\rangle) = \frac{1}{\sqrt{2}} \sum_{y=0}^{1} (-1)^{xy}|y\rangle. \tag{6.24}$$

In fact, this is the defining equation for $n = 1$ QFT as

$$U_{\text{QFT}1}|x\rangle = \frac{1}{\sqrt{2}} \sum_{y=0}^{1} \omega_1^{-xy}|y\rangle = \frac{1}{\sqrt{2}} \sum_{y=0}^{1} (-1)^{xy}|y\rangle. \tag{6.25}$$

It is instructive to demonstrate Eq. (6.7) explicitly here. Let $|\psi\rangle = f(0)|0\rangle + f(1)|1\rangle$ be any one-qubit state. Then

$$\begin{aligned} U_{\text{QFT}1}|\psi\rangle &= f(0)\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) + f(1)\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\ &= \frac{1}{\sqrt{2}}\left(f(0) + f(1)\right)|0\rangle + \frac{1}{\sqrt{2}}\left(f(0) - f(1)\right)|1\rangle = \sum_{y=0}^{1} \tilde{f}(y)|y\rangle. \end{aligned}$$

$\underline{n = 2}$

This case is considerably more complicated than the case $n = 1$. It also gives important insights into implementing QFT with $n \geq 3$. Let us introduce an important gate, the **controlled-$\boldsymbol{B_{jk}}$** gate. The B_{jk} gate is defined by the matrix

$$B_{jk} = \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\theta_{jk}} \end{pmatrix}, \quad \theta_{jk} = \frac{2\pi}{2^{k-j+1}}, \tag{6.26}$$

where $j, k \in \{0, 1, 2, \ldots\}$ and $k \geq j$.

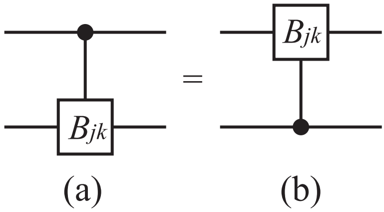

FIGURE 6.1
(a) Controlled-B_{jk} gate. The inverted controlled-B_{jk} gate (b) and the controlled-B_{jk} gate are equivalent (see Lemma 6.1).

LEMMA 6.1 The controlled-B_{jk} gate U_{jk} in Fig. 6.1 (a) acts on $|x\rangle|y\rangle$, $x, y \in \{0, 1\}$, as

$$U_{jk}|x, y\rangle = e^{-i\theta_{jk}xy}|x, y\rangle = \exp\left(-\frac{2\pi i}{2^{k-j+1}}xy\right)|x, y\rangle. \tag{6.27}$$

Proof. The controlled-B_{jk} gate is written as

$$U_{jk} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes B_{jk}, \tag{6.28}$$

and its action on $|x, y\rangle$ is

$$\begin{aligned} U_{jk}|x, y\rangle &= |0\rangle\langle 0|x\rangle \otimes |y\rangle + |1\rangle\langle 1|x\rangle \otimes B_{jk}|y\rangle \\ &= \begin{cases} |x\rangle \otimes |y\rangle & x = 0 \\ |x\rangle \otimes B_{jk}|y\rangle & x = 1. \end{cases} \end{aligned} \tag{6.29}$$

Moreover, when $x = 1$ we have

$$B_{jk}|y\rangle = \begin{cases} |y\rangle & y = 0 \\ e^{-i\theta_{jk}}|y\rangle & y = 1. \end{cases} \tag{6.30}$$

Thus the action of U_{jk} on $|y\rangle$ is trivial if $xy = 0$ and nontrivial if and only if $x = y = 1$. These results may be summarized as Eq. (6.27). ∎

The action of the controlled-B_{jk} gate on a basis vector $|x\rangle|y\rangle$ is detemined by the combination xy and not by x and y independently. Therefore the controlled-B_{jk} gate and the "inverted" controlled-B_{jk} gate are equivalent; see Fig. 6.1.

The DFT for $n = 2$ is defined as

$$\tilde{f}(y) = \frac{1}{2}\sum_{x=0}^{3} \omega_2^{-xy} f(x), \quad \omega_2 = e^{2\pi i/4} = i, \; y \in S_2. \tag{6.31}$$

Equation (6.6) in Proposition 6.2 states that our task is to find a unitary matrix U_{QFT2} such that

$$U_{\text{QFT2}}|x\rangle = \frac{1}{2}\sum_{y=0}^{3} \omega_2^{-xy}|y\rangle. \tag{6.32}$$

Let us write x and y in the binary form as $x = 2x_1 + x_0$ and $y = 2y_1 + y_0$, respectively. The action of U_{QFT2} on $|x\rangle$ is

$$\begin{aligned}
U_{\text{QFT2}}|x_1x_0\rangle &= \frac{1}{2}\sum_{y=0}^{3} e^{-2\pi ixy/2^2}|y\rangle = \frac{1}{2}\sum_{y_0,y_1=0}^{1} e^{-2\pi ix(2y_1+y_0)/2^2}|y_1y_0\rangle \\
&= \frac{1}{2}\sum_{y_1} e^{-2\pi ixy_1/2}|y_1\rangle \otimes \sum_{y_0} e^{-2\pi ixy_0/2^2}|y_0\rangle \\
&= \frac{1}{2}\left(|0\rangle + e^{-2\pi ix/2}|1\rangle\right) \otimes \left(|0\rangle + e^{-2\pi ix/2^2}|1\rangle\right) \\
&= \frac{1}{2}\left(|0\rangle + e^{-2\pi i(2x_1+x_0)/2}|1\rangle\right) \otimes \left(|0\rangle + e^{-2\pi i(2x_1+x_0)/2^2}|1\rangle\right) \\
&= \frac{1}{2}\left(|0\rangle + e^{-\pi ix_0}|1\rangle\right) \otimes \left(|0\rangle + e^{-\pi ix_1}e^{-i(\pi/2)x_0}|1\rangle\right) \\
&= \frac{1}{2}\left(|0\rangle + (-1)^{x_0}|1\rangle\right) \otimes B_{12}^{x_0}\left(|0\rangle + (-1)^{x_1}|1\rangle\right),
\end{aligned} \tag{6.33}$$

where use has been made of the fact $\theta_{12} = 2\pi/2^{2-1+1} = \pi/2$ to obtain the last expression. Note that $B_{12}^{x_0}$ is the controlled-B_{12} gate with the control bit x_0 and the target bit x_1; $B_{12}^0 = I$ while $B_{12}^1 = B_{12}$. Note also that, in spite of its tensor product looking appearance, the last line of Eq. (6.33) is entangled due to this conditional operation. Equation (6.33) suggests that the $n = 2$ QFT are implemented with the Hadamard and the U_{12} gates. Before writing down the quantum circuit realizing Eq. (6.33), we should note that the first qubit has a power $(-1)^{x_0}$, while the second one has $(-1)^{x_1}$, when the input state is $|x_1x_0\rangle$. If we naively applied the Hadamard gate to the second qubit, we would obtain

$$(I \otimes U_{\text{H}})|x_1x_0\rangle = |x_1\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{x_0}|1\rangle).$$

These facts suggest that we need to swap the first and second qubits at the beginning of the implementation so that

$$\begin{aligned}
U_{\text{QFT2}}|x_1x_0\rangle &= \frac{1}{\sqrt{2^2}}\left(|0\rangle + (-1)^{x_0}|1\rangle\right) \otimes B_{12}^{x_0}\left(|0\rangle + (-1)^{x_1}|1\rangle\right) \\
&= (U_{\text{H}} \otimes I)U_{12}(I \otimes U_{\text{H}})|x_0, x_1\rangle \\
&= (U_{\text{H}} \otimes I)U_{12}(I \otimes U_{\text{H}})U_{\text{SWAP}}|x_1x_0\rangle.
\end{aligned} \tag{6.34}$$

Since Eq. (6.34) is true for any $|x_1x_0\rangle$, we should have $U_{\text{QFT2}} = (U_{\text{H}} \otimes I)U_{12}(I \otimes U_{\text{H}})U_{\text{SWAP}}$, which proves the following proposition.

PROPOSITION 6.3 The $n = 2$ QFT gate is implemented as

$$U_{\mathrm{QFT2}} = (U_{\mathrm{H}} \otimes I)U_{12}(I \otimes U_{\mathrm{H}})U_{\mathrm{SWAP}} \tag{6.35}$$

(see Fig. 6.2).

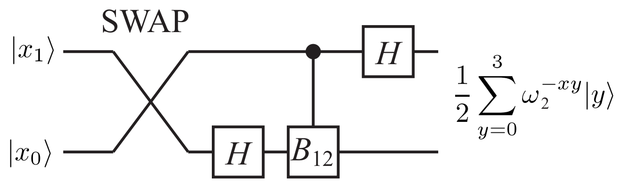

FIGURE 6.2
Implementation of the $n = 2$ QFT, U_{QFT2}.

The reader should verify the above implementation by explicitly writing down the gates as matrices.

EXERCISE 6.4 It is also possible to have the SWAP gate in the very end of the implementation. Design such an $n = 2$ QFT gate.

This construction is easily generalized to $n \geq 3$ as we see next.

$n = 3$ and beyond

It is instructive to rewrite the construction of $n = 2$ QFT in a more generalizable form. The state $|x_1x_0\rangle$ has been transformed as in Eq. (6.33):

$$\begin{aligned}|x_1x_0\rangle &\to \frac{1}{\sqrt{2^2}}\sum_{y=0}^{2^2-1} e^{-2\pi i xy/2^2}|y\rangle \\ &= \frac{1}{\sqrt{2^2}}(|0\rangle + e^{-2\pi i x_0/2}|1\rangle) \otimes (|0\rangle + e^{-2\pi i(x_1/2+x_0/2^2)}|1\rangle).\end{aligned}$$

This observation suggests the following construction of $n = 3$ QFT:

$$\begin{aligned}&U_{\mathrm{QFT3}}|x_2x_1x_0\rangle \\ &= \frac{1}{\sqrt{2^3}}(|0\rangle + e^{-2\pi i x_0/2}|1\rangle) \otimes (|0\rangle + e^{-2\pi i(x_1/2+x_0/2^2)}|1\rangle) \\ &\quad \otimes(|0\rangle + e^{-2\pi i(x_2/2+x_1/2^2+x_0/2^3)}|1\rangle) \\ &= \frac{1}{\sqrt{2^3}}(|0\rangle + (-1)^{x_0}|1\rangle) \otimes B_{01}^{x_0}(|0\rangle + (-1)^{x_1}|1\rangle) \\ &\quad \otimes B_{02}^{x_0}B_{12}^{x_1}(|0\rangle + (-1)^{x_2}|1\rangle)\end{aligned}$$

$$= (U_{\rm H} \otimes I \otimes I)U_{01}(I \otimes U_{\rm H} \otimes I)U_{02}U_{12}(I \otimes I \otimes U_{\rm H})|x_0x_1x_2\rangle$$
$$= (U_{\rm H} \otimes I \otimes I)U_{01}(I \otimes U_{\rm H} \otimes I)U_{02}U_{12}(I \otimes I \otimes U_{\rm H})P|x_2x_1x_0\rangle, \quad (6.36)$$

where U_{jk} is the controlled-B_{jk} gate with the control qubit x_j, and the gate P reverses the order of the qubits as $P|x_2x_1x_0\rangle = |x_0x_1x_2\rangle$. For a three-qubit QFT, P is a SWAP gate between the first qubit (x_2) and the third qubit (x_0). Again note here that we should be careful in ordering the gates so that the control bit x_j acts in U_{jk} before it is acted by a Hadamard gate.

EXERCISE 6.5 Let $x = 2^2x_2 + 2x_1 + x_0$ and $y = 2^2y_2 + 2y_1 + y_0$.
(1) Write down the RHS of

$$U_{\rm QFT3}|x_2x_1x_0\rangle = \frac{1}{\sqrt{2^3}}\sum_{y=0}^{2^3-1} e^{-2\pi ixy/2^3}|y\rangle \quad (6.37)$$

explicitly in terms of x_i and y_i.
(2) Show that the RHS of Eq. (6.37) agrees with the first line of the RHS of Eq. (6.36).

Since Eq. (6.36) is true for any $|x_2x_1x_0\rangle$, we have found

$$U_{\rm QFT3} = (U_{\rm H} \otimes I \otimes I)U_{01}(I \otimes U_{\rm H} \otimes I)U_{02}U_{12}(I \otimes I \otimes U_{\rm H})P. \quad (6.38)$$

Equation (6.38) readily leads us to the quantum circuit in Fig. 6.3.

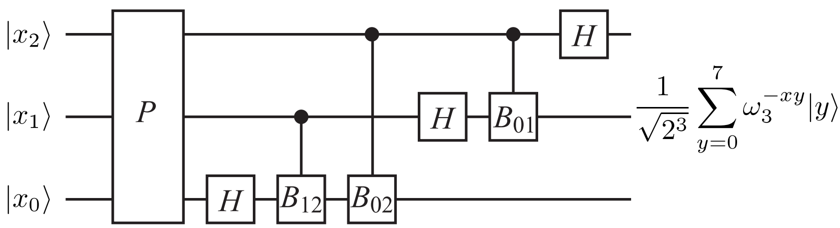

FIGURE 6.3
Implementation of the $n = 3$ QFT.

EXERCISE 6.6 Design a quantum circuit $U_{\rm QFT3}$ in which the permutation gate P is at the very end of the circuit.

Now the generalization of the present construction to $n \geq 4$ should be easy. The equation that generalizes Eq. (6.36) is

$$U_{{\rm QFT}n}|x_{n-1}\ldots x_1x_0\rangle$$

$$
\begin{aligned}
&= \frac{1}{\sqrt{N}}(|0\rangle + e^{-2\pi i x_0/2}|1\rangle) \otimes (|0\rangle + e^{-2\pi i(x_1/2+x_0/2^2)}|1\rangle) \\
&\quad \otimes(|0\rangle + e^{-2\pi i(x_2/2+x_1/2^2+x_0/2^3)}|1\rangle) \otimes \ldots \\
&\quad \ldots \otimes (|0\rangle + e^{-2\pi i(x_{n-1}/2+x_{n-2}/2^2+\ldots x_1/2^{n-1}+x_0/2^n)}|1\rangle) \\
&= (U_{\mathrm{H}} \otimes I \otimes \ldots \otimes I)U_{01}(I \otimes U_{\mathrm{H}} \otimes I \otimes \ldots \otimes I)U_{02}U_{12} \\
&\quad \times(I \otimes I \otimes U_{\mathrm{H}} \otimes \ldots \otimes I) \ldots \\
&\quad \times U_{0,n-1}U_{1,n-1} \ldots U_{n-2,n-1}(I \otimes \ldots \otimes I \otimes U_{\mathrm{H}})|x_0 x_1 \ldots x_{n-1}\rangle \\
&= (U_{\mathrm{H}} \otimes I \otimes \ldots \otimes I)U_{01}(I \otimes U_{\mathrm{H}} \otimes I \otimes \ldots \otimes I)U_{02}U_{12} \\
&\quad \times(I \otimes I \otimes U_{\mathrm{H}} \otimes \ldots \otimes I) \ldots U_{0,n-1}U_{1,n-1} \ldots U_{n-2,n-1} \\
&\quad \times(I \otimes \ldots \otimes I \otimes U_{\mathrm{H}})P|x_{n-1} \ldots x_1 x_0\rangle, \qquad (6.39)
\end{aligned}
$$

where P reverses the order of x_k as $P|x_{n-1} \ldots x_1 x_0\rangle = |x_0 x_1 \ldots x_{n-1}\rangle$.

We finally find the following decompostion of $U_{\mathrm{QFT}n}$:

$$
\begin{aligned}
U_{\mathrm{QFT}n} = &(U_{\mathrm{H}} \otimes I \otimes \ldots \otimes I)U_{01}(I \otimes U_{\mathrm{H}} \otimes I \otimes \ldots \otimes I)U_{02}U_{12} \\
&\times(I \otimes I \otimes U_{\mathrm{H}} \otimes \ldots \otimes I) \ldots \\
&\times U_{0,n-1}U_{1,n-1} \ldots U_{n-2,n-1}(I \otimes \ldots \otimes I \otimes U_{\mathrm{H}})P. \qquad (6.40)
\end{aligned}
$$

A quantum circuit which implements $U_{\mathrm{QFT}n}$ is found from Eq. (6.40) as in Fig. 6.4. It may be proved, by induction, for example, that the circuit in

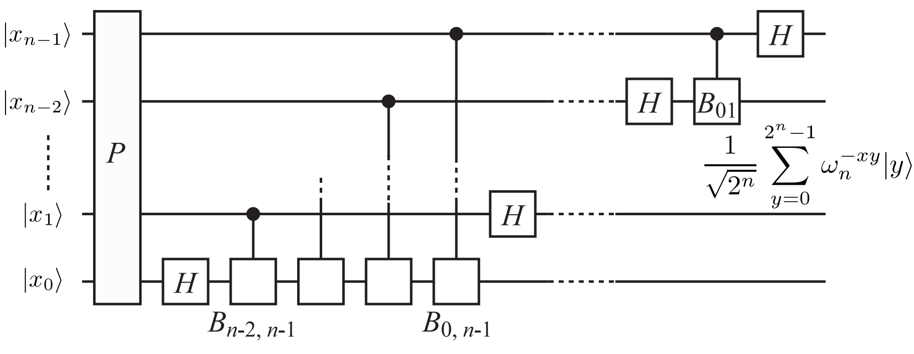

FIGURE 6.4
Implementation of the n-qubit QFT.

Fig. 6.4 really implements the n-qubit QFT.

PROPOSITION 6.4 The n-qubit QFT may be constructed with $\Theta(n^2)$ elementary gates.

Proof. The n-qubit QFT is made of a P gate, n Hadamard gates and $(n -$

$1) + (n-2) + \ldots + 2 + 1 = n(n-1)/2$ controlled-B_{jk} gates (see Fig. 6.4). It has been shown in §4.2.3 that it requres three CNOT gates to construct a SWAP gate. Furthermore, a P gate for n qubits requires $\lfloor n/2 \rfloor$ SWAP gates,† assuming that there exists a SWAP gate for any pair of qubits. Thus a P gate requires $3 \times \lfloor n/2 \rfloor = \Theta(n)$ elementary gates. Proposition 4.1 states that a controlled-B_{ij} gate is constructed with at most six elementary gates. Thus it has been proved that the n-qubit QFT is made of $\Theta(n^2)$ elementary gates. ∎

The above proposition is quite important in estimating the efficiency of quantum algorithms. If we look at the definition

$$\tilde{f}(y) = \frac{1}{\sqrt{N}} \sum_{x=0}^{2^n-1} \omega^{-xy} f(x),$$

we naively expect that $N = 2^n$ steps (including the evaluation of exponential functions followed by multiplication) are required for *each* y and $N \times N$ steps for all y's. In other words, it takes exponentially large steps ($\sim N^2 = e^{2n \ln 2}$) to carry out the QFT. The above proposition states that this is done in $\Theta(n^2)$ steps with the QFT gate if the initial state is a superposition of all x's.

6.5 Walsh-Hadamard Transform

There are two other quantum integral transforms, the Walsh-Hadamard transform and the selective phase rotation transform, which are often employed in quantum computing.

We have already encountered the Walsh-Hadamard transform in §4.2.2 and §5.2. Let $x, y \in S_n = \{0, 1, \ldots, N-1\}$ with binary expressions $x_{n-1}x_{n-2}\ldots x_0$ and $y_{n-1}y_{n-2}\ldots y_0$, where $N = 2^n$. The Walsh-Hadamard transform, written in the form of Eq. (5.7), shows that it is a quantum integral transform with a kernel $W_n : S_n \times S_n \to \mathbb{C}$ defined by

$$W_n(x, y) = \frac{1}{\sqrt{N}}(-1)^{x \cdot y} \quad (x, y \in S_n), \tag{6.41}$$

where $x \cdot y = x_{n-1}y_{n-1} \oplus x_{n-2}y_{n-2} \oplus \ldots \oplus x_0 y_0$. This kernel defines a discrete integral transform

$$\tilde{f}(y) = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} (-1)^{x \cdot y} f(x). \tag{6.42}$$

† $\lfloor x \rfloor$ is the largest integer which is less than or equal to $x \in \mathbb{R}$ and called the floor of x.

6.6 Selective Phase Rotation Transform

DEFINITION 6.2 (**Selective Phase Rotation Transform**) Let us define a kernel

$$K_n(x,y) = e^{i\theta_x}\delta_{xy}, \quad \forall x, y \in S_n, \tag{6.43}$$

where $\theta_x \in \mathbb{R}$. The discrete integral transform

$$\tilde{f}(y) = \sum_{x=0}^{N-1} K(x,y)f(x) = \sum_{x=0}^{N-1} e^{i\theta_x}\delta_{xy} f(x) = e^{i\theta_y} f(y) \tag{6.44}$$

with the kernel K_n is called the **selective phase rotation transform**.

EXERCISE 6.7 Show that K_n defined above is unitary. Write down the inverse transformation K_n^{-1}.

The matrix representations for K_1 and K_2 are

$$K_1 = \begin{pmatrix} e^{i\theta_0} & 0 \\ 0 & e^{i\theta_1} \end{pmatrix}, \quad K_2 = \begin{pmatrix} e^{i\theta_0} & 0 & 0 & 0 \\ 0 & e^{i\theta_1} & 0 & 0 \\ 0 & 0 & e^{i\theta_2} & 0 \\ 0 & 0 & 0 & e^{i\theta_3} \end{pmatrix}.$$

The implementation of K_n is achieved with the universal set of gates as follows. Take $n = 2$, for example. The kernel K_2 has been given above. This is decomposed as a product of two two-level unitary matrices as

$$K_2 = A_0 A_1, \tag{6.45}$$

where

$$A_0 = \begin{pmatrix} e^{i\theta_0} & 0 & 0 & 0 \\ 0 & e^{i\theta_1} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & e^{i\theta_2} & 0 \\ 0 & 0 & 0 & e^{i\theta_3} \end{pmatrix}. \tag{6.46}$$

Note that

$$A_0 = |0\rangle\langle 0| \otimes U_0 + |1\rangle\langle 1| \otimes I, \quad U_0 = \begin{pmatrix} e^{i\theta_0} & 0 \\ 0 & e^{i\theta_1} \end{pmatrix},$$

$$A_1 = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U_1, \quad U_1 = \begin{pmatrix} e^{i\theta_2} & 0 \\ 0 & e^{i\theta_3} \end{pmatrix}.$$

Thus A_1 is realized as an ordinary controlled-U_1 gate while the control bit is negated in A_0. Then what we have to do for A_0 is to negate the control bit first and then to apply ordinary controlled-U_0 gate and finally to negate the control bit back to its input state. In summary, A_0 is implemented as in

Fig. 6.5. In fact, it can be readily verified that the gate in Fig. 6.5 is written as

$$\begin{aligned}&(X \otimes I)(|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U_0)(X \otimes I)\\ &= X|0\rangle\langle 0|X \otimes I + X|1\rangle\langle 1|X \otimes U_0 = |1\rangle\langle 1| \otimes I + |0\rangle\langle 0| \otimes U_0 = A_0.\end{aligned}$$

Thus these gates are implemented with the set of universal gates. In fact, the order of A_i does not matter since $[A_0, A_1] = 0$.

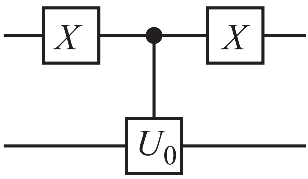

FIGURE 6.5
Implementation of A_0.

EXERCISE 6.8 Repeat the above arguments for $n = 3$. In this case K_3 is written as a product of four two-level unitary matrices. Write down these matrices and find the quantum circuits similar to that in Fig. 6.5 that implements these two-level unitary matrices.

References

[1] Y. Uesaka, *Mathematical Principle of Quantum Computation*, Corona Publishing, Tokyo, in Japanese (2000).

[2] G. P. Berman, G. D. Doolen, R. Mainieri and V. I. Tsifrinovich, *Introduction to Quantum Computers*, World Scientific, Singapore (1998).

7

Grover's Search Algorithm

Suppose there is a stack of N unstructured files and we want to find a particular file (or files) out of the N files. To find someone's phone number in a telephone directory is an easy structured datebase search problem, while to find a person's name who has a particular phone number is a more difficult unstructured database seach problem, with which we are concerned in this chapter. It is required to take $O(N)$ steps on average if a classical algorithm is employed. If we check the files one by one, we will hit the right file with probability $1/2$ after $N/2$ files are examined. It turns out that this takes only $O(\sqrt{N})$ steps with a quantum algorithm, first discovered by Grover [1, 2, 3]. Our presentation in this chapter closely follows [4] and [5].

7.1 Searching for a Single File

Suppose there is a stack of $N = 2^n$ files, randomly placed, that are numbered by $x \in S_n \equiv \{0, 1, \ldots, N-1\}$. Our task is to find an algorithm which picks out a particular file which satisfies a certain condition.

In mathematical language, this is expressed as follows. Let $f : S_n \to \{0, 1\}$ be a function defined by

$$f(x) = \begin{cases} 1 & (x = z) \\ 0 & (x \neq z), \end{cases} \tag{7.1}$$

where z is the address of the file we are looking for. It is assumed that $f(x)$ is *instantaneously* calculable, such that this process does not require any computational steps. A function of this sort is often called an oracle as noted in Chapter 5. Thus, the problem is to find z such that $f(z) = 1$, given a function $f : S_n \to \{0, 1\}$ which assumes the value 1 only at a single point.

Clearly we have to check one file after another in a classical algorithm, and it will take $O(N)$ steps on average. It is shown below that it takes only $O(\sqrt{N})$ steps with Grover's algorithm. This is accomplished by *amplifying* the amplitude of the vector $|z\rangle$ while cancelling that of the vectors $|x\rangle$ $(x \neq z)$.

We describe the algorithm in several steps.

STEP 1 (Selective phase rotation transform; see §6.6.)

Define the kernel of the selective phase rotation transform R_f by

$$K_f(x,y) = e^{i\pi f(x)}\delta_{xy} = (-1)^{f(x)}\delta_{xy}, \tag{7.2}$$

where $x, y \in S_n$. Since R_f maps $|z\rangle \mapsto -|z\rangle$, while leaving all the other vectors unchanged, it can be expressed as

$$R_f = I - 2|z\rangle\langle z|. \tag{7.3}$$

Let us consider a state

$$|\varphi\rangle = \sum_{x=0}^{N-1} w_x|x\rangle, \quad \sum_x |w_x|^2 = 1. \tag{7.4}$$

Then it is easy to verify

$$R_f|\varphi\rangle = w_0|0\rangle + \ldots + (-1)w_z|z\rangle + \ldots + w_{N-1}|N-1\rangle. \tag{7.5}$$

(In other words, R_f changes the sign of w_z while leaving all other coefficients unchanged.)

STEP 2 Define a unitary matrix

$$D = W_n R_0 W_n, \tag{7.6}$$

where W_n is the Walsh-Hadamard transform,

$$W_n(x,y) = \frac{1}{\sqrt{N}}(-1)^{x\cdot y}, \quad (x, y \in S_n) \tag{7.7}$$

and R_0 is the selective phase rotation transform defined by

$$R_0(x,y) = e^{i\pi(1-\delta_{x0})}\delta_{xy} = (-1)^{1-\delta_{x0}}\delta_{xy}. \tag{7.8}$$

PROPOSITION 7.1 Let

$$|\varphi_0\rangle = \frac{1}{\sqrt{N}}\sum_{x=0}^{N-1}|x\rangle. \tag{7.9}$$

Then

$$D = -I + 2|\varphi_0\rangle\langle\varphi_0|. \tag{7.10}$$

Moreover

$$D|\varphi\rangle = \sum_{x=0}^{N-1}\left(\bar{w} - (w_x - \bar{w})\right)|x\rangle, \tag{7.11}$$

where $|\varphi\rangle$ is given in Eq. (7.4) and

$$\bar{w} = \frac{1}{N}\sum_{x=0}^{N-1} w_x \tag{7.12}$$

is the avarage of w_x over S_n.

Proof. Let us evaluate the matrix elements of the RHS of Eq. (7.10). We obtain from

$$-I + 2|\varphi_0\rangle\langle\varphi_0| = -I + \frac{2}{N}\sum_x |x\rangle \sum_y \langle y| = -I + \frac{2}{N}\sum_{x,y} |x\rangle\langle y|$$

that the (x, y)-component of the RHS is

$$\langle x|\mathrm{RHS}|y\rangle = -\delta_{xy} + \frac{2}{N}. \tag{7.13}$$

Let us turn to the LHS next. The (x, y)-component of $D = W_n R_0 W_n$ is

$$\begin{aligned}\langle x|W_n R_0 W_n|y\rangle &= \sum_{u,v} \langle x|W_n|u\rangle\langle u|R_0|v\rangle\langle v|W_n|y\rangle \\ &= \frac{1}{N}\sum_{u,v}(-1)^{x\cdot u} \\ &\quad \times(-1)^{1-\delta_{u0}}\delta_{uv}(-1)^{v\cdot y}.\end{aligned}$$

The summation over u is evaluated as

$$\begin{aligned}&\sum_{u=0}^{N-1}(-1)^{x\cdot u}(-1)^{1-\delta_{u0}}\delta_{uv} \\ &= (-1)^0(-1)^0\delta_{0v} - \sum_{u=1}^{N-1}(-1)^{x\cdot u}\delta_{uv} \\ &= 2\delta_{0v} - \sum_{u=0}^{N-1}(-1)^{x_{n-1}u_{n-1}+\ldots+x_1u_1+x_0u_0}\delta_{u_{n-1}v_{n-1}}\ldots\delta_{u_1v_1}\delta_{u_0v_0} \\ &= 2\delta_{0v} - \left[\sum_{u_{n-1}=0}^{1}(-1)^{x_{n-1}u_{n-1}}\delta_{u_{n-1}v_{n-1}}\right]\ldots\left[\sum_{u_1=0}^{1}(-1)^{x_1u_1}\delta_{u_1v_1}\right] \\ &\quad\times\left[\sum_{u_0=0}^{1}(-1)^{x_0u_0}\delta_{u_0v_0}\right].\end{aligned}$$

Then the LHS becomes

$$\begin{aligned}\langle x|D|y\rangle = \frac{1}{N}\sum_{v=0}^{N-1}&\left[2\delta_{0v} - \left(\sum_{u_{n-1}=0}^{1}(-1)^{x_{n-1}u_{n-1}}\delta_{u_{n-1}v_{n-1}}\right)\right. \\ &\left.\ldots\left(\sum_{u_0=0}^{1}(-1)^{x_0u_0}\delta_{u_0v_0}\right)\right](-1)^{v\cdot y}\end{aligned}$$

$$
\begin{aligned}
&= \frac{2}{N} - \frac{1}{N}\left[\sum_{u_{n-1},v_{n-1}=0}^{1} (-1)^{x_{n-1}u_{n-1}+v_{n-1}y_{n-1}}\delta_{u_{n-1}v_{n-1}}\right] \\
&\quad \ldots \left[\sum_{u_0,v_0=0}^{1} (-1)^{x_0u_0+v_0y_0}\delta_{u_0v_0}\right] \\
&= \frac{2}{N} - \frac{1}{N}\left[1+(-1)^{x_{n-1}+y_{n-1}}\right]\ldots\left[1+(-1)^{x_0+y_0}\right] \\
&= \frac{2}{N} - \frac{2^n}{N}\delta_{x_{n-1}y_{n-1}}\ldots\delta_{x_0y_0} \\
&= \frac{2}{N} - \delta_{xy},
\end{aligned}
$$

which proves Eq. (7.10).

Equation (7.11) is proved as

$$
\begin{aligned}
D|\varphi\rangle &= (-I+2|\varphi_0\rangle\langle\varphi_0|)|\varphi\rangle = \left(-I+\frac{2}{N}\sum_{y,z}|y\rangle\langle z|\right)\sum_x w_x|x\rangle \\
&= -\sum_x w_x|x\rangle + \frac{2}{N}\sum_{x,y,z} w_x|y\rangle\delta_{xz} = -\sum_x w_x|x\rangle + \frac{2}{N}\sum_{y,z} w_z|y\rangle \\
&= -\sum_x w_x|x\rangle + 2\sum_y \bar{w}|y\rangle = \sum_{x=0}^{N-1}\left[\bar{w}-(w_x-\bar{w})\right]|x\rangle.
\end{aligned}
$$

∎

Equation (7.11) shows that D is an operator that produces "inversion about the average" since the quantity $\bar{w}-(w_x-\bar{w}) = 2\bar{w}-w_x$ is obtained by reflecting w_x about $\bar{w}$.

STEP 3 Now let us consider the unitary transformation U_f defined by

$$
U_f = DR_f = (-I+2|\varphi_0\rangle\langle\varphi_0|)\,(I-2|z\rangle\langle z|) \tag{7.14}
$$

and consider its action on $|\varphi\rangle = \sum_x w_x|x\rangle$. Direct application of the results in step 1 and step 2 yields

$$
\begin{aligned}
U_f|\varphi\rangle &= D\left(\sum_{x\neq z} w_x|x\rangle - w_z|z\rangle\right) = \sum_{x\neq z}[\bar{w}-(w_x-\bar{w})]|x\rangle + [\bar{w}+(w_z+\bar{w})]|z\rangle \\
&= \sum_{\substack{x=0 \\ x\neq z}}^{N-1}(2\bar{w}-w_x)|x\rangle + (2\bar{w}+w_z)|z\rangle,
\end{aligned} \tag{7.15}
$$

where $\sum_{x=0}^{N-1} |w_x|^2 = 1$ and

$$\bar{w} = \frac{1}{N}\left(\sum_{x=0, x\neq z}^{N-1} w_x - w_z\right) \tag{7.16}$$

is the average value of the coefficients of the state $R_f|\varphi\rangle$.

This result shows that the amplitude of $|z\rangle$ has increased upon the operation of U_f while that of $|x\rangle$ ($x \neq z$) has decreased, assuming that all the weights w_x are positive. Thus repeated applications of U_f increase the amplitude of $|z\rangle$ so that this particular state is observed with probability close to 1 when the system is measured. Let us find the state obtained after U_f is applied k times on the initial state $|\varphi_0\rangle$.

PROPOSITION 7.2 Let us write

$$U_f^k|\varphi_0\rangle = a_k|z\rangle + b_k \sum_{x\neq z} |x\rangle \tag{7.17}$$

with the initial condition

$$a_0 = b_0 = \frac{1}{\sqrt{N}}.$$

Then the coefficients $\{a_k, b_k\}$ for $k \geq 1$ satisfy the recursion relations

$$a_k = \frac{N-2}{N} a_{k-1} + \frac{2(N-1)}{N} b_{k-1}, \tag{7.18}$$

$$b_k = -\frac{2}{N} a_{k-1} + \frac{N-2}{N} b_{k-1} \tag{7.19}$$

for $k = 1, 2, \ldots$.

Proof. It is easy to see the recursion relations are satified for $k = 1$ by making use of Eqs. (7.15) and (7.16). Let $U_f^{k-1}|\varphi_0\rangle = a_{k-1}|z\rangle + b_{k-1}\sum_{x\neq z}|x\rangle$. Then

$$\begin{aligned}
U_f^k|\varphi_0\rangle &= U_f\left(a_{k-1}|z\rangle + b_{k-1}\sum_{x\neq z}|x\rangle\right) \\
&= (-I + 2|\varphi_0\rangle\langle\varphi_0|)\left(-a_{k-1}|z\rangle + b_{k-1}\sum_{x\neq z}|x\rangle\right) \\
&= -b_{k-1}\sum_{x\neq z}|x\rangle + a_{k-1}|z\rangle + \frac{2}{\sqrt{N}}(N-1)b_{k-1}|\varphi_0\rangle - \frac{2a_{k-1}}{\sqrt{N}}|\varphi_0\rangle \\
&= -b_{k-1}\sum_{x\neq z}|x\rangle + a_{k-1}|z\rangle + \frac{2}{N}(N-1)b_{k-1}\sum_x|x\rangle - \frac{2a_{k-1}}{N}\sum_x|x\rangle \\
&= \left[\frac{N-2}{N}a_{k-1} + \frac{2(N-1)}{N}b_{k-1}\right]|z\rangle
\end{aligned}$$

$$+\left[-\frac{2}{N}a_{k-1}+\frac{N-2}{N}b_{k-1}\right]\sum_{x\neq z}|x\rangle,$$

and proposition is proved. ∎

PROPOSITION 7.3 The solutions of the recursion relations in Proposition 7.2 are explicitly given by

$$a_k = \sin[(2k+1)\theta], \quad b_k = \frac{1}{\sqrt{N-1}}\cos[(2k+1)\theta], \tag{7.20}$$

for $k = 0, 1, 2, \ldots$, where

$$\sin\theta = \sqrt{\frac{1}{N}}, \quad \cos\theta = \sqrt{1-\frac{1}{N}}. \tag{7.21}$$

Proof. Let $c_k = \sqrt{N-1}b_k$. The recursion relations (7.18) and (7.19) are written in a matrix form,

$$\begin{pmatrix} a_k \\ c_k \end{pmatrix} = M\begin{pmatrix} a_{k-1} \\ c_{k-1} \end{pmatrix}, \; M = \begin{pmatrix} (N-2)/N & 2\sqrt{N-1}/N \\ -2\sqrt{N-1}/N & (N-2)/N \end{pmatrix} = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{pmatrix}.$$

Note that M is a rotation matrix in $\mathbb{R}^2$, and its kth power is another rotation matrix corresponding to a rotation angle $2k\theta$. Thus the above recursion relation is easily solved to yield

$$\begin{pmatrix} a_k \\ c_k \end{pmatrix} = M^k\begin{pmatrix} a_0 \\ c_0 \end{pmatrix} = \begin{pmatrix} \cos 2k\theta & \sin 2k\theta \\ -\sin 2k\theta & \cos 2k\theta \end{pmatrix}\begin{pmatrix} \sin\theta \\ \cos\theta \end{pmatrix} = \begin{pmatrix} \sin[(2k+1)\theta] \\ \cos[(2k+1)\theta] \end{pmatrix}.$$

Replacing c_k by b_k proves the proposition. ∎

We have proved that the application of U_f k times on $|\varphi_0\rangle$ results in the state

$$U_f^k|\varphi_0\rangle = \sin[(2k+1)\theta]|z\rangle + \frac{1}{\sqrt{N-1}}\cos[(2k+1)\theta]\sum_{x\neq z}|x\rangle. \tag{7.22}$$

Measurement of the state $U_f^k|\varphi_0\rangle$ yields $|z\rangle$ with the probability

$$P_{z,k} = \sin^2[(2k+1)\theta]. \tag{7.23}$$

It is instructive to visualize what is going on with a simple example. Let us take $n = 4$, for which $N = 2^n = 16$. The probabilities (a_k, b_k) are given by

$$a_0^2 = b_0^2 = 1/16, \quad a_k^2 = \sin^2[(2k+1)\theta], b_k^2 = \frac{\cos^2[(2k+1)\theta]}{16-1},$$

where $\theta = \sin^{-1}(1/4)$. Figure 7.1 shows the probability distributions for $k = 1, 2, 3$ and 4 where we have chosen $z = 10$.

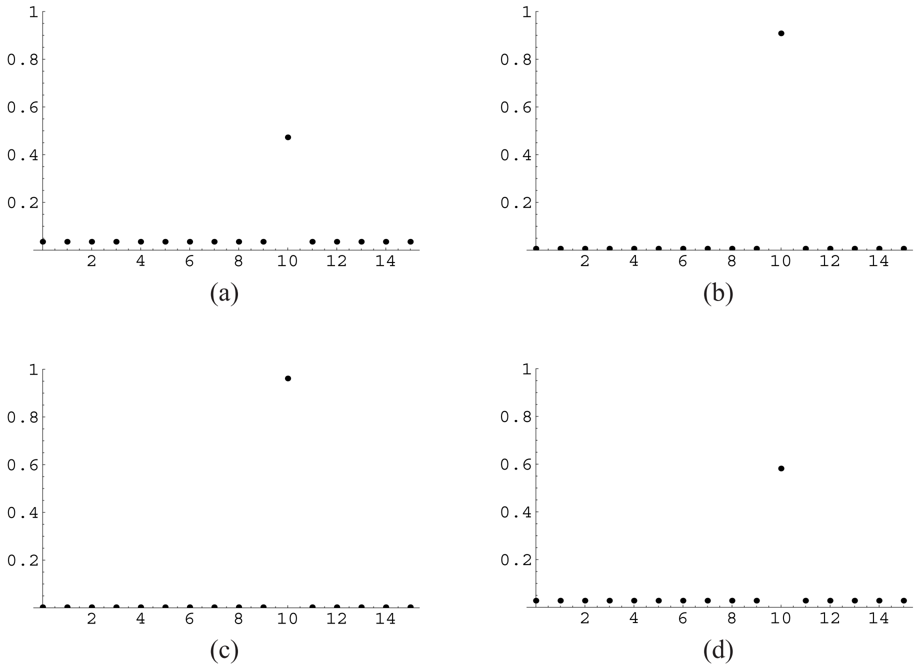

FIGURE 7.1
Probability distribution of $U_f^k|\varphi_0\rangle$, where z is chosen as 10. The number k of iteration is (a) 1, (b) 2, (c) 3 and (d) 4. Observe that $P_{z,k}$ takes its maximum value $\simeq 1$ when $k = 3$.

It should be noted that a_k does not increase monotonically with k, but there is a k ($= 3$ in the present case) that maximizes $P_{z,k} = a_k^2$.

STEP 4 Our final task is to find the k that maximizes $P_{z,k}$. A rough estimate for the maximizing k is obtained by putting

$$(2k+1)\theta = \frac{\pi}{2} \to k = \frac{1}{2}\left(\frac{\pi}{2\theta} - 1\right). \tag{7.24}$$

The previous example gave $k = 3$, which is consistent with this estimate:

$$\theta = \sin^{-1}(1/4) \simeq 0.25268 \to k \simeq 2.6.$$

This can be refined as the following proposition.

PROPOSITION 7.4 Let $N \gg 1$ and let

$$m = \left\lfloor \frac{\pi}{4\theta} \right\rfloor, \tag{7.25}$$

where $\lfloor x \rfloor$ stands for the floor of x. The file we are searching for will be obtained in $U_f^m|\varphi_0\rangle$ with the probability

$$P_{z,m} \geq 1 - \frac{1}{N} \tag{7.26}$$

and

$$m = O(\sqrt{N}). \tag{7.27}$$

Proof. Equation (7.25) leads to the inequality $\pi/4\theta - 1 < m \leq \pi/4\theta$. Let us define $\tilde{m}$ by

$$(2\tilde{m}+1)\theta = \frac{\pi}{2} \to \tilde{m} = \frac{\pi}{4\theta} - \frac{1}{2}.$$

Observe that m and $\tilde{m}$ satisfy

$$|m - \tilde{m}| \leq \frac{1}{2}, \tag{7.28}$$

from which it follows that

$$|(2m+1)\theta - (2\tilde{m}+1)\theta| = \left|(2m+1)\theta - \frac{\pi}{2}\right| \leq \theta. \tag{7.29}$$

Considering that $\theta \sim 1/\sqrt{N}$ is a small number when $N \gg 1$ and $\sin x$ is monotonically increasing in the neighborhood of $x = 0$, we obtain

$$0 < \sin|(2m+1)\theta - \pi/2| < \sin\theta$$

or

$$\cos^2[(2m+1)\theta] \leq \sin^2\theta = \frac{1}{N}. \tag{7.30}$$

Thus it has been shown that

$$P_{m,z} = \sin^2[(2m+1)\theta] = 1 - \cos^2[(2m+1)\theta] \geq 1 - \frac{1}{N}. \tag{7.31}$$

It also follows from $\theta > \sin\theta = 1/\sqrt{N}$ that

$$m = \left\lfloor \frac{\pi}{4\theta} \right\rfloor \leq \frac{\pi}{4\theta} \leq \frac{\pi}{4}\sqrt{N}. \tag{7.32}$$

∎

It is important to note that this quantum algorithm takes only $O(\sqrt{N})$ steps and this is much faster than the classical counterpart which requires $O(N)$ steps.

Figure 7.2 shows the quantum circuit which implements Grover's search algorithm. We gave working space for oracles explicitly.

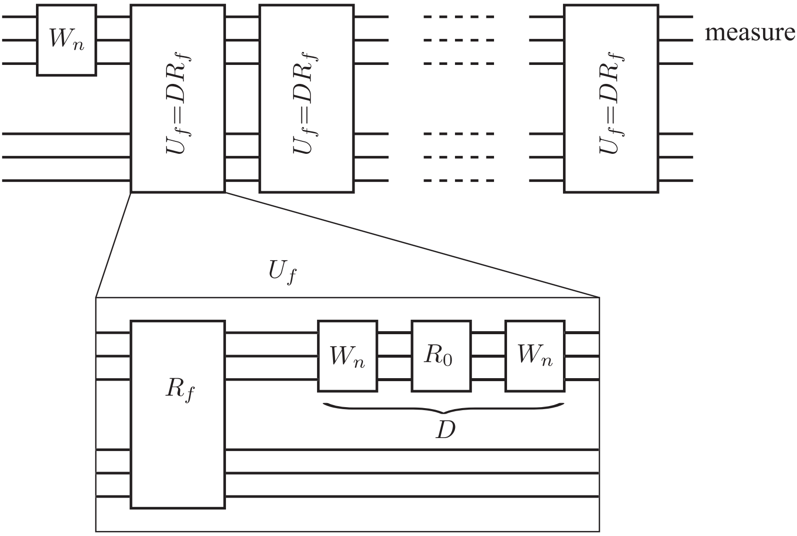

FIGURE 7.2
Implementation of Grover's search algorithm. Details of the box denoted by $U_f = DR_f$ are given in the lower diagram. The box U_f is repeated m times to maximize $P_{z,k}$. The gate R_f is the oracle, and working qubits to implement the oracle are given explicitly.

7.2 Searching for d Files

Suppose there are d $(1 < d \le N)$ files that satisfy a given condition and we are asked to find all of them. This problem is formulated with a help of an oracle

$$f(x) = \begin{cases} 1 & (x \in A) \\ 0 & (x \notin A). \end{cases} \tag{7.33}$$

where A is the subset of S_n, whose elements satisfy the given condition. The subset A is of course unknown to us beforehand.

This problem is solved similarly to the single-file searching problem. Let us define

$$R_f = I - 2\sum_{z \in A} |z\rangle\langle z|. \tag{7.34}$$

Then an application of R_f on $|\varphi\rangle = \sum_{x=0}^{N-1} w_x |x\rangle$ $(\sum_x |w_x|^2 = 1)$ yields

$$R_f|\varphi\rangle = \sum_{x \notin A} w_x|x\rangle - \sum_{z \in A} w_z|z\rangle. \tag{7.35}$$

Consider

$$U_f = DR_f = (-I + 2|\varphi_0\rangle\langle\varphi_0|)\left(I - 2\sum_{z\in A}|z\rangle\langle z|\right), \tag{7.36}$$

where $D = W_n R_0 W_n$ has been defined in Eq. (7.6). Application of U_f on $|\varphi\rangle$ produces the state

$$U_f|\varphi\rangle = \sum_{x\notin A}(2\bar{w} - w_x)|x\rangle + \sum_{z\in A}(2\bar{w} + w_z)|z\rangle, \tag{7.37}$$

where

$$\bar{w} = \frac{1}{N}\left(\sum_{x\notin A} w_x - \sum_{z\in A} w_z\right). \tag{7.38}$$

EXERCISE 7.1 Prove Eq. (7.37).

EXERCISE 7.2 Let $|\varphi_0\rangle = (1/\sqrt{N})\sum_{x=0}^{N-1}|x\rangle$. Show that

$$U_f^k|\varphi_0\rangle = a_k\sum_{z\in A}|z\rangle + b_k\sum_{x\notin A}|x\rangle, \tag{7.39}$$

where $a_0 = b_0 = 1/\sqrt{N}$ and

$$a_k = \frac{N-d}{N}a_{k-1} + \frac{2(N-d)}{N}b_{k-1} \tag{7.40}$$

$$b_k = -\frac{2d}{N}a_{k-1} + \frac{N-2d}{N}b_{k-1}, \tag{7.41}$$

where $d = |A|$.

The above recursion relations are easily solved to yield

$$a_k = \frac{1}{\sqrt{d}}\sin[(2k+1)\theta], \quad b_k = \frac{1}{\sqrt{N-d}}\cos[(2k+1)\theta], \tag{7.42}$$

where

$$\sin\theta = \sqrt{\frac{d}{N}}, \quad \cos\theta = \sqrt{1 - \frac{d}{N}}. \tag{7.43}$$

EXERCISE 7.3 Prove Eq. (7.42).

The above solution shows that the application of U_f on $|\varphi_0\rangle$ k times yields the state

$$U_f^k|\varphi_0\rangle = \frac{1}{\sqrt{d}}\sin[(2k+1)\theta]\sum_{z\in A}|z\rangle + \frac{1}{\sqrt{N-d}}\cos[(2k+1)\theta]\sum_{x\notin A}|x\rangle. \tag{7.44}$$

In order to maximize the probability with which the desired files are observed upon measurements, we have to maximize

$$P_{A,k} = \sum_{z \in A} \left(\frac{1}{\sqrt{d}} \sin[(2k+1)\theta] \right)^2 = \sin^2[(2k+1)\theta]. \tag{7.45}$$

By repeating the arguments in the previous section, we arrive at the following conclusions. Suppose $d \ll N$ and define

$$m = \left\lfloor \frac{\pi}{4\theta} \right\rfloor. \tag{7.46}$$

Then the probability $P_{A,m}$ with which one of the files in A is observed in the state $U_f^m|\varphi_0\rangle$ satisfies

$$P_{A,m} \geq 1 - \frac{d}{N} \tag{7.47}$$

and, moreover,

$$m = O(\sqrt{N/d}). \tag{7.48}$$

EXERCISE 7.4 Prove Eqs. (7.47) and (7.48).

Implementation of the Grover's algorithm with many search files is also given by the quantum circuit in Fig. 7.2 provided that the oracle R_f is properly modified.

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8

Shor's Factorization Algorithm

Shor's factorization algorithm is one of the prime examples in which a quantum computer demonstrates enormous power surpassing its classical counterpart [1, 2]. Although the factorization algorithm may be carried out with a classical computer, it takes an exponentially longer time (i.e., practically impossible) compared to Shor's quantum algorithm. Shor's algorithm is almost identical with the classical one; it consists of a sequence of classical steps with one exception, which is replaced by a quantum algorithm. Our presentation here closely follows [3] and [4].

Let us first consider why factorization of a large number is important.

8.1 The RSA Cryptosystem

The RSA public-key cryptosystem and its variations are widely used to transmit messages over public lines, such as Internet communications, securing the privacy. It is based on the assumption that

"It takes an enormous time to factor a large integer."

The current chapter is written with a PC with a 1.4 GHz Pentium M CPU. The table below shows how long it takes for this PC to factor a large integer N by using `FactorInteger` commmand of Mathematica;

N	time[s]
45878443254366745	0.02
7536576836238936804738907362515346578697687343	3.084
753657683628743673389368047389675407362518902115346578697687	98.88

As the number of digits grows up, it takes more and more time to factorize an integer. Readers who are interested in challenging large number factorization should visit the RSA Security website

`http://www.rsasecurity.com/rsalabs/node.asp?id=2092`

Factorization of a 617-digit decimal number is worth a $200,000 reward!

It should be noted that it is easier to verify whether a number is prime or not [5], but it is very difficult to find the factors of a big number. The **RSA cryptography** [6] makes use of this fact to encode and decode messages.

Let us start with an example. Bob wants to send Alice a message through a public communication channel. He encrypts his message with a key Alice publicizes. Although the key is publicly available, Alice is the only person who can decode the message.

1. Alice prepares two large prime numbers p and q, which she keeps secret and publishes the product N of them. We use the following example here:
$$p = 9281013205404131518475902447276973338969$$
and
$$q = 9591715349237194999547050068718930514279,$$
for which
$$\begin{aligned} N = {} & 8902083681874790795683198927209160030361326460379424 7 \\ & 03263764762563155496163835 1. \end{aligned}$$
It takes quite long (in practical situations with more digits for N) to factor N into p and q. Alice also prepares a number called an **exponent** e $(< N)$, which is relatively prime to $(p-1)(q-1)$. She can easily find such a number:
$$e = 1234567, \ \gcd[e, (p-1)(q-1)] = 1,$$
for example. This number e is also published along with N. She then calculates the modular inverse d of e mod $(p-1)(q-1)$:
$$\begin{aligned} & de \equiv 1 \bmod (p-1)(q-1) \to \\ & \ d = 378539914571696887228359644724123026498967098699116993 \\ & \qquad 554370191326686457372707 99. \end{aligned}$$
Alice keeps d secret.

2. Bob wants to send Alice a message, "hello," for example. This message is transformed into a sequence of decimal numbers less than N under a certain scheme (ASCII etc.). Suppose his scheme transforms the message as
$$\text{hello} \to 12300045600078900012 3,$$
for example. He encodes his message as $\text{hello}^e \bmod N$ and sends Alice the result through an open channel:
$$\begin{aligned} \text{encrypted} &= \text{hello}^e \bmod N \\ &= 378539914571696887228359644724123026498967098699 \\ &\quad 116993554370191326686457372707 99. \end{aligned}$$

3. Alice now decodes the message she received, using d, as

$$\text{encrypted}^d \bmod N = 123000456000789000123.$$

We outline how the RSA system works below. Readers who are eager to proceed to Shor's algorithm may skip this part and jump to the next section. Let us start with **Fermat's little theorem**.

THEOREM 8.1 Let p be an odd prime number and a be any positive integer which is not a multiple of p. Then

$$a^{p-1} \equiv 1 \bmod p. \tag{8.1}$$

Proof. First we prove the congruence

$$m^p - m \equiv 0 \bmod p \tag{8.2}$$

for any $m \in \mathbb{N}$ by induction. Equation (8.2) is true for $m = 1$. Let (8.2) be satisfied with $m = k$. Then we obtain for $m = k + 1$

$$\begin{aligned}(k+1)^p - (k+1) &= k^p + \binom{p}{1} k^{p-1} + \binom{p}{2} k^{p-2} + \ldots + \binom{p}{p-1} k - k \\ &\equiv k^p - k \bmod p,\end{aligned}$$

where we noted that $\binom{p}{k}$ is a multiple of p for any k satisfying $1 \leq k \leq p-1$. Since we assumed $k^p - k \equiv 0 \bmod p$, we obtain $(k+1)^p - (k+1) \equiv 0 \bmod p$.

Now let $m = a$ and write Eq. (8.2) as $a^p - a = a(a^{p-1} - 1) \equiv 0 \bmod p$. Since a is not divisible by p by assumption, $a^{p-1} - 1$ must be divisible by p. In other words, $a^{p-1} - 1 \equiv 0 \bmod p$, and Eq. (8.1) has been proved. ∎

Suppose $N = pq$, p and q being primes, and e $(1 < e < (p-1)(q-1))$ is the encryption exponent which is coprime to $(p-1)(q-1)$ as was assumed previously. The modular inverse d of e satisfies $de \equiv 1 \bmod (p-1)(q-1)$ and $1 < d < (p-1)(q-1)$. Let m $(< N)$ be a message to be encrypted using the public key e as $m_{\text{encrypted}} \equiv m^e \bmod N$. Decryption is possible only with the secret key d since

$$m_{\text{encrypted}}^d = m^{de} \equiv m \bmod N.$$

In fact, the congruence $de \equiv 1 \bmod (p-1)(q-1)$ leads to $de = s(p-1)(q-1) + 1$ $(s \in \mathbb{N})$ and

$$m^{de} = m^{s(p-1)(q-1)+1} = m\left[m^{s(q-1)}\right]^{p-1}.$$

Now suppose m is not a multiple of p. Then Fermat's little theorem asserts that $\left[m^{s(q-1)}\right]^{p-1} \equiv 1 \bmod p$. If m is a multiple of p, then $m^{de} \equiv 0 \bmod p$. By making a trade of p for q, we obtain $[m^{s(p-1)}]^{q-1} \equiv 1 \bmod q$ if q does not divide m, while $m^{de} \equiv 0 \bmod q$ if q divides m. Since p and q are prime, these equalities imply $m^{de} \equiv m \bmod N$.

The RSA cryptosystem depends heavily on the *belief* that factorization of a large number into its prime factors is practically impossible. Shor's algorithm would demolish this myth, as we will see below.

8.2 Factorization Algorithm

Let p and q be prime numbers and let $N = pq$. We want to factor N into a product of p and q. A naive method for the factorization takes $\sqrt{N}$ trials, in the worst case, before p and q are found. Since $\sqrt{N} = e^{(r/2)\ln 2}$ for $N = 2^r$, this method is inefficient. It turns out that the following algorithm is best suited for our purpose.

STEP 1 Take a positive integer m less than N randomly. Calculate the greatest common divisor $\gcd(m, N)$ by the Euclidean algorithm. If $\gcd(m, N) \neq 1$, we are extremely lucky: m is either p or q, and we are done. Suppose $\gcd(m, N) = 1$.

STEP 2 Define $f_N : \mathbb{N} \to \mathbb{N}$ by $a \mapsto m^a \bmod N$. Find the smallest $P \in \mathbb{N}$, such that $m^P \equiv 1 \bmod N$. The number P is called the **order** or **period**. It is known that this takes exponentially large steps in any classical algorithm, but it takes only polynomial steps in Shor's algorithm. A quantum computer is required only in this step, and the rest may be executed in polynomial steps even with a classical computer.

STEP 3 If P is odd, it cannot be used in the following steps. Go back to step 1 and repeat the above steps with different m until an even P is obtained. If P is even, proceed to step 4.

STEP 4 Since P is even, it holds that

$$(m^{P/2} - 1)(m^{P/2} + 1) = m^P - 1 \equiv 0 \bmod N. \tag{8.3}$$

If $m^{P/2} + 1 \equiv 0 \bmod N$, then $\gcd(m^{P/2} - 1, N) = 1$; go back to step 1 and try with different m. If $m^{P/2} + 1 \not\equiv 0 \bmod N$, $m^{P/2} - 1$ contains either p or q, and we proceed to step 5. Note that the number $m^{P/2} - 1$ cannot be a multiple of N in the latter case. If this is the case, it leads to $m^{P/2} \equiv 1 \bmod N$, which contradicts the assumption that P is the smallest number which satisifes $m^P \equiv 1 \bmod N$.

STEP 5 The number

$$d = \gcd(m^{P/2} - 1, N) \tag{8.4}$$

is either p or q, and factorization is done.

EXAMPLE 8.1 An example will clarify the above steps. Let $N = 799 = 17 \cdot 47$.*

STEP 1: The choice $m = 7$ leads to $\gcd(799, 7) = 1$. So this is OK.

STEP 2: It follows from Fig. 8.1 that $7^{368} \equiv 1 \bmod 799$. Thus $P = 368$. Of course we have cheated here and a quantum computer must be used for a large N.

STEP 3: The order thus found is even: $P/2 = 184$. Let us proceed to step 4.

STEP 4: $(7^{184} - 1)(7^{184} + 1) \equiv 0 \bmod 799$. It is easy to see that

$$\gcd(7^{184} + 1, 799) = 17 \neq 1,$$

and we proceed to step 5.

STEP 5: $7^{184} - 1$ and $N = 799$ must have a common prime factor. Indeed, it is found that $d = \gcd(7^{184} - 1, 799) = 47$. It is also found that $799/47 = 17$, which leads to $799 = 47 \cdot 17$.

EXERCISE 8.1 Let $N = 35$. Repeat the above steps to find the factors of N. (There are m whose orders are less than 10. If your m does not give $P < 10$, try another m. Good luck!)

It should be emphasized again that a quantum computation is required only in step 2, where the order P of the function $f : \mathbb{N} \to \mathbb{Z}/N\mathbb{Z}$ $(a \mapsto m^a \bmod N)$ must be found. Here $\mathbb{Z}/N\mathbb{Z}$ stands for the set of equivalence classes in which x and $x + kN$ $(k \in \mathbb{Z})$ are identified. Clearly, we may take x satisfying $0 \leq x \leq N - 1$ as a representative of each equivalence class.†

8.3 Quantum Part of Shor's Algorithm

8.3.1 Settings for STEP 2

Let $N = pq \in \mathbb{N}$ be a number to be factored, where p and q are primes. Find $n \in \mathbb{N}$, such that

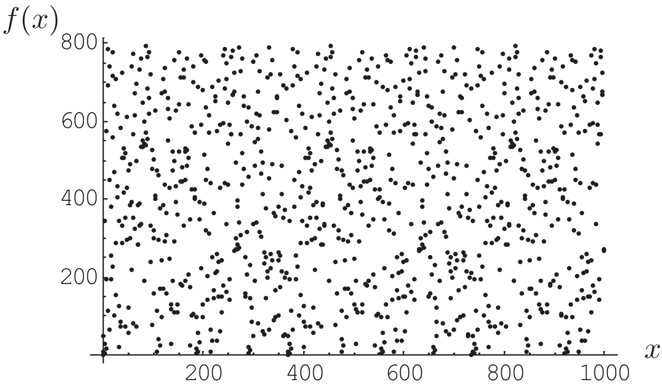
$$N^2 \leq 2^n < 2N^2. \tag{8.5}$$

Let us write $Q = 2^n$ hereafter. Denote $f : a \mapsto m^a \bmod N$ restricted on

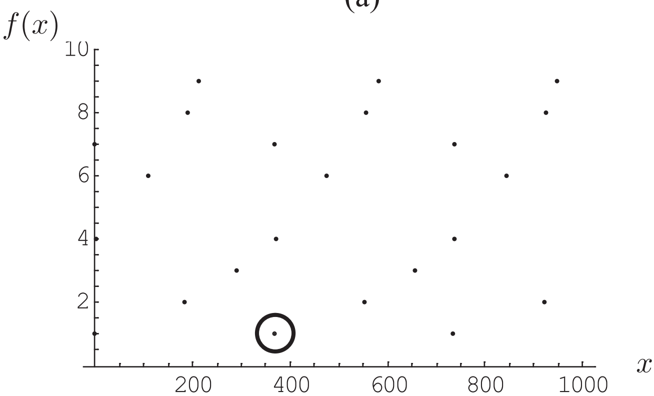
$$S_n = \{0, 1, \ldots, Q - 1\} \tag{8.6}$$

*This example is repeatedly studied in due course.

†In fact $x = 0$ is omitted since m and N are coprime to each other.



(a)



(b)

FIGURE 8.1
(a) Graph of 7^x mod 799. (b) The same graph as (a) with the range $0 \sim 10$. The point encircled is at $x = 368$, which shows that $P = 368$ is the smallest positive integer satisfying $7^P \equiv 1 \bmod 799$.

by the same symbol $f : S_n \to \mathbb{Z}/N\mathbb{Z}$.[‡].

Our quantum computer has two n-qubit registers which we call $|\mathrm{REG1}\rangle$ and $|\mathrm{REG2}\rangle$:

$$|\mathrm{REG1}\rangle|\mathrm{REG2}\rangle = |a\rangle|b\rangle = |a_{n-1}\dots a_1 a_0\rangle|b_{n-1}\dots b_1 b_0\rangle, \tag{8.7}$$

where decimal numbers $a, b \in S_n$ are expressed in binary numbers in the RHS;

$$a = \sum_{j=0}^{n-1} a_j 2^j,\ \ b = \sum_{j=0}^{n-1} b_j 2^j.$$

In the following, extensive use of QFT will be made on an n-qubit system, which is given by

$$|x\rangle \to U_{\mathrm{QFT}n}|x\rangle = \frac{1}{\sqrt{Q}} \sum_{y=0}^{Q-1} \omega_n^{-xy}|y\rangle, \tag{8.8}$$

where $x, y \in S_n$ and $\omega_n = \exp(2\pi i/Q)$. We will denote $U_{\mathrm{QFT}n}$ by $\mathcal{F}$ hereafter.

8.3.2 STEP 2

Let us have a closer look at step 2.

STEP 2.0: Set the registers to the initial state

$$|\psi_0\rangle = |\mathrm{REG1}\rangle|\mathrm{REG2}\rangle = |\underbrace{00\dots0}_{n\text{ qubits}}\rangle|\underbrace{00\dots0}_{n\text{ qubits}}\rangle. \tag{8.9}$$

STEP 2.1: The QFT $\mathcal{F}$ is applied on the first register;

$$|\psi_0\rangle = |0\rangle|0\rangle \overset{\mathcal{F}\otimes I}{\mapsto} |\psi_1\rangle = \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle|0\rangle. \tag{8.10}$$

The first register is in a superposition of all the states $|x\rangle$ $(0 \le x \le Q-1)$, as remarked in Chapter 6.

STEP 2.2: Let us define a function $f : S_n \to \mathbb{Z}/N\mathbb{Z}$ by

$$f(x) = m^x \bmod N, \quad x \in S_n. \tag{8.11}$$

Suppose that the unitary gate U_f realizes the action of f on x in such a way that $U_f|x\rangle|0\rangle = |x\rangle|f(x)\rangle$. Apply U_f on the state prepared in step 2.1 to yield

$$U_f|\psi_1\rangle = |\psi_2\rangle \equiv \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle|f(x)\rangle. \tag{8.12}$$

[‡]It is clear that the range of f is $\mathbb{Z}/N\mathbb{Z}$ since $0 \le f(x) \le N-1 \le \sqrt{Q}-1 < Q-1$, $\forall x \in S_n$.

This shows that the two registers are entangled in general.

STEP 2.3: Apply QFT on $|\text{REG1}\rangle$ again to yield

$$\begin{aligned}|\psi_3\rangle &= (\mathcal{F} \otimes I)|\psi_2\rangle = \frac{1}{Q}\sum_{x=0}^{Q-1}\sum_{y=0}^{Q-1} \omega_n^{-xy}|y\rangle|f(x)\rangle \\ &= \frac{1}{Q}\sum_{y=0}^{Q-1}|y\rangle|\Upsilon(y)\rangle = \frac{1}{Q}\sum_{y=0}^{Q-1}\||\Upsilon(y)\rangle\| \cdot |y\rangle\frac{|\Upsilon(y)\rangle}{\||\Upsilon(y)\rangle\|}, \end{aligned} \tag{8.13}$$

where

$$|\Upsilon(y)\rangle = \sum_{x=0}^{Q-1}\omega_n^{-xy}|f(x)\rangle. \tag{8.14}$$

STEP 2.4: $|\text{REG1}\rangle$ is measured. The result $y \in S_n$ is obtained with the probability

$$\text{Prob}(y) = \frac{\||\Upsilon(y)\rangle\|^2}{Q^2}, \tag{8.15}$$

and, at the same time, the state collapses to

$$|y\rangle\frac{|\Upsilon(y)\rangle}{\||\Upsilon(y)\rangle\|}.$$

The measurement process generates a random variable following a classical probability distribution $\mathcal{S}$ over S_n, in which "symbols" $y \in S_n$ are generated with the probability (8.15).

STEP 2.5: Extract the order P from the measurement outcome.

EXERCISE 8.2 Let $N = 21$ and $m = 11$. Find n which satisifes Eq. (8.5). Find also the order P.[§] Repeat the above steps to find the wave function $|\psi_3\rangle$ and $\text{Prob}(y)$ $(y \in S_n)$.

8.4 Probability Distribution

Let us study the probability distribution $\text{Prob}(y)$ in detail.

[§]The order is less than 10 in this case.

PROPOSITION 8.1 Let $Q = 2^n = Pq + r,\ (0 \le r < P)$, where q and r are uniquely determined non-negative integers. Let $Q_0 = Pq$. Then

$$\mathrm{Prob}(y) = \begin{cases} \dfrac{r\sin^2\left(\frac{\pi P y}{Q}\left(\frac{Q_0}{P}+1\right)\right) + (P-r)\sin^2\left(\frac{\pi P y}{Q}\cdot\frac{Q_0}{P}\right)}{Q^2\sin^2\left(\frac{\pi P y}{Q}\right)} & (Py \not\equiv 0 \bmod Q) \\[2ex] \dfrac{r(Q_0+P)^2 + (P-r)Q_0^2}{Q^2P^2} & (Py \equiv 0 \bmod Q). \end{cases}$$

Proof. It is found from the definition that[¶]

$$\begin{aligned}
|\Upsilon(y)\rangle &= \sum_{x=0}^{Q-1} \omega^{-xy}|f(x)\rangle = \sum_{x=0}^{Q_0-1} \omega^{-xy}|f(x)\rangle + \sum_{x=Q_0}^{Q-1} \omega^{-xy}|f(x)\rangle \\
&= \sum_{x_0=0}^{P-1}\sum_{x_1=0}^{Q_0/P-1} \omega^{-(Px_1+x_0)y}|f(Px_1+x_0)\rangle \\
&\quad + \sum_{x_0=0}^{r-1} \omega^{-[P(Q_0/P)+x_0]y}|f(P(Q_0/P)+x_0)\rangle \\
&= \sum_{x_0=0}^{P-1} \omega^{-x_0y}\left(\sum_{x_1=0}^{Q_0/P-1}\omega^{-Px_1y}\right)|f(x_0)\rangle + \sum_{x_0=0}^{r-1}\omega^{-x_0y}\omega^{-Py(Q_0/P)}|f(x_0)\rangle \\
&= \sum_{x_0=0}^{r-1} \omega^{-x_0y}\sum_{x_1=0}^{Q_0/P-1}\omega^{-Pyx_1}|f(x_0)\rangle \\
&\quad + \sum_{x_0=r}^{P-1} \omega^{-x_0y}\sum_{x_1=0}^{Q_0/P-1}\omega^{-Pyx_1}|f(x_0)\rangle + \sum_{x_0=0}^{r-1}\omega^{-x_0y}\omega^{-Py(Q_0/P)}|f(x_0)\rangle \\
&= \sum_{x_0=0}^{r-1} \omega^{-x_0y}\left(\sum_{x_1=0}^{Q_0/P}\omega^{-Pyx_1}\right)|f(x_0)\rangle \\
&\quad + \sum_{x_0=r}^{P-1} \omega^{-x_0y}\left(\sum_{x_1=0}^{Q_0/P-1}\omega^{-Pyx_1}\right)|f(x_0)\rangle.
\end{aligned}$$

Note that the map $f : a \mapsto m^a \bmod N$ is $1:1$ on $\{0, 1, 2, \ldots, P-1\}$, which we prove now. Suppose $m^a \equiv m^b \bmod N$ $(a > b)$; then $m^b(m^{a-b}-1) \equiv 0 \bmod N$. Since m and N are coprime, so are m^b and N. Then $m^P > m^{a-b} \equiv 1 \bmod N$, which contradicts the assumption that P is the smallest natural number such that $m^P \equiv 1 \bmod N$. This implies that $|f(0)\rangle, |f(1)\rangle, \ldots, |f(P-1)\rangle$ are

[¶]We drop n from ω_n to simplify our notation.

mutually orthogonal. Accordingly

$$\langle \Upsilon(y)|\Upsilon(y)\rangle = r\left|\sum_{x_1=0}^{Q_0/P} \omega^{-Pyx_1}\right|^2 + (P-r)\left|\sum_{x_1=0}^{Q_0/P-1} \omega^{-Pyx_1}\right|^2.$$

In case $Py \equiv 0 \bmod Q$, we put $Py = aQ,\ a \in \mathbb{N}$ and obtain

$$\omega^{-Pyx_1} = e^{-2\pi i(Py/Q)x_1} = e^{-2\pi i a x_1} = 1.$$

Therefore

$$\langle \Upsilon(y)|\Upsilon(y)\rangle = r\cdot\left(\frac{Q_0}{P}+1\right)^2 + (P-r)\left(\frac{Q_0}{P}\right)^2,$$

which leads to the result independent of y,

$$\mathrm{Prob}(y) = \frac{r(Q_0+P)^2 + (P-r)Q_0^2}{P^2Q^2} = \frac{r(q+1)^2 + (P-r)q^2}{Q^2}. \tag{8.16}$$

If $Py \not\equiv 0 \bmod Q$, on the other hand, we obtain

$$\begin{aligned}\langle \Upsilon(y)|\Upsilon(y)\rangle &= r\left|\frac{\omega^{-Py(Q_0/P+1)}-1}{\omega^{-Py}-1}\right|^2 + (P-r)\left|\frac{\omega^{-Py(Q_0/P)}-1}{\omega^{-Py}-1}\right|^2\\ &= r\left|\frac{e^{-(2\pi i/Q)Py(Q_0/P+1)}-1}{e^{-(2\pi i/Q)Py}-1}\right|^2 + (P-r)\left|\frac{e^{-(2\pi i/Q)Py(Q_0/P)}-1}{e^{-(2\pi i/Q)Py}-1}\right|^2.\end{aligned}$$

Here we find from

$$|e^{i\theta}-1|^2 = 2(1-\cos\theta) = 4\sin^2\frac{\theta}{2}$$

that

$$\langle \Upsilon(y)|\Upsilon(y)\rangle = r\frac{\sin^2\dfrac{\pi}{Q}Py\left(\dfrac{Q_0}{P}+1\right)}{\sin^2\dfrac{\pi}{Q}Py} + (P-r)\frac{\sin^2\dfrac{\pi}{Q}Py\dfrac{Q_0}{P}}{\sin^2\dfrac{\pi}{Q}Py}.$$

Therefore, the probability distribution is given by

$$\mathrm{Prob}(y) = \frac{\||\Upsilon(y)\rangle\|^2}{Q^2} = \frac{r\sin^2\left[\dfrac{\pi}{Q}Py\left(\dfrac{Q_0}{P}+1\right)\right] + (P-r)\sin^2\left[\dfrac{\pi}{Q}Py\dfrac{Q_0}{P}\right]}{Q^2\sin^2\dfrac{\pi}{Q}Py}, \tag{8.17}$$

which proves the proposition. ∎

COROLLARY 8.1 Suppose $Q/P \in \mathbb{Z}$ (namely $Q_0 = Q$). Then the probability of obtaining a measurement outcome y is

$$\mathrm{Prob}(y) = \begin{cases} 0 & (Py \not\equiv 0 \bmod Q) \\ \dfrac{1}{P} & (Py \equiv 0 \bmod Q) \end{cases}$$

Proof. When $Py \not\equiv 0 \bmod Q$, $r = 0$ implies $Q = Pq$. Therefore

$$\mathrm{Prob}(y) = \frac{P \sin^2 \pi y}{Q^2 \sin^2 \frac{\pi y}{q}} = 0.$$

In case $Py \equiv 0 \bmod Q$, we obtain

$$\mathrm{Prob}(y) = \frac{PQ^2}{Q^2 P^2} = \frac{1}{P}.$$

∎

Figure 8.2 shows the probability distribution $\mathrm{Prob}(y)$ with the parameters $N = 799 = 17 \cdot 47, P = 368, Q = 2^{20} = 1,048,576$. Note that $N^2 = 638,401$ and $2N^2 = 1,276,802$ so that they satisfy $N^2 \le Q < 2N^2$. Then $Q \equiv 144 \bmod 368 \to r = 144, Q_0 = Q - r = 1,048,432, \to q = Q_0/P = 2849$. Accordingly, $\mathrm{Prob}(y)$ exhibits sharp peaks at integral multiples of $q = 2849$. Figure 8.3 shows $\mathrm{Prob}(y)$ for $8,520 < y < 8,580$. Observe that it has a sharp peak at $y = 8548$, for which $8548/2849 = 3.00035$. Compare the numbers

$$\mathrm{Prob}(8547) = 0.00005393,\ \mathrm{Prob}(8548) = 0.00245753,\ \mathrm{Prob}(8549) = 0.00010892.$$

For neighboring numbers, we obtain $8547/2849 = 3$ and $8549/2849 = 3.0007$. Note that there are $P = 368$ sharp peaks, and $\mathrm{Prob}(y)$ at each peak is roughly on the order of $1/386 \sim 0.00272$.

Since y is restricted within the range $0 \le y \le Q-1$, repeated measurements reveal that the minimal distance between the peaks is ~ 2849, which yields the approximate order $P = Q/2849 \sim 368.0505$. The order thus obtained is probabilistic, and its plausibility must be checked by carrying out step 3 $\sim$ step 5. Needless to say, this strategy is not practical when N is considerably large. There is a powerful method of continued fraction expansion by which we find the order P with a single measurement of the first register, which is the subject of §8.5.

It will be shown that factorization of a number $N = pq$ is carried out efficiently by a quantum computer. A quantum algorithm is employed to find the order of the function $f(x) = m^x \bmod N$, and the other steps are done with classical algorithms. The quantum circuit in Fig. 8.4 implements the quantum part of the algorithm where U_f and $\mathcal{F}$ stand for the map

$$U_f|x\rangle|0\rangle = |x\rangle|m^x \bmod N\rangle$$

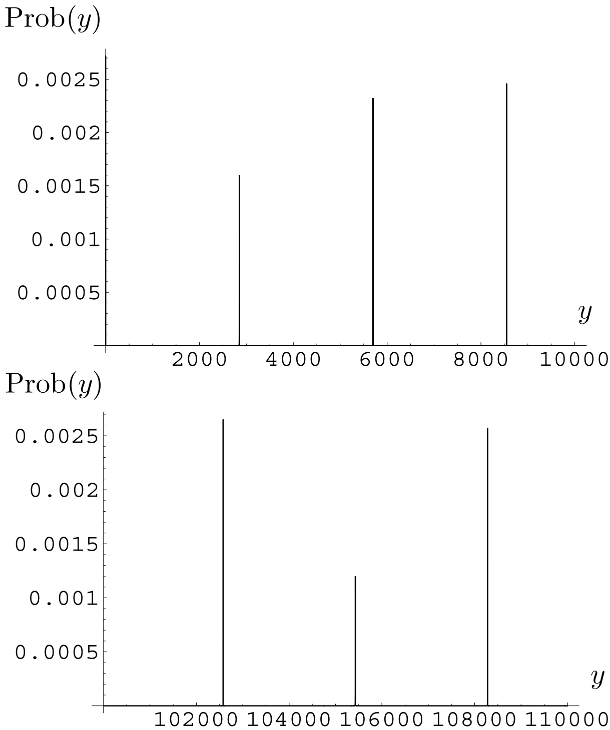


FIGURE 8.2
(a) Probability distribution Prob(y) for $0 \le y \le 10,000$. (b) Same graph for the range $100,000 \le y \le 110,000$.

and the QFT, respectively.

It is instructive to recollect step 2 with our example of $799 = 17 \cdot 47$, for which $n = 20$. We take $m = 7$ as before.
STEP 2.0: The initial state is

$$|\psi_0\rangle = |0\rangle|0\rangle. \tag{8.18}$$

STEP 2.1: The QFT on the first register results in

$$|\psi_1\rangle = \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle|0\rangle, \tag{8.19}$$

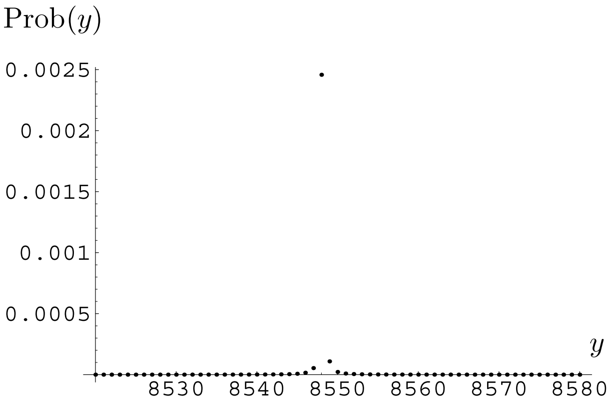


FIGURE 8.3
Probability distribution Prob(y) for $8,520 \leq y \leq 8,580$.

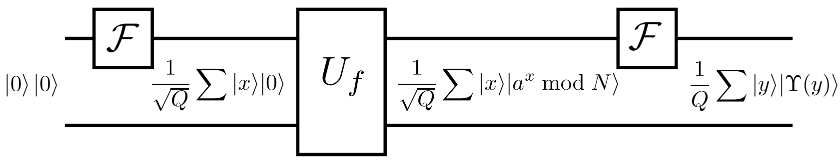


FIGURE 8.4
Quantum circuit to find the order of $f(x) = m^x \bmod N$.

where $Q = 2^{20} = 1048576$.

STEP 2.2: Application of U_f on $|\psi_1\rangle$ produces

$$\begin{aligned}|\psi_2\rangle &= \frac{1}{\sqrt{Q}}\sum_{x=0}^{Q-1}|x\rangle|7^x \bmod 799\rangle \\ &= \frac{1}{\sqrt{Q}}\Big[|0\rangle|1\rangle + |1\rangle|7\rangle + |2\rangle|49\rangle + |3\rangle|343\rangle + |4\rangle|4\rangle + |5\rangle|28\rangle \\ &\quad + \ldots + |368\rangle|1\rangle + |369\rangle|7\rangle + |370\rangle|49\rangle + \ldots \\ &\quad + |Q-2\rangle|756\rangle + |Q-1\rangle|498\rangle\Big]. \end{aligned} \tag{8.20}$$

Note that there are only $P = 368$ different states in the second register.

STEP 2.3: The QFT with $\omega = e^{2\pi i/Q}$, $Q = 2^n$, is applied to the first register. This results in

$$|\psi_3\rangle = \frac{1}{\sqrt{Q}}\sum_{x=0}^{Q-1}\frac{1}{\sqrt{Q}}\sum_{y=0}^{Q-1}\omega^{-xy}|y\rangle|7^x \bmod 799\rangle$$

$$\equiv \frac{1}{Q}\sum_{y=0}^{Q-1}|y\rangle|\Upsilon(y)\rangle, \tag{8.21}$$

where

$$\begin{aligned}
|\Upsilon(y)\rangle &= \sum_{x=0}^{Q-1}\omega^{-xy}|7^x \bmod 799\rangle \\
&= \sum_{x=0}^{Q-1}e^{-2\pi ixy/Q}|7^x \bmod 799\rangle \\
&= |1\rangle+\omega^{-y}|7\rangle+\omega^{-2y}|49\rangle+\omega^{-3y}|343\rangle+\ldots \\
&\quad +\omega^{-368y}|1\rangle+\omega^{-369y}|7\rangle+\omega^{-370y}|49\rangle+\omega^{-371y}|343\rangle+\ldots \\
&\qquad\qquad +\ldots+ \\
&\quad +\omega^{-736y}|1\rangle+\omega^{-737y}|7\rangle+\omega^{-738y}|49\rangle+\omega^{-739y}|343\rangle+\ldots \\
&\qquad\qquad +\ldots+ \\
&\quad +\omega^{-1048432y}|1\rangle+\omega^{-1048433y}|7\rangle+\omega^{-1048434y}|49\rangle+\omega^{-1048435y}|343\rangle \\
&\qquad \ldots+\omega^{-1048575y}|498\rangle \\
&= (1+\omega^{-368y}+\omega^{-736y}+\ldots+\omega^{-1048432y})|1\rangle \\
&\quad +(\omega^{-y}+\omega^{-369y}+\omega^{-737y}+\ldots+\omega^{-1048433y})|7\rangle \\
&\quad +(\omega^{-2y}+\omega^{-370y}+\omega^{-738y}+\ldots+\omega^{-1048434y})|49\rangle \\
&\quad +(\omega^{-3y}+\omega^{-371y}+\omega^{-739y}+\ldots+\omega^{-1048435y})|343\rangle \\
&\qquad\qquad +\ldots \\
&\quad +(\omega^{-87y}+\omega^{-455y}+\omega^{-823y}+\ldots)|794\rangle.
\end{aligned} \tag{8.22}$$

There are 368 ket vectors in the above expansion. The coefficient of each vector becomes sizeable when and only when y is approximately a multiple of 2849. For example,

$$\sum_{k=0}^{2849}\omega^{-368ky}=0.608696+0.000262611i,\quad \left|\sum_{k=0}^{2849}\omega^{-368ky}\right|=0.608696$$

for $y=1$ while

$$\sum_{k=0}^{2849}\omega^{-368ky}=2315.79+1408.03i,\quad \left|\sum_{k=0}^{2849}\omega^{-368ky}\right|=2710.25$$

for $y=8548$. The previous result is recovered as

$$\mathrm{Prob}(8548)=368\left(\frac{2710.25}{Q}\right)^2=0.00245848. \tag{8.23}$$

The order P may be inferred by repeating measurements. However, the number of measurements required to guess P grows rapidly as N becomes larger and larger. We certainly need a technique with which we may find P with a single measurement, which is the subject of the next section.

8.5 Continued Fractions and Order Finding

We introduce a few symbols. For $\forall x \in \mathbb{R}$, we define the **ceiling** $\lceil x \rceil = \inf\{n \in \mathbb{Z} | x \le n\}$. In other words, $\lceil x \rceil = n$, where $n - 1 < x \le n$. For example, $\lceil 2 \rceil = 2$, $\lceil 2.6 \rceil = 3$, $\lceil -4.5 \rceil = -4$ and $\lceil -5 \rceil = -5$. Similarly the **floor** is defined as $\lfloor x \rfloor = \sup\{n \in \mathbb{Z} | n \le x\}$ for $\forall x \in \mathbb{R}$. The floor function $\lfloor x \rfloor$ is also called the integer part of x. For example, $\lfloor 4.5 \rfloor = 4$, $\lfloor 2 \rfloor = 2$ and $\lfloor -4.7 \rfloor = -5$. The floor function has been already introduced in §6.4.

Let us consider **continued fraction expansion** of a rational number. Continued fraction expansion exists also for an irrational number, but it does not terminate. The continued fraction expansion of $x \in \mathbb{Q}$ is

$$x = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ldots + \cfrac{1}{a_q}}}}, \tag{8.24}$$

where $a_i \in \mathbb{N}$ for $j \ge 1$. This number is also written as

$$x = [a_0, a_1, \ldots, a_q]. \tag{8.25}$$

Let us consider $x = 17/47$, for example. x is written in a continued fraction form as

$$\begin{aligned}\frac{17}{47} &= 0 + \cfrac{1}{\cfrac{47}{17}} = 0 + \cfrac{1}{2 + \cfrac{13}{17}} \\ &= 0 + \cfrac{1}{2 + \cfrac{1}{1 + \cfrac{4}{13}}} = 0 + \cfrac{1}{2 + \cfrac{1}{1 + \cfrac{1}{3 + \cfrac{1}{4}}}} = [0, 2, 1, 3, 4].\end{aligned}$$

Let us summarize what we have done to obtain this expansion. We first find the integer part of x as $a_0 = \lfloor 17/47 \rfloor = 0$ and the fractional part as $r_0 = x - a_0 = 17/47$. We invert 17/47 and find that the integral part of $1/r_0$ is $a_1 = \lfloor 1/r_0 \rfloor = \lfloor 47/17 \rfloor = 2$, and the fractional part is $r_1 = 1/r_0 - \lfloor 1/r_0 \rfloor = 13/17$. The interger part of $1/r_1$ is $a_2 = \lfloor 1/r_1 \rfloor = 1$, and the fractional part is $r_2 = 1/r_1 - \lfloor 1/r_1 \rfloor = 4/13$. The integer part of $1/r_2$ is $\lfloor 1/r_2 \rfloor = 3$, and the fractional part is $r_3 = 1/r_2 - \lfloor 1/r_2 \rfloor = 1/4$. The expansion terminates when r_j has the numerator 1 ($j = 3$ in the present example).

EXERCISE 8.3 Find the continued fraction expansions of $x = 61/45$ and $x = 121/13$.

The algorithm to obtain a continued fraction expansion is summarized as

1. Let x be a rational number to be expanded. Let $a_0 = \lfloor x \rfloor$ and $r_0 = 1/x - a_0$.

2. Let $m = 1$ and set $a_m = \left\lfloor \dfrac{1}{r_{m-1}} \right\rfloor$ and $r_m = \dfrac{1}{r_{m-1}} - a_m$. Repeat this step until $r_M = 0$ is reached. M is always finite when x is a rational number.

3. The continued fraction expansion of x is

$$x = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots \cfrac{1}{a_{M-1} + \cfrac{1}{a_M}}}}} = [a_0, a_1, \ldots, a_{M-1}, a_M].$$

Given $x = [a_0, a_1, \ldots, a_M]$, the continued fraction $[a_0, a_1, \ldots, a_j]$ $(j \le M)$ is called the jth **convergent** of x. The Mth convergent is x itself. Note that $[a_0, a_1, \ldots, a_M] = [a_0, a_1, \ldots, a_M - 1, 1]$. Thus the number M may be made either even or odd.

With the above preliminaries, we come back to Shor's algorithm. Suppose y is obtained upon the measurement of the first register in Shor's algorithm. Then y/Q is a rational number close to n/P with some $n \in \mathbb{N}$. Now we show that the following algorithm finds the order P of $m^x \bmod N$.

1. Find the continued fraction expansion $[a_0, a_1, \ldots, a_M]$ of y/Q. We always have $a_0 = 0$ since $y/Q < 1$.

2. Let $p_0 = a_0$ and $q_0 = 1$.

3. Let $p_1 = a_1 p_0 + 1$ and $q_1 = a_1 q_0$.

4. Let $p_i = a_i p_{i-1} + p_{i-2}$ and $q_i = a_i q_{i-1} + q_{i-2}$ for $2 \le i \le M$. We obtain the sequence $(p_0, q_0), (p_1, q_1), \ldots, (p_M, q_M)$. It is shown below that p_j/q_j is the jth convergent of y/Q.

5. Find the smallest k $(0 \le k \le M)$ such that $|p_k/q_k - y/Q| \le 1/(2Q)$. Such k is unique.

6. The order is found as $P = q_k$.

EXAMPLE 8.2 Let us consider our favorite example. Let $N = 799$, $Q = 2^{20} = 1048576$ and $m = 7$. The error bound is $1/(2Q) = 4.76837 \times 10^{-7}$. Suppose we obtain $y = 8548$ as a measurement outcome of the first register. We expect that y/Q is an approximation of n/P for some $n \in \mathbb{N}$.

1. The continued fraction expansion of $8548/1048576$ is $[0, 122, 1, 2, 44, 5, 3]$ and $M = 6$.

2. Let $p_0 = a_0 = 0$ and $q_0 = 1$.

3. We obtain $p_1 = a_1 p_0 + 1 = 122 \times 0 + 1 = 1$ and $q_1 = a_1 q_0 = 122 \times 1 = 122$. We find $|p_1/q_1 - y/Q| = |1/122 - 8548/1048576| = 4.47133 \times 10^{-5} > 1/(2Q)$.

4. $p_2 = a_2 p_1 + p_0 = 1$, $q_2 = a_2 q_1 + q_0 = 123$ and $|p_2/q_2 - y/Q| = |1/123 - 8548/1048576| = 2.19 \times 10^{-5} > 1/(2Q)$.

5. $p_3 = a_3 p_2 + p_1 = 3$, $q_3 = a_3 q_2 + q_1 = 368$ and $|p_3/q_3 - y/Q| = |3/368 - 8548/1048576| = 1.65856 \times 10^{-7} \le 1/(2Q)$. We have obtained $k = 3$.

6. The order is found to be $P = q_3 = 368$.

EXERCISE 8.4 Suppose $y = 37042$ is the measurement outcome in the above example. Find the order P by repeating the above algorithm. Suppose $y = 65536$ has been obtained in the next measurement. Apply the above algorithm. What is the "order" you find?

We must know when we obtain the correct order and when not. Here is the sufficient condition.

The correct order P is obtained in the above algorithm when the measurement outcome y belongs to the set

$$\mathcal{C} = \left\{ y \,\middle|\, \exists d \in \{1, 2, \dots, P-1\},\ \left| \frac{d}{P} - \frac{y}{Q} \right| \le \frac{1}{2Q},\ \gcd(P, d) = 1 \right\}. \tag{8.26}$$

The set $\mathcal{C}$ is not an empty set. In fact, for any $P < Q$ there always exists $y \in \{0, 1, 2, \dots, Q-1\}$ such that $-P/2 < Q - yP \le P/2$, from which we obtain

$$\left| \frac{1}{P} - \frac{y}{Q} \right| \le \frac{1}{2Q}.$$

Therefore, the set $\mathcal{C}$ contains at least one element y, for which $d/P = 1/P$. It is important to note that it is impossible to tell whether a particular outcome y is in $\mathcal{C}$ or not since P is not known in advance.

Let us look at the second case $y = 65536$ in the exercise above. Although we verify that $|23/368 - 65536/Q| = 0$, we also have $\gcd(368, 23) = 23$. The next smallest $|d/P - y/Q|$ is attained when $d = 22$ and $d = 24$, for which case $|d/368 - 65536/Q| = 0.00271 > 1/(2Q)$. Therefore an integer $d \in \mathcal{C}$ does not exist for this case.

We outline the proof of the factor-finding algorithm based on the continued fraction expansion. We have $y/Q = [a_0, a_1, \dots, a_M]$ in our mind in the following lemmas.

LEMMA 8.1 The sequences $a_0, a_1, \ldots, a_M$ and $(p_0, q_0), (p_1, q_1), \ldots, (p_M, q_M)$ obtained in the algorithm introduced in this section satisfy

$$[a_0] = \frac{p_0}{q_0},\ [a_0, a_1] = \frac{p_1}{q_1}, \ldots, [a_0, a_1, \ldots, a_M] = \frac{p_M}{q_M}. \tag{8.27}$$

Proof. We prove this by induction. It is easily verified that $[a_0] = p_0/q_0$ and $[a_0, a_1] = p_1/q_1$. Suppose $[a_0, a_1, \ldots, a_k] = p_k/q_k$. Then

$$\begin{aligned}
[a_0, a_1, \ldots, a_k, a_{k+1}] &= [a_0, a_1, \ldots, a_k + 1/a_{k+1}] \\
&= \frac{(a_k + 1/a_{k+1})p_{k-1} + p_{k-2}}{(a_k + 1/a_{k+1})q_{k-1} + q_{k-2}} \\
&= \frac{a_{k+1}(a_k p_{k-1} + p_{k-2}) + p_{k-1}}{a_{k+1}(a_k q_{k-1} + q_{k-2}) + q_{k-1}} \\
&= \frac{a_{k+1}p_k + p_{k-1}}{a_{k+1}q_k + q_{k-1}} = \frac{p_{k+1}}{q_{k+1}}.
\end{aligned}$$

∎

LEMMA 8.2 All the fractions $p_1/q_1, p_2/q_2, \ldots, p_M/q_M$ are irreducible.

Proof. We first prove

$$p_{k-1}q_k - q_{k-1}p_k = (-1)^k \tag{8.28}$$

for $1 \le k \le M$ by induction. This is obviously satisfied for $k = 1$ since $p_0 q_1 - q_0 p_1 = 0 - 1$ and for $k = 2$ since $p_1 q_2 - q_1 p_2 = (a_1 a_2 + 1) - a_1 a_2 = 1$. Suppose $p_{k-2}q_{k-1} - q_{k-2}p_{k-1} = (-1)^{k-1}$ is satisfied. Then

$$\begin{aligned}
p_{k-1}q_k - q_{k-1}p_k &= p_{k-1}(a_k q_{k-1} + q_{k-2}) - q_{k-1}(a_k p_{k-1} + p_{k-2}) \\
&= p_{k-1}q_{k-2} - q_{k-1}p_{k-2} = (-1)^k.
\end{aligned}$$

Now suppose p_k and q_k are not coprime and let $\gcd(p_k, q_k) = d_k \ge 2$. Put $p_k = d_k p'_k$ and $q_k = d_k q'_k$, where $\gcd(p'_k, q'_k) = 1$. Then $p_{k-1}q_k - q_{k-1}p_k = d_k(p_{k-1}q'_k - q_{k-1}p'_k) = (-1)^k$. The second equality is a contradiction since $d_k \ge 2$. ∎

LEMMA 8.3 Let p and q be positive integers such that $\gcd(p, q) = 1$ and x be a positive rational number. The rational number p/q is a convergent of x if they satisfy the inequality

$$\left| \frac{p}{q} - x \right| \le \frac{1}{2q^2}. \tag{8.29}$$

Proof. Let $p/q = [a_0, a_1, \ldots, a_m]$. We assume, without loss of generality, that m is even. Note that $p = p_m$ and $q = q_m$.

Let $\delta = 2q^2(x - p/q)$. We find $|\delta| \le 2q^2/(2q^2) = 1$ by assumption. The statement of the lemma is trivial when $\delta = 0$. Suppose $0 < \delta \le 1$. (Redefine δ by $-\delta$ if $\delta < 0$.) There is a rational number α such that

$$x = \frac{\alpha p_m + q_{m-1}}{\alpha q_m + q_{m-1}} = [a_0, a_1, \ldots, a_m, \alpha].$$

The condition $\alpha \ge 1$ must be satisfied for $[a_0, a_1, \ldots, a_m, \alpha]$ to be a continued fraction expansion of x. By inverting the above relation, we obtain

$$\begin{aligned} \alpha &= \frac{p_{m-1} - xq_{m-1}}{xq_m - p_m} = \frac{2q_m(p_{m-1} - xq_{m-1})}{\delta} \\ &= \frac{2(p_{m-1}q_m - q_{m-1}p_m)}{\delta} - \frac{q_{m-1}}{q_m}. \end{aligned}$$

It has been shown in the proof of Lemma 8.2 that $p_{m-1}q_m - q_{m-1}p_m = 1$ for an even m, from which we obtain $\alpha = 2/\delta - q_{m-1}/q_m > 1$, where we noted $0 < \delta \le 1$ and $q_m > q_{m-1}$. Therefore α has a continued fraction expansion $[b_0, \ldots, b_n]$ with $b_0 \ge 1$ and we obtain $x = [a_0, \ldots, a_m, b_0, \ldots, b_n]$, from which $p/q = [a_0, \ldots, a_m]$ is shown to be a convergent of x. ∎

We note that the above lemma does not claim the uniqueness of the convergent $p/q = p_m/q_m$. There may be many convergents p_k/q_k of x, which satisify

$$\left|\frac{p_k}{q_k} - x\right| \le \frac{1}{2q_k^2}.$$

It simply claims that a rational number p/q satisfying $|p/q - x| \le 1/2q^2$ is one of the convergents of x. It should be also noted that there may be convergents of x, which do not satisfy the above inequality.

LEMMA 8.4 For a given N, a measurement outcome y and Q, such that $N^2 \le Q < 2N^2$, there exists a unique rational number d/P such that $0 < P < N$ and

$$\left|\frac{d}{P} - \frac{y}{Q}\right| \le \frac{1}{2Q}. \tag{8.30}$$

Proof. Suppose there are two sets of (d, P) satisfying this condition, which we call (d_1, P_1) and (d_2, P_2). Then we find

$$\begin{aligned} \left|\frac{d_1}{P_1} - \frac{d_2}{P_2}\right| &= \left|\frac{d_1}{P_1} - \frac{y}{Q} + \frac{y}{Q} - \frac{d_2}{P_2}\right| \\ &\le \left|\frac{d_1}{P_1} - \frac{y}{Q}\right| + \left|\frac{y}{Q} - \frac{d_2}{P_2}\right| \le \frac{1}{2Q} + \frac{1}{2Q} = \frac{1}{Q} \le \frac{1}{N^2}. \end{aligned}$$

It follows from $P_i < N$ that

$$\frac{1}{N^2} \ge \left|\frac{d_1}{P_1} - \frac{d_2}{P_2}\right| = \frac{|d_1P_2 - d_2P_1|}{P_1P_2} > \frac{|d_1P_2 - d_2P_1|}{N^2},$$

from which we obtain $|d_1P_2 - d_2P_1| < 1$, namely $d_1P_2 = d_2P_1 = 0$. ∎

Now we are ready to justify the order-finding algorithm.

PROPOSITION 8.2 Suppose $y \in \mathcal{C}$. Then the number P obtained by the algorithm in this section is the correct order of the modular exponential function $m^x \bmod N$.

Proof. Let $[a_0, a_1, \ldots, a_M]$ be the continued fraction expansion of y/Q. Since $y \in \mathcal{C}$, there exists $d \in \{1, 2, \ldots, P-1\}$ such that

$$\gcd(P, d) = 1, \quad \left| \frac{d}{P} - \frac{y}{Q} \right| \leq \frac{1}{2Q}.$$

Such d must be unique due to Lemma 8.4. By noticing that $P < N$ and $P^2 < N^2 \leq Q$, we also obtain

$$\left| \frac{d}{P} - \frac{y}{Q} \right| \leq \frac{1}{2Q} \leq \frac{1}{2N^2} < \frac{1}{2P^2}.$$

It then follows from Lemma 8.3 that d/P is one of the convergents of $y/Q = [a_0, a_1, \ldots, a_M]$. Let us call this convergent p_m/q_m.

Now we show that this m is the smallest k which satisfies

$$\left| \frac{p_k}{q_k} - \frac{y}{Q} \right| \leq \frac{1}{2Q}.$$

It is obvious that this inequality is satisfied for any p_k/q_k such that $m \leq k \leq M$. However, Lemma 8.4 tells us that there is only one p_k/q_k which satisfies $q_k < N$ and the above inequality. Therefore $q_k > N$ for $k > m$ and we have shown $d/P = p_m/q_m$. Since $\gcd(d, P) = \gcd(p_m, q_m) = 1$ due to Lemma 8.2, we must have $q_m = P$. ∎

In summary, a single measurement of the first register provides the correct order if the measurement outcome y belongs to the set $\mathcal{C}$. We have seen in Example 8.2 that $y = 8548$ satisfies $|3/P - y/Q| \leq 1/(2Q)$ and hence $y \in \mathcal{C}$. It should be kept in mind that P, and hence $\mathcal{C}$, is not known in advance.

8.6 Modular Exponential Function

The block diagram of the quantum circuit to find the order of $f(x) = m^x \bmod N$ is depicted in Fig. 8.4. It has been shown in §6.4 that an n-qubit QFT circuit may be implemented with $\Theta(n^2)$ elementary gates. We now work out the implementation of the other component in Shor's algorithm, the modular exponential function,

$$U_f|x\rangle|0\rangle = |x\rangle|m^x \bmod N\rangle.$$

There are several proposals for the implementation of this function; some save computational steps while the other saves the number of qubits required. Here we follow the standard implementation given in [7] and [4].

Implementation of a modular exponential function is divided into several steps. We need to implement

1. Adder, which outputs $a + b$ given non-negative integers a and b.

2. Modular adder, which outputs $a + b \bmod N$.

3. Modular multiplexer, which outputs $ab \bmod N$.

4. Modular exponential function, which outputs $m^x \bmod N$.

8.6.1 Adder

Let $a = a_{n-1}2^{n-1} + a_{n-2}2^{n-2} + \ldots + a_1 2 + a_0$ and $b = b_{n-1}2^{n-1} + b_{n-2}2^{n-2} + \ldots + b_1 2 + b_0$ be non-negative integers and $s = s_n 2^n + s_{n-1}2^{n-1} + \ldots + s_1 2 + s_0$ be their sum. Let c_k denote the carry bit. The binary numbers s_k satisfy the following recursion relations,

$$\begin{aligned} s_0 = a_0 \oplus b_0,\ c_0 = a_0 b_0,\ s_k = a_k \oplus b_k \oplus c_{k-1}, \\ c_k = a_k b_k \oplus a_k c_{k-1} \oplus b_k c_{k-1}, \quad (1 \le k \le n), \end{aligned} \tag{8.31}$$

where $\oplus$ is addition mod 2, $a_k b_k = a_k \wedge b_k$ and we formally put $a_n = b_n = 0$. Note that $s_n = c_{n-1}$. Carry bit c_k is 1 only when at least two of a_k, b_k and c_{k-1} are 1. This condition is expressed in the final relation in Eq. (8.31).

It is instructive to examine how two-digit numbers are added quantum mechanically. Let $a = a_1 2 + a_0$ and $b = b_1 2 + b_0$, for which $s = s_2 2^2 + s_1 2 + s_0$. It follows from Eq. (8.31) that

$$\begin{aligned} s_0 = a_0 \oplus b_0,\ c_0 = a_0 b_0,\ s_1 = a_1 \oplus b_1 \oplus a_0 b_0, \\ s_2 = c_1 = a_1 b_1 \oplus a_0 b_0 (a_1 \oplus b_1). \end{aligned} \tag{8.32}$$

Now we want to implement a quantum circuit, which we call ADD(2), carrying out the above algebra. Our implementation is generalized to an n-qubit adder ADD(n) subsequently. We have to align qubits in such a way that the gate acts on them in a nice way. It turns out that the following order is the most convenient one in our implementation;

$$\mathrm{ADD}(2)|0, a_0, b_0, 0, a_1, b_1, 0\rangle = |0, a_0, s_0, 0, a_1, s_1, s_2\rangle, \tag{8.33}$$

where the first and the second 0 are scratch qubits to deal with carry bits, while the third 0 is ultimately replaced by the sum s_2. We choose a_i to remain their input values while b_i ($i = 0, 1$) are updated to s_i. We further decompose ADD(2) into a one-bit adder SUM and carry bit gate CARRY.

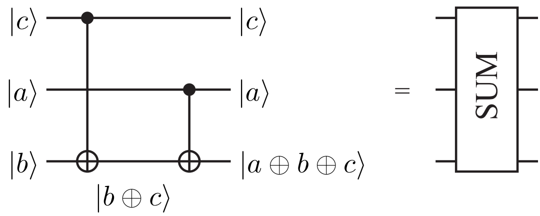


FIGURE 8.5
Quantum circuit to sum two binary bits and a carry bit. It will be also denoted as a black box called SUM.

The action of SUM is defined as

$$\mathrm{SUM}|c_{k-1}, a_k, b_k\rangle = |c_{k-1}, a_k, a_k \oplus b_k \oplus c_{k-1}\rangle, \tag{8.34}$$

where c_{k-1} is the carry bit from the last bit. We drop all the subscripts to simplify our notation hereafter. By recalling that the CNOT gate, whose action is $U_{\mathrm{CNOT}}|a, b\rangle = |a, a \oplus b\rangle$, works as a mod 2 adder, we immediately find the circuit given in Fig. 8.5 implements the SUM gate. Let us make sure it works OK. We abuse the notation so that CNOT_{ij} denotes the CNOT gate with the control bit i and the target bit j whose action on a qubit $k(\neq i, j)$ is trivial. For example CNOT_{12} really means $\mathrm{CNOT}_{12} \otimes I$. According to this notation, we obtain

$$\begin{aligned}\mathrm{SUM}|c, a, b\rangle &= \mathrm{CNOT}_{23}\mathrm{CNOT}_{13}|c, a, b\rangle = \mathrm{CNOT}_{23}|c, a, b \oplus c\rangle \\ &= |c, a, a \oplus b \oplus c\rangle.\end{aligned}$$

Next we consider the CARRY gate whose action is defined as

$$\mathrm{CARRY}|c_{k-1}, a_k, b_k, 0\rangle = |c_{k-1}, a_k, a_k \oplus b_k, c_k = a_k b_k \oplus a_k c_{k-1} \oplus b_k c_{k-1}\rangle. \tag{8.35}$$

By considering that the carry bit c_k is 1 if and only if two or more of a_k, b_k and c_{k-1} are 1, we find that the quantum circuit given in Fig. 8.6 implements the CARRY gate. In fact, we verify

$$\begin{aligned}&\mathrm{CCNOT}_{13;4}\mathrm{CNOT}_{23}\mathrm{CCNOT}_{23;4}|c, a, b, 0\rangle \\ &= \mathrm{CCNOT}_{13;4}\mathrm{CNOT}_{23}|c, a, b, ab\rangle = \mathrm{CCNOT}_{13;4}|c, a, a \oplus b, ab\rangle \\ &= |c, a, a \oplus b, ab \oplus c(a \oplus b)\rangle = |c, a, a \oplus b, ab \oplus ac \oplus bc\rangle \\ &= \mathrm{CARRY}|c, a, b, 0\rangle,\end{aligned}$$

where we have dropped the subscripts for a, b and c for simplicity. Here $\mathrm{CCNOT}_{ij;k}$ stands for the CCNOT gate with the control bits i and j and the target bit k. We have explicitly written nontrivial gates only, and all the other qubits are acted by the unit matrix I as before.

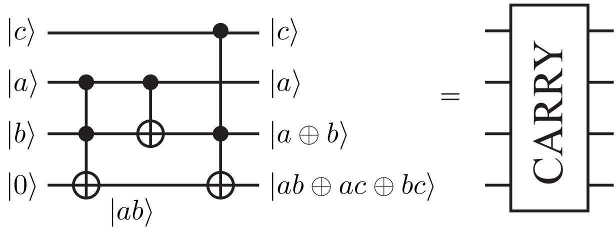


FIGURE 8.6
Quantum circuit which implements the CARRY gate.

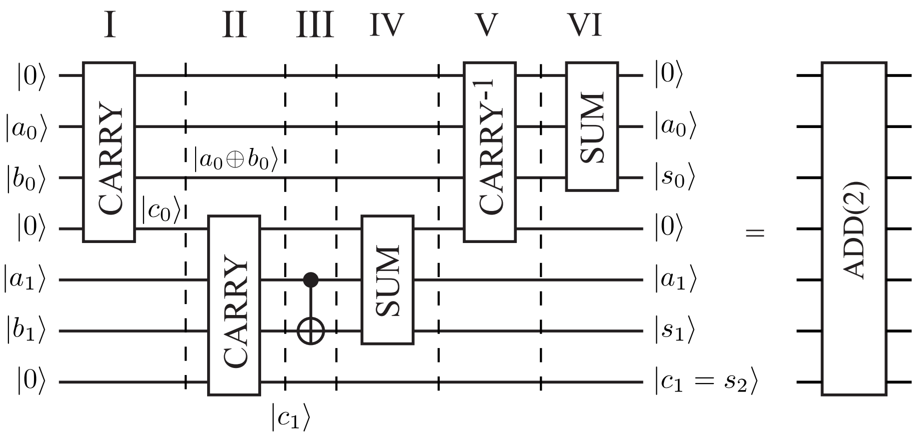


FIGURE 8.7
Quantum circuit to implement the ADD(2) gate, which adds two 2-bit numbers.

EXERCISE 8.5 Suppose the fourth input qubit is not 0 but a binary number c'. Show that the output state of the CARRY gate shown in Fig. 8.6 on this state is

$$\text{CARRY}|c, a, b, c'\rangle = |c, a, a \oplus b, ab \oplus ac \oplus bc \oplus c'\rangle. \tag{8.36}$$

Now we are ready to implement ADD(2) with these gates, which is explicitly shown in Fig. 8.7. Note that carry bits are required to compute s_i. Therefore we evaluate c_0 and c_1 first to compute s_1 and s_2 The bit b_0 is updated to $a_0 \oplus b_0$ during this process. We need to apply the inverse gate CARRY^{-1} on $|0, a_0, a_0 \oplus b_0, c_0\rangle$ to put it back to $|0, a_0, b_0, 0\rangle$ so that the state produces $|0, a_0, s_0, 0\rangle$ after applying the SUM gate of the layer VI. Thus the prescription is

Layer I Compute $c_0 = a_0 b_0$ from a_0 and b_0 with the first CARRY.

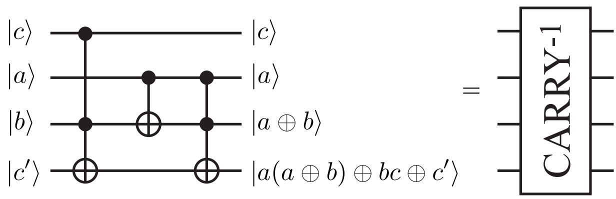


FIGURE 8.8
Inverse of the CARRY gate.

- Layer II Compute $c_1 = s_2$ from a_1, b_1 and c_0 with the second CARRY.
- Layer III If follows from Eq. (8.35) that the input b_1 is updated to $a_1 \oplus b_1$ in the layer II and we need to put this bit back to b_1 to evaluate s_1. This is done by the CNOT gate as $\text{CNOT}|a_1, a_1 \oplus b_1\rangle = |a_1, a_1 \oplus b_1 \oplus a_1\rangle = |a_1, b_1\rangle$. We need b_1 for the next step.
- Layer IV Apply the SUM gate on $|c_0, a_1, b_1\rangle$ to obtain $s_1 = a_1 \oplus b_1 \oplus c_0$.
- Layer V We need to compute $s_0 = a_0 \oplus b_0$, for which we have to retrieve b_0. (Note that b_0 is mapped to $a_0 \oplus b_0$ in the second layer.) This is done by applying the CARRY^{-1} gate. This gate also puts the carry bit c_0 back to the initial state $|0\rangle$ for further use.
- Layer VI Finally the SUM gate is applied on the input bits $|0, a_0, b_0\rangle$ to produce $|0, a_0, s_0\rangle$.

Before we verify the circuit in Fig. 8.7 indeed implements the ADD(2) gate, we look at the newly introduced CARRY $^{-1}$ gate in some details. We obtain, from the implementation of the CARRY gate in Fig. 8.6, that

$$\begin{aligned}\text{CARRY}^{-1} &= (\text{CCNOT}_{13;4}\text{CNOT}_{23}\text{CCNOT}_{23;4})^\dagger \\ &= \text{CCNOT}^\dagger_{23;4}\text{CNOT}^\dagger_{23}\text{CCNOT}^\dagger_{13;4} \\ &= \text{CCNOT}_{23;4}\text{CNOT}_{23}\text{CCNOT}_{13;4},\end{aligned} \tag{8.37}$$

where we noted that $\text{CCNOT}^\dagger = \text{CCNOT}$ and $\text{CNOT}^\dagger = \text{CNOT}$. Therefore CARRY^{-1} is obtained by reversing the order of the constituent controlled gates in the CARRY gate as depicted in Fig. 8.8.

EXERCISE 8.6 Show explicitly that

$$\text{CARRY}^{-1}|c, a, b, c'\rangle = |c, a, a \oplus b, a(a \oplus b) \oplus (bc) \oplus c'\rangle. \tag{8.38}$$

Now we are ready to verify the implementation of the ADD(2) in Fig. 8.7. We denote the unitary matrix correspoinding to the kth layer by U_k. We have

$$\text{ADD}(2)|0, a_0, b_0, 0, a_1, b_1, 0\rangle$$

$$
\begin{aligned}
&= U_{\mathrm{VI}}U_{\mathrm{V}}U_{\mathrm{IV}}U_{\mathrm{III}}U_{\mathrm{II}}U_{\mathrm{I}}|0, a_0, b_0, 0, a_1, b_1, 0\rangle \\
&= U_{\mathrm{VI}}U_{\mathrm{V}}U_{\mathrm{IV}}U_{\mathrm{III}}U_{\mathrm{II}}|0, a_0, a_0 \oplus b_0, a_0b_0, a_1, b_1, 0\rangle \\
&= U_{\mathrm{VI}}U_{\mathrm{V}}U_{\mathrm{IV}}U_{\mathrm{III}}|0, a_0, a_0 \oplus b_0, a_0b_0, a_1, a_1 \oplus b_1, a_1b_1 \oplus a_1a_0b_0 \oplus b_1a_0b_0\rangle \\
&= U_{\mathrm{VI}}U_{\mathrm{V}}U_{\mathrm{IV}}|0, a_0, a_0 \oplus b_0, a_0b_0, a_1, b_1, a_1b_1 \oplus a_1a_0b_0 \oplus b_1a_0b_0\rangle \\
&= U_{\mathrm{VI}}U_{\mathrm{V}}|0, a_0, a_0 \oplus b_0, a_0b_0, a_1, a_1 \oplus b_1 \oplus a_0b_0, a_1b_1 \oplus a_1a_0b_0 \oplus b_1a_0b_0\rangle \\
&= U_{\mathrm{VI}}|0, a_0, b_0, 0, a_1, a_1 \oplus b_1 \oplus a_0b_0, a_1b_1 \oplus a_1a_0b_0 \oplus b_1a_0b_0\rangle \\
&= |0, a_0, a_0 \oplus b_0, 0, a_1, a_1 \oplus b_1 \oplus a_0b_0, a_1b_1 \oplus a_1a_0b_0 \oplus b_1a_0b_0\rangle.
\end{aligned}
$$

The last line of the above equation is identified with $|0, a_0, s_0, 0, a_1, s_1, s_2\rangle$.

The adder for n-bit numbers, which we call ADD(n), is obtained immediately by generalizing the above implementation. Figure 8.9 shows the ADD(n) gate. The most significant sum bit s_n of $a + b$ is evaluated first. We need to calculate all the carry bits c_i for this purpose, and the left layers of the gates before s_n is obtained are devoted for calculating the carry bits. Then we reverse all the operations, except for the most significant bit, to restore $\{a_i\}$ and $\{b_i\}$ with which $\{s_i\}$ are evaluated as $b_i \mapsto s_i = a_i \oplus b_i \oplus c_{i-1}$. The readers should verify the circuit indeed implements n-bit addition.

8.6.2 Modular Adder

Let us consider the **modular adder**, denoted MODADD(n), which evaluates $a + b \bmod N$ for given inputs a and b, where $a, b < N$. We need to introduce an n-qubit subtraction circuit to this end. It is shown that ADD$(n)^{-1}$, the inverse of ADD(n) does the job. Instead of giving a general proof, we will be satisfed with demonstration of this statement for the simplest case, in which both a and b are 2-bit numbers, $a = a_1 2 + a_0, b = b_1 2 + b_0$.

Subtraction is carried out by introducing a **two's complement** in a similar way as in a classical computer. Let $a = a_{n-1}2^{n-1} + \ldots a_1 2 + a_0$ be a positive n-bit number. A negative number $-a$ is stored in a computer memory as its two's complement, which is defined as $2^{n+1} - a$. By noting that $2^{n+1} = (2^n + 2^{n-1} + \ldots + 2 + 1) + 1$, we obtain

$$2^{n+1} - a = 2^n + (1 - a_{n-1})2^{n-1} + \ldots (1 - a_1)2 + (1 - a_0) + 1. \tag{8.39}$$

Note that $2^{n+1} - a$ is an $(n + 1)$-bit number for a postive number a. In summary, the two's complement of a number $a = a_{n-1}a_{n-2} \ldots a_1a_0$ is obtained by flipping each bit as $a_k \to 1 - a_k$, adding 2^n and finally adding 1. Let $a = 2$, in decimal notation, for example. Then its binary expression is 10 and the two's complement of -2 is $101 + 1 = 110$.

EXERCISE 8.7 Let $n = 3$. Find the two's complements of negative numbers from -7 to -1.

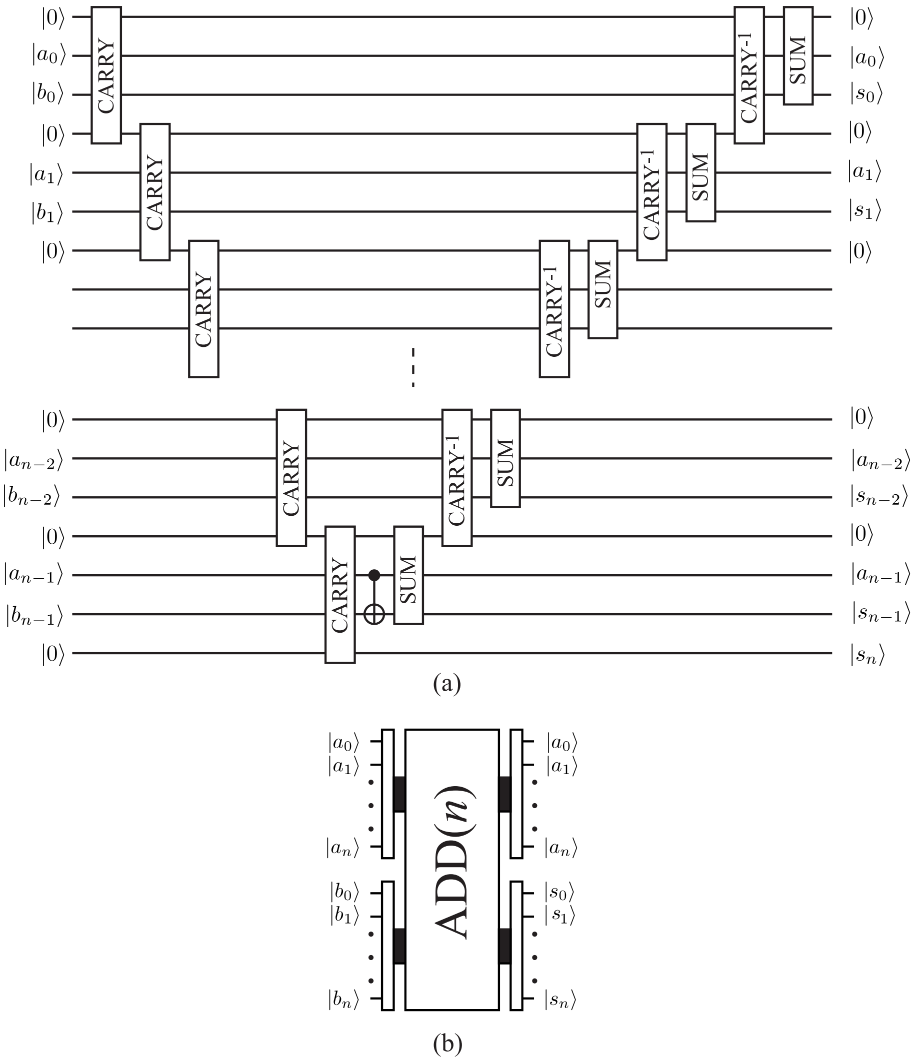


FIGURE 8.9

(a) Quantum circuit to implement the ADD(n) gate, which adds two n-bit numbers a and b. The result is encoded in the qubits $\{|s_k\rangle\}$. (b) The black box representation of ADD(n). Note the order of the input bits and the output bits. We have explicitly added $|a_n\rangle = |b_n\rangle = |0\rangle$.

Let us concentrate on the case in which $n = 2$. Then the two's complement of $-a$ is $2^3 - a = 2^2 + (1 - a_1)2 + (1 - a_0) + 1$. It is also written as

$$\begin{aligned} 2^3 - a &= 2^2 + (a_0 \oplus a_1)2 + a_0 \\ &= (a_0 \oplus a_1 \oplus a_0 a_1)2^2 + (a_0 \oplus a_1)2 + a_0, \end{aligned} \tag{8.40}$$

where we noted that $a > 0$ and $a_0 \oplus a_1 \oplus (a_0 a_1) = 1$. Let us evaluate $b - a$, where both a and b are 2-bit numbers. We carry out this subtraction as $b + (2^3 - a)$ by making use of the two's complement of $-a$. We find

$$\begin{aligned} b + (2^3 - a) &= (b_1 2 + b_0) + \left[(a_0 \oplus a_1 \oplus a_0 a_1)2^2 + (a_0 \oplus a_1)2 + a_0\right] \\ &= [(a_0 \oplus a_1 \oplus a_0 a_1) \oplus (a_0 \oplus a_1)b_1 \oplus a_0 b_0 (a_0 \oplus a_1 \oplus b_1)]2^2 \\ &\quad + (a_0 \oplus a_1 \oplus b_1 \oplus a_0 b_0)2 + (a_0 \oplus b_0) \\ &= s_2 2^2 + s_1 2 + s_0, \end{aligned} \tag{8.41}$$

where

$$\begin{aligned} s_2 &= (a_0 \oplus a_1 \oplus a_0 a_1) \oplus (a_0 \oplus a_1)b_1 \oplus a_0 b_0 (a_0 \oplus a_1 \oplus b_1) \\ &= a_0 \oplus a_1 \oplus a_0 a_1 \oplus a_0 b_1 \oplus a_1 b_1 \oplus a_0 b_0 \oplus a_0 a_1 b_1 \oplus a_0 b_0 b_1, \\ s_1 &= a_0 \oplus a_1 \oplus b_1 \oplus a_0 b_0, \\ s_0 &= a_0 \oplus b_0. \end{aligned} \tag{8.42}$$

Now we show that

$$\mathrm{ADD}(2)^{-1}|0, a_0, b_0, 0, a_1, b_1, 0\rangle = |0, a_0, s_0, 0, a_1, s_1, s_2\rangle. \tag{8.43}$$

In fact, we verify

$$\begin{aligned} &\mathrm{ADD}(2)^{-1}|0, a_0, b_0, 0, a_1, b_1, 0\rangle \\ &= U_{\mathrm{I}} U_{\mathrm{II}} U_{\mathrm{III}} U_{\mathrm{IV}} U_{\mathrm{V}} U_{\mathrm{VI}} |0, a_0, b_0, 0, a_1, b_1, 0\rangle \\ &= U_{\mathrm{I}} U_{\mathrm{II}} U_{\mathrm{III}} U_{\mathrm{IV}} U_{\mathrm{V}} |0, a_0, a_0 \oplus b_0, 0, a_1, b_1, 0\rangle \\ &= U_{\mathrm{I}} U_{\mathrm{II}} U_{\mathrm{III}} U_{\mathrm{IV}} |0, a_0, b_0, a_0 \oplus a_0 b_0, a_1, b_1, 0\rangle \\ &= U_{\mathrm{I}} U_{\mathrm{II}} U_{\mathrm{III}} |0, a_0, b_0, a_0 \oplus a_0 b_0, a_1, a_1 \oplus b_1 \oplus a_0 \oplus a_0 b_0, 0\rangle \\ &= U_{\mathrm{I}} U_{\mathrm{II}} |0, a_0, b_0, a_0 \oplus a_0 b_0, a_1, b_1 \oplus a_0 \oplus a_0 b_0, 0\rangle \\ &= U_{\mathrm{I}} |0, a_0, b_0, a_0 \oplus a_0 b_0, a_1, a_1 \oplus b_1 \oplus a_0 b_0, \\ &\quad a_0 \oplus a_1 \oplus a_0 a_1 \oplus a_0 b_0 \oplus a_0 b_1 \oplus a_1 b_1 \oplus a_0 b_0 a_1 \oplus a_0 b_0 b_1, 0\rangle \\ &= |0, a_0, a_0 \oplus b_0, 0, a_1, a_0 \oplus a_1 \oplus b_1 \oplus a_0 b_0, \\ &\quad a_0 \oplus a_1 \oplus a_0 a_1 \oplus a_0 b_1 \oplus a_1 b_1 \oplus a_0 b_0 \oplus a_0 a_1 b_0 \oplus a_0 b_0 b_1\rangle \\ &= |0, a_0, s_0, 0, a_1, s_1, s_2\rangle \end{aligned}$$

where we noticed that $U_k^\dagger = U_k$ in all the layers in the circuit. This shows that the action of $\mathrm{ADD}(2)^{-1}$ yields $b + (2^{n+1} - a)$ as promised.

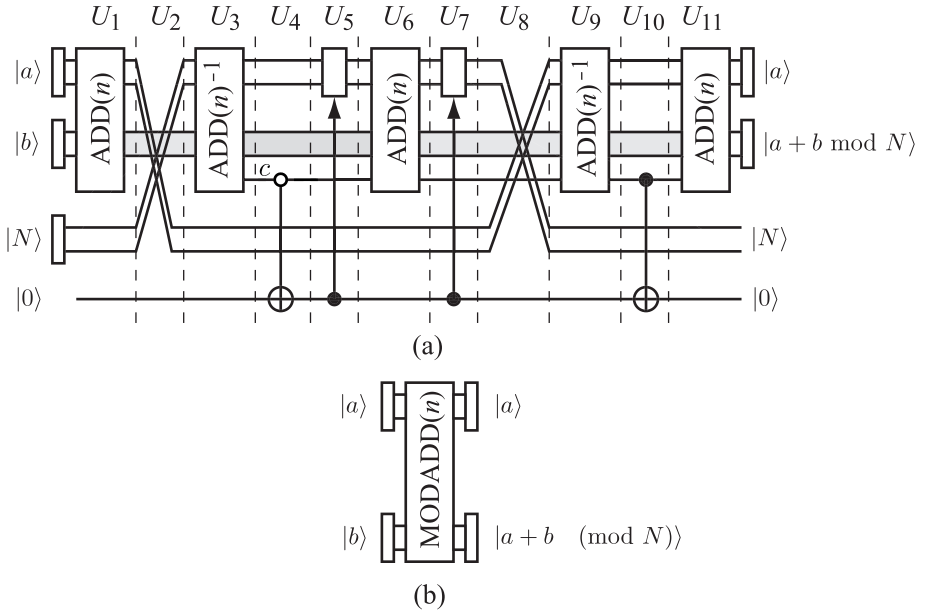


FIGURE 8.10
(a) Modular adder which computes $a + b \bmod N$. The gray line shows the information flow of the second register, corresponding to the input b and the output $a + b \bmod N$. U_k denotes the unitary operation of the kth layer. The white circle in U_4 denotes a negated control node, which flips the bottom qubit when the control bit is 0, while does nothing when it is 1. (b) Modular adder is symbolically denoted as MODADD(n). The third register $|N\rangle$ and the carry qubit $|0\rangle$ are omitted.

The n-qubit subtraction is similarly obtained by inverting ADD(n) as

$$\begin{aligned} &\text{ADD}(n)^{-1}|0, a_0, b_0, 0, a_1, b_1, 0, \ldots, a_{n-1}, b_{n-1}, 0\rangle \\ &= |0, a_0, s_0, 0, a_1, s_1, 0, \ldots, a_{n-1}, s_{n-1}, s_n\rangle. \end{aligned} \tag{8.44}$$

The output bits $s_0 \sim s_{n-1}$ represent $b - a$, while the last digit s_n is 1 if $b - a < 0$ and 0 if $b - a > 0$.

EXERCISE 8.8 Let a and b be 2-bit numbers. Verify that s_2 in Eq. (8.42) is 1 when $b < a$, while 0 when $b > a$.

Now we are ready to implement the modular adder. We find the quantum circuit depicted in Fig. 8.10 indeed performs modular addition $a + b \bmod N$, where a, b and N are n-bit numbers satisfying $0 < a, b < N$. Therefore, we have either $0 < a + b < N$ or $N \leq a + b < 2N$. We write the input state as $|a, b, N, 0\rangle$, where the last qubit $|0\rangle$ will be used as a scratch space. Let us verify its operations.

- The gate U_1 is an ordinary adder;

$$U_1|a, b, N, 0\rangle = |a, a + b, N, 0\rangle. \tag{8.45}$$

- The gate U_2 swaps the first and the third registers;

$$U_2|a, a + b, N, 0\rangle = |N, a + b, a, 0\rangle. \tag{8.46}$$

- The gate U_3 subtracts N from $a + b$;

$$U_3|N, a + b, a, 0\rangle = |N, a + b - N, c, a, 0\rangle. \tag{8.47}$$

  The $(n + 1)$st output bit c of the second register is 1 when $a + b < N$ and 0 when $a + b \geq N$.

- The gate U_4 changes the scratch bit $|0\rangle$ to $|1\rangle$ if $c = 0$, while it is left unchanged when $c = 1$;

$$U_4|N, a + b - N, c, a, 0\rangle = |N, a + b - N, c, a, 1 - c\rangle. \tag{8.48}$$

- The gate U_5 resets the first register $|N\rangle$ to $|0\rangle$ when the temporary qubit is $|1\rangle$ while it remains in the state $|N\rangle$ when it is $|0\rangle$;

$$\begin{aligned} U_5|N, a + b - N, c, a, 1 - c\rangle &= \begin{cases} |N, a + b - N, c, a, 0\rangle & (c = 1) \\ |0, a + b - N, c, a, 1\rangle & (c = 0) \end{cases} \\ &= |cN, a + b - N, c, a, 1 - c\rangle. \end{aligned} \tag{8.49}$$

  Unitarity of U_5 requires that $U_5|0, a+b-N, c, a, 1\rangle = |N, a+b-N, c, a, 1\rangle$.

- The gate U_6 is a simple adder;

$$U_6|cN, a+b-N, c, a, 1-c\rangle = |cN, a+b-(1-c)N, 1-c, a, 1-c\rangle. \tag{8.50}$$

  Note that the carry bit c has been flipped.

- The gate U_7 is the same as the layer 5. The first register is mapped to $|N\rangle$ when $c = 0$, and it remains in $|N\rangle$ when $c = 1$;

$$U_7|cN, a+b-(1-c)N, 1-c, a, 1-c\rangle = |N, a+b-(1-c)N, 1-c, a, 1-c\rangle. \tag{8.51}$$

- The gate U_8 swaps $|a\rangle$ and $|N\rangle$ so that

$$U_8|N, a+b-(1-c)N, 1-c, a, 1-c\rangle = |a, a+b-(1-c)N, 1-c, N, 1-c\rangle. \tag{8.52}$$

- The gate U_9 subtracts a from $a+b-(1-c)N$;

$$U_9|a, a+b-(1-c)N, 1-c, N, 1-c\rangle = |a, b-(1-c)N, 1-c, N, 1-c\rangle. \tag{8.53}$$

- The gate U_{10} transforms the temporary qubit to $(1-c) \oplus (1-c) = 0$;

$$U_{10}|a, b-(1-c)N, 1-c, N, 1-c\rangle = |a, b-(1-c)N, 1-c, N, 0\rangle. \tag{8.54}$$

- The final gate U_{11} adds the first and the second registers;

$$U_{11}|a, b-(1-c)N, 1-c, N, 0\rangle = |a, a+b-(1-c)N, 1-c, N, 0\rangle. \tag{8.55}$$

  In case $a+b \geq N$, for which $c=0$, the second register in the output state is $|a+b-N\rangle$. If, in contrast, $a+b < N$, for which $c=1$, the second register is $|a+b\rangle$. These results are conveniently written as $|a+b \bmod N\rangle$.

We call this quantum circuit an n-qubit modular adder, denoted MODADD(n); see Fig. 8.10 (b).

8.6.3 Modular Multiplexer

It turnes out that a **controlled-modular multiplexer**

$$\mathrm{CMODMULTI}(n)|c, x, 0, 0\rangle = \begin{cases} |c, x, 0, ax \bmod N\rangle & (c=1) \\ |c, x, 0, x\rangle & (c=0) \end{cases} \tag{8.56}$$

is requried to construct the modular exponential gate, which computes a^x mod N, rather than the ordinary modular multiplexer circuit.

Fig. 8.11 depicts the **controlled-modular multiplexer circuit**. The control bit is denoted as $|c\rangle$, while two registers are initially set to $|x\rangle$ and $|0\rangle$. There is also a temporary register which is also set initially to $|0\rangle$. We need to evaluate ax mod N for various x with the numbers a and N fixed. Therefore a and N are hardwired as parts of the circuit, while x is one of the input parameters. Let us verify it works as expected.

Suppose $|c\rangle = |1\rangle$ first. Let $x = x_{n-1}2^{n-1} + \ldots x_1 2 + x_0$ and $a = a_{n-1}2^{n-1} + \ldots a_1 2 + a_0$ be binary expressions of positive integers x and a, respectively. The product of these numbers is

$$ax = ax_{n-1}2^{n-1} + \ldots + ax_1 2 + ax_0 = \sum_{x_k=1} a2^k.$$

The RHS tells us that the ax mod N is obtained by adding $a2^k$ mod N with respect to those k for which $x_k = 1$. The modular adder MODADD(n) in Fig. 8.11 adds $a2^i$ mod N for such i as $x_i = 1$ to the second register whose initial state is $|0\rangle$. Each MODADD(n) is accompanied with a pair of controlled

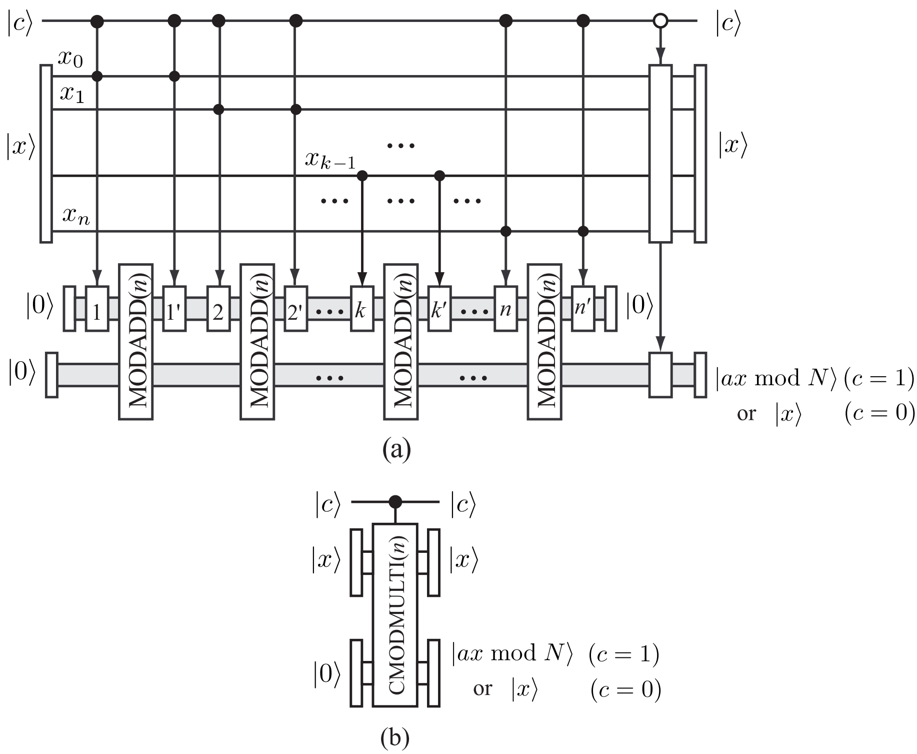


FIGURE 8.11

(a) Quantum circuit of an n-qubit controlled-modular multiplexer CMODMULTI(n). There are n layers of MODADD(n), where the kth layer adds $a2^{k-1} \bmod N$ to the second register when $c = 1$ and $x_{k-1} = 1$. The numbers a and N are fixed and are hardwired. The output of the second register is $ax \bmod N$. This circuit is denoted as (b), where the temporary register has no external input and output ports and is not shown explicitly.

gates. Let us concentrate on the kth MODADD(n) gate. The first controlled gate, denoted as k in Fig. 8.11 associated with the kth MODADD(n) gate stores $a2^{k-1}$ in the temporary register if $x_{k-1} = 1$, while it stores 0 if $x_{k-1} = 0$. MODADD(n) adds $x_{k-1}2^{k-1}$ to $\sum_{i=0}^{k-2} ax_i 2^i$ and updates the second register to $\sum_{i=0}^{k-1} ax_i 2^i$. The second controlled gate immediately after the kth MODADD(n), denoted as k', undoes the operation of the first controlled gate when $x_{k-1} = 1$, so that the first register is now put back to 0 for recycled use for the ($k+1$)st MODADD(n). Both controlled gates leave the temporary gate to $|0\rangle$ when $x_{k-1} = 0$. In this way, we finally obtain $ax \bmod N$ in the second register.

Let $c = 0$ next. Then the temporary register remains in $|0\rangle$ and each MODADD(n) just adds 0 to the second register. We will obtain $|0, x, 0\rangle$ after the nth MODADD(n) has been applied. We need to act a set of controlled gates to copy the first register $|x\rangle$ to the second register so that the output is $|0, x, x\rangle$. This is done by n CNOT gates, which act on each pair of qubits of the first and the second registers as $U_{\text{CNOT}}|0, x_i, 0_i\rangle = |0, x_i, 0_i \oplus x_i\rangle = |0, x_i, x_i\rangle$.

In summary, the CMODMULTI gate works as

$$\text{CMODMULTI}|c, x, 0\rangle = \begin{cases} |1, x, ax \bmod N\rangle & (c = 1) \\ |0, x, x\rangle & (c = 0). \end{cases} \tag{8.57}$$

8.6.4 Modular Exponential Function

Let $a = a_{n-1}2^{n-1} + \ldots + a_1 2 + a_0$ and $x = x_{n-1}2^{n-1} + \ldots + x_1 2 + x_0$ be two natural numbers. We need to implement a quantum circuit which outputs $a^x \bmod N$.

Figure 8.12 shows the quantum circuit which implements the modular exponential function. Let us verify how it works to produce the correct result.

Let $x = x_{n-1}2^{n-1} + x_{n-2}2^{n-2} + \ldots + x_1 2 + x_0$. Then the power a^x is expressed as

$$a^x = a^{x_{n-1}2^{n-1}} \times a^{x_{n-2}2^{n-2}} \times \ldots a^{x_1 2} \times a^{x_0} = \prod_{\substack{k=0 \\ x_k=1}}^{n-1} a^{2^k}. \tag{8.58}$$

Namely, terms of the form a^{2^k} are multiplied with respect to k for which $x_k = 1$ to evaluate a^x. The modular exponential $a^x \bmod N$ is obtained by multiplying a^{2^k} from $k = 0$ to $k = n - 1 \bmod N$ for those k satisfying $x_k = 1$.

- Let us look at the first layer composed of gates I and II in Fig. 8.12. The output of the gate I is

$$\text{CMODMULTI}|x, 1, 0\rangle = |x, 1, a^{x_0} \bmod N\rangle.$$

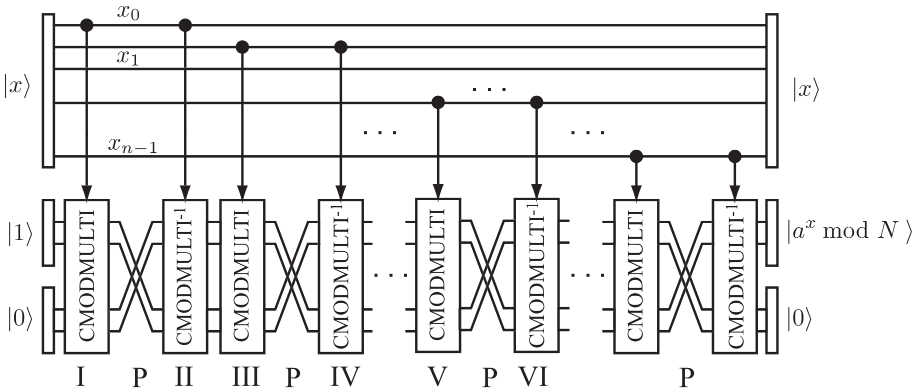


FIGURE 8.12
Quantum cirucuit which implements the modular exponential function $a^x \bmod N$. The numbers a and N are hardwired. The number x is initially input to the first register, while the second register is set to $|1\rangle$. The result of the modular exponential function is stored in the second register. There is an extra temporary register which is set initially to $|0\rangle$.

The output states $|1\rangle$ and $|a^{x_0} \bmod N\rangle$ are exchanged by the permutation gate P before they are fed to the gate II so that

$$P|x, 1, a^{x_0} \bmod N\rangle = |x, a^{x_0} \bmod N, 1\rangle.$$

Now the gate II acts on this state is

$$\text{CMODMULTI}^{-1}|x, a^{x_0} \bmod N, 1\rangle = |x, a^{x_0} \bmod N, 0\rangle.$$

- The next layer is comprised of two gates: III and IV. The output of the gate III is

$$\text{CMODMULTI}|x, a^{x_0} \bmod N, 0\rangle = |x, 1, a^{x_1 2}a^{x_0} \bmod N\rangle,$$

which is followed by P and then CMODMULTI^{-1} to produce

$$\text{CMODMULTI}^{-1}P|x, 1, a^{x_1 2}a^{x_0} \bmod N\rangle = |x, a^{x_1 2}a^{x_0} \bmod N, 0\rangle.$$

- The factor $a^{x_k 2^k}$ is multiplied to $\prod_{j=0}^{k-1} a^{x_j 2^j} \bmod N$ each time a new layer comprised of a pair of gates CMODMULTI and CMODMULTI^{-1} is applied. Eventually we obtain the output

$$\text{MODEXP}|x, 1, 0\rangle = |x, a^x \bmod N, 0\rangle.$$

8.6.5 Computational Complexity of Modular Exponential Circuit

We evaluate the complexity of the modular exponential circuit before we close this chapter. The evaluation is divided into several steps.

- Let us first look at ADD(n) given in Fig. 8.9. It follows from this figure that ADD(n) requires n SUM, $n-1$ CARRY^{-1}, n CARRY and one CNOT. Therefore the complexity $T_{\mathrm{ADD}(n)}$ of ADD(n) is

$$T_{\mathrm{ADD}(n)} = nT_S + (2n-1)T_C + 1, \tag{8.59}$$

  where T_S and T_C are numbers of elementary gates in SUM and CARRY, which are clearly independent of n.

- Figure 8.10 shows that the modular adder MODADD(n) is composed of three ADD(n), two ADD(n)$^{-1}$, two permutation gates, two controlled gates and four CNOT gates. Therefore the complexity is

$$T_{\mathrm{MODADD}(n)} = 5T_{\mathrm{ADD}(n)} + 2T_P + 2T_U + 4, \tag{8.60}$$

  where T_P and T_U are numbers of elementary gates in P and the controlled-U gate, respectively. It can be shown that these gates can be implemented with the Toffoli gates and each Toffoli gate is implemented with a polynomial number of elementary gates. The power of polynomial depends on the actual implementation. Implementation with a minimal number of qubits requires $O(n^2)$ elementary gates, but it may be reduced to $O(n)$ if some extra qubits are added.

- The controlled modular multiplexer depicted in Fig. 8.11 requires n MODADD(n), n controlled gates of type k (see Fig. 8.11) and n controlled gates of type k', two CNOT gates and one "copy" gate which works as $|x0\rangle \to |xx\rangle$ when the control bit is $c = 0$. The copy gate, controlled gates of types k and k' are all implemented with Toffoli gates. The number of elementary gates required to implement CMODMULTI(n) is therefore

$$T_{\mathrm{CMODMULT}(n)} = nT_{\mathrm{MODADD}(n)} + nT_k + nT_{k'} + T_{\mathrm{copy}} + 2, \tag{8.61}$$

  where T_k and $T_{k'}$ are the numbers of elementary gates in the controlled gates of types k and k', respectively, and T_{copy} is the number of elementary gates in the copy gate.

- The modular exponential gate MODEXP(n) depicted in Fig. 8.12 requires n CMODMULTI(n), n CMODMULTI(n)$^{-1}$ and n exchange gates P. The complexity is

$$T_{\mathrm{MODEXP}(n)} = 2nT_{\mathrm{CMODMULTI}(n)} + nT_P. \tag{8.62}$$

In summary, the number of elementary gates in the modular exponential circuit MODEXP(n) is of polynomial order in n. The maximum power of the polynomial depends on actual circuit implementation and algorithms employed to design the circuit. It may happen that adding extra qubits makes the maximum power smaller. Alternatively, the number of qubits required to implement a given algorithm may be reduced by sacrificing the number of elementary gates.

It was also shown in Chapter 6 that the quantum Fourier transform circuit is implemented with $O(n^2)$ number of elementary gates. Therefore prime number factorization for a number $N \sim 2^n$ is carried out with a polynomial number of steps in n in contrast with a classical algorithm which requires an exponentially large number of steps.

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9

Decoherence

A quantum system is always in interaction with its environment. This interaction inevitably alters the state of the quantum system, which causes loss of information encoded in the system. The system under consideration is not a *closed* system any more when interaction with the outside world is in action. We formulate the theory of an *open* quantum system in this chapter by regarding the combined system of the quantum system and its environment as a closed system and subsequently trace out the environmental degrees of freedom. Let ρ_S and ρ_E be the density matrices of the system and the environment, respectively. Even when the initial state is an uncorrelated state $\rho_S \otimes \rho_E$, the system-environment interaction entangles the total system so that the total state develops to an inseparable entangled state in general. Decoherence is a process in which environment causes various changes in the quantum system, which manifests itself as undesirable noise.

We study quantum error correcting codes (QECC), which is one of the strategies to fight against decoherence, in the next chapter.

9.1 Open Quantum System

Let us start our exposition with some mathematical background materials. We will closely follow [1] and [2].

We deal with general quantum states described by density matrices. We are interested in a general evolution of a quantum system, which is described by a powerful tool called a **quantum operation**. One of the simplest quantum operations is a unitary time evolution of a closed system. Let ρ_S be a density matrix of a closed system at $t = 0$ and let $U(t)$ be the time evolution operator. Then the corresponding quantum map $\mathcal{E}$ is defined as

$$\mathcal{E}(\rho_S) = U(t)\rho_S U(t)^\dagger. \tag{9.1}$$

One of our primary aims in this section is to generalize this map to cases of open quantum systems.

9.1.1 Quantum Operations and Kraus Operators

Suppose a system of interest is coupled with its **environment**. We must specify the details of the environment and the coupling between the system and the environment to study the effect of the environment on the behavior of the system. Let H_S, H_E and H_{SE} be the system Hamiltonian, the environment Hamiltonian and their interaction Hamiltonian, respectively. We assume the system-environment interaction is weak enough so that this separation into the system and its environment makes sense. To avoid confusion, we often call the system of interest a **principal system**. The total Hamiltonian H_T is then

$$H_T = H_S + H_E + H_{SE}. \tag{9.2}$$

Correspondingly, we denote the system Hilbert space and the environment Hilbert space as $\mathcal{H}_S$ and $\mathcal{H}_E$, respectively, and the total Hilbert space as $\mathcal{H}_T = \mathcal{H}_S \otimes \mathcal{H}_E$. The condition of weak system-environment interaction may be lifted in some cases. Let us consider a qubit propagating through a noisy quantum channel, for example. "Propagating" does not necessarily mean propagating in space. The qubit may be spatially fixed and subject to time-dependent noise. When the noise is localized in space and time, the input and the output qubit states belong to a well-defined Hilbert space $\mathcal{H}_S$, and the above separation of the Hamiltonian is perfectly acceptable even for strongly interacting cases. We consider, in the following, how the principal system state ρ_S at $t = 0$ evolves in time in the presence of its environment. A map which describes a general change of the state from ρ_S to $\mathcal{E}(\rho_S)$ is called a **quantum operation**. We have already noted that the unitary time evolution is an example of a quantum operation. Other quantum operations include state change associated with measurement and state change due to noise. The latter quantum map is our primary interest in this chapter.

The state of the total system is described by a density matrix ρ. Suppose ρ is uncorrelated initally at time $t = 0$,

$$\rho(0) = \rho_S \otimes \rho_E, \tag{9.3}$$

where ρ_S (ρ_E) is the initial density matrix of the principal system (environment). The total system is assumed to be closed and to evolve with a unitary matrix $U(t)$ as

$$\rho(t) = U(t)(\rho_S \otimes \rho_E)U(t)^\dagger. \tag{9.4}$$

Note that the resulting state is not a tensor product state in general. We are interested in extracting information on the state of the system at some later time $t > 0$.

Even under these circumstances, however, we may still define the system density matrix $\rho_S(t)$ by taking partial trace of $\rho(t)$ over the environment Hilbert space as

$$\rho_S(t) = \mathrm{tr}_E[U(t)(\rho_S \otimes \rho_E)U(t)^\dagger]. \tag{9.5}$$

We may forget about the environment by taking a trace over $\mathcal{H}_E$. This is an example of a quantum operation, $\mathcal{E}(\rho_S) = \rho_S(t)$. Details of the environment dynamics are made irrelevant at this stage. Let $\{|e_j\rangle\}$ be a basis of the system Hilbert space and $\{|\varepsilon_a\rangle\}$ be that of the environment Hilbert space. We may take the basis of $\mathcal{H}_T$ to be $\{|e_j\rangle \otimes |\varepsilon_a\rangle\}$. The initial density matrices may be written as

$$\rho_S = \sum_j p_j |e_j\rangle\langle e_j|, \;\; \rho_E = \sum_a r_a |\varepsilon_a\rangle\langle\varepsilon_a|.$$

Action of the time evolution operator on a basis vector of $\mathcal{H}_T$ is explicitly written as

$$U(t)|e_j, \varepsilon_a\rangle = \sum_{k,b} U_{kb;ja}|e_k, \varepsilon_b\rangle, \tag{9.6}$$

where $|e_j, \varepsilon_a\rangle = |e_j\rangle \otimes |\varepsilon_a\rangle$, for example. Using this expression, the density matrix $\rho(t)$ is written as

$$\begin{aligned} U(t)(\rho_S \otimes \rho_E)U(t)^\dagger &= \sum_{j,a} p_j r_a U(t)|e_j, \varepsilon_a\rangle\langle e_j, \varepsilon_a|U(t)^\dagger \\ &= \sum_{j,a,k,b,l,c} p_j r_a U_{kb;ja}|e_k, \varepsilon_b\rangle\langle e_l, \varepsilon_c|U^*_{lc;ja}. \end{aligned} \tag{9.7}$$

The partial trace over $\mathcal{H}_E$ is carried out to yield

$$\begin{aligned} \rho_S(t) &= \mathrm{tr}_E[U(t)(\rho_S \otimes \rho_E)U(t)^\dagger] \\ &= \sum_{j,a,k,b,l} p_j r_a U_{kb;ja}|e_k\rangle\langle e_l|U^*_{lb;ja} \\ &= \sum_{j,a,b} p_j \left(\sum_k \sqrt{r_a} U_{kb;ja}|e_k\rangle\right)\left(\sum_l \sqrt{r_a}\langle e_l|U^*_{lb;ja}\right). \end{aligned} \tag{9.8}$$

To write down the quantum operation in a closed form, we assume the initial environment state is a pure state, which we take, without loss of generality, $\rho_E = |\varepsilon_0\rangle\langle\varepsilon_0|$. Even when ρ_E is a mixed state, we may always complement $\mathcal{H}_E$ with a fictitious Hilbert space to "purify" ρ_E (see §2.4.2). With this assumption, $\rho_S(t)$ is written as

$$\begin{aligned} \rho_S(t) &= \mathrm{tr}_E[U(t)(\rho_S \otimes |\varepsilon_0\rangle\langle\varepsilon_0|)U(t)^\dagger] \\ &= \sum_a (I \otimes \langle\varepsilon_a|)U(t)(\rho_S \otimes |\varepsilon_0\rangle\langle\varepsilon_0|)|U(t)^\dagger(I \otimes |\varepsilon_a\rangle) \\ &= \sum_a (I \otimes \langle\varepsilon_a|)U(t)(I \otimes |\varepsilon_0\rangle)\rho_S(I \otimes \langle\varepsilon_0|)U(t)^\dagger(I \otimes |\varepsilon_a\rangle). \end{aligned}$$

We will drop $I\otimes$ from $I \otimes \langle\varepsilon_a|$ hereafter, whenever it does not cause confusion. Let us define the **Kraus operator** $E_a(t) : \mathcal{H}_S \to \mathcal{H}_S$ by

$$E_a(t) = \langle\varepsilon_a|U(t)|\varepsilon_0\rangle. \tag{9.9}$$

Then we may write

$$\mathcal{E}(\rho_S) = \rho_S(t) = \sum_a E_a(t)\rho_S E_a(t)^\dagger. \tag{9.10}$$

This is called the **operator-sum representation** (**OSR**) of a quantum operation $\mathcal{E}$. Note that $\{E_a\}$ satisfies the **completeness relation**

$$\begin{aligned}\left[\sum_a E_a(t)^\dagger E_a(t)\right]_{kl} &= \left[\sum_a \langle\varepsilon_0|U(t)^\dagger|\varepsilon_a\rangle\langle\varepsilon_a|U(t)|\varepsilon_0\rangle\right]_{kl} \\ &= \langle\varepsilon_0|U(t)^\dagger U(t)|\varepsilon_0\rangle_{lk} = I_{kl} = \delta_{kl},\end{aligned} \tag{9.11}$$

where I is the unit matrix in $\mathcal{H}_S$. This is equivalent with the trace-preserving property of $\mathcal{E}$ as

$$1 = \mathrm{tr}_S\rho_S(t) = \mathrm{tr}_S(\mathcal{E}(\rho_S)) = \mathrm{tr}_S\left(\sum_a E_a^\dagger E_a \rho_S\right)$$

for any $\rho_S \in \mathcal{S}(\mathcal{H}_S)$. The completeness relation and trace-preserving property are satisfied since our total system is a closed system. A general quantum map does not necessarily satisfy these properties [3].

At this stage, it turns out to be useful to relax the condition that $U(t)$ be a time evolution operator. Instead, we assume U to be any operator including an arbitrary unitary gate. Let us consider a two-qubit system on which the CNOT gate acts. Suppose the principal system is the control qubit while the environment is the target qubit. Then we find

$$E_0 = (I \otimes \langle 0|)U_{\mathrm{CNOT}}(I \otimes |0\rangle) = P_0,\ E_1 = (I \otimes \langle 1|)U_{\mathrm{CNOT}}(I \otimes |0\rangle) = P_1,$$

where $P_i = |i\rangle\langle i|$, and consequently

$$\mathcal{E}(\rho_S) = P_0\rho_S P_0 + P_1\rho_S P_1 = \rho_{00}P_0 + \rho_{11}P_1 = \begin{pmatrix} \rho_{00} & 0 \\ 0 & \rho_{11} \end{pmatrix}, \tag{9.12}$$

where

$$\rho_S = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix}.$$

We interchangeably regard U as a time evolution operator or as a black box unitary operator. The unitarity condition may be relaxed when measurements are included as quantum operations, for example.

Another generalization of a quantum operation is a map $\rho_E \to \rho_S(t)$, for example. Let us take the initial state $|e_0\rangle\langle e_0| \otimes \rho_E$ at $t = 0$. Then after application of $U(t)$ on the total system, we obtain

$$\rho_S(t) = \mathrm{tr}_E\left[U(t)(|e_0\rangle\langle e_0| \otimes \rho_E)U(t)^\dagger\right]. \tag{9.13}$$

Therefore quantum operations do not necessarily map density matrices to density matrices belonging to the same state space.

Tracing out the extra degrees of freedom makes it impossible to invert a quantum operation. Given an initial principal system state ρ_S, there are infinitely many U that yield the same $\mathcal{E}(\rho_S)$. Therefore even though it is possible to compose two quantum operations, the set of quantum operations is not a group but merely a **semigroup**.*

9.1.2 Operator-Sum Representation and Noisy Quantum Channel

Operator-sum representation (OSR) introduced in the previous subsection seems to be rather abstract. Here we give an interpretation of OSR as a noisy quantum channel. Suppose we have a set of unitary matrices $\{U_a\}$ and a set of non-negative real numbers $\{p_a\}$ such that $\sum_a p_a = 1$. By choosing U_a randomly with probability p_a and applying it to ρ_S, we introduce the expectation value of the resulting density matrix as

$$\mathcal{M}(\rho_S) = \sum_a p_a U_a \rho_S U_a^\dagger, \tag{9.14}$$

which we call a **mixing process** [4]. This takes place, for example, when a flying qubit is sent through a noisy quantum channel which transforms the qubit density matrix by U_a with probability p_a. Note that no environment has been introduced in the above definition, and hence no partial trace is involved.

Now the correspondence between $\mathcal{E}(\rho_S)$ and $\mathcal{M}(\rho_S)$ should be clear. Let us define

$$E_a \equiv \sqrt{p_a} U_a. \tag{9.15}$$

Then Eq. (9.14) is rewritten as

$$\mathcal{M}(\rho_S) = \sum_a E_a \rho_S E_a^\dagger, \tag{9.16}$$

and the equivalence has been shown. Operators E_a are identified with the Kraus operators. The system transforms, under the action of U_a, as

$$\rho_S \to \frac{E_a \rho_S E_a^\dagger}{\mathrm{tr}\left(E_a \rho_S E_a^\dagger\right)}. \tag{9.17}$$

Conversely, given a noisy quantum channel $\{U_a, p_a\}$ we may introduce an "environment" with the Hilbert space $\mathcal{H}_E$ as follows. Let $\mathcal{H}_E = \mathrm{Span}(|\varepsilon_a\rangle)$

*A set S is called a semigroup if S is closed under a product satisfying associativity $(ab)c = a(bc)$. If S has a unit element e, such that $ea = ae = a, \forall a \in S$, it is called a monoid.

be a Hilbert space with the dimension equal to the number of the unitary matrices $\{U_a\}$, where $\{|\varepsilon_a\rangle\}$ is an orthonormal basis. Define formally the environment density matrix $\rho_E = \sum_a p_a |\varepsilon_a\rangle\langle\varepsilon_a|$ and

$$U \equiv \sum_a U_a \otimes |\varepsilon_a\rangle\langle\varepsilon_a| \tag{9.18}$$

which acts on $\mathcal{H}_S \otimes \mathcal{H}_E$. It is easily verified from the orthonormality of $\{|\varepsilon_a\rangle\}$ that U is indeed a unitary matrix. Now a partial trace over the fictitious $\mathcal{H}_E$ yields

$$\begin{aligned}
\mathcal{E}(\rho_S) &= \mathrm{tr}_E[U(\rho_S \otimes \rho_E)U^\dagger] \\
&= \sum_a (I \otimes \langle\varepsilon_a|) \left(\sum_b U_b \otimes |\varepsilon_b\rangle\langle\varepsilon_b|\right) \left(\rho_S \otimes \sum_c p_c |\varepsilon_c\rangle\langle\varepsilon_c|\right) \\
&\quad \times \left(\sum_d U_d \otimes |\varepsilon_d\rangle\langle\varepsilon_d|\right) (I \otimes |\varepsilon_a\rangle) \\
&= \sum_a p_a U_a \rho_S U_a^\dagger = \mathcal{M}(\rho_S),
\end{aligned} \tag{9.19}$$

showing that the mixing process is also decribed by a quantum operation with a fictitious environment.

EXERCISE 9.1 Show that U defined as Eq. (9.18) is unitary.

9.1.3 Completely Positive Maps

All linear operators we have encountered so far map vectors to vectors. A quantum operation maps a density matrix to another density matrix linearly.† A linear operator of this kind is called a **superoperator**. Let Λ be a superoperator acting on the system density matrices, $\Lambda : \mathcal{S}(\mathcal{H}_S) \to \mathcal{S}(\mathcal{H}_S)$. The operator Λ is easily extended to an operator acting on $\mathcal{H}_T$ by $\Lambda_T = \Lambda \otimes I_E$, which acts on $\mathcal{S}(\mathcal{H}_S \otimes \mathcal{H}_E)$. Note, however, that Λ_T is not necessarily a map $\mathcal{S}(\mathcal{H}_T) \to \mathcal{S}(\mathcal{H}_T)$. It may happen that $\Lambda_T(\rho)$ is not a density matrix any more. We have already encountered this situation when we introduced partial transpose operation in §2.4.1. Let $\mathcal{H}_T = \mathcal{H}_1 \otimes \mathcal{H}_2$ be a two-qubit Hilbert space, where $\mathcal{H}_k$ is the kth qubit Hilbert space. It is clear that the transpose operation $\Lambda_t : \rho_1 \to \rho_1^t$ on a single-qubit state ρ_1 preserves the density matrix properties, i.e., non-negativity, Hermiticity and unit trace property. For a two-qubit density matrix ρ_{12}, however, this is not always the case. In fact,

†Of course, the space of density matrices is not a linear vector space. What is meant here is a linear operator, acting on the vector space of Hermitian matrices, also acts on the space of density matrices and maps a density matrix to another density matrix.

we have seen that $\Lambda_t \otimes I : \rho_{12} \to \rho_{12}^{\mathrm{pt}}$ defined by Eq. (2.42) maps a density matrix to a matrix which is not a density matrix when ρ_{12} is inseparable.

A map Λ which maps a positive operator acting on $\mathcal{H}_S$ to another positive operator acting on $\mathcal{H}_S$ is said to be positive. Moreover, it is called a **completely positive map** (**CP map**), if its extension $\Lambda_T = \Lambda \otimes I_n$ remains a positive operator for an arbitrary $n \in \mathbb{N}$.

THEOREM 9.1 A linear map Λ is CP if and only if there exists a set of operators $\{E_a\}$ such that $\Lambda(\rho)$ can be written as

$$\Lambda(\rho_S) = \sum_a E_a \rho_S E_a^\dagger. \tag{9.20}$$

We require not only that Λ be CP but also that $\Lambda(\rho)$ be a density matrix, i.e.,

$$\operatorname{tr} \Lambda(\rho_S) = \operatorname{tr}\left(\sum_a E_a \rho E_a^\dagger\right) = \operatorname{tr}\left(\sum_a E_a^\dagger E_a \rho\right) = 1. \tag{9.21}$$

This condition is satisfied for any ρ if and only if

$$\sum_a E_a^\dagger E_a = I_S. \tag{9.22}$$

Therefore, any quantum operation obtained by tracing out the environment degrees of freedom is CP and preserves trace.

9.2 Measurements as Quantum Operations

We have already seen that a unitary evoluation $\rho_S \to U \rho_S U^\dagger$ is a quantum operation and that the mixing process $\rho_S \to \sum_i p_i U_i \rho_S U_i^\dagger$ is a quantum operation. We will see further examples of quantum operations in this section and the next. This section deals with measurements as quantum operations.

9.2.1 Projective Measurements

Suppose we measure an observable $A = \sum_i \lambda_i P_i$, where $P_i = |\lambda_i\rangle\langle\lambda_i|$ is the projection operator corresponding to the eigenvector $|\lambda_i\rangle$. We have seen in Chapter 2 that the probability of observing λ_i upon a measurement of A in a state ρ is

$$p(i) = \langle\lambda_i|\rho|\lambda_i\rangle = \operatorname{tr}(P_i \rho), \tag{9.23}$$

and the state changes as $\rho \to P_i \rho P_i / p(i)$. This process happens with a probability $p(i)$. Thus we may regard the measurement process as a quantum operation

$$\rho_S \to \sum_i p(i) \frac{P_i \rho_S P_i}{p(i)} = \sum_i P_i \rho_S P_i, \tag{9.24}$$

where the set $\{P_i\}$ satisifes the completeness relation $\sum_i P_i P_i^\dagger = I$.

The projective measurement is a special case of a quantum operation in which the Kraus operators are $E_i = P_i$.

9.2.2 POVM

We have been concerned with projective measurements so far. However, it should be noted that they are not a unique type of measurements. Here we will be concerned with the most general framework of measurement and show that it is a quantum operation.

Suppose a system and an environment, prepared in a product state $|\psi\rangle|e_0\rangle$, are acted by a unitary operator U, which applies an operator M_i on the system and, at the same time, put the environment to $|e_i\rangle$ for various i. It is written explicitly as

$$|\Psi\rangle = U|\psi\rangle|e_0\rangle = \sum_i M_i|\psi\rangle|e_i\rangle. \tag{9.25}$$

The system and its environment are correlated in this way. This state must satisfy the normalization condition since U is unitary;

$$\begin{aligned} \langle\psi|\langle e_0|U^\dagger U|\psi\rangle|e_0\rangle &= \sum_{i,j} \langle\psi|\langle e_i|M_i^\dagger M_j \otimes I|\psi\rangle|e_j\rangle \\ &= \langle\psi| \sum_i M_i^\dagger M_i|\psi\rangle = 1. \end{aligned} \tag{9.26}$$

Since $|\psi\rangle$ is arbitrary, we must have

$$\sum_i M_i^\dagger M_i = I_S, \tag{9.27}$$

where I_S is the unit matrix acting on the system Hilbert space $\mathcal{H}_S$. Operators $\{M_i^\dagger M_i\}$ are said to form a **POVM** (**positive operator-valued measure**).

Suppose we measure the environment with a measurement operator

$$O = I_S \otimes \sum_i \lambda_i |e_i\rangle\langle e_i| = \sum_i \lambda_i \left(I_S \otimes |e_i\rangle\langle e_i|\right).$$

We obtain a measurement outcome λ_k with a probability

$$\begin{aligned} p(k) &= \langle\Psi|(I_S \otimes |e_k\rangle\langle e_k|)|\Psi\rangle \\ &= \sum_{i,j} \langle\psi|\langle e_i|M_i^\dagger (I_S \otimes |e_k\rangle\langle e_k|) M_j|\psi\rangle|e_j\rangle \\ &= \langle\psi|M_k^\dagger M_k|\psi\rangle, \end{aligned} \tag{9.28}$$

where $|\Psi\rangle = U|\psi\rangle|e_0\rangle$. The combined system immediately after the measurement is

$$\begin{aligned}\frac{1}{\sqrt{p(k)}}(I_S \otimes |e_k\rangle\langle e_k|)U|\psi\rangle|e_0\rangle &= \frac{1}{\sqrt{p(k)}}(I_S \otimes |e_k\rangle\langle e_k|)\sum_i M_i|\psi\rangle|e_i\rangle \\ &= \frac{1}{\sqrt{p(k)}}M_k|\psi\rangle|e_k\rangle. \qquad (9.29)\end{aligned}$$

Let $\rho_S = \sum_i p_i|\psi_i\rangle\langle\psi_i|$ be an arbitrary density matrix of the principal system. It follows from the above observation for a pure state $|\psi\rangle\langle\psi|$ that the reduced density matrix immediately after the measurement is

$$\sum_k p(k)\frac{M_k\rho_S M_k^\dagger}{p(k)} = \sum_k M_k\rho_S M_k^\dagger. \qquad (9.30)$$

This shows that POVM measurement is a quantum operation in which the Kraus operators are given by the generalized measurement operators $\{M_i\}$. The projective measurement is a special class of POVM, in which $\{M_i\}$ are the projection operators.

9.3 Examples

Now we examine several important examples which have relevance in quantum information theory. Decoherence appears as an error in quantum information processing. The next chapter is devoted to strategies to fight against some errors introduced in this section.

9.3.1 Bit-Flip Channel

A bit-flip channel is defined by a quantum operation

$$\mathcal{E}(\rho_S) = (1-p)\rho_S + p\sigma_x\rho_S\sigma_x,\ 0 \le p \le 1. \qquad (9.31)$$

In other words, the input ρ_S is bit-flipped ($|0\rangle \mapsto |1\rangle$ and $|1\rangle \mapsto |0\rangle$) with a probability p while it remains in its input state with a probability $1-p$. The Kraus operators are easily read off as

$$E_0 = \sqrt{1-p}I,\ E_1 = \sqrt{p}\sigma_x. \qquad (9.32)$$

Let us next consider a quantum circuit which models this channel. Consider a closed two-qubit system with a Hilbert space $\mathbb{C}^2\otimes\mathbb{C}^2$. We call the first qubit the "(principal) system" and the second qubit the "environment." We show that the circuit depicted in Fig. 9.1 indeed models the bit-flip channel provided

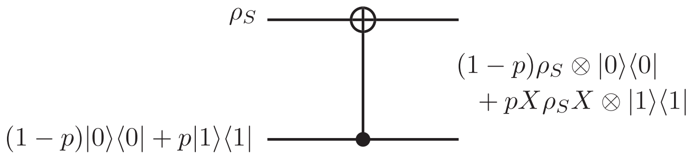

FIGURE 9.1
Quantum circuit modelling a bit-flip channel. The gate is the inverted CNOT gate $I \otimes |0\rangle\langle 0| + \sigma_x \otimes |1\rangle\langle 1|$.

that the second qubit is in a mixed state $(1-p)|0\rangle\langle 0| + p|1\rangle\langle 1|$. The circuit is nothing but the inverted CNOT gate

$$V = I \otimes |0\rangle\langle 0| + \sigma_x \otimes |1\rangle\langle 1|.$$

The output of this circuit is

$$\begin{aligned} &V\left(\rho_S \otimes [(1-p)|0\rangle\langle 0| + p|1\rangle\langle 1|]\right) V^\dagger \\ &= (1-p)\rho_S \otimes |0\rangle\langle 0| + p\sigma_x\rho_S\sigma_x|1\rangle\langle 1|, \end{aligned} \tag{9.33}$$

from which we obtain

$$\mathcal{E}(\rho_S) = (1-p)\rho_S + p\sigma_x\rho_S\sigma_x \tag{9.34}$$

after tracing over the environment Hilbert space.

The choice of the second qubit input state is far from unique and so is the choice of the circuit. Suppose the initial state of the environment is a pure state

$$|\psi_E\rangle = \sqrt{1-p}|0\rangle + \sqrt{p}|1\rangle, \tag{9.35}$$

for example. Then the output of the circuit in Fig. 9.1 is

$$\mathcal{E}(\rho_S) = \mathrm{tr}_E[V\rho_S \otimes |\psi_E\rangle\langle\psi_E|V^\dagger] = (1-p)\rho_S + p\sigma_x\rho_S\sigma_x, \tag{9.36}$$

which is the same result as before.

Let us see what transformation this quantum operation brings about in ρ_S. We parametrize ρ_S using the Bloch vector as

$$\rho_S = \frac{1}{2}\left(I + \sum_{k=x,y,z} c_k\sigma_k\right), \quad (c_k \in \mathbb{R}), \tag{9.37}$$

where $\sum_k c_k^2 \le 1$. We obtain

$$\begin{aligned} \mathcal{E}(\rho_S) &= (1-p)\rho_S + p\sigma_x\rho_S\sigma_x \\ &= \frac{1-p}{2}(I + c_x\sigma_x + c_y\sigma_y + c_z\sigma_z) + \frac{p}{2}(I + c_x\sigma_x - c_y\sigma_y - c_z\sigma_z) \\ &= \frac{1}{2}\begin{pmatrix} 1+(1-2p)c_z & c_x - i(1-2p)c_y \\ c_x + i(1-2p)c_y & 1-(1-2p)c_z \end{pmatrix}. \end{aligned} \tag{9.38}$$

Observe that the radius of the Bloch sphere is reduced along the y- and the z-axes so that the radius in these directions is $|1-2p|$. Equation (9.38) shows that the quantum operation has produced a mixture of the Bloch vector states (c_x, c_y, c_z) and $(c_x, -c_y, -c_z)$ with weights $1-p$ and p, respectively. Figure 9.2 (a) shows the Bloch sphere which represents the input qubit states. The Bloch sphere shrinks along the y- and z-axes, which results in the ellipsoid shown in Fig. 9.2 (b).

9.3.2 Phase-Flip Channel

The phase-flip channel is defined by a quantum operation

$$\mathcal{E}(\rho_S) = (1-p)\rho_S + p\sigma_z\rho_S\sigma_z, \ 0 \le p \le 1. \tag{9.39}$$

In other words, the input ρ_S is phase-flipped ($|0\rangle \mapsto |0\rangle$ and $|1\rangle \mapsto -|1\rangle$) with a probability p while it remains in its input state with a probability $1-p$. The corresponding Kraus operators are

$$E_0 = \sqrt{1-p}I, \ E_1 = \sqrt{p}\sigma_z. \tag{9.40}$$

Consider again a closed two-qubit system, whose first qubit is called the "(principal) system" and the second qubit its "environment." A quantum circuit which models the phase-flip channel is shown in Fig. 9.3. Let ρ_S be the first qubit input state and $(1-p)|0\rangle\langle 0| + p|1\rangle\langle 1|$ be the second qubit input state. The circuit is the inverted controlled-σ_z gate

$$V = I \otimes |0\rangle\langle 0| + \sigma_z \otimes |1\rangle\langle 1|.$$

The output of this circuit is

$$\begin{aligned} &V\left(\rho_S \otimes [(1-p)|0\rangle\langle 0| + p|1\rangle\langle 1|]\right)V^\dagger \\ &\quad = (1-p)\rho_S \otimes |0\rangle\langle 0| + p\sigma_z\rho_S\sigma_z \otimes |1\rangle\langle 1|, \end{aligned} \tag{9.41}$$

from which we obtain

$$\mathcal{E}(\rho_S) = (1-p)\rho_S + p\sigma_z\rho_S\sigma_z. \tag{9.42}$$

The second qubit input state may be a pure state

$$|\psi_E\rangle = \sqrt{1-p}|0\rangle + \sqrt{p}|1\rangle, \tag{9.43}$$

for example. Then we find

$$\begin{aligned} \mathcal{E}(\rho_S) &= \mathrm{tr}_E[V\rho_S \otimes |\psi_E\rangle\langle\psi_E|V^\dagger] \\ &= E_0\rho_S E_0^\dagger + E_1\rho_S E_1^\dagger, \end{aligned} \tag{9.44}$$

where the Kraus operators are

$$E_0 = \langle 0|V|\psi_E\rangle = \sqrt{1-p}I, \ E_1 = \langle 1|V|\psi_E\rangle = \sqrt{p}\sigma_z. \tag{9.45}$$

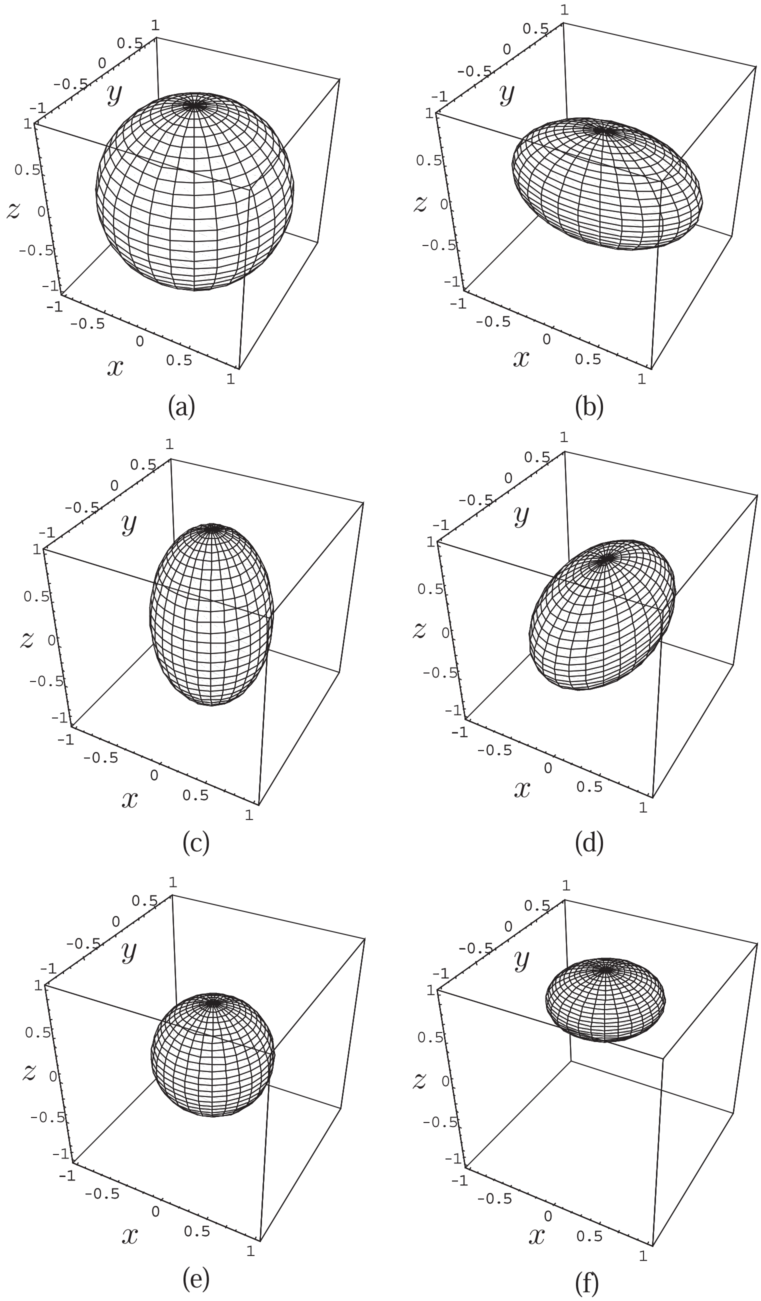

FIGURE 9.2
Bloch sphere of the input state ρ_S (a) and output states (b) $\sim$ (f) of various noisy channels. (b) Bit-flip channel, (c) phase-flip channel, (d) bit-phase flip channel, (e) depolarizing channel and (f) amplitude damping channel. The probability $p = 0.2$ is common to all the noisy channels except in (f), for which $p = 0.7$.

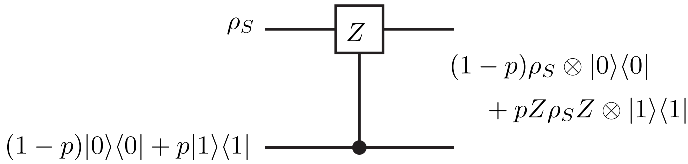

FIGURE 9.3
Quantum circuit modelling a phase-flip channel. The gate is the inverted controlled-σ_z gate.

Let us work out the transformation this quantum operation brings about to ρ_S. We parametrize ρ_S using the Bloch sphere as before. We obtain

$$\begin{aligned}
\mathcal{E}(\rho_S) &= (1-p)\rho_S + p\sigma_z\rho_S\sigma_z \\
&= \frac{1-p}{2}(I + c_x\sigma_x + c_y\sigma_y + c_z\sigma_z) + \frac{p}{2}(I - c_x\sigma_x - c_y\sigma_y + c_z\sigma_z) \\
&= \frac{1}{2}\begin{pmatrix} 1+c_z & (1-2p)(-c_x - ic_y) \\ (1-2p)(c_x + ic_y) & 1-c_z \end{pmatrix}.
\end{aligned} \tag{9.46}$$

Observe that the off-diagonal components decay while the diagonal components remain the same. Equation (9.46) shows that the quantum operation has produced a mixture of the Bloch vector states (c_x, c_y, c_z) and $(-c_x, -c_y, c_z)$ with weights $1-p$ and p, respectively. The initial state has a definite phase $\phi = \tan^{-1}(c_y/c_x)$ in the off-diagonal components. The phase after the quantum operation is applied is a mixture of states with ϕ and $\phi + \pi$. This process is called the **phase relaxation process**, or the T_2 process in the context of NMR. The radius of the Bloch sphere is reduced along the x- and the y-axes to $|1-2p|$. Figure 9.2 (c) shows the effect of the phase-flip channel on the Bloch sphere for $p = 0.2$.

EXERCISE 9.2 Let us define the bit-phase flip channel by a quantum operation

$$\mathcal{E}(\rho_S) = (1-p)\rho_S + pY\rho_S Y, \tag{9.47}$$

where $Y = -i\sigma_y$. It follows from the observation $XZ = Y$ that bit-phase flip may be considered as a combination of bit-flip and phase-flip.

Find a quantum circuit which models the bit-phase flip channel. Show that the effect of the channel on the Bloch sphere is given by Fig. 9.2 (d).

9.3.3 Depolarizing Channel

A depolarizing channel maps the input state ρ to a maximally mixed state with a probability p and leaves as it is with a probability $1-p$;

$$\mathcal{E}(\rho_S) = (1-p)\rho_S + p\frac{I}{2}. \tag{9.48}$$

We introduce the decomposition $\rho_S = (I + c_x\sigma_x + c_y\sigma_y + c_z\sigma_z)/2$ to write the OSR form of the channel. We observe first that

$$\begin{aligned}
\sigma_x\rho_S\sigma_x &= \frac{1}{2}(I + c_x\sigma_x - c_y\sigma_y - c_z\sigma_z) \\
\sigma_y\rho_S\sigma_y &= \frac{1}{2}(I - c_x\sigma_x + c_y\sigma_y - c_z\sigma_z) \\
\sigma_z\rho_S\sigma_z &= \frac{1}{2}(I - c_x\sigma_x - c_y\sigma_y + c_z\sigma_z),
\end{aligned}$$

from which, we obtain

$$2I = \rho_S + \sum_{k=x,y,z} \sigma_k\rho_S\sigma_k. \tag{9.49}$$

Substituting this into Eq. (9.48), we obtain the OSR of the depolarizing channel

$$\mathcal{E}(\rho_S) = \left(1 - \frac{3}{4}p\right)\rho_S + \frac{p}{4}\sum_k \sigma_k\rho_S\sigma_k. \tag{9.50}$$

The Kraus operators are read off as

$$E_0 = \sqrt{1 - \frac{3}{4}p}I, E_k = \sqrt{\frac{p}{4}}\sigma_k \ (k = x, y, z). \tag{9.51}$$

It is convenient to rescale the probability as $p' = 3p/4$ $(0 \le p' \le 3/4)$ to obtain

$$\mathcal{E}(\rho_S) = (1 - p')\,\rho_S + \frac{p'}{3}\sum_k \sigma_k\rho_S\sigma_k. \tag{9.52}$$

The channel transports ρ_S without any change with probability $1 - p'$ while X, Y and Z acts each with probability $p'/3$.

The fact that there are four Kraus operators suggests that the environment Hilbert space is at least four-dimensional. In fact, we can construct a quantum circuit model shown in Fig. 9.4. It is a Fredkin gate with the bottom control bit. The input state is

$$\rho_S \otimes \frac{I}{2} \otimes [(1-p)|0\rangle\langle 0| + p|1\rangle\langle 1|].$$

The action of the Fredkin gate on the input state yields the output

$$\begin{aligned}
\rho &= (I_4 \otimes |0\rangle\langle 0| + U_{\text{SWAP}} \otimes |1\rangle\langle 1|)\left(\rho_S \otimes \frac{I}{2} \otimes [(1-p)|0\rangle\langle 0| + p|1\rangle\langle 1|]\right) \\
&\quad (I_4 \otimes |0\rangle\langle 0| + U_{\text{SWAP}} \otimes |1\rangle\langle 1|) \\
&= (1-p)\rho_S \otimes \frac{I}{2} \otimes |0\rangle\langle 0| + p\frac{I}{2} \otimes \rho_S \otimes |1\rangle\langle 1|,
\end{aligned} \tag{9.53}$$

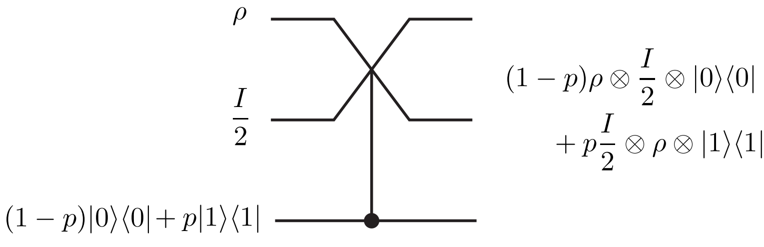

FIGURE 9.4
Quantum circuit which models a depolarizing channel. The gate is an inverted Fredkin gate, in which the control bit is the third qubit.

where use has been made of the relation

$$U_{\text{SWAP}}(\rho_1 \otimes \rho_2)U_{\text{SWAP}} = \rho_2 \otimes \rho_1. \tag{9.54}$$

Tracing out the two-qubit environment produces

$$\text{tr}_E \rho = (1-p)\rho_S + p\frac{I}{2} \tag{9.55}$$

as promised.

The effect of this channel on the Bloch sphere of the qubit is obtained from the expression

$$\mathcal{E}(\rho_S) = p\frac{I}{2} + \frac{1-p}{2}(I + \sum_k c_k \sigma_k) = \frac{I}{2} + \frac{1-p}{2}\sum_k c_k \sigma_k. \tag{9.56}$$

It is found that the radius is uniformly reduced from 1 to $1-p$. Figure 9.2 (e) shows the Bloch sphere of the output state (9.56) with $p = 0.2$.

9.3.4 Amplitude-Damping Channel

Our final example is the amplitude-damping channel, which is described by a one-way decay only. A qubit decays from $|1\rangle$ to $|0\rangle$ with a probability p but not in the other way around. This process describes the T_1 process in NMR, for example.

Let

$$\mathcal{E}(\rho_S) = E_0 \rho_S E_0^\dagger + E_1 \rho_S E_1^\dagger \tag{9.57}$$

be the OSR of the quantum operation. This downward decay process is naturally described by a Kraus operator

$$E_1 = \sqrt{p}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \tag{9.58}$$

The other Kraus operator E_0 is fixed by the requirement $\sum_k E_k^\dagger E_k = I$ as

$$E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}. \tag{9.59}$$

Let us consider the effect of this channel on the Bloch sphere. By substituting the Bloch sphere representation of ρ_S, we obtain

$$\begin{aligned}\mathcal{E}(\rho_S) &= p \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \rho_S \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix} \rho_S \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix} \\ &= \begin{pmatrix} \rho_{00} + p\rho_{11} & \sqrt{1-p}\rho_{01} \\ \sqrt{1-p}\rho_{10} & \rho_{11} - p\rho_{11} \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 + [p + (1-p)c_z] & \sqrt{1-p}(c_x - ic_y) \\ \sqrt{1-p}(c_x + ic_y) & 1 - [p + (1-p)c_z] \end{pmatrix}. \end{aligned} \tag{9.60}$$

Note that the center of the Bloch sphere is shifted toward the north pole ($|0\rangle$) by p under the influence of the channel and, at the same time, the radius in the x- and y-directions are reduced by $\sqrt{1-p}$ and the radius in the z-direction by $1-p$. Figure 9.2 (f) shows this effect graphically.

9.4 Lindblad Equation

Let $\mathcal{H}_S$ be a closed system with the Hamiltonian H_S. The system density matrix ρ_S evolves according to the Liouville-von Neumann equation

$$\frac{\partial \rho_S}{\partial t} = \frac{1}{i}[H, \rho_S]. \tag{9.61}$$

What is the corresponding equation for an open quantum system? We have found in §9.1 that the principal system density matrix $\rho_S(t)$ at later time $t > 0$ is obtained by tracing out the environment degrees of freedom as

$$\rho_S(t) = \mathrm{tr}_E[U(t)(\rho_S \otimes \rho_E)U(t)^\dagger], \tag{9.62}$$

where $U(t)$ is the time evolution operator of the total system and the initial state is assumed to be a tensor product state. Then the dynamics of $\rho_S(t)$ is

$$\frac{\partial \rho_S(t)}{\partial t} = \frac{\partial}{\partial t}\mathrm{tr}_E[U(t)(\rho_S \otimes \rho_E)U(t)^\dagger] = \frac{1}{i}\mathrm{tr}_E[H, U(t)(\rho_S \otimes \rho_E)U(t)^\dagger], \tag{9.63}$$

where $H = H_S + H_E + H_{SE}$ is the total Hamiltonian and we noted that time derivative commutes with tr_E. This equation, albeit exact, is not closed and is of little use in actual applications. Now we must introduce an approximation in the above exact equation, with which we may extract useful information on the system dynamics.

One of the most popular assumptions is the **Markovian approximation**. The state $\rho_S(t)$ depends on the previous history of the system in general. For some problems, however, this memory effect may be negligible, and the dynamics of $\rho_S(t)$ is determined by $\rho_S(t)$ itself. For example we may assume that the behavior of the system is Markovian, if we are interested in a time scale much longer than the environment correlation times. If this is the case, the master equation should look like

$$\frac{\partial}{\partial t}\rho_S(t) = \mathcal{L}\rho_S(t) \tag{9.64}$$

with a superoperator $\mathcal{L}$.

9.4.1 Quantum Dynamical Semigroup

Let

$$\Phi_t : \rho_S \to \rho_S(t), \quad \Phi_0 = I \tag{9.65}$$

be the quantum operation associated with a time evolution operator $U(t)$. We require that Φ_t be trace-preserving and completely positive. The map Φ_t has the OSR

$$\Phi_t(\rho_S) = \sum_a E_a(t)\rho_S E_a^\dagger(t), \quad \sum_a E_a^\dagger(t)E_a(t) = I. \tag{9.66}$$

Suppose the reduced Liouville-von Neumann equation satisfies the Markovian property. Then it is formally solved to yield

$$\Phi_t(\rho_S) = e^{\mathcal{L}t}\rho_S, t \in [0, \infty). \tag{9.67}$$

This map satisfies the semigroup property

$$\Phi_{t_2}(\Phi_{t_1}(\rho_S)) = \Phi_{t_1+t_2}(\rho_S). \tag{9.68}$$

The set of maps $\{\Phi_t\}$ is called the **quantum dynamical semigroup**.

9.4.2 Lindblad Equation

Let $\dim\mathcal{H}_S = d$. A superoperator $\mathcal{L}$ has components $\mathcal{L}_{ij,kl}$ with $1 \le i, j, k, l \le d$ showing that it acts on a d^2-dimensional vector space. Let us introduce the Hilbert-Schmidt scalar product in the space of matrices as

$$(A, B)_{\mathrm{HS}} = \mathrm{tr}\,(A^\dagger B), \tag{9.69}$$

where A and B are $d \times d$ matrices. Verify that this definition satisfies all the axioms of a scalar product.

The basis of the vector space of matrices may be $\{e_{ij} = |i\rangle\langle j|\}$ where

$$(e_{ij})_{pq} = \delta_{ip}\delta_{jq}.$$

Instead of the bi-index notation $\{e_{ij}\}$, we properly assign a single index to the basis vectors to write $\{e_J\}$ $(1 \le J \le d^2)$. It turns out to be convenient to introduce a special basis vector

$$e_{d^2} = \frac{1}{\sqrt{d}} I_d \tag{9.70}$$

and take the other $d^2 - 1$ basis vectors traceless. The basis vectors satisfy

$$(e_J, e_K)_{\mathrm{HS}} = \delta_{JK}, \quad \mathrm{tr}\, e_J = \begin{cases} 0, & 1 \le J \le d^2 - 1 \\ \sqrt{d}, & J = d^2. \end{cases} \tag{9.71}$$

Let us write the OSR of Φ_t as

$$\Phi_t(\rho_S) = \sum_a E_a(t)\rho_S E_a^\dagger(t), \tag{9.72}$$

where we have written the time dependence of the Kraus operator E_a explicitly. Now we expand E_a in terms of $\{e_J\}$ as

$$E_a(t) = \sum_J e_J (e_J, E_a(t))_{\mathrm{HS}} \tag{9.73}$$

and substitute this into Eq. (9.72) to obtain

$$\Phi_t(\rho_S) = \sum_{J,K} c_{JK}(t) e_J \rho_S e_K^\dagger, \tag{9.74}$$

where

$$c_{JK}(t) = \sum_a (e_J, E_a(t))_{\mathrm{HS}} (e_K, E_a(t))_{\mathrm{HS}}^*. \tag{9.75}$$

The coefficient matrix (c_{JK}) is positive and Hermitian.

Now we are ready to derive the Lindblad equation by employing these tools. By separating the summation over J and K in Eq. (9.74) into $1 \le J, K \le d^2 - 1$ and $J, K = d^2$, we obtain

$$\begin{aligned}\mathcal{L}\rho &= \lim_{\tau \to 0} \frac{\Phi_\tau(\rho_S) - \rho_S}{\tau} \\ &= \lim_{\tau \to 0} \frac{c_{d^2 d^2}(\tau)/d - 1}{\tau}\rho_S + \left(\lim_{\tau \to 0} \sum_{J=1}^{d^2-1} \frac{c_{Jd^2}(\tau)}{\sqrt{d}\tau} e_J \right) \rho_S \\ &\quad + \rho_S \left(\lim_{\tau \to 0} \sum_{K=1}^{d^2-1} \frac{c_{d^2 K}(\tau)}{\sqrt{d}\tau} e_K^\dagger \right) + \lim_{\tau \to 0} \sum_{J,K=1}^{d^2-1} \frac{c_{JK}(\tau)}{\tau} e_J \rho_S e_K^\dagger. \end{aligned} \tag{9.76}$$

Now we introduce notations

$$c_0 = \lim_{\tau\to 0}\frac{c_{d^2d^2}(\tau)/d-1}{\tau},\ B = \lim_{\tau\to 0}\sum_{J=1}^{d^2-1}\frac{c_{Jd^2}(\tau)}{\sqrt{d}\tau}e_J,$$

$$B^\dagger = \lim_{\tau\to 0}\sum_{K=1}^{d^2-1}\frac{c_{d^2K}(\tau)}{\sqrt{d}\tau}e_K^\dagger,\ \alpha_{JK} = \lim_{\tau\to 0}\frac{c_{JK}(\tau)}{\tau} \tag{9.77}$$

to rewrite Eq. (9.76) as

$$\begin{aligned}\mathcal{L}\rho_S &= c_0\rho_S + B\rho_S + \rho_S B^\dagger + \sum_{J,K=1}^{d^2-1}\alpha_{JK}e_J\rho_S e_K^\dagger \\ &= \frac{1}{i}[\tilde{H},\rho_S] + (G\rho_S + \rho_S G) + \sum_{J,K}\alpha_{JK}e_J\rho_S e_K^\dagger,\end{aligned} \tag{9.78}$$

where

$$\tilde{H} = \frac{1}{2i}(B - B^\dagger),\ G = \frac{1}{2}(B + B^\dagger + c_0). \tag{9.79}$$

Although we are tempted to identify $\tilde{H}$ with the Hamiltonian in the von-Neumann equation, it is not necessarily justifiable in the present case. It is possible to eliminate G in favor of α_{st} if we notice the trace-preserving property $\mathrm{tr}\,\rho_S = \mathrm{tr}\,(\Phi_t\rho_S) = 1$. In fact, it follows from $\mathrm{tr}\,(\mathcal{L}\rho_S) = \mathrm{tr}\,\partial_t\Phi_t(\rho_S) = \partial_t\mathrm{tr}\,\Phi_t(\rho_S) = 0$ that

$$0 = \mathrm{tr}\,(\mathcal{L}\rho_S) = \mathrm{tr}\left[\left(2G + \sum_{J,K=1}^{d^2-1}\alpha_{JK}e_K^\dagger e_J\right)\rho_S\right] \tag{9.80}$$

for an arbitrary ρ_S. Now G is solved as

$$G = -\frac{1}{2}\sum_{J,K}\alpha_{JK}e_K^\dagger e_J. \tag{9.81}$$

Substituting this into Eq. (9.78), we obtain

$$\mathcal{L}\rho_S = \frac{1}{i}[\tilde{H},\rho_S] + \sum_{J,K=1}^{d^2-1}\alpha_{JK}\left(e_J\rho_S e_K^\dagger - \frac{1}{2}e_K^\dagger e_J\rho_S - \frac{1}{2}\rho_S e_K^\dagger e_J\right). \tag{9.82}$$

The first term in the right hand side generates the unitary time evolution, while the second term brings about incoherent time evolution in the system dynamics.

The matrix α is positive and Hermitian and can be diagonalized with a properly chosen unitary matrix W. Let $W\alpha W^\dagger = D = \mathrm{diag}(\gamma_1,\ldots\gamma_{d^2})$, where γ_i is a positive real number. Let us define operators

$$L_K = \sum_{J=1}^{d^2-1}e_J W_{JK}^\dagger \tag{9.83}$$

so that $e_J = \sum_K L_K W_{KJ}$. Then Eq. (9.82) is put in the form

$$\mathcal{L}\rho_S = \frac{1}{i}[\tilde{H}, \rho_S] + \sum_{K=1}^{N} \gamma_K \left(L_K \rho_S L_K^\dagger - \frac{1}{2} L_K^\dagger L_K \rho_S - \frac{1}{2} \rho_S L_K^\dagger L_K \right), \quad (9.84)$$

where $N \leq d^2 - 1$. Note that if some elements γ_K vanish, then the number of L_K involved reduces from $d^2 - 1$ by this amount. The operators L_K are called the **Lindblad operators**, while Eq. (9.84) is called the **Lindblad equation** [5, 6]. Let us compare the term $\gamma_K L_K \rho_S L_K^\dagger$ in Eq. (9.84) with Eq. (9.14). The operator L_K may be compared with U_a and γ_K with p_a.

Note that the Lindblad equation is invariant under the transformation

$$L_K \to L_K + c_K, \ \tilde{H} \to \tilde{H} + \frac{1}{2i} \sum_K \left(c_K^* L_K - c_K L_K^\dagger \right) + c_0. \quad (9.85)$$

By making use of this freedom, it is always possible to make L_K traceless. Moreover, if we absorb γ_K in the definition of L_K as $\sqrt{\gamma_K} L_K$, the diagonal matrix $W \alpha W^\dagger$ reduces to the unit matrix, and the remaining freedom

$$\sqrt{\gamma_K} L_K \to \sum_J U_{KJ} \sqrt{\gamma_J} L_J \quad (9.86)$$

becomes manifest.

9.4.3 Examples

We consider two examples taken from [7].

The first example is **spontaneous emission**. Let us consider a two-level atom whose Hamiltonian is

$$H = -\frac{\omega_0}{2} \sigma_z, \quad (9.87)$$

where ω_0 is the energy difference between the ground state $|0\rangle = (1, 0)^t$ and the excited state $|1\rangle = (0, 1)^t$. Suppose there is a single Lindblad operator

$$L = \sqrt{\Gamma} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad (9.88)$$

which corresponds to relaxation process $|1\rangle \to |0\rangle$. It is reasonable to assume $\tilde{H}$ in Eq. (9.84) reduces to H when the system-environment coupling is small, for which case the Lindblad equation should reduce to the Liouville-von Neumann equation. The Lindblad equation is then

$$\begin{aligned} \frac{\partial}{\partial t} \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} &= -i[H, \rho] + L \rho L^\dagger - \frac{1}{2} \left(L^\dagger L \rho + \rho L^\dagger L \right) \\ &= i\omega_0 \begin{pmatrix} 0 & \rho_{01} \\ -\rho_{10} & 0 \end{pmatrix} + \Gamma \begin{pmatrix} \rho_{11} & -\frac{1}{2}\rho_{01} \\ -\frac{1}{2}\rho_{10} & -\rho_{11} \end{pmatrix}, \end{aligned} \quad (9.89)$$

where we have dropped the subscript S to simplify the notation. The above equation is solved with the initial condition $\rho(0)$ as

$$\begin{gathered}\rho_{00}(t) = \rho_{00}(0) + \rho_{11}(0)(1 - e^{-\Gamma t}),\ \rho_{01}(t) = \rho_{01}(0)e^{(i\omega_0 - \Gamma/2)t}, \\ \rho_{10}(t) = \rho_{10}(0)e^{(-i\omega_0 - \Gamma/2)t},\ \rho_{11}(t) = \rho_{11}(0)e^{-\Gamma t}.\end{gathered} \tag{9.90}$$

The population ρ_{11} in the excited state $|1\rangle$ decays with the characteristic time $T_1 = 1/\Gamma$, while the off-diagonal components (coherence) ρ_{01} and ρ_{10} decay with the time constant $T_2 = 2/\Gamma$. This shows that the coherence decay process is slower than the amplitude damping process.

The second example is the **Bloch equation** in NMR. Let us take the Hamiltonian (9.87) again. We introduce three Lindblad operators

$$L_+ = \sqrt{\Gamma_+}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad L_- = \sqrt{\Gamma_-}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad L_z = \sqrt{\Gamma_z}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{9.91}$$

where L_+ describes relaxation process $|1\rangle \to |0\rangle$, L_- to excitation process $|0\rangle \to |1\rangle$ and L_z to dephasing process without energy transfer between the spin and the environment.

The Lindblad equation is

$$\begin{aligned}\frac{\partial}{\partial t}\begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} &= i\omega_0\begin{pmatrix} 0 & \rho_{01} \\ -\rho_{10} & 0 \end{pmatrix} + \Gamma_+\begin{pmatrix} \rho_{11} & -\rho_{01}/2 \\ -\rho_{10}/2 & -\rho_{11} \end{pmatrix} \\ &+\Gamma_-\begin{pmatrix} -\rho_{00} & -\rho_{01}/2 \\ -\rho_{10}/2 & \rho_{00} \end{pmatrix} + \Gamma_z\begin{pmatrix} 0 & -2\rho_{01} \\ -2\rho_{10} & 0 \end{pmatrix}.\end{aligned} \tag{9.92}$$

Let us find the equilibrium values ρ^{eq} of $\rho(t)$. It should be clear that the off-diagonal components (the coherence) disappear as $t \to \infty$. As for the diagonal components, the condition $\partial\rho(t)/\partial t = 0$ leads to $\Gamma_+\rho_{11}^{\text{eq}} - \Gamma_-\rho_{00}^{\text{eq}} = 0$, namely,

$$\frac{\rho_{11}^{\text{eq}}}{\rho_{00}^{\text{eq}}} = \frac{\Gamma_-}{\Gamma_+} = e^{-\omega_0/k_B T}, \tag{9.93}$$

where we noted that the diagonal components of $\rho(t)$ represent the state populations, and the equilibrium population ratio follows the Boltzmann distribution. Straightforward but tedious calculations yield the solution

$$\begin{gathered}\rho_{00}(t) = \rho_{00}^{\text{eq}} + (\rho_{00} - \rho_{00}^{\text{eq}})\, e^{-t/T_1}, \\ \rho_{01}(t) = \rho_{01}e^{(i\omega_0 - 1/T_2)t}, \quad \rho_{10}(t) = \rho_{10}e^{(-i\omega_0 - 1/T_2)t}, \\ \rho_{11}(t) = \rho_{11}^{\text{eq}} + (\rho_{11} - \rho_{11}^{\text{eq}})\, e^{-t/T_1},\end{gathered} \tag{9.94}$$

where

$$T_1^{-1} = \Gamma_+ + \Gamma_-, \quad T_2^{-1} = \frac{\Gamma_+ + \Gamma_-}{2} + 2\Gamma_z. \tag{9.95}$$

It is exepcted that the L_+ process involves the emission of rf photons (see Chapter 12) and the L_- process the absorption of photons. Therefore we assume

$$\Gamma_+ = \gamma[1 + n(\omega_0)], \quad \Gamma_- = \gamma n(\omega_0). \tag{9.96}$$

The function $n(\omega)$ is found by making use of Eq. (9.93) as

$$n(\omega_0) = \frac{1}{1 - e^{-\omega_0/k_B T}}. \tag{9.97}$$

Then T_1 and T_2 are given as

$$T_1^{-1} = \gamma e^{\omega_0/k_B T} \coth\left(\frac{\omega_0}{2k_B T}\right), \quad T_2^{-1} = \frac{T_1^{-1}}{2} + 2\Gamma_z. \tag{9.98}$$

EXERCISE 9.3 Let $\rho(t) = (I + \sum_{k=x,y,z} c_k(t)\sigma_k)/2$ be the density matrix of a qubit, where $\{c_k(t)\}$ are the components of the Bloch vector.

Write down the Lindblad equation for $\{c_k(t)\}$ with the Hamiltonian (9.87) and the Lindblad operators (9.91). Find the solution $\{c_k(t)\}$ of the Lindblad equation.

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10

Quantum Error Correcting Codes

10.1 Introduction

It has been shown in the previous chapter that interactions between a quantum system with environment cause undesirable changes in the state of the quantum system. In the case of qubits, they appear as bit-flip and phase-flip errors, for example. To reduce such errors, we must build in some sort of error correcting mechanism in the algorithm.

Before we introduce quantum error correcting codes, we have a brief look at the simplest version of error correcting code in classical bits. Suppose we transmit a series of 0's and 1's through a noisy classical channel. Each bit is assumed to flip independently with a probability p. Thus a bit 0 sent through the channel will be received as 0 with probability $1 - p$ and as 1 with probability p. To reduce channel errors, we may invoke the majority vote. Namely, we encode logical 0 by 000 and 1 by 111, for example. When 000 is sent through this channel, it will be received as 000 with probability $(1-p)^3$, as 100, 010 or 001 with probability $3p(1-p)^2$, as 011, 101 or 110 with probability $3p^2(1-p)$ and finally as 111 with probability p^3. Note that the summation of all the probabilities is 1 as it should be. By taking the majority vote, we correctly reproduce the desired result 0 with probability $p_0 = (1-p)^3 + 3p(1-p)^2 = (1-p)^2(1+2p)$ and fail with probability $p_1 = 3p^2(1-p) + p^3 = (3-2p)p^2$. We obtain $p_0 \gg p_1$ for sufficiently small $p \geq 0$. In fact, we find $p_0 = 0.972$ and $p_1 = 0.028$ for $p = 0.1$. The success probability p_0 increases as p approaches 0, or alternatively, if we use more bits to encode 0 or 1.

EXERCISE 10.1 Suppose five bits are employed to encode a bit. Find the success probability when the bit is sent through a noisy channel, one by one, in which a bit is flipped with probability p. Assume the received bits are decoded according to majority vote.

This method cannot be applicable to qubits, however, due to the no-cloning theorem. We have to somehow think out the way to overcome this theorem.

General references for this chapter are [1, 2, 3] and [4].

10.2 Three-Qubit Bit-Flip Code and Phase-Flip Code

It is instructive to introduce a simple example of **quantum error correcting codes** (**QECC**) before we consider more general theory. We closely follow Steane [2] here.

10.2.1 Bit-Flip QECC

Suppose Alice wants to send a qubit or a serise of qubits to Bob through a noisy quantum channel. Let $|\psi\rangle = a|0\rangle + b|1\rangle$ be the state she wants to send. If she is to transmit a serise of qubits, she sends them one by one and the following argument applies to each of the qubits. Let p be the probability with which a qubit is flipped ($|0\rangle \leftrightarrow |1\rangle$), and we assume there are no other types of errors in the channel. In other words, the operator X is applied to the qubit with probability p, and consequently the state is mapped to

$$|\psi\rangle \to |\psi'\rangle = X|\psi\rangle = a|1\rangle + b|0\rangle. \tag{10.1}$$

We have already seen in the previous section that this channel is described by a quantum operation (9.31).

10.2.1.1 Encoding

To reduce the error probability, we want to mimic somehow the classical counterpart without using a clone machine. Let us recall that the action of a CNOT gate is

$$\text{CNOT} : |j0\rangle \to |jj\rangle, \quad j \in \{0, 1\} \tag{10.2}$$

and therefore it duplicates the control bit $j \in \{0, 1\}$ when the initial target bit is set to $|0\rangle$. We use this fact to *triplicate* the basis vectors as

$$|\psi\rangle|00\rangle = (a|0\rangle + b|1\rangle)|00\rangle \to |\psi\rangle_L = a|000\rangle + b|111\rangle, \tag{10.3}$$

where $|\psi\rangle_L$ denotes the encoded state. The state $|\psi\rangle_L$ is called the **logical qubit**, while each constituent qubit is called the **physical qubit**. We borrow terminologies from classical error correcting code (ECC) and call the set

$$C = \{a|000\rangle + b|111\rangle | a, b \in \mathbb{C}, |a|^2 + |b|^2 = 1\} \tag{10.4}$$

the **code** and call each member of C a **codeword**. It is important to note that the state $|\psi\rangle$ is not triplicated but only the basis vectors are triplicated. This redundancy makes it possible to detect errors in $|\psi\rangle_L$ and correct them as we see below.

A quantum circuit which implements the encoding (10.3) is easily found from our experience in the CNOT gate. Let us consider the circuit shown in

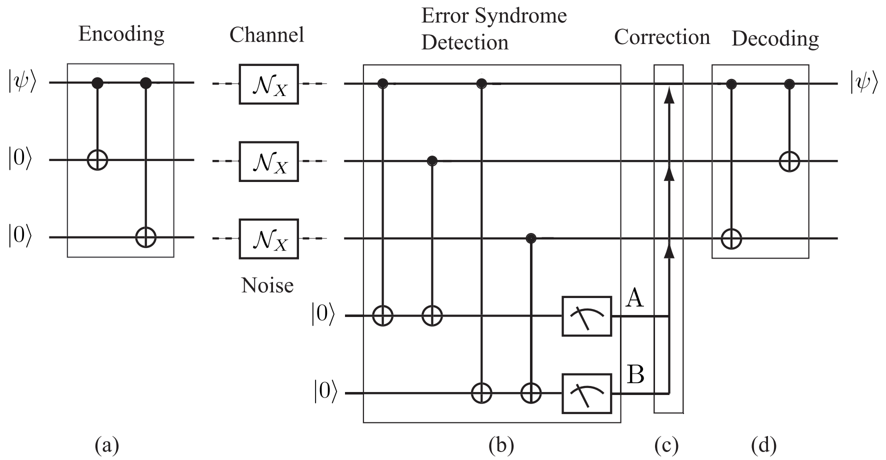

FIGURE 10.1
Quantum circuits to (a) encode, (b) detect bit-flip error syndrome, (c) make correction to a relevant qubit and (d) decode. The gate $\mathcal{N}_X$ stands for the bit-flip noise. The circuit (a) belongs to Alice, while the circuits (b), (c) and (d) belong to Bob.

Fig. 10.1 (a) whose inputs are the state $|\psi\rangle$ and two ancillary qubits in the state $|00\rangle$. When the target bit is $|\psi\rangle = |0\rangle$, the ancillary qubits are left unchanged and the output is $|000\rangle$, while when the input is $|\psi\rangle = |1\rangle$, the output is $|111\rangle$. By superposition principle, the input $|\psi\rangle|00\rangle = (a|0\rangle + b|1\rangle)|00\rangle$ results in the output $|\psi\rangle_L = a|000\rangle + b|111\rangle$ as promised.

10.2.1.2 Transmission

Now the state $|\psi\rangle_L$ is sent through a quantum channel which introduces bit-flip error with a rate p for each qubit independently. We assume p is sufficiently small so that not many errors occur during the qubit transmission. The received state depends on in which physical qubit(s) the bit-flip occurred. Table 10.1 lists possible received states and the probabilities with which these states are received.

10.2.1.3 Error Syndrome Detection and Correction

Now Bob has to extract from the received state which error occurred during the qubit transmission. For this purpose, Bob prepares two ancillary qubits in the state $|00\rangle$ as depicted in Fig. 10.1 (b) and applies four CNOT operations whose control bits are the encoded qubits while the target qubits are Bob's two ancillary qubits. Let $|x_1x_2x_3\rangle$ be a basis vector Bob has received and let A (B) be the output state of the first (second) ancilla qubit. It is seen from

TABLE 10.1
Bob receives the following states with given probabilities.

Received state	Probability
$a\|000\rangle + b\|111\rangle$	$(1-p)^3$
$a\|100\rangle + b\|011\rangle$	$p(1-p)^2$
$a\|010\rangle + b\|101\rangle$	$p(1-p)^2$
$a\|001\rangle + b\|110\rangle$	$p(1-p)^2$
$a\|110\rangle + b\|001\rangle$	$p^2(1-p)$
$a\|101\rangle + b\|010\rangle$	$p^2(1-p)$
$a\|011\rangle + b\|100\rangle$	$p^2(1-p)$
$a\|111\rangle + b\|000\rangle$	p^3

TABLE 10.2
States after error extraction is made and the probabilities with which these states are produced.

State after error syndrome extraction	Probability
$(a\|000\rangle + b\|111\rangle)\|00\rangle$	$(1-p)^3$
$(a\|100\rangle + b\|011\rangle)\|11\rangle$	$p(1-p)^2$
$(a\|010\rangle + b\|101\rangle)\|10\rangle$	$p(1-p)^2$
$(a\|001\rangle + b\|110\rangle)\|01\rangle$	$p(1-p)^2$
$(a\|110\rangle + b\|001\rangle)\|01\rangle$	$p^2(1-p)$
$(a\|101\rangle + b\|010\rangle)\|10\rangle$	$p^2(1-p)$
$(a\|011\rangle + b\|100\rangle)\|11\rangle$	$p^2(1-p)$
$(a\|111\rangle + b\|000\rangle)\|00\rangle$	p^3

Fig. 10.1 (b) that $A = x_1 \oplus x_2$ and $B = x_1 \oplus x_3$. Let $a|100\rangle + b|011\rangle$ be the received logical qubit, for example. Note that the first qubit state in each of the basis vectors is different from the second and the third qubit states. These differences are detected by the pairs of CNOT gates in Fig. 10.1 (b). The error extracting sequence transforms the ancillary qubits as

$$(a|100\rangle + b|011\rangle)|00\rangle \to a|10011\rangle + b|01111\rangle = (a|100\rangle + b|011\rangle)|11\rangle.$$

Both of the ancillary qubits are flipped since $x_1 \oplus x_2 = x_1 \oplus x_3 = 1$ for both $|100\rangle$ and $|011\rangle$. The set of two bits is called the **(error) syndrome**, and it tells Bob in which physical qubit the error occurred during transmission. It is important to note that (i) the syndrome is independent of a and b and (ii) the received state $a|100\rangle + b|011\rangle$ is left unchanged; we have detected an error without measuring the received state! These features are common to all QECC.

We list the results of other cases in Table 10.2. Note that among eight possible states, there are exactly two states with the same ancilla state. Does it mean this error extraction scheme does not work? Now let us compare the probabilities associated with the same ancillary state. When the ancillary

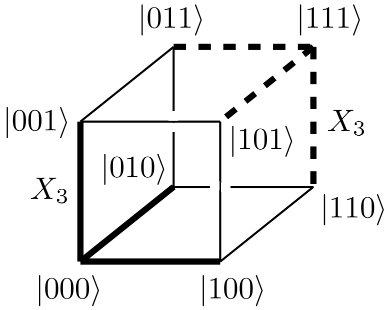

FIGURE 10.2
Encoded basis vectors $|000\rangle$ and $|111\rangle$. The thick solid line shows the action of $X = \sigma_x$ on the vector $|000\rangle$, while the thick broken line shows the action of X on $|111\rangle$. The action of $X_3 = I \otimes I \otimes X$ takes $|000\rangle$ to $|001\rangle$ and $|111\rangle$ to $|110\rangle$ as depicted. Observe that the intersection between thick solid lines and thick broken lines is an empty set. The Hamming distance, to be defined in §10.4.1, between 000 and 111 is 3, which guarantees any single bit-flip error may be corrected.

state is $|10\rangle$, for example, there are two possible received states $a|010\rangle + b|101\rangle$ and $a|101\rangle + b|010\rangle$. The former is received with probability $p(1-p)^2$ and the latter with $p^2(1-p)$. Therefore the latter probability is negligible compared to the former for sufficiently small p.

It is instructive to visualize what errors do to the encoded basis vectors as depicted in Fig. 10.2. The encoded basis vectors $|000\rangle$ and $|111\rangle$ are assigned to the opposite vertices. An action of X_i, the operator $X = \sigma_x$ acting on the ith qubit, takes these basis vectors to the nearest neighbor vetices, which differ from the correct basis vectors in the ith position. Note that the intersection of the sets of vectors obtained by a single action of X_i on $|000\rangle$ and $|111\rangle$ is an empty set. Therefore an action of a single error operator X can be corrected with no ambiguity. It should be clear from this observation that two-qubit encoding cannot correct single qubit error.

Now Bob measures his ancillary qubits and obtains two bits of classical information (syndrome). Bob applies correcting procedure to the received state according to the error syndrome he has obtained. Ignoring multiple error states with small probabilities, we immediately find that the following action must be taken:

Error syndrome	Correction to be made	
(00)	identity operation (nothing is required)	
(01)	apply X_3	(10.5)
(10)	apply X_2	
(11)	apply X_1	

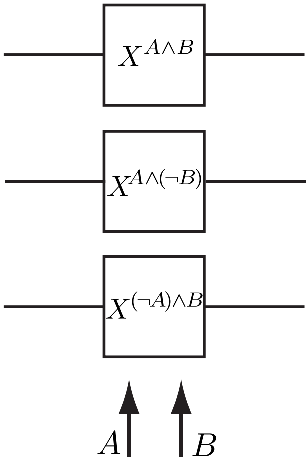

FIGURE 10.3
Explicit form of the bit-flip error correction circuit. It is understood that $X^0 = I$ and $X^1 = \sigma_x$. $\neg A$ stands for the negation of the bit A.

Suppose the syndrome is 01, for example. The state Bob received is likely to be $a|001\rangle + b|110\rangle$. Bob recovers the initial state Alice has sent by applying $X_3 = I \otimes I \otimes X$ on the received state:

$$X_3(a|001\rangle + b|110\rangle) = a|000\rangle + b|111\rangle.$$

Figure 10.3 depicts the explicit form of the error correction circuit.

If Bob receives the state $a|110\rangle + b|001\rangle$, unfortunately, he will obtain

$$X_3(a|110\rangle + b|001\rangle) = a|111\rangle + b|000\rangle.$$

In fact, for any error syndrome, Bob obtains either $a|000\rangle + b|111\rangle$ or $a|111\rangle + b|000\rangle$. The latter case occurs when and only when more than one qubit is flipped, and hence it is less likely to happen for small error rate p.

10.2.1.4 Decoding

Now that Bob has corrected an error, what is left for him is to decode the encoded state. This is nothing but the inverse transformation of the encoding (10.3). It can be seen from Fig. 10.1 (d) that

$$\mathrm{CNOT}_{12}\mathrm{CNOT}_{13}(a|000\rangle+b|111\rangle) = a|000\rangle+b|100\rangle = (a|0\rangle+b|1\rangle)|00\rangle. \quad (10.6)$$

Suppose Bob has received a state with more than one qubit flipped; we will obtain

$$\mathrm{CNOT}_{12}\mathrm{CNOT}_{13}(a|111\rangle+b|000\rangle) = a|100\rangle+b|000\rangle = (a|1\rangle+b|0\rangle)|00\rangle. \quad (10.7)$$

The probability with which this error happens is found from Table 10.1 as

$$P(\text{error}) = 3p^2(1-p) + p^3 = 3p^2 - 2p^3. \tag{10.8}$$

This error rate is less than p, the error probability of a single channel, provided that $p < 1/2$. In contrast, success probability has been enhanced from $1-p$ to $1 - P(\text{error}) = 1 - 3p^2 + 2p^3$. Let $p = 0.1$, for example. Then the error rate is lowered to $P(\text{error}) = 0.028$, while the success probability is enhanced from 0.9 to 0.972.

10.2.1.5 Miracle of Entanglement

This example, albeit simple, contains almost all fundamental ingredients of QECC. We prepare some redundant qubits which somehow "triplicate" the original qubit state to be sent without violating the no-cloning theorem. Then the encoded qubits are sent through a noisy channel, which causes a bit-flip in at most one of the qubits. The received state, which may be subject to an error, is then entangled with ancillary qubits which detect what kind of an error occurred during the state transmission. This results in an entangled state

$$\sum_k |\text{A bit-flip error in the } k\text{th qubit}\rangle \otimes |\text{corresponding error syndrome}\rangle, \tag{10.9}$$

set aside the state without errors. The wave function, upon the measurement of the ancillary qubits, collapses to a state with a bit-flip error corresponding to the observed error syndrome. In a sense, syndrome measurement singles out a particular error state which produces the observed syndrome. This principle "measure the syndrome and single out an error" will be used repeatedly in the rest of this chapter.

Once the syndrome is found, it is an easy task to transform the received state back to the original state. Note that everything is done without knowing what the origial state is.

10.2.1.6 Continuous Rotations

We have considered noise X so far. Suppose noise in the channel is characterized by a continuous paramter α as

$$U_\alpha = e^{i\alpha X} = \cos\alpha I + i \sin\alpha X, \tag{10.10}$$

which maps a state $|\psi\rangle$ to

$$U_\alpha|\psi\rangle = \cos\alpha|\psi\rangle + i\sin\alpha X|\psi\rangle. \tag{10.11}$$

It is assumed that U_α acts on each qubit of the logical qubit independently with probability p. The probability with which the logical qubit is received without affected by U_α is $(1-p)^3$ as before.

Suppose U_α acts on the first qubit, for example. Bob then receives

$$\begin{aligned}&(U_\alpha \otimes I \otimes I)(a|000\rangle + b|111\rangle)\\&= \cos\alpha(a|000\rangle + b|111\rangle) + i\sin\alpha(a|100\rangle + b|011\rangle).\end{aligned}$$

The output of the error syndrome detection circuit, before the syndrome measurement is made, is an entangled state

$$\cos\alpha(a|000\rangle + b|111\rangle)|00\rangle + i\sin\alpha(a|100\rangle + b|111\rangle)|11\rangle \tag{10.12}$$

(see Table 10.2). Measurement of the error syndrome yields either (00) or (11). In the former case the state collapses to $|\psi\rangle = a|000\rangle + b|111\rangle$, and this happens with a probability $p\cos^2\alpha$. In the latter case, on the other hand, the received state collapses to $X|\psi\rangle = a|100\rangle + b|011\rangle$, and this happens with a probability $p\sin^2\alpha$. Bob applies I (X) to the first qubit to correct the error when the syndrome readout is 00 (11). Now the error probability is given by $P(\text{error}) = p\sin^2\alpha$.

It is clear that error U_α may act on the second or the third qubit. In this way, continuous rotation U_α for any α may be corrected. In general, linearity of a quantum circuit guarantees that any QECC, which corrects the bit-flip error X, corrects continuous error U_α too.

10.2.2 Phase-Flip QECC

Let us consider a phase-flip channel. Phase flip $Z : |x\rangle \mapsto (-1)^x|x\rangle$, ($x \in \{0,1\}$) occurs with probability p for each qubit independently when it is sent through a channel. We consider how to correct phase-flip error mimicking the prescription given for a bit-flip error.

EXERCISE 10.2 (1) Show that $U_{\rm H} Z U_{\rm H} = X$, where $U_{\rm H}$ is the matrix representation of the Hadamard gate.
(2) Show that $Z = \sigma_z$ maps $|+\rangle$ to $|-\rangle$ and $|-\rangle$ to $|+\rangle$, where

$$|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle).$$

(3) Show that $X|+\rangle = |+\rangle$ and $X|-\rangle = -|-\rangle$.

What we have learned from the above exercise is that the phase-flip error in the basis $\{|0\rangle, |1\rangle\}$ is nothing but the bit-flip error in the basis $\{|+\rangle, |-\rangle\}$. This observation suggests that we encode $|\psi\rangle = a|0\rangle + b|1\rangle$ into $|\Psi_1\rangle = a|+++\rangle + b|---\rangle$ by applying $U_{\rm H}^{\otimes 3}$ on the state $|\Psi_0\rangle = a|000\rangle + b|111\rangle$. Now $|\Psi_1\rangle$ is sent through a noisy channel with possible phase-flip errors. We assume at most a single qubit is subject to the error.

Suppose the phase-flip error acts on the first physical qubit, for example, when $|\Psi_1\rangle$ is sent through the channel. Bob then receives a state $|\Psi_1'\rangle =$

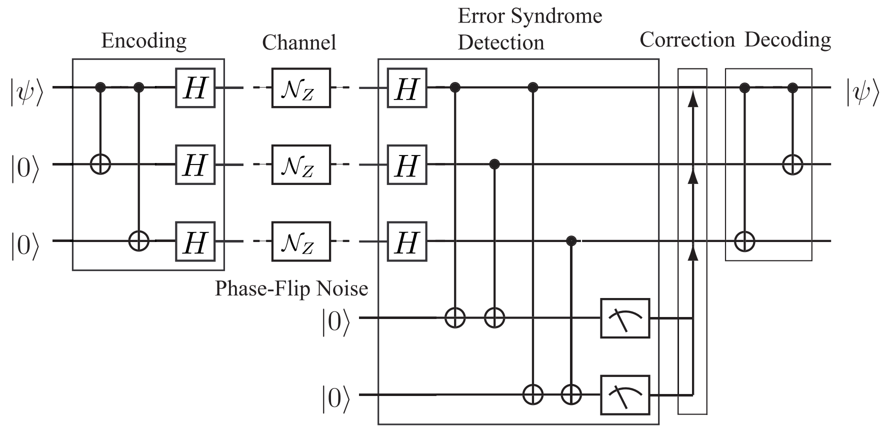

FIGURE 10.4
Three-qubit phase-flip error QECC.

$a|-++\rangle + b|+--\rangle$. We found in the above exercise that a phase-flip error acts as a bit-flip error in the basis $|\pm\rangle$; $U_{\rm H} Z U_{\rm H} = X$. Therefore, we need to put the basis back from $|\pm\rangle$ to $\{|0\rangle, |1\rangle\}$ by applying the second Walsh-Hadamard transform $U_{\rm H}^{\otimes 3}$ so that a phase-flip error is recognized as a bit-flip error and we can employ the same error syndrome detection circuit as well as the error correction circuit as those used for the bit-flip error QECC.

Collecting these results, the quantum circuit for the phase-flip QECC is constructed as shown in Fig. 10.4.

EXERCISE 10.3 Show that the circuit depicted in Fig. 10.4 is able to correct also a continuous error $U_\beta = e^{i\beta Z}$ acting on one of the qubits.

10.3 Shor's Nine-Qubit Code

Let us consider a more general noisy channel in which all possible single-qubit errors occur. Namely, the following errors are now active in the channel;

$$\begin{array}{rl} \text{Bit-Flip Error} & X : \begin{pmatrix} a \\ b \end{pmatrix} \mapsto \begin{pmatrix} b \\ a \end{pmatrix} \\ \text{Phase-Flip Error} & Z : \begin{pmatrix} a \\ b \end{pmatrix} \mapsto \begin{pmatrix} a \\ -b \end{pmatrix} \\ \text{Phase- and Bit-Flip Error} & Y : \begin{pmatrix} a \\ b \end{pmatrix} \mapsto \begin{pmatrix} -b \\ a \end{pmatrix}. \end{array} \tag{10.13}$$

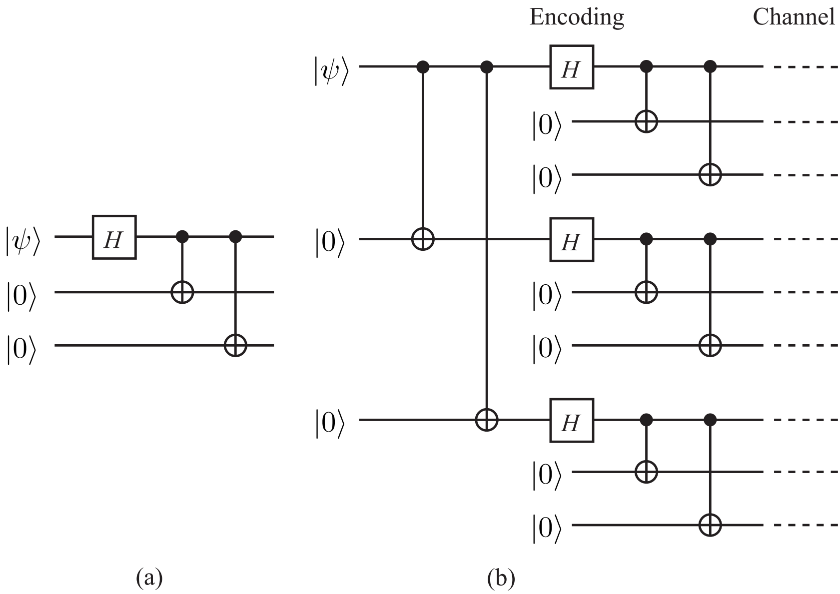

FIGURE 10.5
(a) Quantum circuit to transform $|0\rangle \to |+\rangle$ and $|1\rangle \to |-\rangle$. (b) Encoding circuit for Shor's nine-qubit QECC, which maps $|\psi\rangle = a|0\rangle + b|1\rangle$ to $a|0\rangle_L + b|1\rangle_L$.

Shor's nine-qubit QECC which corrects the above three types of noise is constructed by concatenating the phase-flip QECC into the bit-flip QECC, which were introduced in the previous section [5].

10.3.1 Encoding

We encode $|0\rangle$ and $|1\rangle$ as

$$|0\rangle \to |0\rangle_L \equiv |+++\rangle,\ |1\rangle \to |1\rangle_L \equiv |---\rangle, \tag{10.14}$$

where

$$|+\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle),\ |-\rangle = \frac{1}{\sqrt{2}}(|000\rangle - |111\rangle). \tag{10.15}$$

The encoding circuit for a codeword $a|0\rangle_L + b|1\rangle_L$ is shown in Fig. 10.5.

EXERCISE 10.4 Show that the circuit in Fig. 10.5 (a) maps

$$|0\rangle|00\rangle \mapsto |+\rangle, \quad |1\rangle|00\rangle \mapsto |-\rangle. \tag{10.16}$$

It follows from Exercise 10.4 that the quantum circuit given in Fig. 10.5 (b) maps $(a|0\rangle + b|1\rangle)|00000000\rangle$ to $a|+++\rangle + b|---\rangle$. In fact

$$\begin{aligned}(a|0\rangle + b|1\rangle)|00\rangle &\to a|000\rangle + b|111\rangle \\ &\to \frac{a}{\sqrt{8}}[(|0\rangle + |1\rangle)|00\rangle]^{\otimes 3} + \frac{b}{\sqrt{8}}[(|0\rangle - |1\rangle)|00\rangle]^{\otimes 3} \\ &\to \frac{a}{\sqrt{8}}(|000\rangle + |111\rangle)^{\otimes 3} + \frac{b}{\sqrt{8}}(|000\rangle - |111\rangle)^{\otimes 3} \\ &= a|+++\rangle + b|---\rangle. \end{aligned} \tag{10.17}$$

Now encoded logical qubits are sent through the noisy channel which causes errors (10.13).

10.3.2 Transmission

Noise depicted by Eq. (10.13) is in action while the encoded qubit is sent through the quantum channel. Let p be the probability with which each physical qubit is flipped by one of the error operators in (10.13).

The logical qubit is sent intact with probability $(1-p)^9$. The probability with which one of the physical qubits is affected is $9p(1-p)^8$, which may be corrected by the following process. Then the probability that more than one qubit are flipped is

$$1 - (1-p)^9 - 9p(1-p)^8 = 1 - (1+8p)(1-p)^8 \simeq 36p^2, \tag{10.18}$$

where $p \ll 1$ is assumed.

10.3.3 Error Syndrome Detection and Correction

Thanks to the superposition principle, we may consider errors in the basis vectors $|0\rangle_L = |+++\rangle$ and $|1\rangle_L = |---\rangle$ separately. We concentrate on $|0\rangle_L$ first and consider the following example. Suppose a bit-flip error occurs in the fifth qubit during transmission. Then the received state will be

$$|+\rangle\frac{1}{\sqrt{2}}(|010\rangle + |101\rangle)|+\rangle$$

instead of the expected $|+++\rangle$. To correct this error we have to identify at which qubit the error occurred during transmission. To this end, we introduce two ancilla qubits to each group of three qubits as shown in Fig. 10.6. These ancillary qubits are all set to the initial state $|0\rangle$ in the beginning. Suppose the input state to the ith group of three qubits is $|x_1x_2x_3\rangle$. Then the ancilla output states $|A_i\rangle$ and $|B_i\rangle$ are given as $A_i = x_1 \oplus x_2$ and $B_i = x_1 \oplus x_3$. Let's come back to our example. Error did not occur in the first group of three qubits, and hence we measure $A_1 = B_1 = 0$ for both input states $|000\rangle$ and $|111\rangle$. Therefore we obtain $A_1 = B_1 = 0$. Similarly we find $A_3 = B_3 = 0$ for the third group. For the second group of three qubits, on the other hand, we find $A_2 = 1$ and $B_2 = 0$, from which we find a bit-flip error occurred in the fifth qubit (the second qubit in the second group of three qubits). We simply need to apply the X gate to the fifth qubit to correct this error.

EXERCISE 10.5 Suppose bit-flip error occurred in (1) the first, (2) the second or (3) the third qubit of the first group. What is the syndrome (A_1, B_1) in each case?

Next, let us look at the phase-flip errors. The following exercise suggests that the syndrome $(A_i, B_i)_{1\leq i\leq 3}$ is not able to detect phase-flip errors.

EXERCISE 10.6 Suppose phase-flip error occurred in the second qubit:

$$|0\rangle_L \to \frac{1}{\sqrt{2}}(|000\rangle - |111\rangle)|+\rangle|+\rangle,$$

$$|1\rangle_L \to \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)|-\rangle|-\rangle.$$

Show that the output bits A_i and B_i are both 0 for all $1 \leq i \leq 3$.

Let us recall the phase-flip error correcting code introduced in the previous section, in which Z acts as a bit-flip operator in the basis $\{|\pm\rangle\}$. Let us check the syndrome (A_4, B_4). Suppose phase-flip error acted on one of the qubits in the second group of three qubits. Now $|0\rangle_L$ is received as $(|+\rangle(|000\rangle - |111\rangle)|+\rangle)/\sqrt{2} = |+-+\rangle$ while $|1\rangle_L$ as $(|+\rangle(|000\rangle + |111\rangle)|+\rangle)/\sqrt{2} = |-+-\rangle$. Each group of three qubits, after it goes through the Hadamard gates in the middle layer of Fig. 10.6, transforms as

$$|+\rangle \mapsto \frac{1}{2}\left(|000\rangle + |011\rangle + |101\rangle + |110\rangle\right)$$

$$|-\rangle \mapsto \frac{1}{2}\left(|001\rangle + |010\rangle + |100\rangle + |111\rangle\right).$$

It is important to note that $|+\rangle$ is mapped to a superposition of states with even number of 1 and $|-\rangle$ to one with odd number of 1. Let $|x_1x_2x_3\rangle$ be a particular component of a state after the first Hadamard gates operate on the input state. Then $x_1 \oplus x_2 \oplus x_3 = 0$ when the input state is $|+\rangle$, while $x_1 \oplus x_2 \oplus x_3 = 1$ when the input state is $|-\rangle$. Therefore the syndrome (A_4, B_4) for the example in our hands take values $A_4 = 1$ and $B_4 = 0$, from which we find a phase-flip occurred in the second group of three qubits.

Readers should verify the phase-flip error syndrome is summarized as shown in Table 10.3. By measuring the syndrome (A_4, B_4), Bob knows for sure on which group of qubits the phase-flip operator acted, assuming there exists only a single error.

Now Bob transforms the above state to $|\pm\rangle$ by applying the rightmost layer of the Hadamard gates in Fig. 10.6. Error syndrome has been extracted between two successive operations of the Hadamard gates. For our example, Bob has syndrome (10), from which he knows phase-flip error occurred in the second group. Now he operates Z on one of the three qubits in the group to recover the input state $a|+++\rangle + b|---\rangle$.

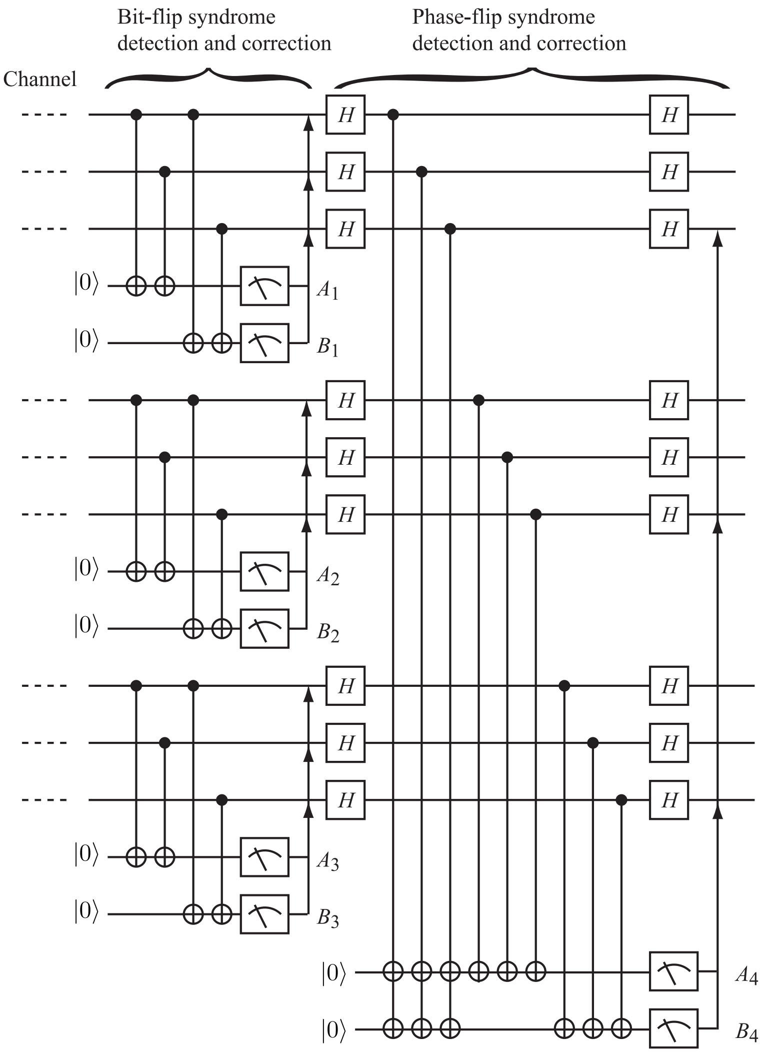

FIGURE 10.6
Quantum circuit to detect bit-flip and phase-flip error syndrome and correct the error.

TABLE 10.3
States after error extraction is made.

Input to the detecting circuit	Ancillas state $\lvert A_4, B_4\rangle$
$a\lvert +++\rangle + b\lvert ---\rangle$	$\lvert 00\rangle$
$a\lvert -++\rangle + b\lvert +--\rangle$	$\lvert 11\rangle$
$a\lvert +-+\rangle + b\lvert -+-\rangle$	$\lvert 10\rangle$
$a\lvert ++-\rangle + b\lvert --+\rangle$	$\lvert 01\rangle$
$a\lvert --+\rangle + b\lvert ++-\rangle$	$\lvert 01\rangle$
$a\lvert -+-\rangle + b\lvert +-+\rangle$	$\lvert 10\rangle$
$a\lvert +--\rangle + b\lvert -++\rangle$	$\lvert 11\rangle$
$a\lvert ---\rangle + b\lvert +++\rangle$	$\lvert 00\rangle$

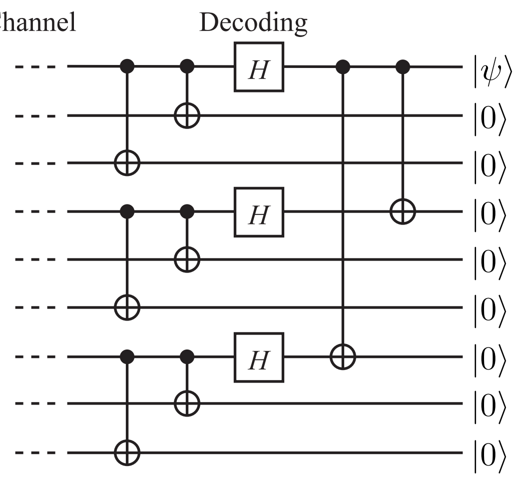

FIGURE 10.7
Decoding circuit for Shor's nine-qubit QECC, which maps $a\lvert +++\rangle + b\lvert ---\rangle$ to $(a\lvert 0\rangle + b\lvert 1\rangle)\lvert 00000000\rangle$.

10.3.4 Decoding

The decoding circuit given in Fig. 10.7 is the inverse of the encoding circuit in Fig. 10.5 (a). Readers should verify that it indeed maps $a\lvert +++\rangle + b\lvert ---\rangle$ to $(a\lvert 0\rangle + b\lvert 1\rangle)\lvert 00000000\rangle$.

Although we have not proved that Shor's nine-qubit QECC also corrects Y-errors, its validity should be clear from the identity $Y = XZ$ (see the following exercise).

EXERCISE 10.7 Suppose the Y-error occurs in the first qubit of the third group of three qubits. The state is fed into Bob's circuit in Fig. 10.6.
(1) What is the reading of the syndrome $(A_i, B_i)_{1\le i\le 3}$?
(2) Find the state after the first layer of the Hadamard gates is applied to the

received state. What is the syndrome (A_4, B_4)?
(3) Repeat (1) and (2) when the error is a continuous rotation $e^{-\gamma Y}$.

The results of this section show that Shor's nine-bit QECC corrects any one-qubit errors, so far as it acts on a single qubit only. In general a QECC which corrects I, X, Y and Z corrects every single-qubit error.

10.4 Seven-Qubit QECC

Shor's nine-qubit QECC is by no means the most efficient QECC. Logical qubits may be implemented with fewer physical qubits as we show in the following sections. We need to summarize classical error correcting codes before we introduce general theory of QECC. See [6] for more extensive accounts of the classical error correcting codes. The set of bit states has been denoted simply as $\{0, 1\}$ so far. Here we want to look at the same set with its field operations, multiplication and addition mod 2. For this reason, we will often use the symbol GF(2) to denote the same set equipped with the field operations.

Our presentation closely follows [3, 4, 1] in this and the next sections.

10.4.1 Classical Theory of Error Correcting Codes

Let us recall the simplest classical error correcting scheme, in which the bit 0 is encoded as 000 and 1 as 111. Three bits ($n = 3$) are used to send a single logical bit ($k = 1$), in which case we say the redundancy is $n - k = 2$ bits.

We assume the noisy transmission channel is symmetric, i.e., $i \in \mathrm{GF}(2)$ is received as i with probability $1-p$, while i is received as $1 \oplus i$ with probability p.

Supppose Alice encodes k bits of information into a code c which is made of n $(> k)$ bits and sends it to a receiver Bob through a noisy channel. Bob checks if there were errors in the transmission process by using a **parity check matrix** H, which is designed in such a way that

$$Hc^t = 0 \tag{10.19}$$

when there are no errors during transmission. In contrast, at least one of the bits is flipped during transmission if $Hc^t \neq 0$. The output Hc^t is called the **syndrome**. Bit-flip occurred if a syndrome does not vanish.

Let us consider sending $k = 4$ bits of information by using a code of the length $n = 7$. It is assumed that at most a single bit error occurs during transmission of a codeword. We need to identify which bit has erroneously transmitted, for which three bits are required. We assign 000 for error-free transmission, 001 for error in x_1, 010 for error in x_2 and so forth. Let us write the list of three bits, except for the error-free case 000, as

$$\begin{array}{c} 001 \\ 010 \\ \cdots \\ 111 \end{array}$$

and take the transpose of the resulting matix to obtain the parity check matrix

$$H = \begin{pmatrix} 0\,0\,0\,1\,1\,1\,1 \\ 0\,1\,1\,0\,0\,1\,1 \\ 1\,0\,1\,0\,1\,0\,1 \end{pmatrix}. \tag{10.20}$$

Let $c_1 = (0,1,1,0,0,1,1)$ be a code Bob receives. He applies H on c^t to obtain

$$Hc_1^t = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

If $c_2 = (0,1,1,1,0,1,1)$ is received, on the other hand, the output of the parity check matrix is

$$Hc_2^t = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

and Bob knows there is an error in the code he received. Moreover, he knows from the syndrome that the bit flip error acted on the fourth bit x_4, as we show now. Let $c = (x_1, x_2, \ldots, x_7)$ be the bit string Bob has received. Application of H produces the syndrome

$$Hc^t = \begin{pmatrix} x_4 \oplus x_5 \oplus x_6 \oplus x_7 \\ x_2 \oplus x_3 \oplus x_6 \oplus x_7 \\ x_1 \oplus x_3 \oplus x_5 \oplus x_7 \end{pmatrix}. \tag{10.21}$$

All the components in the RHS should be even numbers if all the bits are received without error. It is found from the syndrome Bob obtained for c_2 that x_1, x_2, x_3, x_5, x_6 and x_7 are received without error because at most a single error is assumed. Error must have occurred in a bit x_4, which is in the first component of the RHS of Eq. (10.21) but not in the second nor the third components. Bob applies a NOT gate on the fourth bit of the codeword he receives to recover the correct codeword.

EXERCISE 10.8 Suppose Bob receives a code $(0,0,0,1,1,1,0)$. What is the syndrome he will obtain after applying the parity check matrix H. Recover the correct code by making error correction. Repeat this for codes $(1,1,0,1,0,0,0)$ and $(1,1,0,0,1,1,1)$.

It should be noted that the syndrome depends only on which bit has suffered from error and not on the original code Alice has sent.

It should be clear that the kernel (mod 2) of H, denoted $C = \ker(H)$, is the set of correct codewords. The condition $Hc^t = 0$ introduces three relations among seven variables $\{x_i\}$, from which we obtain $\dim C = 7 - 3 = 4$, in agreement with the fact $k = 4$. The order of the set C is therefore $2^4 = 16$. The code space C is generated by the matrix

$$M = \begin{pmatrix} 0\,0\,0\,1\,1\,1\,1 \\ 0\,1\,1\,0\,0\,1\,1 \\ 1\,0\,1\,0\,1\,0\,1 \\ 1\,1\,1\,1\,1\,1\,1 \end{pmatrix}. \tag{10.22}$$

The matrix M is called the **generating matrix** of the error correcting code C. This particular code generated by the above M is called the **Hamming code**. Note that M is obtained from H by adding the fourth row, with all the components unity. Vectors in C are written as vM, where $v = (i_1, i_2, i_3, i_4),\ i_k \in \mathrm{GF}(2)$. The table for v and vM is

$$\begin{array}{llll} v & vM & v & vM \\ (0000) & (0000000) & (1000) & (0001111) \\ (0001) & (1111111) & (1001) & (1110000) \\ (0010) & (1010101) & (1010) & (1011010) \\ (0011) & (0101010) & (1011) & (0100101) \\ (0100) & (0110011) & (1100) & (0111100) \\ (0101) & (1001100) & (1101) & (1000011) \\ (0110) & (1100110) & (1110) & (1101001) \\ (0111) & (0011001) & (1111) & (0010110). \end{array} \tag{10.23}$$

Let us pick out an element of the code C, which contains even number of 1. They are

$$\begin{array}{l} \{(0000000),\ (1010101),\ (0110011),\ (1100110), \\ (0001111),\ (1011010),\ (0111100),\ (1101001)\}. \end{array} \tag{10.24}$$

Observe that they are (mod 2)-orthogonal to all the members of C. For example,

$$\begin{aligned} (1010101) \cdot (0111100) &= 0 \bmod 2, \\ (1010101) \cdot (0010110) &= 0 \bmod 2. \end{aligned}$$

The set of eight elements in (10.24) is denoted as $C^\perp$. Note that they are generated by v of the form $(i_1, i_2, i_3, 0)$ as $(i_1, i_2, i_3, 0)M$, which justifies $\dim C^\perp = 8$. Note also that $C^\perp \subset C$ in the present case.

Our choice of M as a generating matrix for the code is clear by now. Let us write H and M as

$$H = \begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix},\ M = \begin{pmatrix} h_1 \\ h_2 \\ h_3 \\ \mathbb{I} \end{pmatrix},$$

where h_k is a row vector with even number of 1 and $\mathbb{I}$ is a row vector whose all entries are 1. It follows from the orthogonality of $C^\perp$ with C that $HM^t = 0$ and hence $Hc^t = HM^tv^t = 0$.

The elements in $C - C^\perp$ contain odd number of 1 and are explicitly given as

$$\begin{array}{l}\{(1111111),\ (0101010),\ (1001100),\ (0011001),\\ (1110000),\ (0100101),\ (1000011),\ (0010110)\}.\end{array} \tag{10.25}$$

Suppose $c_1 = (x_1, x_2, \ldots, x_n)$ and $c_2 = (y_1, y_2, \ldots, y_n)$ are codes in C. The **Hamming distance** $d_H(c_1, c_2)$ between c_1 and c_2 is defined as the number of indices i such that $x_i \neq y_i$. In other words, the Hamming distance measures how many bits are different in c_1 and c_2. Let $c_1 = (0101010)$ and $c_2 = (0111100)$, for example. Their Hamming distance is $d_H(c_1, c_2) = 3$. It is easy to see that the Hamming distance satisfies the axioms of distance (see the next exercise).

EXERCISE 10.9 Let d_H be the Hamming distance for a k-bit code space C. Show that

(1) $d_H(c, c) \geq 0$ for any $c \in C$.
(2) $d_H(c_1, c_2) = d_H(c_2, c_1)$ for any $c_1, c_2 \in C$.
(3) $d_H(c_1, c_3) \leq d_H(c_1, c_2) + d_H(c_2, c_3)$ for any $c_1, c_2, c_3 \in C$.

The minimal distance $d_H(C)$ of the code C is defined as the minimum Hamming distance between different elements of C:

$$d_H(C) = \min_{c, c' \in C} d_H(c, c'). \tag{10.26}$$

EXERCISE 10.10 Show that the minimal distance of the code (10.23) is 3.

A code which encodes k bits into a bit string (codeword) with length n and having the minimal distance d is denoted as (n, k, d). The Hamming code is thus characterized as $(7, 4, 3)$.

A single bit flip error generates a bit string whose Hamming distance from the original codeword is 1. Therefore we can tell for sure which codeword is closest to the bit string containing an error (see Fig. 10.8). The original codeword is recovered by applying a NOT gate on an appropriate bit in the string.

It is clear that a code with the minimal distance $d_H(C)$ is able to correct $\lfloor (d_H(C) - 1)/2 \rfloor$ bits of errors, where $\lfloor x \rfloor$ denotes the floor function [6]. It is known that the minimal distance $d_H(C)$ satisfies an inequality (**the Singleton bound**)

$$d_H(C) \leq n - k + 1, \tag{10.27}$$

where n is the codeword length and k is the number of bits to be encoded. It is also known for an error correcting code based on GF(2) that the bound is

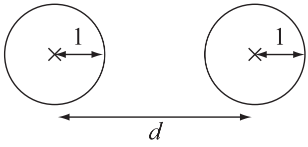

FIGURE 10.8
The crosses ($\times$) denote the closest codewords in a code C, and d is the minimal distance $d_H(C)$. A single bit error sends the codeword to a point on a circle with the radius 1, which does not belong to the code C. The minimal distance d must be at least 3 for single-error detection to make sense. Similarly, n-bit error correction requires the inequality $d-1 \geq 2n$. Compare this figure with Fig. 10.2.

never saturated and the above equality is further limited as

$$d_H(C) \leq n-k. \tag{10.28}$$

Exercise 10.10 shows that the Hamming code saturates the inequality (10.28). We need to introduce three redundancy bits to attain the minimal distance of 3 in the code space C.

10.4.2 Seven-Qubit QECC

Steane [7] and Calderbank and Shor [8] proposed a seven-qubit QECC scheme inspired by the wisdom of classical error correcting code $(7,4,3)$. We index the seven qubits as $i = 0, 1, \ldots, 6$.

A possible error is the bit-flip error X and the phase-flip error Z acting independently on one of seven qubits. They act as $Y = XZ$ if they heppen to operate on a common qubit. Therefore there are $7 + 7 + 7 \times 7 + 1 = 64 = 2^6$ error types, including 1 for no errors, and at least six bits are required to classify the syndrome.

10.4.2.1 Encoding

Let us start with some algebra. Let

$$M_0 = X_4X_3X_2X_1,\ M_1 = X_5X_3X_2X_0,\ M_2 = X_6X_3X_1X_0 \tag{10.29}$$

and

$$N_0 = Z_4Z_3Z_2Z_1,\ N_1 = Z_5Z_3Z_2Z_0,\ N_2 = Z_6Z_3Z_1Z_0. \tag{10.30}$$

It is obvious that they satisfy $M_i^2 = N_i^2 = I$ and

$$M_i(I + M_i) = I + M_i,\ [M_i, M_j] = [N_i, N_j] = 0. \tag{10.31}$$

A little extra work is required to show

$$[M_i, N_j] = 0 \tag{10.32}$$

by noting that $[X_iX_j, Z_iZ_j] = 0$ in spite of noncommutativity $X_iZ_i = -Z_iX_i$. We also define the following matrices,

$$\bar{X} \equiv X^{\otimes 7} = X_0X_1X_2X_3X_4X_5X_6,\ \bar{Z} \equiv Z^{\otimes 7} = Z_0Z_1Z_2Z_3Z_4Z_5Z_6. \quad (10.33)$$

It is important to note that

$$[\bar{X}, M_i] = [\bar{X}, N_i] = 0,\ [\bar{Z}, M_i] = [\bar{Z}, N_i] = 0. \quad (10.34)$$

We encode $|0\rangle$ and $|1\rangle$ into superpositions of seven-qubit states

$$\begin{aligned}|0\rangle_L &= \frac{1}{\sqrt{8}}(I + M_0)(I + M_1)(I + M_2)|0\rangle^{\otimes 7} \\ &= \frac{1}{\sqrt{8}}(|0000000\rangle + |1010101\rangle + |0110011\rangle + |1100110\rangle \\ &\quad +|0001111\rangle + |1011010\rangle + |0111100\rangle + |1101001\rangle) \end{aligned} \quad (10.35)$$

and

$$\begin{aligned}|1\rangle_L &= \frac{1}{\sqrt{8}}(I + M_0)(I + M_1)(I + M_2)|1\rangle^{\otimes 7} \\ &= \frac{1}{\sqrt{8}}(I + M_0)(I + M_1)(I + M_2)\bar{X}|0\rangle^{\otimes 7} \\ &= \frac{1}{\sqrt{8}}(|1111111\rangle + |0101010\rangle + |1001100\rangle + |0011001\rangle \\ &\quad +|1110000\rangle + |0100101\rangle + |1000011\rangle + |0010110\rangle). \end{aligned} \quad (10.36)$$

Observe that they are normalized and the binary basis vectors which comprise $|0\rangle_L$ have even number of 1, while those which comprise $|1\rangle_L$ have odd number of 1, and hence they are orthogonal. Therefore they satisfy ${}_L\langle x|y\rangle_L = \delta_{xy}$. It is important to note, using the commutation relations among $\{M_i, N_i, \bar{X}, \bar{Z}\}$, that $|0\rangle_L$ and $|1\rangle_L$ are eigenvectors of M_i and N_i with all the eigenvalues $+1$;

$$M_i|x\rangle_L = N_i|x\rangle_L = |x\rangle_L \quad (x \in \mathrm{GF}(2)). \quad (10.37)$$

It should be also noted that basis vectors comprises $|0\rangle_L$ correspond to the codewords given in Eq. (10.24) and those comprises $|1\rangle_L$ are the rest of the codewords in Eq. (10.23). Seven qubits must be required to encode a single logical qubit since there are more types of errors compared to the classical counterpart.

Let us consider the encoding circuit for the coding (10.35) and (10.36). Let us analyze the circuit shown in Fig. 10.9 to begin with. The action of the circuit on $|0\rangle|\psi\rangle$, where $|\psi\rangle$ is an arbitrary n-qubit state, is

$$\frac{1}{\sqrt{2}}\left[|0\rangle|\psi\rangle + |1\rangle(U|\psi\rangle)\right] = \frac{1}{\sqrt{2}}\left(I + X \otimes U\right)|0\rangle|\psi\rangle. \quad (10.38)$$

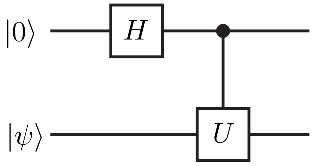

FIGURE 10.9
Hadamard gate $U_{\rm H} \otimes I$ followed by the controlled-U gate acting on $|0\rangle|\psi\rangle$ produces $\frac{1}{\sqrt{2}}(I + X \otimes U)|0\rangle|\psi\rangle$, where $|\psi\rangle$ is an arbitrary n-qubit state and U is an arbitrary n-qubit gate.

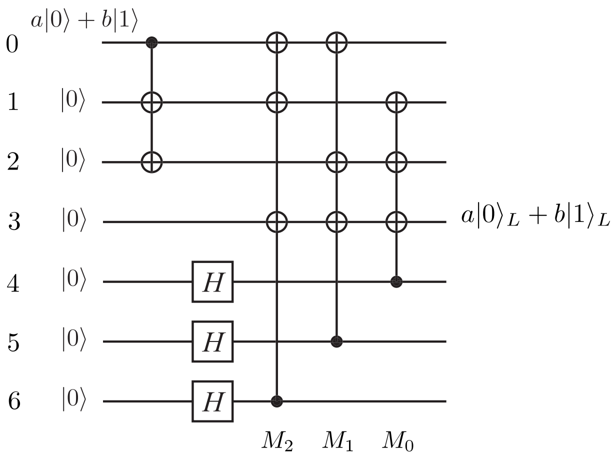

FIGURE 10.10
Encoding circuit for seven-qubit QECC [3]. H stands for the Hadamard gate.

This shows that

$$\frac{1}{\sqrt{2}}(I + M_0) = \frac{1}{\sqrt{2}}(I + X_4X_3X_2X_1),$$

for example, is implemented as a controlled-$X_3X_2X_1$ gate where the control bit is the fourth qubit in the state $U_{\rm H}|0\rangle$.

Collecting these facts, we construct the encoding circuit depicted in Fig. 10.10. The right three controlled-$X^{\otimes 3}$ gates together with the three Hadamard gates in the left implement $(I+M_0)(I+M_1)(I+M_2)$, which maps $|0\rangle^{\otimes 7}$ to $|0\rangle_L$. There is a controlled-X_1X_2 gate in the very beginning of the circuit, which is inert when the input (the 0th qubit) is $|0\rangle$. Suppose the input is $|\psi\rangle = |1\rangle$. Then the input state of the first controlled-$X^{\otimes 3}$ and the Hadamard gates is $X_0X_1X_2|0\rangle^{\otimes 7}$. Since X_i commutes with all M_i's, we may move $X_0X_1X_2$ to the very end of the circuit to find the output state of this circuit is $X_0X_1X_2|0\rangle_L$. However, we find $M_0M_1M_2 = X_3X_4X_5X_6$, and hence

$$X_0X_1X_2 = M_0M_1M_2\bar{X}.$$

Now we show

$$\begin{aligned} X_0X_1X_2|0\rangle_L &= M_0M_1M_2\bar{X}\frac{1}{\sqrt{8}}(I+M_0)(I+M_1)(I+M_2)|0\rangle^{\otimes 7} \\ &= \frac{1}{\sqrt{8}}(I+M_0)(I+M_1)(I+M_2)\bar{X}|0\rangle^{\otimes 7} \\ &= |1\rangle_L, \end{aligned} \tag{10.39}$$

which proves the circuit is indeed the seven-qubit QECC encoder.

10.4.2.2 Error Detection

Now our task is to identify what error occured during transmission by manipulating the received qubits. Note first that X_i, Y_i, Z_i commute with some of $\{M_j, N_j\}$ and anticommute with the rest. Suppose noise X_0 acts on the state $|0\rangle_L$. Then the resulting state is an eigenstate of N_1 with the eigenvalue -1,

$$N_1X_0|0\rangle_L = -X_0N_1|0\rangle_L = -X_0|0\rangle_L.$$

It is also verified that

$$\begin{aligned} &N_2X_0|0\rangle_L = -X_0|0\rangle_L,\ N_0X_0|0\rangle_L = X_0|0\rangle_L, \\ &M_iX_0|0\rangle_L = X_0|0\rangle_L\ (i = 0,1,2). \end{aligned}$$

Therefore the disrupted state $X_0|0\rangle_L$ is characterized by the eigenvalues $(1,1,1;1,-1,-1)$ of the operators $(M_0,M_1,M_2;N_0,N_1,N_2)$. This fact suggests that we may use the set of these operators as the syndrome to identify the error occured in the received qubit. To justify this conjecture, we have to show that the same eigenvalues are assigned to $X_0|1\rangle_L$. It is easy to show from $[\bar{X}, N_i] = 0$ this is indeed the case since

$$N_1X_0|1\rangle_L = N_1X_0\bar{X}|0\rangle_L = -X_0\bar{X}N_1|0\rangle_L = -X_0|1\rangle_L,$$

where we noted that $|1\rangle_L = \bar{X}|0\rangle_L$. These observations show that $X_0|\Psi\rangle$ with $|\Psi\rangle = a|0\rangle_L + b|1\rangle_L$ $(\forall a, b \in \mathbb{C})$ is an eigenvector of $(M_0,M_1,M_2;N_0,N_1,N_2)$ with the eigenvalues $(1,1,1;1,-1,-1)$.

The eigenvalues associated with $X_i|\Psi\rangle$ are $+1$ for all M_i's while

$$\begin{array}{cccccccc} & X_0 & X_1 & X_2 & X_3 & X_4 & X_5 & X_6 \\ N_0 & & * & * & * & * & & \\ N_1 & * & & * & * & & * & \\ N_2 & * & * & & * & & & * \end{array} \tag{10.40}$$

where $*$ stands for the eigenvalue -1 while the empty entry denotes the eigenvalue $+1$. Similarly the eigenvalues associated with $Z_i|\Psi\rangle$ are $+1$ for all N_i's while

$$
\begin{array}{cccccccc}
 & Z_0 & Z_1 & Z_2 & Z_3 & Z_4 & Z_5 & Z_6 \\
M_0 & & * & * & * & * & & \\
M_1 & * & & * & * & & * & \\
M_2 & * & * & & * & & & *
\end{array}
\tag{10.41}
$$

It follows from $Y = XZ$ that the eigenvalues associated with $Y_i|\Psi\rangle$ are

$$
\begin{array}{cccccccc}
 & Y_0 & Y_1 & Y_2 & Y_3 & Y_4 & Y_5 & Y_6 \\
M_0 & & * & * & * & * & & \\
M_1 & * & & * & * & & * & \\
M_2 & * & * & & * & & & * \\
N_0 & & * & * & * & * & & \\
N_1 & * & & * & * & & * & \\
N_2 & * & * & & * & & & *
\end{array}
\tag{10.42}
$$

where use has been made of the relation $Y_i = X_i Z_i$. Note that M_i has an eigenvalue -1 whenever N_i has -1 and *vice versa*. Suppose both N_i and M_j $(i \neq j)$ have eigenvalues -1. This case corresponds to a two-qubit error $X_i Z_j$, with which we will not be concerned.

It was shown above that the set of operators $(M_0, M_1, M_2; N_0, N_1, N_2)$ reveals the syndrome. Now let us consider how they are measured. Let us first consider a simpler case. The syndromes of the bit-flip QECC are measured as shown in Fig. 10.1 (b). It is easy to see the measurement outcome A corresponds to the eigenvalue of the operator $Z_1 Z_2$, while B corresponds to that of $Z_1 Z_3$. It turns out to be convenient to switch the control qubit and the target qubit by making use of the result of Exercise 4.5. Figure 10.11 shows

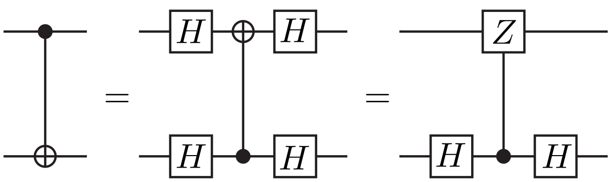

FIGURE 10.11
Switching the control qubits and the target qubit.

the equivalence of the two circuits;

$$U_{\rm CNOT} = (I \otimes U_{\rm H})(I \otimes |0\rangle\langle 0| + Z \otimes |1\rangle\langle 1|)(I \otimes U_{\rm H}),$$

where use has been made of the identity $U_{\rm H} X U_{\rm H} = Z$. Now the error syndrome detection circuit in Fig. 10.1 (b) takes a suggestive form given in

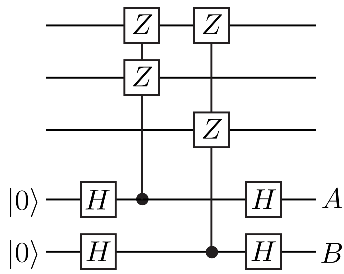

FIGURE 10.12
Error detection circuit of the bit-flip QECC with inverted controlled gates. Note that the controlled unitary gates are Z_1Z_2 and Z_1Z_3, whose eigenvalues fix the syndrome.

Fig. 10.12. Note that the first gate is the controlled-Z_1Z_2 gate, while the second one is the controlled-Z_1Z_3 gate.

The above observation leads to the seven-qubit error detection circuit shown in Fig. 10.13. The column with a product of Z's corresponds to the operator N_i as was shown for the bit-flip QECC. Let us examine the controlled-$X^{\otimes 4}$ gates next. We susupect that they might corresponds to the operators M_i. Let us examine the first controlled gate in Fig. 10.13. The relevant part of the circuit is

$$\begin{aligned}&(I \otimes U_{\mathrm{H}})(I \otimes |0\rangle\langle 0| + M_0 \otimes |1\rangle\langle 1|)(I \otimes U_{\mathrm{H}})|\Psi\rangle|0\rangle \\ &= \frac{1}{2}(I + M_0)|\Psi\rangle|0\rangle + \frac{1}{2}(I - M_0)|\Psi\rangle|1\rangle,\end{aligned} \tag{10.43}$$

where $|\Psi\rangle$ is an encoded seven-qubit state with a possible sigle-qubit error and the last qubit is the ancilla. Recall that $|\Psi\rangle$ is an eigenvector of M_i with the eigenvalue $+1$ or -1. In case the eigenvalue is $+1$, the output of this part is $|\Psi\rangle|0\rangle$, while it is $|\Psi\rangle|1\rangle$ if the eigenvalue is -1. Accordingly the measurement of the ancilla reveals the eigenvalue of M_0. Similarly other two gates evaluate the eigenvalues of M_1 and M_2. Now we have justified that the set of eigenvalues (i.e., the syndrome) of the received state is evaluated by the measurement of the six ancillary qubits in Fig. 10.13.

Correction of the disrupted encoded state can be done by consulting with Eqs. (10.40), (10.41) and (10.42).

It was shown above that the code space is a subspace of the seven-qubit Hilbert space, in which all the eigenvalues of the six operators $\{M_i, N_i\}$ are $+1$. In other words, encoded state $|\Psi\rangle$ is left invaraint under the action of these operators; $M_i|\Psi\rangle = N_i|\Psi\rangle = |\Psi\rangle$. An error operator E_k, which disrupts the code, is a tensor product of Pauli matrices, which anticommutes with some of M_i and N_i. Therefore the disrupted state has eigenvalue -1 for some operators in $\{M_i, N_i\}$ that anticommute with E_k. The set of the

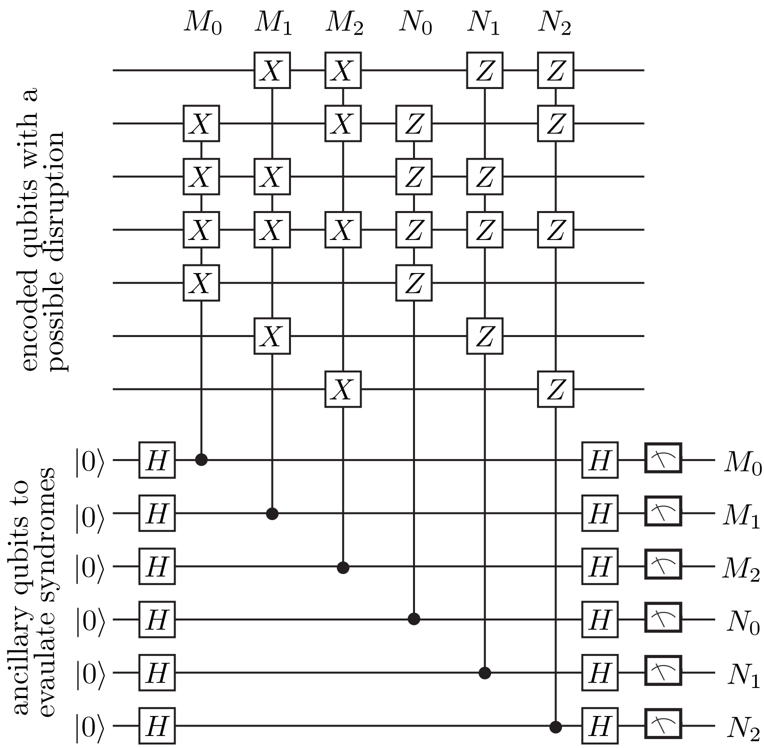

FIGURE 10.13
Error detecting circuit of the seven-qubit QECC. The unitary gates controlled by the ancillary qubits correspond to $\{M_i, N_i\}$, as denoted at the top of the circuit. The readout of the six ancillary qubits fixes the set of syndromes.

measurement outcomes of $\{M_i, N_i\}$ (syndrome) tells us which error operator acted on the encoded state. The Abelian (commuting) group $\{M_i, N_i\}$ is called the **stabilizer** [9, 10], while the subspace $C(S) = \{|\psi\rangle \in \mathbb{C}^{2^n} \mid M_i|\Psi\rangle = N_i|\Psi\rangle = |\Psi\rangle\}$ is called the **stabilizer code** associated with S.

10.4.2.3 Decoding

Once syndromes are detected and an appropriate correction is made, we have to decode the logical qubit so as to extract the input state $|\psi\rangle$. This is done by applying gates in Fig. 10.10 one by one in the reversed order and the initial state will be reproduced in the 0th qubit.

10.4.3 Gate Operations for Seven-Qubit QECC

It is shown in the next section that there exists more efficient encoding scheme, in which merely five qubits are required to encode a single qubit state. In spite of this, the seven-qubit QECC is advantageous over the five-qubit QECC in that gate operations for encoded qubits are easy to implement. Let us look at a few examples.

10.4.3.1 One-Qubit Gates

Let us start with simple examples. Suppose we introduce the $\tilde{X}$ gate which acts on $|0\rangle_L$ and $|1\rangle_L$ as

$$\tilde{X}|0\rangle_L = |1\rangle_L,\ \tilde{X}|1\rangle_L = |0\rangle_L. \tag{10.44}$$

This gate is implemented by $\bar{X} = X_0X_1X_2X_3X_4X_5X_6$ since

$$\begin{aligned}\bar{X}|0\rangle_L &= \bar{X}\frac{1}{\sqrt{8}}(I + M_0)(I + M_1)(I + M_2)|0\rangle^{\otimes 7}\\ &= \frac{1}{\sqrt{8}}(I + M_0)(I + M_1)(I + M_2)\bar{X}|0\rangle^{\otimes 7}\\ &= |1\rangle_L,\end{aligned} \tag{10.45}$$

where use has been made of the commutativity $[\bar{X}, M_i] = 0$. Application of $\bar{X}$ on Eq. (10.45) proves $\bar{X}|1\rangle_L = |0\rangle_L$.

Let us find $\tilde{Z}$, which satisfies

$$\tilde{Z}|0\rangle_L = |0\rangle_L,\ \tilde{Z}|1\rangle_L = -|1\rangle_L. \tag{10.46}$$

Again this is implemented by $\bar{Z} = Z_0Z_1Z_2Z_3Z_4Z_5Z_6$ since

$$\begin{aligned}\bar{Z}|0\rangle_L &= \frac{1}{\sqrt{8}}(I + M_0)(I + M_1)(I + M_2)\bar{Z}|0\rangle^{\otimes 7}\\ &= |0\rangle_L\end{aligned} \tag{10.47}$$

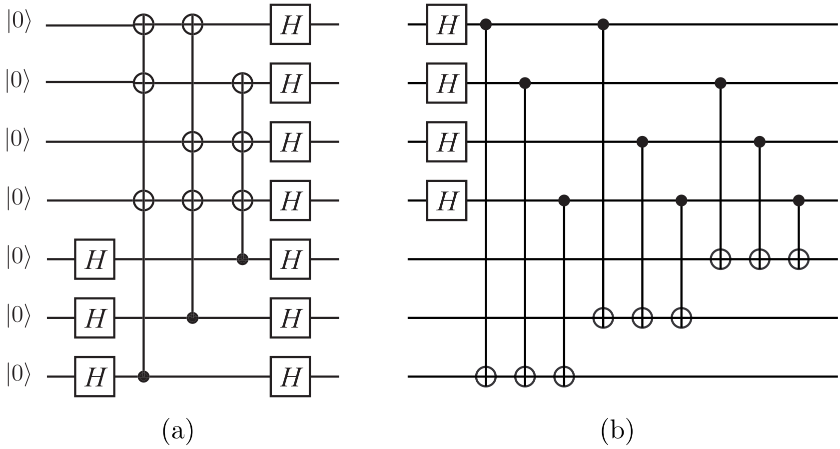

FIGURE 10.14
(a) Seven Hadamard gates applied on the logical qubit state $|x\rangle_L$. (b) A circuit equivalent with (a).

and

$$\begin{aligned}\bar{Z}|1\rangle_L &= \frac{1}{\sqrt{8}}(I+M_0)(I+M_1)(I+M_2)\bar{Z}|1\rangle^{\otimes 7} \\ &= -|1\rangle_L, \end{aligned} \tag{10.48}$$

where we noted that $[\bar{Z}, M_i] = 0$ and $(-1)^7 = -1$.

Next, we consider the Haramard gate $\tilde{H}$ acting on the encoded states as

$$\tilde{U}_{\mathrm{H}}|0\rangle_L = \frac{1}{\sqrt{2}}(|0\rangle_L + |1\rangle_L),\ \tilde{U}_{\mathrm{H}}|1\rangle_L = \frac{1}{\sqrt{2}}(|0\rangle_L - |1\rangle_L). \tag{10.49}$$

Surprisingly, this gate is also implemented by a tensor product of seven Hadamard gates,

$$\tilde{U}_{\mathrm{H}} = U_{\mathrm{H}}^{\otimes 7} = W_7, \tag{10.50}$$

where W_7 is the Walsh-Hadamard transform acting on seven qubits.

Let us consider the action of W_7 on $|0\rangle_L$ first. The circuit

$$W_7|0\rangle_L = W_7\frac{1}{\sqrt{8}}(I+M_0)(I+M_1)(I+M_2)|0\rangle^{\otimes 7}$$

given in Fig. 10.14 (a) is put in the form given in (b) by making use of the trick introduced in Fig. 10.9 and $U_{\mathrm{H}}^2 = I$. The resulting circuit is written as

$$\begin{aligned} W_7|0\rangle_L &= [\mathrm{C}_0(X_6X_5)][\mathrm{C}_1(X_6X_4)][\mathrm{C}_2(X_5X_4)] \\ &\quad \times[\mathrm{C}_3(X_6X_5X_4)]U_{\mathrm{H}0}U_{\mathrm{H}1}U_{\mathrm{H}2}U_{\mathrm{H}3}|0\rangle^{\otimes 7} \\ &= [\mathrm{C}_0(X_6X_5)U_{\mathrm{H}0}][\mathrm{C}_1(X_6X_4)U_{\mathrm{H}1}][\mathrm{C}_2(X_5X_4)U_{\mathrm{H}2}] \\ &\quad \times[\mathrm{C}_3(X_6X_5X_4)U_{\mathrm{H}3}]|0\rangle^{\otimes 7}, \end{aligned} \tag{10.51}$$

where

$$\mathrm{C}_i U = |0_i\rangle\langle 0_i| \otimes I + |1_i\rangle\langle 1_i| \otimes U$$

is the controlled-U gate with the control bit i. (The vector $|x_i\rangle$ denotes the state $|x\rangle$ of the ith qubit.) By noting the identities $M_0M_1M_2 = X_3X_4X_5X_6$ and (10.38), we obtain

$$\begin{aligned}
W_7|0\rangle_L &= \frac{1}{4}(I + X_0X_5X_6)(I + X_1X_4X_6) \\
&\quad \times (I + X_2X_4X_5)(I + M_0M_1M_2)|0\rangle^{\otimes 7} \\
&= \frac{1}{4}(I + M_0\bar{X})(I + M_1\bar{X})(I + M_2\bar{X}) \\
&\quad \times (I + M_0M_1M_2)|0\rangle^{\otimes 7}.
\end{aligned} \tag{10.52}$$

After expansion and factorization of the last expression, we obtain

$$\begin{aligned}
W_7|0\rangle_L &= \frac{1}{\sqrt{8}}(I + M_0)(I + M_1)(I + M_2)\frac{1}{\sqrt{2}}(I + \bar{X})|0\rangle^{\otimes 7} \\
&= \frac{1}{\sqrt{2}}(|0\rangle_L + |1\rangle_L).
\end{aligned} \tag{10.53}$$

The action of W_7 on $|1\rangle_L$ is easily obtained by applying W_7 on Eq. (10.53) as

$$W_7^2|0\rangle_L = |0\rangle_L = \frac{1}{\sqrt{2}}(W_7|0\rangle_L + W_7|1\rangle_L),$$

from which we obtain

$$W_7|1\rangle_L = \sqrt{2}|0\rangle_L - \frac{1}{\sqrt{2}}(|0\rangle_L + |1\rangle_L) = \frac{1}{\sqrt{2}}(|0\rangle_L - |1\rangle_L). \tag{10.54}$$

10.4.3.2 CNOT Gate

Another surprise is the CNOT gate acting on two logical qubits is also easily implmenented with the seven-qubit QECC. One simply introduces seven CNOT gates, $U_{\mathrm{CNOT}}^{\otimes 7}$, connecting corresponding physical qubits in two logical qubits as shown in Fig. 10.15. Suppose the control bit is in the state

$$|0\rangle_L = \frac{1}{\sqrt{8}}\left(I + \sum_i M_i + \sum_{i<j} M_iM_j + M_0M_1M_2\right)|0\rangle^{\otimes 7}.$$

It follows from the linearity of $U_{\mathrm{CNOT}}^{\otimes 7}$ that each term of the above expansion acts on $|0\rangle_L|x\rangle_L$ independently and outputs the same state since $|x\rangle_L$ is a simultaneous eigenvector of all M_i's with all the eigenvalues $+1$. Let us take $M_0 = X_4X_3X_2X_1$ for example. We find

$$\begin{aligned}
U_{\mathrm{CNOT}}^{\otimes 7}M_0|0\rangle^{\otimes 7}|x\rangle_L &= U_{\mathrm{CNOT}}^{\otimes 7}|0111100\rangle|x\rangle_L \\
&= |0111100\rangle \otimes X_4X_3X_2X_1|x\rangle_L \\
&= M_0|0\rangle^{\otimes 7} \otimes M_0|x\rangle_L = M_0|0\rangle^{\otimes 7} \otimes |x\rangle_L.
\end{aligned}$$

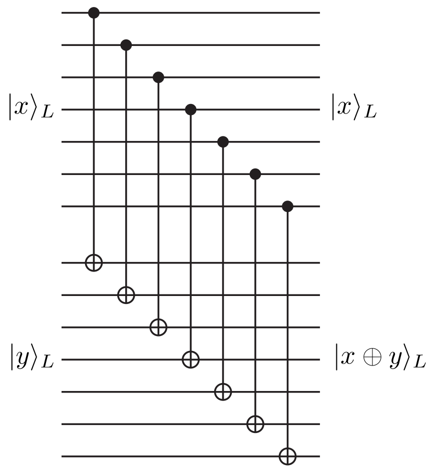

FIGURE 10.15
The CNOT gate acting on two logical qubits is simply a product of seven CNOT gates acting on pairs of physical qubits.

On the other hand, the target state $|x\rangle_L$ is flipped if the control bit is in the state

$$|1\rangle_L = \frac{1}{\sqrt{8}}\left(I + \sum_i M_i + \sum_{i<j} M_i M_j + M_0 M_1 M_2\right)|1\rangle^{\otimes 7}.$$

Take the above example of M_0 again. We obtain for the component $M_0|1111111\rangle$, the output

$$\begin{aligned} U_{\mathrm{CNOT}}^{\otimes 7} M_0|1111111\rangle|x\rangle_L &= U_{\mathrm{CNOT}}^{\otimes 7}|1000011\rangle|x\rangle_L \\ &= |1000011\rangle \otimes X_6 X_5 X_0|x\rangle_L \\ &= M_0|1111111\rangle \otimes \bar{X} M_0|x\rangle_L \\ &= M_0|1111111\rangle \otimes \bar{X}|x\rangle_L. \end{aligned}$$

Clearly the similar output is obtained for other components in the expansion of $|1\rangle_L$.

In summary, we have proved that

$$U_{\mathrm{CNOT}}^{\otimes 7} = |0\rangle_{LL}\langle 0| \otimes I + |1\rangle_{LL}\langle 1| \otimes \bar{X}. \tag{10.55}$$

Note that logical one-qubit gates are made of seven single-qubit gates acting on each physical qubit independently and the logical CNOT gates are made of seven CNOT gates acting on pairs of two qubits, one from the logical control bit and the other from the logical target bit. Since there are only one qubit involved in each logical qubit, any single qubit error during the gate operation may be corrected by the QECC. This desirable property of non-propagating error is called the **fault tolerance**, which is a key component in reliable quantum computing.

10.5 Five-Qubit QECC

DiVincenzo and Shor further reduced the number of qubits required for QECC [11].

Suppose n qubits are used to implement one logical qubit with the basis states $|0\rangle_L$ and $|1\rangle_L$. There are $3n$ operators X_i, Y_i and Z_i $(0 \le i \le n-1)$ and correspondingly $3n$ single-qubit errors. Each operator maps the basis $\{|0\rangle_L, |1\rangle_L\}$ to one of $\{X_i|0\rangle_L, X_i|1\rangle_L\}$, $\{Y_i|0\rangle_L, Y_i|1\rangle_L\}$ and $\{Z_i|0\rangle_L, Z_i|1\rangle_L\}$, each of which defines a two-dimensional subspace of 2^n-dimensional vector space associated with the n-qubit system. Therefore n must satisfy $2^n \ge 2(3n+1)$, 1 for the nondisrupted basis, or equivalently

$$2^{n-1} \ge 3n+1. \tag{10.56}$$

This inequality is saturated with $n = 5$. In other words, $n = 5$ is the optimal number to correct all types of single-qubit errors.

10.5.1 Encoding

Our exposition goes almost parallel to the previous seven-qubit encoding. Let us consider the algebra of the following operators,

$$\begin{aligned} M_0 &= X_2X_3Z_1Z_4,\ M_1 = X_3X_4Z_2Z_0 \\ M_2 &= X_4X_0Z_3Z_1\ M_3 = X_0X_1Z_4Z_2. \end{aligned} \tag{10.57}$$

EXERCISE 10.11 (1) Show that $M_i^2 = I$. The eigenvalues of these operators are therefore ± 1.
(2) Show that $[M_i, M_j] = 0\ \forall i, j$.
(3) Let $M_4 = X_1X_2Z_0Z_3$. Show that $M_4 = M_0M_1M_2M_3$. (M_4 is not an independent operator. The eigenvalues of M_0, M_1, M_2 and M_3 fix the eigenvalue of M_4 uniquely.)

Now we introduce the following encoding of logical qubit basis vectors,

$$\begin{aligned} |0\rangle_L &= \frac{1}{4}(I+M_0)(I+M_1)(I+M_2)(I+M_3)|00000\rangle \\ &= \frac{1}{4}\big[|00000\rangle + |11000\rangle + |01100\rangle + |00110\rangle + |00011\rangle + |10001\rangle \end{aligned}$$

$$-|10100\rangle - |01010\rangle - |00101\rangle - |10010\rangle - |01001\rangle$$
$$-|11110\rangle - |01111\rangle - |10111\rangle - |11011\rangle - |11101\rangle\big] \tag{10.58}$$

and

$$\begin{aligned}|1\rangle_L &= \frac{1}{4}(I+M_0)(I+M_1)(I+M_2)(I+M_3)|11111\rangle \\ &= \frac{1}{4}\big[|11111\rangle + |00111\rangle + |10011\rangle + |11001\rangle + |11100\rangle + |01110\rangle \\ &\quad -|01011\rangle - |10101\rangle - |11010\rangle - |01101\rangle - |10110\rangle \\ &\quad -|00001\rangle - |10000\rangle - |01000\rangle - |00100\rangle - |00010\rangle\big].\end{aligned} \tag{10.59}$$

It is easy to see that these vectors are orthonormal. Moreover they are simultaneous eigenvectors of $\{M_i\}$ with all the eigenvalues $+1$; $M_i|x\rangle_L = |x\rangle_L$. An input state $|\psi\rangle = a|0\rangle + b|1\rangle$ is encoded into the five-qubit state $|\Psi\rangle = a|0\rangle_L + b|1\rangle_L$.

Next, let us work out a quantum circuit which implements this encoding. We again use the identity

$$(|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U)(U_{\rm H} \otimes I)|0\rangle|\Psi\rangle = \frac{1}{\sqrt{2}}(I + X \otimes U)|0\rangle|\Psi\rangle \tag{10.60}$$

depicted in Fig. 10.9. Let us consider encoding $|0\rangle_L$ first. We rewrite

$$\begin{aligned}\frac{1}{\sqrt{2}}(I+M_3)|0\rangle^{\otimes 5} &= \frac{1}{\sqrt{2}}(I + X_0X_1Z_2Z_4)|0\rangle^{\otimes 5} \\ &= \frac{1}{\sqrt{2}}(|0_1\rangle\langle 0_1| \otimes I + |1_1\rangle\langle 1_1| \otimes X_0Z_2Z_4)U_{\rm H1}|0\rangle^{\otimes 5}\end{aligned} \tag{10.61}$$

by making use of Eq. (10.60). Here $U_{\rm H1}$ is the Hadamard gate acting on qubit 1. Now $\frac{1}{\sqrt{2}}(I + M_3)|0\rangle^{\otimes 5}$ has been implemented as the controlled-$X_0Z_2Z_4$ gate with the control qubit 1.

Let us consider the action of $\frac{1}{\sqrt{2}}(I + M_2)$ next. We write the state $\frac{1}{\sqrt{2}}(I + M_3)|0\rangle^{\otimes 5}$ as $|0_4\rangle|\phi\rangle$, where we noted that the state of qubit 4 is $|0\rangle$ and wrote the state of the other qubits collectively as $|\phi\rangle$. It is also implemented as a controlled gate as

$$\begin{aligned}\frac{1}{2}(I+M_2)(I+M_3)|0\rangle^{\otimes 5} &= \frac{1}{2}(I+M_2)|0_4\rangle|\phi\rangle \\ &= \frac{1}{2}(I + X_0X_4Z_1Z_3)|0_4\rangle|\phi\rangle \\ &= (|0_4\rangle\langle 0_4| \otimes I + |1_4\rangle\langle 1_4| \otimes X_0Z_1Z_3) \\ &\quad \times U_{\rm H4}|0_4\rangle|\phi\rangle.\end{aligned} \tag{10.62}$$

Note that the fourth qubit has been chosen to be the control bit since the initial control qubit state must be $|0\rangle$ for the identity (10.60) to be applicable.

We write the resulting state of this gate operation as $|0_3\rangle|\phi'\rangle$ by noting that the third qubit state is $|0\rangle$. Here $|\phi'\rangle$ collectively denotes the state of the other qubits.

Now we repeat the above substitution further for $\frac{1}{\sqrt{2}}(I + M_1)$ and $\frac{1}{\sqrt{2}}(I + M_0)$. The first operation is replaced by

$$\frac{1}{\sqrt{2}}(|0_3\rangle\langle 0_3| \otimes I + |1_3\rangle\langle 1_3| \otimes X_4 Z_0 Z_2)U_{\mathrm{H3}}|0_3\rangle|\phi'\rangle \tag{10.63}$$

by taking the qubit 3 as the control bit. We write the state after this operation as $|0_2\rangle|\phi''\rangle$ by noting that the second qubit state is $|0\rangle$. Now the second operation is replaced by

$$\frac{1}{\sqrt{2}}(|0_2\rangle\langle 0_2| \otimes I + |1_2\rangle\langle 1_2| \otimes X_3 Z_1 Z_4)U_{\mathrm{H2}}|0_2\rangle|\phi''\rangle. \tag{10.64}$$

Now we have shown how $|0\rangle_L$ is obtained from $|0\rangle^{\otimes 5}$ by applying four controlled gates.

We use the identity

$$M_3 M_2 M_0 = -X_1 X_2 X_3 X_4 Z_2 Z_3 \tag{10.65}$$

to generate $|1\rangle_L$ when the qubit 0 is in the state $|1\rangle$. We note that

$$|11111\rangle = -M_3 M_2 M_0|10000\rangle = M_3 M_2 M_0 Z_0|10000\rangle. \tag{10.66}$$

Suppose the encoding circuit for $|0\rangle_L$ acts on a state

$$Z_0(a|0\rangle + b|1\rangle)|0000\rangle = a|00000\rangle - b|10000\rangle.$$

Then the circuit outputs

$$\begin{aligned}
&\frac{1}{4}(I + M_0)(I + M_1)(I + M_2)(I + M_3)(a|0\rangle - b|1\rangle)|0\rangle^{\otimes 4} \\
&= \frac{1}{4}(I + M_0)(I + M_1)(I + M_2)(I + M_3)(a|00000\rangle + bM_3M_2M_0|11111\rangle) \\
&= a\frac{1}{4}(I + M_0)(I + M_1)(I + M_2)(I + M_3)|00000\rangle \\
&\quad + bM_3M_2M_0\frac{1}{4}(I + M_0)(I + M_1)(I + M_2)(I + M_3)|11111\rangle \\
&= a|0\rangle_L + bM_3M_2M_0|1\rangle_L = a|0\rangle_L + b|1\rangle_L,
\end{aligned} \tag{10.67}$$

where we used the facts $[M_i, M_j] = 0$ and $|1\rangle_L$ is an simultaneous eigenvector of M_i with all the eigenvalues 1.

This shows that the circuit

$$U_{5\text{-encoding}} = \frac{1}{4}(I + M_0)(I + M_1)(I + M_2)(I + M_3)Z_0, \tag{10.68}$$

depicted in Fig. 10.16 is the correct encoding circuit which acts as

$$U_{5\text{-encoding}}(a|0\rangle + b|1\rangle)|0000\rangle = a|0\rangle_L + b|1\rangle_L. \tag{10.69}$$

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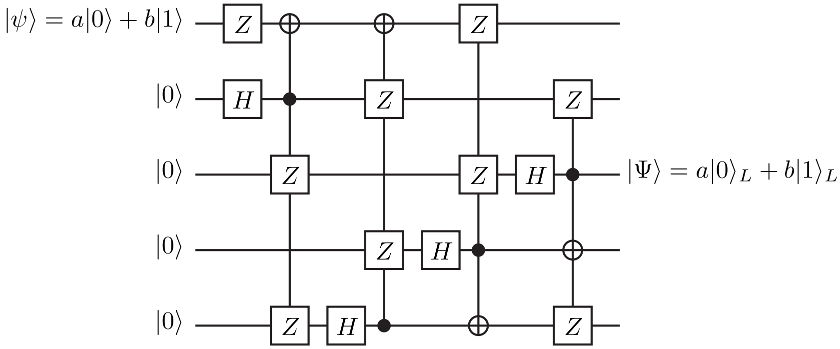

FIGURE 10.16
Encoding circuit for five-qubit QECC.

10.5.2 Error Syndrome Detection

The eigenvalues of four operators $\{M_i\}$ are used to discriminate possible single-qubit disruptions in codewords. Let us recall that $|0\rangle_L$, $|1\rangle_L$ and linear combinations thereof are the eigenvectors of M_i with all the eigenvalues $+1$. The eigenvalues of $\{M_i\}$ corresponding to single-qubit errors are [11, 3]

$$\begin{array}{c|cccccccccccccccc} & I & X_0 & X_1 & X_2 & X_3 & X_4 & Y_0 & Y_1 & Y_2 & Y_3 & Y_4 & Z_0 & Z_1 & Z_2 & Z_3 & Z_4 \\ M_0 & & & * & & & * & & * & * & * & * & & & * & * & \\ M_1 & & * & & * & & & * & & * & * & * & & & & * & * \\ M_2 & & & * & & * & & * & * & & * & * & * & & & & * \\ M_3 & & & & * & & * & * & * & * & & * & * & * & & & \end{array} \tag{10.70}$$

where $*$ denotes the eigenvalue -1, while the eigenvalue $+1$ is denoted by an empty entry. The disrupted state $X_0(a|0\rangle_L + b|1\rangle_L)$, for example, produces the syndrome $(M_0, M_1, M_2, M_3) = (+1, -1, +1, +1)$.

The error detection circuit is constructed following the strategy employed in the seve-qubit QECC. Figure 10.17 shows the error detection circuit associated with our encoding scheme.

Correction of the disrupted encoded state can be done by consulting with Eq. (10.70).

10.5.2.1 Decoding

Once syndrome is determined and an appropriate correction is made, we have to decode the logical qubit so as to extract the input state $|\psi\rangle$. This is done by applying gates in Fig. 10.16 one by one in the reversed order and the initial state will be reproduced in the 0th qubit.

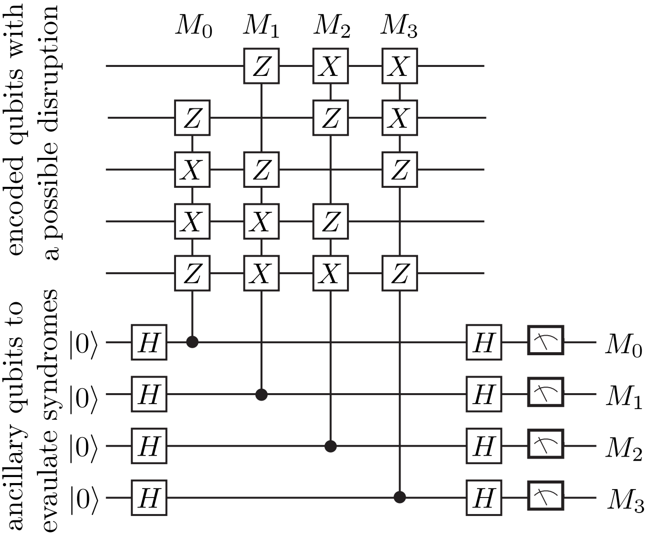

FIGURE 10.17
Error syndrome detecting circuit of the five-qubit QECC. The unitary gates controlled by the ancillary qubits correspond to $\{M_i\}$, as denoted at the top of the circuit. The readout of the four ancillary qubits detects the syndrome.

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